



## **Specification for M-PHY®**

**Version 5.0 – 23 June 2021**

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## **Specification for M-PHY®**

**Version 5.0  
23 June 2021**

MIPI Board Adopted 29 September 2021

Further technical changes to this document are expected as work continues in the M-PHY Working Group.

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## Release History

| Date       | Version  | Description             |
|------------|----------|-------------------------|
| 2008 08 29 | v0.10.00 | Initial release         |
| 2010 08 13 | v0.80.00 | Board approved release. |
| 2011 05 03 | v1.00.00 | Board approved release. |
| 2012 07 09 | v2.0     | Board approved release. |
| 2013 09 30 | v3.0     | Board approved release. |
| 2014 06 17 | v3.1     | Board approved release. |
| 2015 08 03 | v4.0     | Board approved release. |
| 2017-03-28 | v4.1     | Board-approved release. |
| 2021-09-29 | v5.0     | Board-approved release. |

## 1 Introduction

This document describes a serial interface technology with high bandwidth capabilities, which is particularly developed for mobile applications to obtain low pin count combined with very good power efficiency. It is targeted to be suitable for multiple protocols, including UniPro<sup>SM</sup> and DigRF<sup>SM</sup> v4, and for a wide range of applications.

The M-PHY Specification features the following aspects:

- BURST mode operation for improved power efficiency
- Multiple transmission modes with different bit-signaling and clocking schemes intended for different bandwidth ranges to enable better power efficiency over a huge range of data rates
- Multiple transmission speed ranges and rates per BURST mode to further scale bandwidth to application needs, and for mitigation of interference problems. Rates for high-speed mode are fixed, for low-speed modes they are flexible within ranges
- Multiple power saving modes, where power consumption can be traded-off against recovery time
- Symbol coding (8b10b) for spectral conditioning, clock recovery, and in-band control options for both PHY and Protocol Layer.
- Clocking: designed to exploit the benefits of a shared reference clock
- Distance: optimized for short interconnect (<10 cm)
- Configurability: differences in supported functionality (to reduce cost) and tune for best performance (implementation) without hampering interoperability
- Debuggability: receiver has optional eye monitor function

### 1.1 Scope

This document specifies unidirectional LANEs and its individual parts, as building blocks for composition of a dual-simplex LINK by application protocols. An M-PHY implementation allows one or more LANEs in each direction, allows differences in optional functionality between LANEs, allows different momentary operating modes between LANEs, and allows asymmetry in amount of LANEs and LANE properties for the two directions of the dual-simplex LINK. Protocols applying M-PHY technology may have different LANE constraints, and choose different operation control, or data striping and merging solutions. Therefore, this document provides the features to enable LINK composition, but does not specify how multiple transmitters and receivers are combined into a PHY-unit for a certain LINK composition. Each LANE has its own interface to the Protocol Layer.

A MODULE can disclose its capabilities, and contains several configurable parameters in order to allow differentiation on supported functionality and tune for best performance without hampering interoperability. Therefore, protocols need to support some configuration mechanism to determine and define the operational settings. Most flexible is an auto-discovery negotiation protocol to determine the commonly-supported settings of the Physical Layer which are most desirable for running the application. M-PHY supports this, but does not include the configuration protocol itself. Alternatively, the protocol may directly program the required settings if there is predetermined higher system knowledge about which MODULEs are present at both ends of that LINK.

The M-PHY specification shall always be used in combination with a higher layer MIPI specification that references this specification. Any other use of the M-PHY specification is strictly prohibited, unless approved in advance by the MIPI Board of Directors.

### 1.2 Purpose

Mobile devices face increasing bandwidth demands for each of its functions as well as an increase of the number of functions integrated into the system. This requires wide bandwidth, low-pin count (serial) and

- 42 highly power-efficient (network) interfaces that provides sufficient flexibility to be attractive for multiple  
43 applications, but which can also be covered with one physical layer technology.



## 2 Terminology

### 2.1 Use of Special Terms

The MIPI Alliance has adopted Section 13.1 of the *IEEE Standards Style Manual*, which dictates use of the words “shall”, “should”, “may”, and “can” in the development of documentation, as follows:

The word *shall* is used to indicate mandatory requirements strictly to be followed in order to conform to the Specification and from which no deviation is permitted (*shall* equals *is required to*).

The use of the word *must* is deprecated and shall not be used when stating mandatory requirements; *must* is used only to describe unavoidable situations.

The use of the word *will* is deprecated and shall not be used when stating mandatory requirements; *will* is only used in statements of fact.

The word *should* is used to indicate that among several possibilities one is recommended as particularly suitable, without mentioning or excluding others; or that a certain course of action is preferred but not necessarily required; or that (in the negative form) a certain course of action is deprecated but not prohibited (*should* equals *is recommended that*).

The word *may* is used to indicate a course of action permissible within the limits of the Specification (*may* equals *is permitted to*).

The word *can* is used for statements of possibility and capability, whether material, physical, or causal (*can* equals *is able to*).

All sections are normative, unless they are explicitly indicated to be informative.

### 2.2 Definitions

**ACTIVATED:** The combined states within HS-MODE or LS-MODE.

**BER:** Bit Error Rate.

**BURST:** Sequence of 8b10b encoded data transmission delimited by and including a HEAD-OF-BURST and TAIL-OF-BURST.

**COMMA:** Non-data symbol which can not be found at any bit position within any combination of other valid symbols.

**CRPAT:** Compliant Random Pattern, see [CTS01].

**CJTPAT:** Compliant Jitter Tolerance Pattern, see [CTS01].

**DIF-N:** Logical LINE state, driven by the M-TX, corresponding with a negative differential LINE voltage. Voltage levels and signal transition timing specifications for the M-TX as well as detection requirements for the M-RX are defined in **Section 5**.

**DIF-P:** Logical LINE state, driven by the M-TX, corresponding with a positive differential LINE voltage. Voltage levels and signal transition timing specifications for the M-TX as well as detection requirement for the M-RX are defined in **Section 5**.

**DIF-Q:** LINE state when the M-RX can be high-impedance resulting in undriven lines with an undefined LINE state.

**DIF-X:** Indication that LINE state can be either DIF-P or DIF-N, but nothing else.

**DIF-Z:** Logical LINE state, driven by the M-RX, corresponding with almost zero differential LINE voltage. Voltage levels and signal transition timing specifications for M-TX and M-RX are defined in **Section 5**.

**DISABLED:** MODULE state when the MODULE is powered, but not enabled.

- 82 **FILLER:** Non-data symbol(s) inserted when no data is provided by the protocol during a BURST.
- 83 **FLAG:** Control signal that indicates the occurrence of a certain event.
- 84 **FRAME:** Series of symbols separated by MARKERs.
- 85 **GEAR:** Speed range (PWM) or fixed RATEs (HS) of communication in LS or HS mode. Each HS GEAR
- 86 includes two RATEs which differ about 15% for mitigation of EMI.
- 87 **HEAD-OF-BURST:** Period between exiting STALL state or SLEEP state and the first MARKER0 in a
- 88 BURST, indicating start of PAYLOAD data.
- 89 **HIBERN8:** Deepest low-power state without loss of configuration information.
- 90 **HS-BURST:** High speed state including PREPARE, SYNC, MARKERs, and data.
- 91 **HS-GEAR:** GEAR in HS-MODE.
- 92 **HS-MODE:** High-Speed operation loop consisting of STALL and HS-BURST.
- 93 **LANE:** A LANE is a unidirectional, point-to-point, differential serial connection, consisting of an M-PHY
- 94 transmit MODULE (M-TX), an M-PHY receive MODULE (M-RX), and a LINE.
- 95 **LINE:** Differential point-to-point interconnect between the PINs of M-TX and M-RX.
- 96 **LINE-RESET:** Reset via the LINE by means of the exceptional signal condition of a long DIF-P.
- 97 **LINK:** One or more PHY LANEs in each direction plus an additional LANE management layer that provides
- 98 a bidirectional data transport means, agnostic to the actual LANE composition.
- 99 **LS-BURST:** Low speed state including PREPARE, MARKERs, and data.
- 100 **LS-MODE:** Type-I: Combination of SLEEP, PWM-BURST, INIT states.
- 101 Type-II: Combination of SLEEP and SYS-BURST states.
- 102 **MARKER:** Non-data symbol, used for protocol related control purposes.
- 103 **MODE:** Indicates either HS-MODE or LS-MODE.
- 104 **MODULE:** Indication for either an M-TX or M-RX.
- 105 **M-PORT:** Combination of MODULEs at one side of a LINK.
- 106 **PAYLOAD:** BURST without HOB and TOB. PAYLOAD may consist of multiple FRAMEs.
- 107 **PIN:** A point of external physical electrical connection for a component. Examples of a “PIN” may include
- 108 (but are not limited to) a BGA ball, QFP lead, or solder pad.
- 109 **POWERED:** Any LANE or MODULE state when power supply is available.
- 110 **PREPARE:** First part of the HOB after exiting STALL or SLEEP up to but not including the SYNC sequence.
- 111 **PWM:** Bit modulation scheme carrying the data information in the duty-cycle, and explicit clock information
- 112 in the period.
- 113 **PWM-BURST:** Transmission of an LS-BURST in pulse-width modulated bit format and using 8b10b
- 114 coding.
- 115 **RATE:** Exact speed of communication in a certain mode in kbps, Mbps, or Gbps.
- 116 **SAVE:** Set of power saving states STALL, SLEEP, HIBERN8, DISABLED, and UNPOWERED.
- 117 **SLEEP:** Power saving state used between LS-BURSTs.
- 118 **STALL:** Power saving state between HS-BURSTs with fast recovery time.
- 119 **SUB-LINK:** All LANEs in the same direction as a fraction of a LINK.
- 120 **SYMBOL-INTERVAL:** 10-bit period for the transmission of one symbol. SYMBOL-INTERVAL scales
- 121 with data rate.
- 122 **SYNC:** An 8b10b symbol sequence with high edge-density intended for fast phase alignment.

- 123 **SYS-BURST:** Transmission of an LS-BURST synchronous at the SysClk rate. Only possible for shared  
124 SysClk applications.
- 125 **TAIL-OF-BURST:** Run-length violating constant bit sequence used to return a MODULE to a SAVE state.
- 126 **UNIT-INTERVAL:** Nominal length of one bit.
- 127 **UNPOWERED:** MODULE state when the power supply is removed.

## 2.3 Abbreviations

- 128 e.g. For example (Latin: *exempli gratia*)
- 129 i.e. That is (Latin: *id est*)

## 2.4 Acronyms

- 130 b0, b1 Bit with logical value “0” or “1”, respectively. The signaling format depends on operating  
131 MODE. A prefix indicating the MODE is occasionally used for clarification, e.g. PWM b0.
- 132 CFG Configuration
- 133 EMI Electromagnetic Interference
- 134 EYEMON Eye Monitor function in receiver
- 135 FLR FILLER symbol
- 136 FSM Finite State Machine
- 137 HOB HEAD-OF-BURST
- 138 HS High-Speed
- 139 LS Low-Speed
- 140 LSb Least Significant bit
- 141 MIB Management Information Base
- 142 MK# Short indicator for MARKER symbols
- 143 MSb Most Significant bit
- 144 M RX M-PHY electrical Receiver
- 145 M TX M-PHY electrical Transmitter
- 146 NRZ Non-Return-to-Zero
- 147 PIF Protocol InterFace
- 148 PWM Pulse-Width-Modulation
- 149 RCT Re-Configuration Trigger
- 150 RD Running Disparity
- 151 RMMI Reference M-PHY MODULE Interface
- 152 SAP Service Access Point (defining interactions with Protocol Layer)
- 153 SECDED Single Error Correction, Double Error Detection
- 154 SI Symbol Interval
- 155 SYS System-clock Synchronous

|     |     |               |
|-----|-----|---------------|
| 156 | TOB | TAIL-OF-BURST |
| 157 | UI  | Unit Interval |

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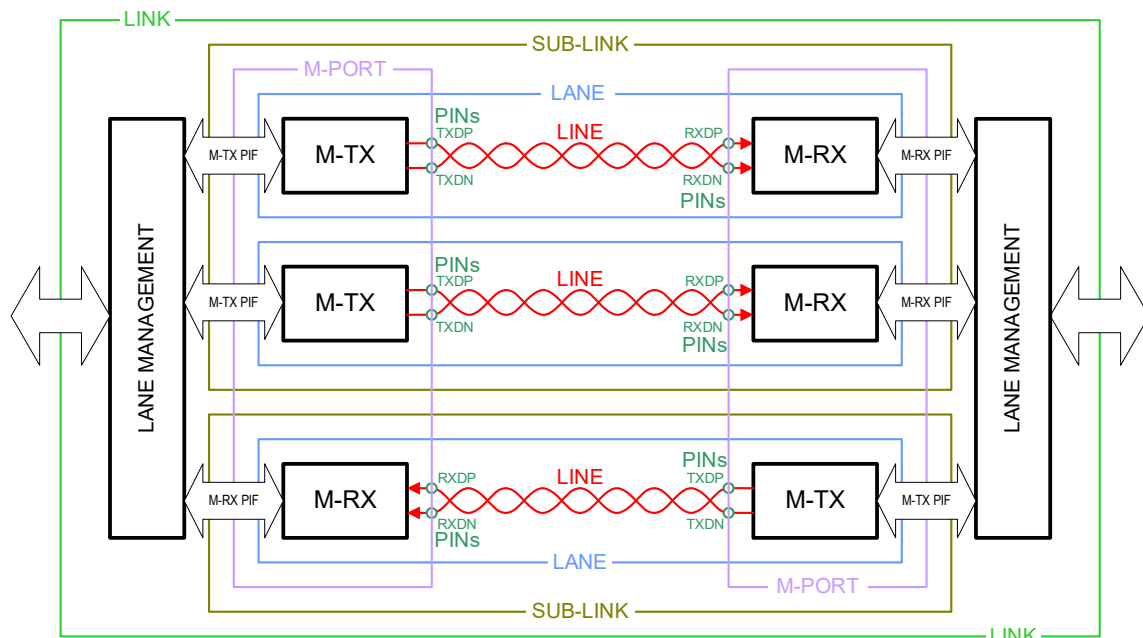
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## 4 Architecture and Operation

This section specifies the concept, communication principles, signaling schemes, interface structure and operation of M-PHY interfaces.

### 4.1 PIN, LINE, LANE, SUB-LINK, LINK, and M-PORT

A LANE is a unidirectional, single-signal, physical transmission channel used to transport information from point A to point B. A LANE consists of an M-PHY transmit MODULE (M-TX), an M-PHY receive MODULE (M-RX), and a LINE, which is the point-to-point interconnect between the M-TX and M-RX. An M-TX or M-RX has only one differential electrical output or input LINE interface, respectively, which corresponds with two signaling PINs for each MODULE. The PINs are individually denoted as DP and DN, where DP is defined as the positive node of the differential signal. An optional prefix, TX or RX, can be used to indicate the M-TX or M-RX PINs, respectively. Specifications in this document are defined at the PINs of the M-TX and M-RX, and PINs-to-PINs through the LINE. **Figure 1** illustrates the relationship between different parts of an M-PHY LINK.



**Figure 1 M-PHY Lane Example**

The LINE consists of two differentially-routed wires connecting the LINE interface PINs of the M-TX and M-RX. Typically, these wires are transmission lines. Guidelines for LINE characteristics are described in **Section 6**. For data transfer purposes, such a LINE might be considered as a black box with end-to-end signal transfer requirements defined at the PINs.

An interface based on M-PHY technology shall contain at least one LANE in each direction. There are no symmetry requirements from an M-PHY perspective for the number of LANEs in each direction.

All LANEs in the same direction within a LINK are denoted as a SUB-LINK. Two SUB-LINKs with opposite directions plus additional LANE management, which provides bidirectional data transport functionality agnostic to the actual LANE composition, is called a LINK. A set of M-TXs and M-RXs in a device that compose one interface port is denoted as an M-PORT.

This document specifies LANEs and their individual parts including M-TX, M-RX, interconnect. Furthermore, this specification sets some boundary conditions for M-TX and M-RX inside a single M-PORT,

which puts some constraints for the usage of LANEs within SUB-LINKs. This document does not specify the LANE management function in order to allow maximum flexibility of LANE exploitation by protocols. Therefore, the composition of LANEs in the two SUB-LINKs and the specification of LANE management, which completes the LINK, is left to protocols applying M-PHY technology.

## 4.2 LINE States

M-PHY technology exploits only differential signaling. A LINE can show the following states:

- A positive differential voltage, driven by the M-TX, which is denoted by LINE state DIF-P
- A negative differential voltage, driven by the M-TX, which is denoted by LINE state DIF-N
- A weak zero differential voltage, maintained by M-RX, which is denoted by LINE state DIF-Z
- An unknown, floating LINE voltage, or no LINE drive, which is denoted by LINE state DIF-Q

**Table 1** list all possible LINE conditions with the resulting LINE state

**Table 1 LINE Conditions and Resulting LINE States**

| Differential LINE Voltage | M-TX Output Impedance | M-RX Input Impedance | LINE State Set by | LINE State Name |
|---------------------------|-----------------------|----------------------|-------------------|-----------------|
| Positive                  | Low                   | Any                  | M-TX              | DIF-P           |
| Negative                  | Low                   | Any                  | M-TX              | DIF-N           |
| Zero                      | High                  | Medium               | M-RX              | DIF-Z           |
| Unknown or floating       | High                  | High                 | None              | DIF-Q           |

For data transmission, only DIF-P and DIF-N are exploited. DIF-Z can only occur during power-up and power-saving states.

The transition point between DIF-Z and DIF-N is defined by the squelch threshold level, which is positioned between the DIF-N and DIF-Z electrical LINE levels (**Section 5.2.5**). The transition point between DIF-P and DIF-N is defined at the zero crossing of the differential signal.



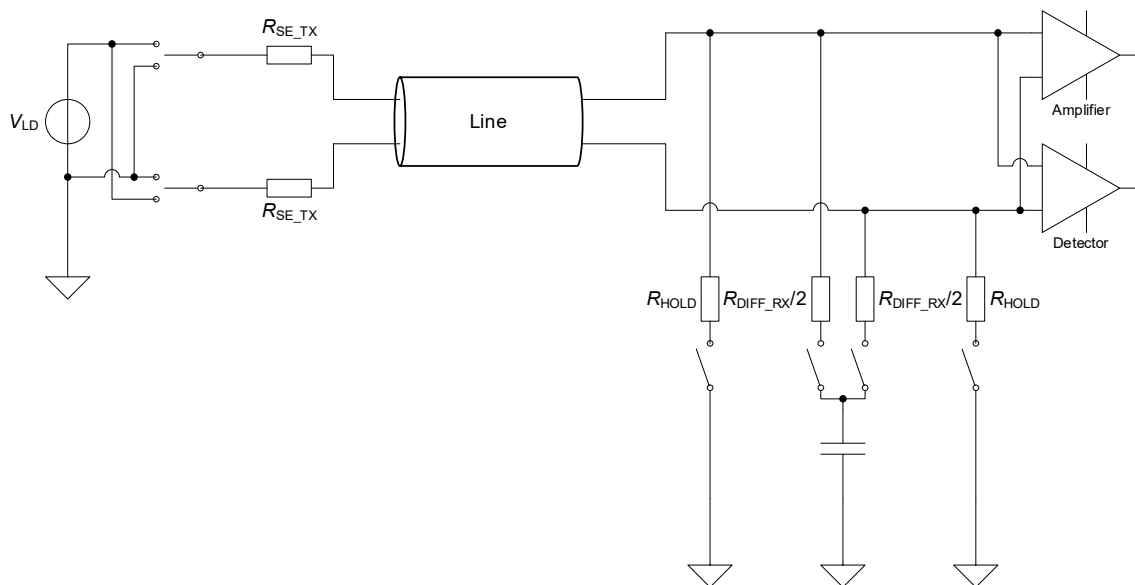
#### 4.2.1 Termination Scheme

An M-TX shall terminate both wires in the LINE with a characteristic impedance  $R_{SE\_TX}$  during any DIF-P or DIF-N state, both differentially as well as common-mode with respect to ground. The M-TX can have a larger resistance during SLEEP and STALL as described in **Section 5.1.1.3** as  $R_{SE\_PO\_TX}$ .

An M-RX does not always terminate the LINE, but certain options such as HS-MODE require support for terminated operation. Therefore, an M-RX including these options shall include a switchable differential LINE termination.

The M-RX termination condition are optionally indicated in the electrical parameter and LINE state name by a subscript RT (Resistively Terminated) or NT (Not Terminated). For example, DIF-PR<sub>RT</sub> is a DIF-P state with receiver termination enabled.

**Figure 2** shows an example of a LINE termination scheme. The electrical characteristics of LINE states and terminations are specified in **Section 5**.



**Figure 2 Example I/O Termination**

#### 4.2.2 Signal Amplitudes

All communication is based on low-swing, DC-coupled, differential signaling. The LINE driver in an M-TX may support two drive strengths, Large Amplitude (LA) and Small Amplitude (SA), resulting in different signal amplitudes. Detailed electrical level specifications are provided in **Section 5**. Drivers can support either one of these two, or both, amplitudes. If both amplitudes are supported, Large Amplitude shall be the default configuration setting. An M-RX is able to receive both amplitudes if an appropriate interconnect is used according to the guidelines in **Section 6**. Signal amplitudes are optionally indicated in parameter names by an “LA” or an “SA” subscript.

### 4.3 Signaling Schemes

M-PHY technology exploits two different signaling schemes for transmission of bits, which are conceptually described in the following sections. Detailed parameter value specifications are provided in **Section 5**.

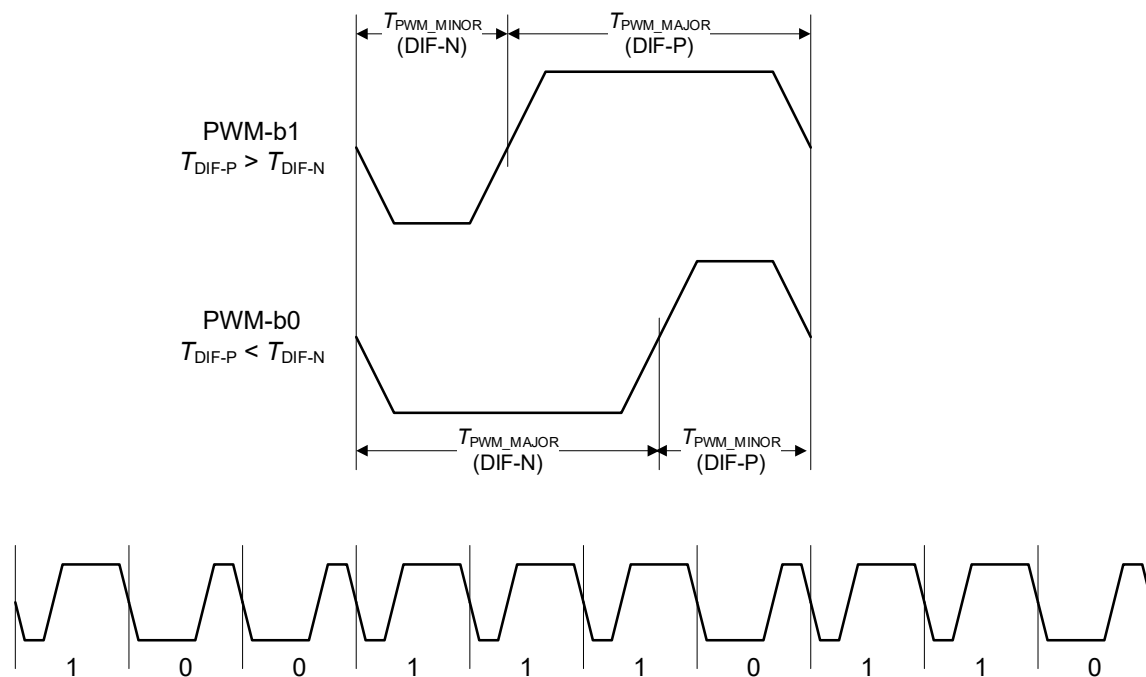
#### 4.3.1 Non-Return-to-Zero (NRZ)

For NRZ, each bit is represented by a period of either DIF-P or DIF-N, corresponding to a binary one or a binary zero, respectively. All bits are directly concatenated and have equal length.

#### 4.3.2 Pulse Width Modulation

The Pulse Width Modulation (PWM) scheme has self-clocking properties. Each bit consists of a combination of two sub-phases, a DIF-N followed by a DIF-P. One of the two sub-phases is longer than the other:  $T_{PWM\_MAJOR} > T_{PWM\_MINOR}$ , depending upon whether a binary one, or binary zero is being sent. The binary information is in the ratio of the duration of the DIF-N and DIF-P states. If the LINE state is DIF-P for the majority of the bit period, the bit is a binary one (PWM-b1). If the LINE state is DIF-N for the majority of the bit period, the bit is a binary zero (PWM-b0).

Each bit period contains two edges, where the falling edge is at a fixed position and the rising edge position is modulated. This means that the PWM bit stream explicitly contains a bit clock with period  $T_{PWM}$ , which equals the duration of one bit.  $T_{PWM}$  may vary from bit to bit during a transmission within the limits specified in **Section 5**. The bit waveforms for this signaling technique are shown in **Figure 3**.



**Figure 3 PWM Bit Waveforms and Bit Stream Example**

M-PHY technology utilizes PWM signaling with FIXED-RATIO and FIXED-MINOR format. For the FIXED-RATIO format, the durations of  $T_{PWM\_MAJOR}$  and  $T_{PWM\_MINOR}$  are ideally two-thirds and one-third of the bit period, respectively. For the FIXED-MINOR format, the duration of  $T_{PWM\_MINOR}$  is specified as an absolute time duration, while  $T_{PWM\_MAJOR}$  scales with the bit period.

## 4.4 Overview of Concept, Features, and Options

This document encompasses the full specification of LANEs, including transmitters (M-TX) and receivers (M-RX), and interconnect (LINE), to support the required set of data transmission, power saving, and control states. Furthermore, this document defines some constraints on options and operation between transmitter and receiver MODULEs within a single M-PORT.

A MODULE is specified by the characteristics that can be observed on its PINs. Therefore, M-TX and M-RX operation is fully characterized by the sequence of LINE states. All allowed sequences of LINE states are structured into MODULE states and modes, which are specified by means of state machines in subsequent sections. Detailed electrical characteristics of a MODULE are covered in **Section 5**.

Data transfer occurs in BURSTs, which can be either in High-Speed mode (HS-MODE) or Low-Speed mode (LS-MODE).

There are two fundamentally different types of MODULEs, denoted as Type-I and Type-II, depending on the signaling scheme used in LS-MODE. A Type-I MODULE employs PWM signaling, while a Type-II MODULE uses system-clock synchronous, NRZ signaling (denoted by “SYS”). This implies differences in the sequence of LINE states and state machines for an M-TX and an M-RX, as well as in the LINE performance constraints. Therefore, PWM and SYS signaling are mutually exclusive, and only one of the two signaling schemes shall be selected for an application. Note that both Type-I and Type-II MODULEs require a shared reference clock between the two ends of the LINE. Note that Type-I and Type-II MODULEs are not interoperable. However, implementations may support both types of MODULEs in order to enable hardware reuse.

All MODULEs in an M-PORT shall support LS-MODE, utilizing either the PWM or SYS signaling scheme depending on the M-PORT type. For PWM signaling (Type-I), there is only one supported (mandatory) GEAR which is PWM-G1, ranging from 3 to 9 Mbps.

All MODULEs in an M-PORT shall support HS-MODE. HS-MODE includes a default GEAR (HS-G1) and four optional GEARs (HS-G2, HS-G3, HS-G4, and HS-G5) at incremental 2x higher rates.

Each GEAR includes two data rates for EMI mitigation reasons, e.g. HS-G1 supports 1.25 Gbps and 1.45 Gbps. For the two M-PORT types, HS-MODEs are functionally equal, and very similar regarding signal specifications.

The HS unit interval is defined as  $UI_{HS} = \frac{1}{DR_{HS}}$ , where  $UI_{HS}$  is the HS unit interval and  $DR_{HS}$  is the high speed data rate.

Support for an optional GEAR in HS-MODE requires support for all GEARs below it, down to the default GEAR.

In the default configuration, M-RX shall terminate the LINE in HS-BURST and in all other states shall leave the LINE unterminated. Optionally, LS-BURST may be operated with termination. Capabilities and settings for each GEAR are handled by configuration, which is specified in **Section 4.8**. During power-saving states the M-RX shall leave the LINE unterminated.

An M-TX can have two different drive strengths, which implies a large amplitude or a small amplitude on the PINs. An M-TX shall support at least one of the two possible drive strengths. The drive strength setting holds for all operating states simultaneously, so changing it adapts the signaling levels of all LINE states. An M-TX that supports both drive strengths shall use Large Amplitude as the default setting.

The different options are depicted in **Figure 4**, where the selected set of options of every M-TX and M-RX shall map onto a contingent part of the figure. The different types result in two option diagrams (and two state machines) intended for different applications.

The functional options like supported modes, GEARs, and I/O settings shall be available for read-out in a capability registry for configuration purposes. In combination with a configuration protocol of a higher level specification, this enables interoperability between M-PORTs of the same type, while allowing operation up

to the highest commonly supported GEAR and the most optimal commonly supported settings. This configuration process is conceptually specified in **Section 4.8.1**.

Besides functional options, there are also a number of programmable parameters. These parameters shall not be mandated or defined at a fixed value by the protocol or application specifications. They are meant only for design and performance optimizations. Examples of this are programmable Slew-Rate-Control for HS-MODE and programmable timer intervals to optimize timing for actual LINE length, and PHY hardware capabilities. The complete list of options and programmable parameters can be found in **Section 7.4**.

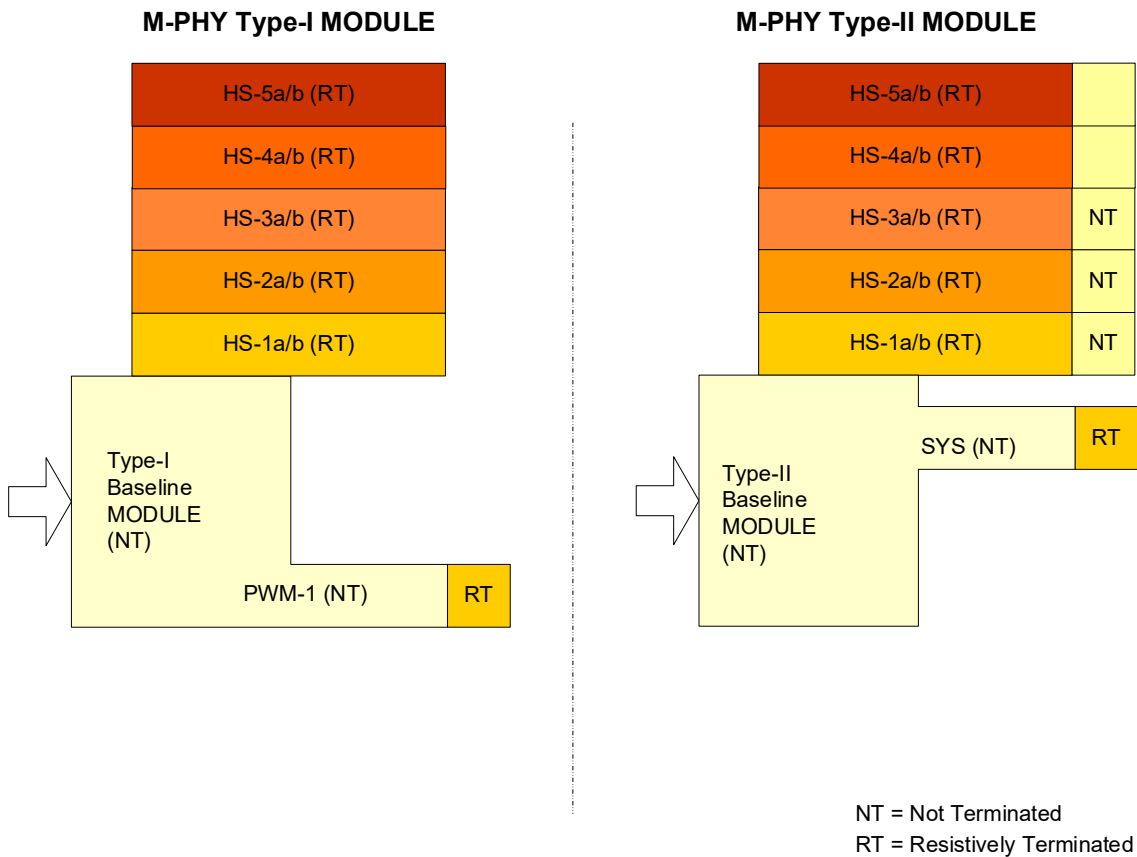


Figure 4 Functional Options for MODULEs in Type I and Type II M PORTs

23-Jun-2021

## 4.5 Line Coding

305 All information communicated inside BURST states shall be 8b10b encoded *[IBM01]* according to the data  
306 and control symbols assignments prescribed in this section.

### 4.5.1 Data Symbols

307 The coding of each byte consist of a 5b6b and a 3b4b sub-block encoding. The bits in a data byte are indicated  
308 by the capital letters HGFEDCBA. The five data bits “EDCBA” shall encode into a 6-bit sub-block “abcdei”,  
309 according to **Table 2**. The three data bits “HGF” shall encode into the 4-bit sub-block “fghj”, according to  
310 **Table 3**. For D.x.7 there is a Primary (D.x.P7) and an Alternate (D.x.A7) coding as shown in the table. The  
311 Alternate encoding shall be selected if the Primary coding combined with the preceding 5b/6b code results  
312 in five or more consecutive zeroes or ones. This implies that D.x.A7 shall only be used for x=17, x=18, and  
313 x=20 when RD=-1 and for x=11, x=13, and x=14 when RD=+1. With x=23, x=27, x=29, and x=30, the  
314 Alternate code represents the control symbol K.x.7. Any other x.A7 code cannot be used as it would result in  
315 chances for misaligned comma sequences.

316 Several 5b and 3b sub-blocks have two complimentary encoded representations with opposite disparity. The  
317 representation with the disparity sign opposite to the running disparity shall be applied for DC balance. For  
318 more information on disparity control, see **Section 4.5.3.1**. For selection of the correct 3b4b sub-block  
319 representation, the RD shall be evaluated including the preceding 5b6b sub-block, which is part of the same  
320 symbol.

321

**Table 2 5b6b Sub-Block Data Encoding**

| Input Data |       | RD = -1 | RD = +1 | Input Data |       | RD = -1 | RD = +1 |
|------------|-------|---------|---------|------------|-------|---------|---------|
| Symbol     | EDCBA | abcdei  |         | Symbol     | EDCBA | abcdei  |         |
| D.00       | 00000 | 100111  | 011000  | D.16       | 10000 | 011011  | 100100  |
| D.01       | 00001 | 011101  | 100010  | D.17       | 10001 | 100011  | –       |
| D.02       | 00010 | 101101  | 010010  | D.18       | 10010 | 010011  | –       |
| D.03       | 00011 | 110001  | –       | D.19       | 10011 | 110010  | –       |
| D.04       | 00100 | 110101  | 001010  | D.20       | 10100 | 001011  | –       |
| D.05       | 00101 | 101001  | –       | D.21       | 10101 | 101010  | –       |
| D.06       | 00110 | 011001  | –       | D.22       | 10110 | 011010  | –       |
| D.07       | 00111 | 111000  | 000111  | D/K.23     | 10111 | 111010  | 000101  |
| D.08       | 01000 | 111001  | 000110  | D.24       | 11000 | 110011  | 001100  |
| D.09       | 01001 | 100101  | –       | D.25       | 11001 | 100110  | –       |
| D.10       | 01010 | 010101  | –       | D.26       | 11010 | 010110  | –       |
| D.11       | 01011 | 110100  | –       | D/K.27     | 11011 | 110110  | 001001  |
| D.12       | 01100 | 001101  | –       | D.28       | 11100 | 001110  | –       |
| D.13       | 01101 | 101100  | –       | K.28       | 11100 | 001111  | 110000  |
| D.14       | 01110 | 011100  | –       | D/K.29     | 11101 | 101110  | 010001  |
| D.15       | 01111 | 010111  | 101000  | D/K.30     | 11110 | 011110  | 100001  |
| –          | –     | –       | –       | D.31       | 11111 | 101011  | 010100  |

322

**Table 3 3b4b Sub-Block Data Encoding**

| Input  |     | RD = -1 | RD = +1 | Input              |     | RD = -1 | RD = +1 |
|--------|-----|---------|---------|--------------------|-----|---------|---------|
| Symbol | HGF | fghj    |         | Symbol             | HGF | fghj    |         |
| D.x.0  | 000 | 1011    | 0100    | K.x.0              | 000 | 1011    | 0100    |
| D.x.1  | 001 | 1001    | –       | K.x.1 <sup>1</sup> | 001 | 0110    | 1001    |
| D.x.2  | 010 | 0101    | –       | K.x.2 <sup>1</sup> | 010 | 1010    | 0101    |
| D.x.3  | 011 | 1100    | 0011    | K.x.3              | 011 | 1100    | 0011    |
| D.x.4  | 100 | 1101    | 0010    | K.x.4              | 100 | 1101    | 0010    |
| D.x.5  | 101 | 1010    | –       | K.x.5 <sup>1</sup> | 101 | 0101    | 1010    |
| D.x.6  | 110 | 0110    | –       | K.x.6 <sup>1</sup> | 110 | 1001    | 0110    |
| D.x.P7 | 111 | 1110    | 0001    | –                  | –   | –       | –       |
| D.x.A7 | 111 | 0111    | 1000    | K.x.7 <sup>1</sup> | 111 | 0111    | 1000    |

<sup>1</sup> The alternate encoding for the K.x.y codes with disparity 0 allow for K.28.1, K.28.5, and K.28.7 to be “comma” codes that contain a bit sequence that can't be found elsewhere in the data stream.

#### 4.5.2 Control Symbols

Control symbols are special symbols that do not occur in the data symbol set, that can be used for embedded control features during BURSTs. **Table 4** lists all control symbols of the 8b10b code set. M-PHY technology exploits four control symbols, namely K28.1, K28.3, K28.5, and K28.6. Their functions are briefly mentioned in the table. Symbol K28.5 has comma properties, and shall be detected anywhere in the bitstream for symbol alignment. Details on usage of symbols can be found in **Section 4.7**. An M-PHY shall not use a reserved control symbol.

**Table 4 Control Symbols**

| Input               |           | RD = -1     | RD = +1     | Name     | Function                                              |
|---------------------|-----------|-------------|-------------|----------|-------------------------------------------------------|
| Symbol              | HGF EDCBA | abcdei fghj | abcdei fghj |          |                                                       |
| K.28.0              | 000 11100 | 001111 0100 | 110000 1011 | Reserved | –                                                     |
| K.28.1 <sup>1</sup> | 001 11100 | 001111 1001 | 110000 0110 | FILLER   | NOP                                                   |
| K.28.2              | 010 11100 | 001111 0101 | 110000 1010 | Reserved | –                                                     |
| K.28.3              | 011 11100 | 001111 0011 | 110000 1100 | MARKER1  | Protocol Separator                                    |
| K.28.4              | 100 11100 | 001111 0010 | 110000 1101 | Reserved | –                                                     |
| K.28.5 <sup>1</sup> | 101 11100 | 001111 1010 | 110000 0101 | MARKER0  | HEAD-OF-BURST;<br>Start-of-FRAME;<br>Symbol Alignment |
| K.28.6              | 110 11100 | 001111 0110 | 110000 1001 | MARKER2  | Protocol Separator                                    |
| K.28.7 <sup>2</sup> | 111 11100 | 001111 1000 | 110000 0111 | Reserved | –                                                     |
| K.23.7              | 111 10111 | 111010 1000 | 000101 0111 | MARKER3  | Defined in protocol specification                     |
| K.27.7              | 111 11011 | 110110 1000 | 001001 0111 | MARKER4  | Defined in protocol specification                     |
| K.29.7              | 111 11101 | 101110 1000 | 010001 0111 | MARKER5  | Defined in protocol specification                     |
| K.30.7              | 111 11110 | 011110 1000 | 100001 0111 | MARKER6  | Defined in protocol specification                     |

**Note:**

1. Within the control symbols, K.28.1, K.28.5 are comma symbols. Comma symbols are used for synchronization (finding the alignment of the 8b and 10b codes within a bit-stream). K.28.7 has also comma properties, but sets constraints on the symbols around it. Because K.28.7 is not used, the unique comma sequences 0011111 or 1100000 cannot be found at any bit position within any combination of normal codes.
2. See note 1 for Table 3.



### 4.5.3 Running Disparity

The applied 8b10b transmission coding is a DC-balanced coding scheme. The Running-Disparity (RD) is the disparity between the number of ones and zeroes in the proceeding part of the BURST, where each one is counted as +1 and each zero is counted as -1. RD tracking is necessary for correct encoding in the M-TX and error checking in the M-RX.

#### 4.5.3.1 RD Characteristics and M-TX Coding Rules

In the absence of transmission errors, the RD stays within -3 and +3, while it always equals -1 or +1 at any of the 6b and 4b sub-block boundaries. All sub-blocks have a disparity of 0, -2, or +2. Sub-blocks with non-zero disparity have complementary representations with positive and negative disparity. In these cases, the representation with the disparity polarity opposite to the RD shall be used such that RD changes from -1 to +1 or vice versa at sub-block boundaries, and accumulation of disparity cannot occur. The starting value of the RD may be +1 or -1 for any BURST. The M-TX shall follow the RD rules for a BURST, from the first SYNC symbol up to, and including, the last 8b10b symbol preceding TAIL-OF-BURST.

#### 4.5.3.2 M-RX Disparity Handling

Although decoding 8b10b does not require RD information, it is useful for error checking purposes. Therefore, the M-RX shall track the RD and flag per symbol to the protocol if an  $|RD| > 1$  condition is observed at any sub-block boundary. An erroneous RD shall be clipped immediately to +1 or -1, which in most cases corresponds to the correct value, such that the RD tracking is immediately capable of detecting further RD errors in subsequent symbols (see *Figure 5*).

Normally, bit errors occur during bit synchronization, and the M-RX is not symbol synchronized until the first MARKER0. Therefore, the M-RX shall not report RD errors during the HEAD-OF-BURST to the protocol. The M-RX shall begin a new RD tracking sequence after receipt of a MARKER0 inside a BURST.

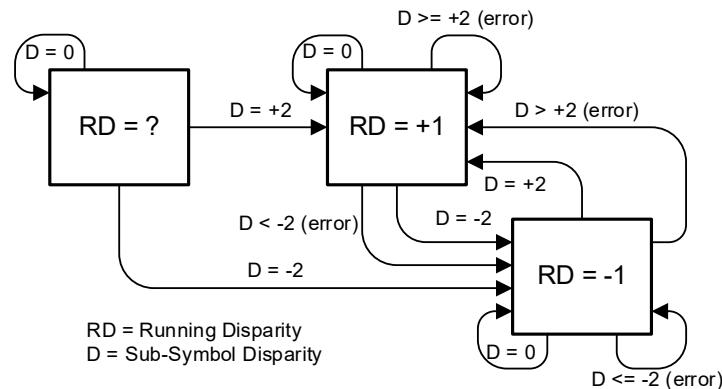


Figure 5 Running Disparity (RD) State Diagram

#### 4.5.4 Bit Order and Binary Value

350 Throughout this document, the chronology for serial binary sequences and timing diagrams is from left (first  
351 in time) to right (last in time). Therefore, the notation for 8b10b symbols is “abcdeifghj”, where the “a” bit  
352 is transmitted first. When 8b10b encoding is bypassed, the “j” bit is transmitted first.

353 The notation of binary data values is MSb to LSb when reading from left to right. Data bytes are therefore  
354 indicated by “HGFEDCBA” where “H” is the MSb and “A” is the LSb. This notation is used for PAYLOAD  
355 data bytes as well as for configuration parameter values. When 8b10b encoding is bypassed, LSb of  
356 DataValue in M-LANE-SYMBOL.request is transmitted first.

357 Protocol shall enable 8b10b encoding/decoding for normal protocol operation. The protocol may bypass  
358 8b10b encoding/decoding for testing purposes. The behavior of a MODULE whether to include PREPARE,  
359 SYNC, TOB, or PIF is implementation specific when 8b10b encoding/decoding is bypassed.

## 4.6 State Machines

The two types of MODULEs result in two alternate state machines intended for different applications with different application boundary conditions. M-PORTs of different type are not interoperable.

Both state machines allow for LS-MODE and HS-MODE operation, each including a BURST data transmission and power saving state. Performance scalability can be achieved by use of these modes combined with GEARS within modes.

The main differences between the two state machines are the following:

- Signaling scheme for LS-BURST (PWM versus SYS)
- Assumptions about availability of auxiliary signals (e.g. reference clock, reset)

High level commonalities between the two state-machines are the following:

- LS-MODE for transmission in the Mbps speed range
- HS-MODE for transmission at Gbps rates
- Individual power saving states SLEEP and STALL in LS-MODE and HS-MODE, respectively
- Ultra-low power state HIBERN8
- LINE controlled state switching between BURSTs and its power saving state
- Protocol assisted configuration mechanism

Despite the high-level commonalities these aspects are not identical for the two MODULE types, and sometimes not even similar, e.g. LS-MODE with PWM (Type-I) versus SYS (Type-II) signaling.

The state-machines in a LANE are similar for M-TX and M-RX, however the state transition conditions are different from both perspectives. Therefore, separate state machines are provided for M-TX and M-RX.

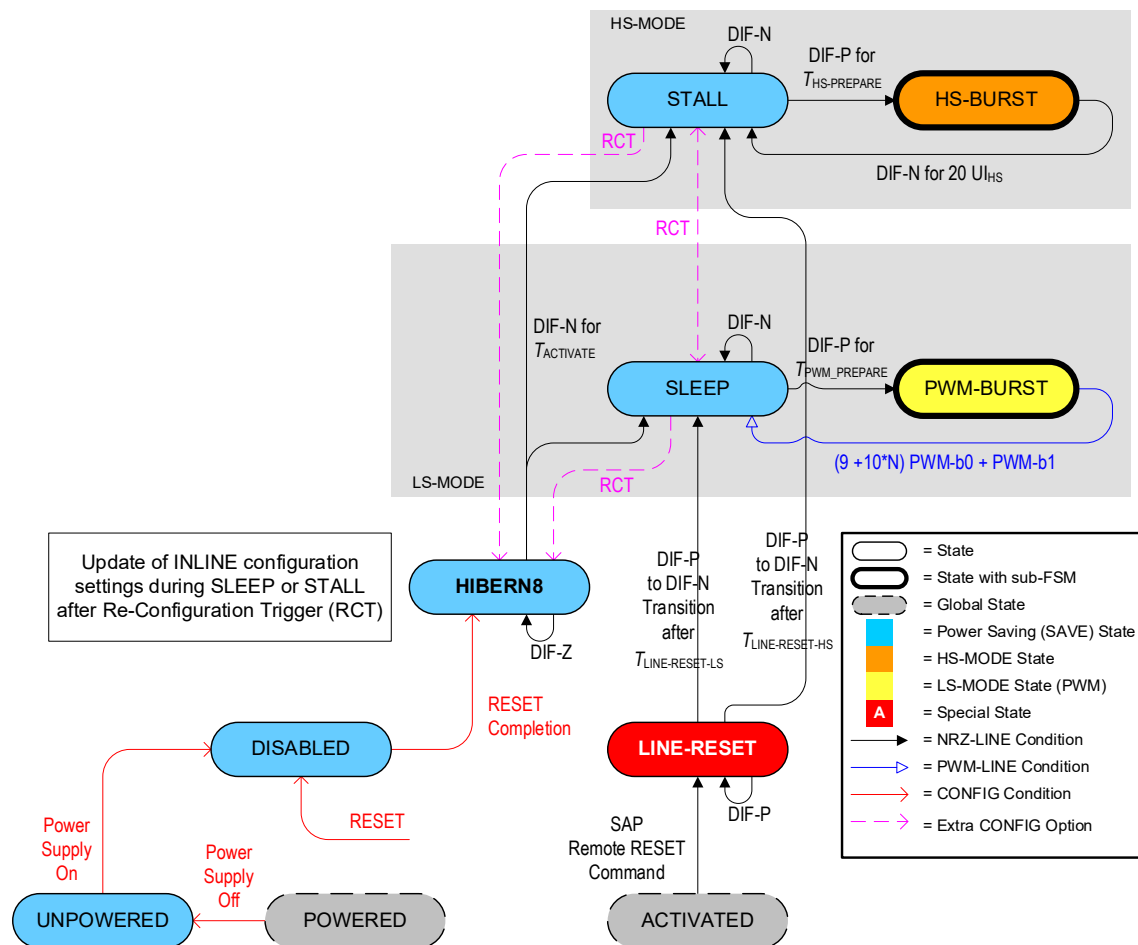
LINE-CFG state is no longer supported starting from M-PHY v5.0. This is related to dropping support for optical media converters.

#### 4.6.1 State Machine for a Type-I MODULE

Specific features of a Type-I MODULE include the following:

- PWM self-clocked LS signaling
- Operation with shared reference clock
- Fully embedded control within the LANE (additional auxiliary signals are not required)

State machines for Type-I M-TX and M-RX are shown in **Figure 6** and **Figure 7**, respectively, and explained in the sections that follow.



**Figure 6 State Diagram for Type I M-TX**

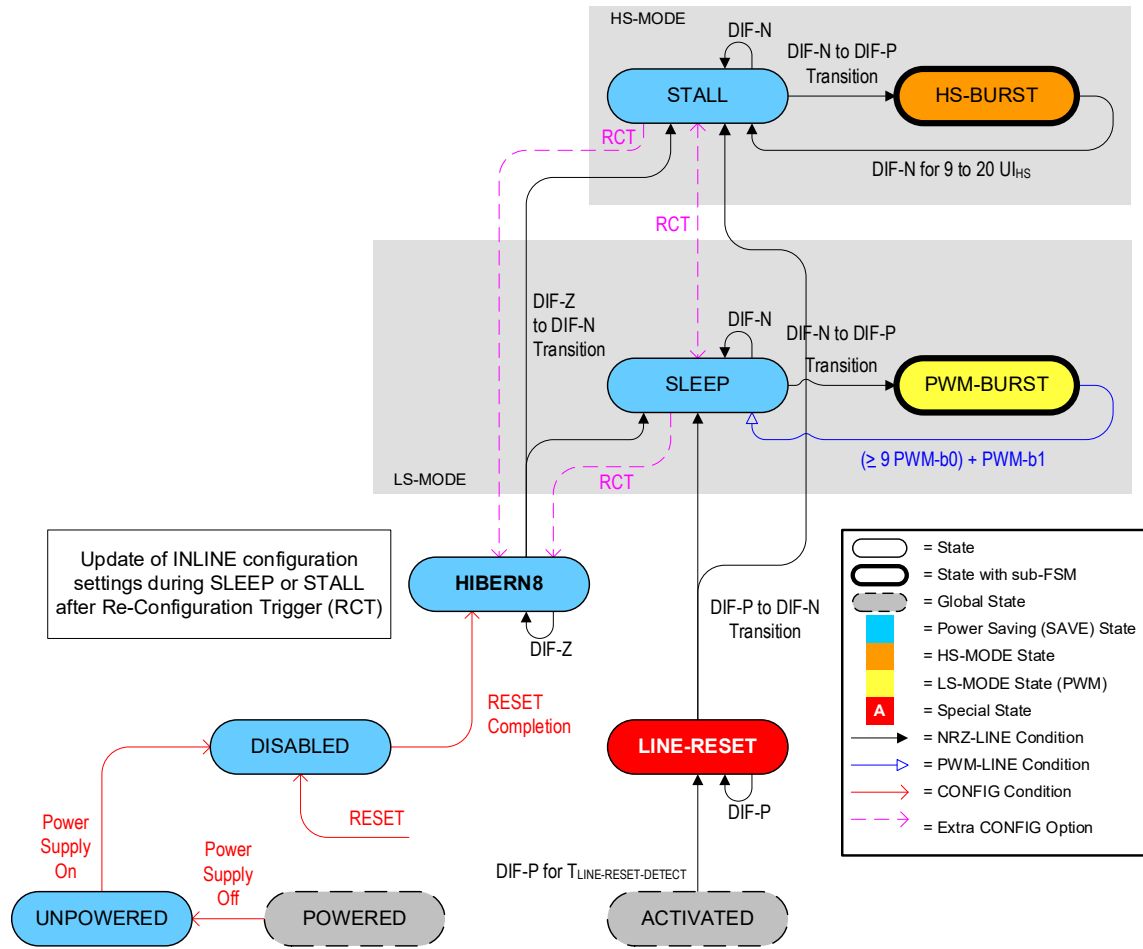


Figure 7 State Diagram for Type I M-RX

#### 4.6.2 State Machine for a Type-II MODULE

Specific features of a Type-II MODULE include the following:

- System-Clock-Synchronous LS signaling (SYS)
- Requires availability of a shared reference clock
- Partially embedded control within the LANE (some state transitions require additional auxiliary control signals)

State machines for Type-II M-TX and M-RX are shown in **Figure 8** and **Figure 9**, respectively, and explained in the sections that follow.

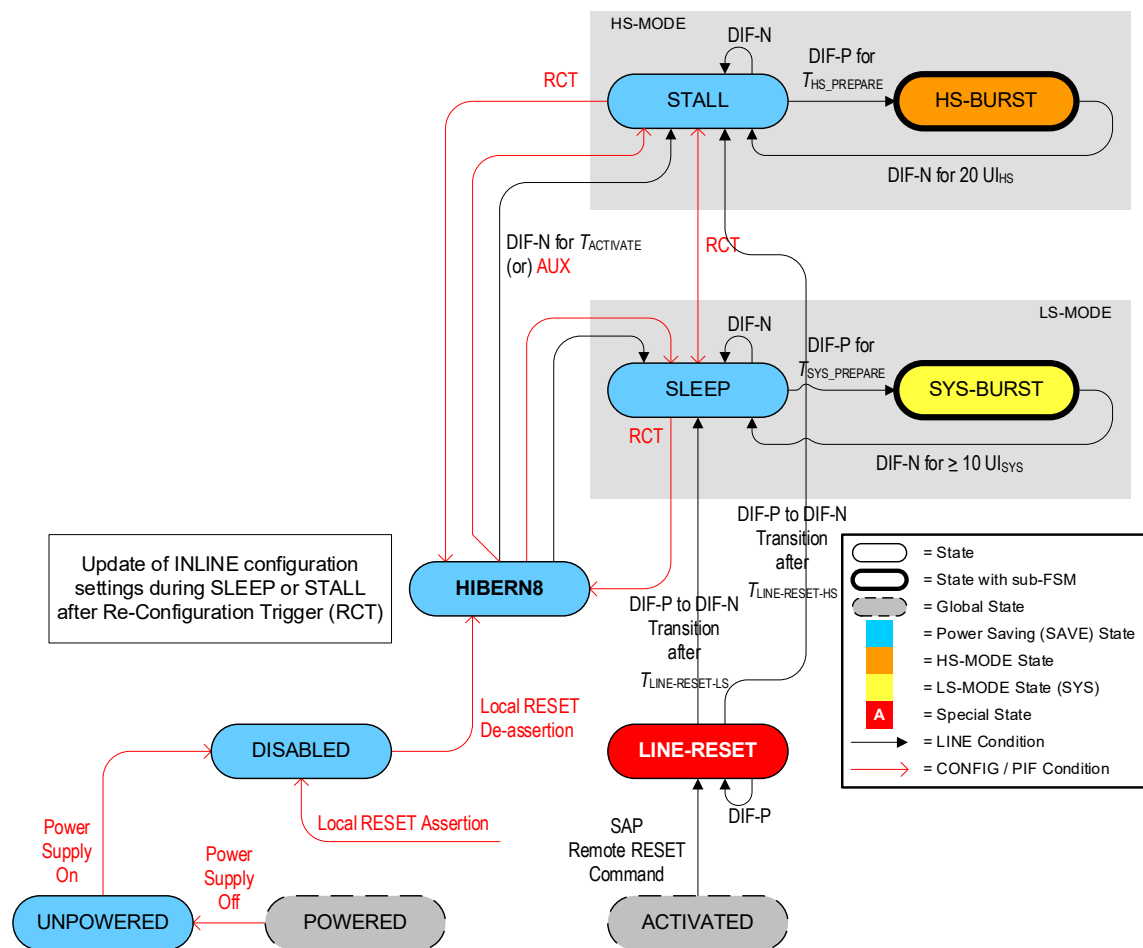


Figure 8 State Diagram for Type II M-TX

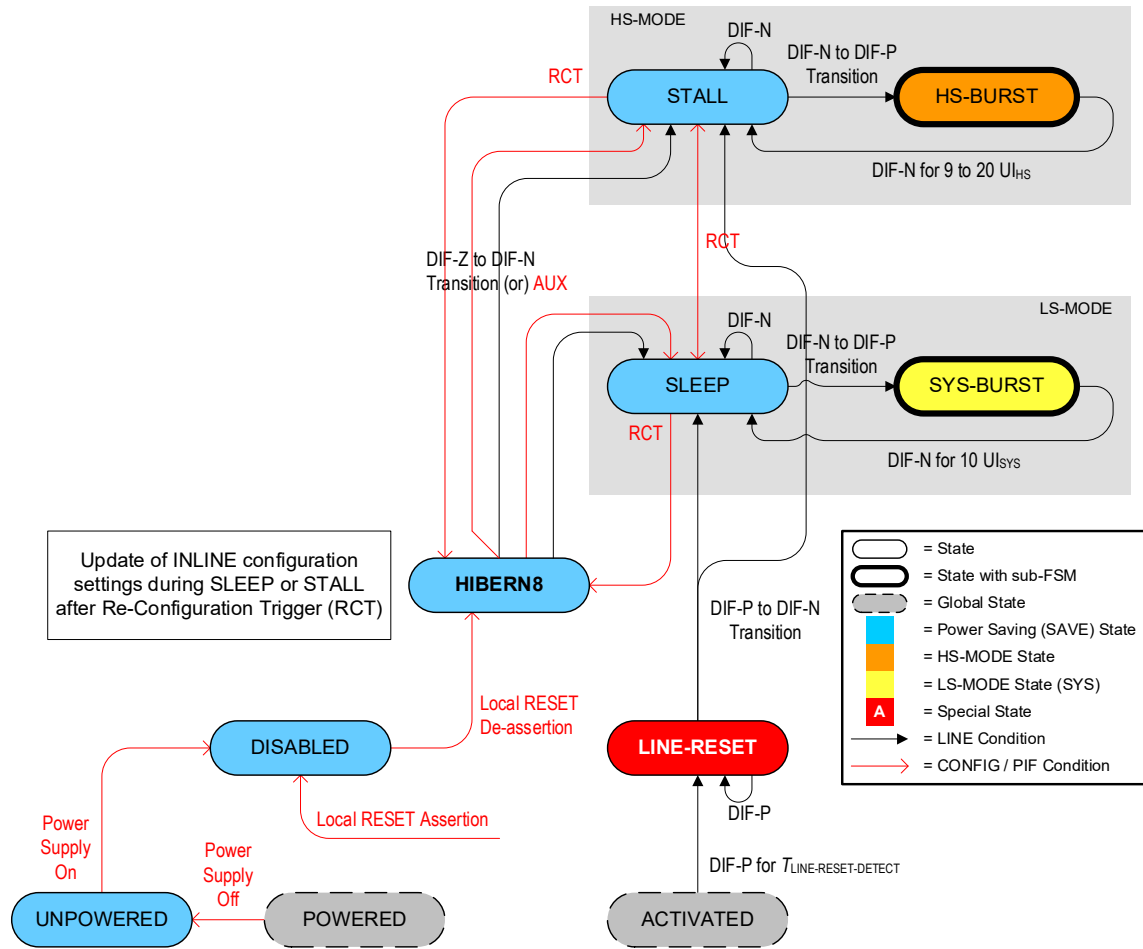


Figure 9 State Diagram for Type II M-RX

### 4.6.3 State Machine Structure and State Categories

Each state machine encompasses two operating modes, HS-MODE and LS-MODE, which include a data transmission (BURST) state and a MODE-specific power saving (SAVE) state.

STALL is the SAVE state of HS-MODE, and SLEEP of LS-MODE. The BURST state of LS-MODE is denoted as PWM-BURST for a Type-I MODULE, and SYS-BURST for a Type-II MODULE, in alignment with the signaling scheme. Each mode has the following states:

- HS-MODE: STALL, HS-BURST
- LS-MODE (Type-I MODULE): SLEEP, PWM-BURST
- LS-MODE (Type-II MODULE): SLEEP, SYS-BURST

Therefore, each state machine includes only two BURST states. A MODULE may support LS-MODE only. BURST states for each MODULE type are as follows:

- PWM-BURST (Type-I MODULE only)
- SYS-BURST (Type-II MODULE only)
- HS-BURST (Type-I and Type-II MODULEs, optional)

BURST states contain sub-FSMs, which are specified in **Section 4.7.2**.

Each state machine contains five SAVE states with a stationary LINE state. There is a specific SAVE state for each operating MODE, an ultra-low power state (HIBERN8), and two system-controlled power saving states for which the interface is no longer functional.

- STALL (HS-MODE)
- SLEEP (LS-MODE)
- HIBERN8 (Ultra-low power state where configuration is retained)
- DISABLED (POWERED, but not enabled due to a Power-on Reset, or a local RESET via the Protocol Interface (Type-II MODULE only))
- UNPOWERED (No power supply)

Furthermore, the following state is special purposes BREAK states:

- LINE-RESET (Embedded remote reset via the LINE)

Finally, there are some global state names that are not additional unique states, but are aliases for a subset of the states according to common characteristics.

The following names are global state names:

- POWERED (any state in the state machine, except UNPOWERED)
- ACTIVATED (all states within HS-MODE or LS-MODE taken together)

An M-RX state transition is triggered by either a LINE or Protocol Interface (PIF) event. A LINE event is either a LINE state transition, LINE state sequence or a bit sequence in the applied signaling format. Some trigger events are also conditional on configuration settings.



## 4.7 FSM State Descriptions

431 This section specifies the purpose and operation for each of the SAVE, BURST, and BREAK states.

### 4.7.1 SAVE States

432 This section specifies the five power-saving states, STALL, SLEEP, HIBERN8, DISABLED, and  
433 UNPOWERED.

#### 4.7.1.1 STALL

434 STALL is the power saving state in HS-MODE. STALL is mandatory for a MODULE that supports  
435 HS-MODE. In this state, the M-RX shall not be terminated, while the M-TX shall drive DIF-N. This  
436 ACTIVATED state is intended for power savings without a severe penalty on HS-BURST start-up time, in  
437 order to enable fast and efficient BURST cycles.

438 This state is exited to HS-BURST by a LINE transition to DIF-P. Entering STALL can occur from HIBERN8,  
439 LINE-RESET, or SLEEP. See *Section 4.7.1.3*, *Section 4.7.4.2*, *Section 4.7.4.1*, and *Section 4.7.1.2*,  
440 respectively. A MODULE shall disclose, via a capability attribute, the minimum time it requires in STALL  
441 prior to starting a new BURST. See *Section 7.4*.

442 The output resistance of the M-TX shall be  $R_{SE\_TX}$  until the end of the M-RX termination disable time.  
443 Afterwards, the M-TX output resistance can be switched from  $R_{SE\_TX}$  to  $R_{SE\_PO\_TX}$ . Leaving STALL state, the  
444 M-TX output resistance shall be  $R_{SE\_TX}$  before the transition to DIF-P. See *Section 5.1.1.3*.

#### 4.7.1.2 SLEEP

445 SLEEP is the power saving state of LS-MODE. SLEEP is mandatory for a MODULE. The M-RX shall not  
446 be terminated, and the M-TX shall drive DIF-N. This state allows the lowest power consumption of all  
447 ACTIVATED states. This state is exited to LS-BURST by a LINE transition to DIF-P. Entering SLEEP can  
448 occur from HIBERN8, LINE-RESET, or STALL. See *Section 4.7.1.3*, *Section 4.7.4.2*, *Section 4.7.4.1*, and  
449 *Section 4.7.1.1*, respectively. A MODULE shall disclose to the protocol, via a capability attribute, the  
450 minimum time it requires in SLEEP prior to starting a new BURST. See *Section 7.4*.

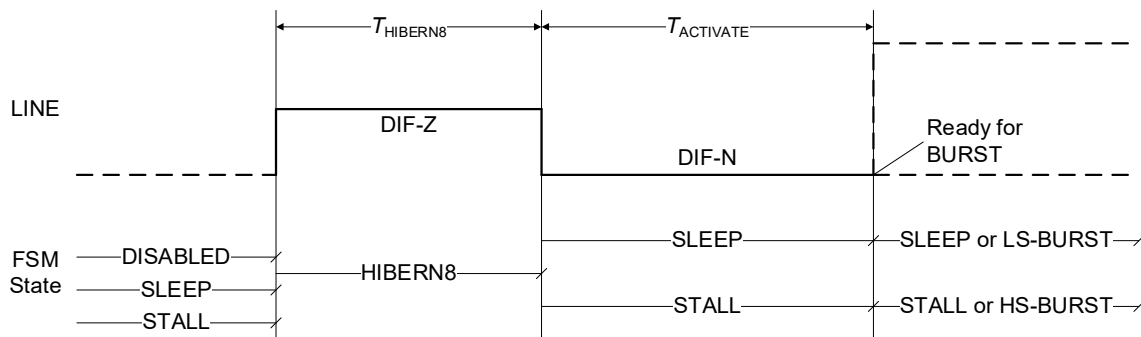
451 The output resistance of the M-TX shall be  $R_{SE\_TX}$  until the end of the M-RX termination disable time.  
452 Afterwards, the M-TX output resistance can be switched from  $R_{SE\_TX}$  to  $R_{SE\_PO\_TX}$  by evaluation of the  
453 optional TX\_Min\_SLEEP\_NoConfig\_Time or TX\_Min\_STALL\_NoConfig\_Time configuration attributes.  
454 Leaving SLEEP state, the M-TX output resistance shall be  $R_{SE\_TX}$  before the transition to DIF-P. See  
455 *Section 5.1.1.3*.

### 4.7.1.3 HIBERN8

HIBERN8 state enables ultra-low power consumption, while maintaining the configuration settings. A MODULE shall support HIBERN8. The M-TX shall be high-impedance in HIBERN8, while the M-RX shall hold the LINE at DIF-Z. Under these conditions, the M-RX is considered to be in squelch. When entering HIBERN8 from LS-MODE or HS-MODE, the Protocol Layer shall not request a MODULE exit HIBERN8 before a minimum period in HIBERN8 of  $T_{\text{HIBERN8}}$ , which is defined as the larger of local TX\_Hibern8Time\_Capability and remote RX\_Hibern8Time\_Capability. If the local M-TX supports the Advanced Granularity Capability, then  $T_{\text{HIBERN8}}$  shall be calculated as described in **Table 5**. If the remote M-RX supports for the Advanced Granularity Capability and the RX\_Advanced\_Hibern8Time\_Capability is smaller than the RX\_Hibern8Time\_Capability, it shall be used for the calculation of  $T_{\text{HIBERN8}}$  as shown in **Table 5**.

Upon state transition from SLEEP/STALL to HIBERN8, the LINE state(s) before observing DIF-Z shall not be interpreted as a HIBERN8 exit condition. For each LANE entering HIBERN8 from ACTIVATED, the protocol shall ensure M-RX enters HIBERN8 before M-TX.

The local M-TX shall drive DIF-N for a period of  $T_{\text{ACTIVATE}}$  on exit of HIBERN8 with de-emphasis disabled. The output resistance of the M-TX shall be  $R_{\text{SE\_TX}}$  during this period.  $T_{\text{ACTIVATE}}$  shall conform to RX\_Min\_ActivateTime\_Capability of the remote M-RX, if the Advanced Granularity Capability is not supported. If the remote M-RX supports for the Advanced Granularity Capability and the RX\_Advanced\_Min\_ActivateTime\_Capability is smaller than the RX\_Min\_ActivateTime\_Capability, it shall be used for the calculation of  $T_{\text{ACTIVATE}}$  as shown in **Table 5**. For embedded HIBERN8 exit control, the M-RX needs to detect a non-squelch state for a LINE transition to DIF-N. A Type-I MODULE shall use embedded HIBERN8 exit control. For a Type-II MODULE, HIBERN8 exit control can be embedded or, alternatively, by use of auxiliary control signals. Note that squelch detection is only utilized in HIBERN8, so this function can be disabled for all other states. A LANE MODULE becomes ACTIVATED on exit of HIBERN8, and shall return to the power saving state of the configured operating mode and be ready for a BURST within  $T_{\text{ACTIVATE}}$ .



Entry to HIBERN8 from SLEEP or STALL only with RCT for Type-II, or Type-I

**Figure 10 Entry and Exit of HIBERN8**

Entering HIBERN8 can occur from STALL, SLEEP, and DISABLED states. Entry of HIBERN8 from STALL or SLEEP state is controlled via configuration (see **Table 40**). The mechanism is specified in **Section 4.7.4.2**. Therefore, HIBERN8 is always entered from a DIF-N LINE state. Entering HIBERN8 from DISABLED does not typically happen simultaneously for the M-TX and the M-RX in a LANE because it depends on independent timings of RESET signals on each side of the LANE. Signals and states before, during, and after HIBERN8 state are illustrated in **Figure 10**.

When entering HIBERN8 is requested by an RCT immediately following a transition from a BURST state to a SAVE state, additional timings apply. After issuing TOB, the M-TX shall drive the LINE with DIF-N for  $T_{\text{HIBERN8\_ENTER\_TX}}$ . The M-RX shall begin driving the LINE to DIF-Z within  $T_{\text{HIBERN8\_ENTER\_RX}}$  after detection of TOB.

**Table 5  $T_{\text{HIBERN8}}$  and  $T_{\text{ACTIVATE}}$  Capabilities and Parameters**

| Attribute or Parameter                  | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Units             |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| RX_Advanced_Granularity_Capability      | 4, 8, 16, 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\mu\text{s}$     |
| RX_Advanced_Hibern8Time_Capability      | 1 to 128                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | n/a               |
| RX_Advanced_Min_ActivateTime_Capability | 1 to 14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | n/a               |
| RX_Min_ActivateTime_Capability          | 1 to 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 100 $\mu\text{s}$ |
| TX_Advanced_Granularity                 | 1 to 15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | n/a               |
| TX_Advanced_Granularity_Capability      | 4, 8, 16, 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\mu\text{s}$     |
| TX_Advanced_Hibern8Time_Capability      | 1 to 128                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | n/a               |
| TX_Min_ActivateTime                     | IF (Tx_ResetMode = HS_MODE)<br>1 to 4<br>ELSE<br>1 to 15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 100 $\mu\text{s}$ |
| $T_{\text{ACTIVATE}}$                   | IF (RX_Advanced_Granularity_Capability[0] = 1)<br>$T_{\text{ACTIVATE\_RX}} =$<br>$\text{MIN}(\text{RX\_Min\_ActivateTime\_Capability} * 100,$<br>$\text{RX\_Advanced\_Min\_ActivateTime\_Capability} * 2^{(2^{\text{b10}} + \text{RX\_Advanced\_Granularity\_Capability}[2:1])})$<br>ELSE<br>$T_{\text{ACTIVATE\_RX}} =$<br>$\text{RX\_Min\_ActivateTime\_Capability} * 100$<br>END<br>$T_{\text{ACTIVATE}} \geq T_{\text{ACTIVATE\_RX}}$<br>Set TX_Min_ActivateTime or<br>TX_Advanced_Granularity such that $T_{\text{ACTIVATE}}$ condition holds true:<br>IF (TX_Advanced_Granularity_Step[0] = 1)<br>$T_{\text{ACTIVATE}} = \text{TX\_Advanced\_Granularity} * 2^{(2^{\text{b10}} + \text{TX\_Advanced\_Granularity\_Step}[2:1])}$<br>ELSE<br>$T_{\text{ACTIVATE}} = \text{TX\_Min\_ActivateTime} * 100$<br>END | $\mu\text{s}$     |
| $T_{\text{HIBERN8}}$                    | IF (RX_Advanced_Granularity_Capability[0] = 1)<br>$T_{\text{HIBERN8\_RX}} =$<br>$\text{MIN}(\text{RX\_Hibern8Time\_Capability} * 100,$<br>$\text{RX\_Advanced\_Hibern8Time\_Capability} * 2^{(2^{\text{b10}} + \text{RX\_Advanced\_Granularity\_Capability}[2:1])})$<br>ELSE<br>$T_{\text{HIBERN8\_RX}} = \text{RX\_Hibern8Time\_Capability} * 100$<br>END<br>IF (TX_Advanced_Granularity_Capability[0] = 1)<br>$T_{\text{HIBERN8\_TX}} =$<br>$\text{MIN}(\text{TX\_Hibern8Time\_Capability} * 100,$<br>$\text{TX\_Advanced\_Hibern8Time\_Capability} * 2^{(2^{\text{b10}} + \text{TX\_Advanced\_Granularity\_Capability}[2:1])})$<br>ELSE<br>$T_{\text{HIBERN8\_TX}} = \text{TX\_Hibern8Time\_Capability} * 100$<br>END<br>$T_{\text{HIBERN8}} \geq \text{MAX}(T_{\text{HIBERN8\_RX}}, T_{\text{HIBERN8\_TX}})$   | $\mu\text{s}$     |

#### 4.7.1.4 DISABLED

DISABLED is a POWERED state, while MODULE operation is disabled by a RESET signal. When DISABLED, an M-TX shall be high impedance, and an M-RX shall keep the LINE at DIF-Z. All configuration settings shall be reset to default values. LANE operation cannot be (re-)established via LINE signaling. For a Type I state machine, entry into and exit from DISABLED state occurs with RESET, which is typically a Power-on Reset (POR). For a Type II MODULE, entry and exit of DISABLED state are controlled by asserting or de-asserting the local RESET with a POR signal or through the Protocol Interface.

#### 4.7.1.5 UNPOWERED

UNPOWERED is the state of a MODULE when the power supply is withdrawn. Both M-TX and M-RX shall be high-impedance while UNPOWERED. During UNPOWERED state the LINE level is undefined, except that the LINE voltages shall not exceed the safe operation voltage window,  $V_{PIN}$ . All configuration settings are lost. During powering-up, a MODULE shall exit into DISABLED state on the assertion of a RESET signal. This is typically a Power-on Reset signal.

##### 4.7.1.5.1 Power-Up Cycle

When the power supply comes up on a MODULE, the RESET signal shall drive the MODULE into DISABLED. This RESET is typically derived from the system POR. During power-up, until shortly after the assertion of RESET, an M-TX may temporarily expose a lower impedance. However, a Type-I M-TX shall not cause a differential level exceeding the squelch threshold until it drives a DIF-N to signal exit of HIBERN8.

The LINE state becomes defined when the M-RX is POWERED and enters DISABLED state. In DISABLED state the M-TX is high-impedance, while the M-RX pulls the LINE state to DIF-Z. A MODULE remains DISABLED while the RESET signal is asserted. Default operating mode shall be chosen during DISABLED state. When the RESET signal is de-asserted following power-up, the MODULE shall enter HIBERN8.

After Power On Reset, the M-TX enters HIBERN8. When the protocol changes the TX\_HIBERN8\_Control value to "EXIT" and issues a M-CTRL-CFGREADY.request to validate the newly set value of TX\_HIBERN8\_Control, the M-TX exits HIBERN8.

For a Type II MODULE, using a local RESET through the Protocol InterFace, the local RESET shall not be de-asserted before the complementary LANE MODULE at the other side of the LINE has been DISABLED.

The procedure for a Type I MODULE to exit from HIBERN8 following power-up is illustrated in **Figure 11** and **Figure 12**. Before starting a data BURST, the M-TX initiating exit from HIBERN8 drives a DIF-N, and continues to drive DIF-N, until an M-RX of the same M-PORT detects DIF-N. Exit of HIBERN8 on the remote side remains a decision of the remote Protocol Layer, but is triggered by detection of HIBERN8 exit on the remote side M-RX. The remote M-RX in the initiating LANE shall not exit HIBERN8 until local RESET is de-asserted, and the M-RX has transitioned from DISABLED to HIBERN8. Note that the minimum  $T_{HIBERN8}$  time does not apply for an M-RX following power-up. Detecting DIF-N on the local M-RX indicates that both ends of the LINK are operational, and have exited HIBERN8. Boundary conditions for multi-LANE behavior are provided in **Section 4.9**.

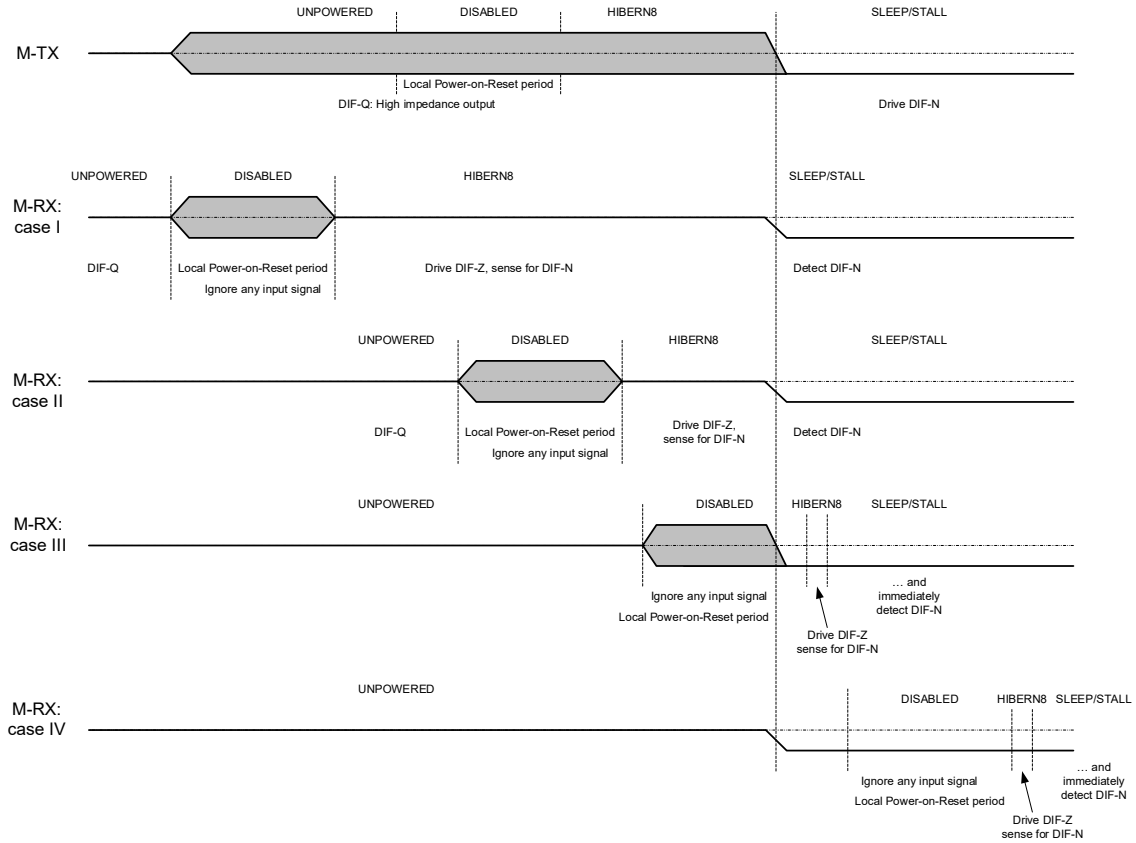
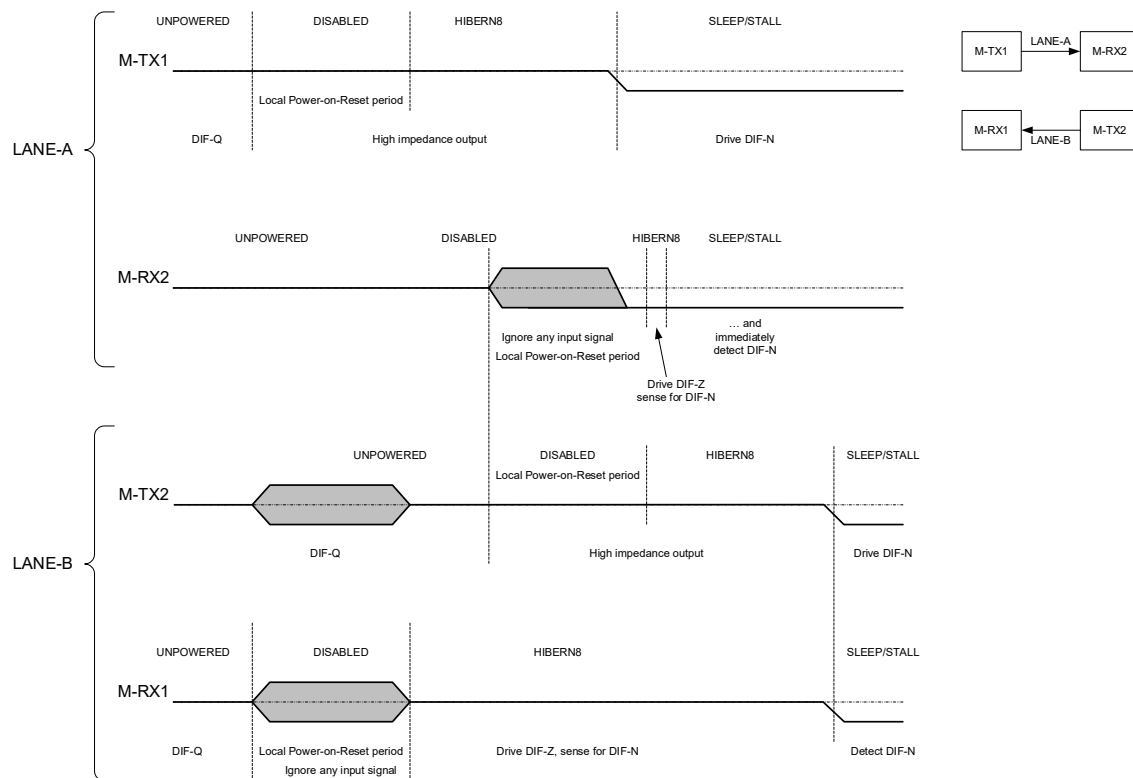


Figure 11 Lane Power Up



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#### 4.7.2 BURST States

Data transmission occurs in BURSTS with power saving states between BURSTS. BURSTS can be transferred in HS-MODE or LS-MODE, HS-BURST in HS-MODE, and LS-BURST in LS-MODE. There are two variants of LS-BURSTS depending on the applied signaling scheme, PWM-BURST for a Type-I MODULE, and SYS-BURST for a Type-II MODULE. This section specifies the sequence of events during BURST states.

- The Min\_SAVE\_Config\_Time\_Capability attribute includes all implementation specific timings required to prepare for the reception of the next BURST after configuration during SAVE.
  - The RX\_Min\_SAVE\_Config\_Time\_Capability attribute (or alternatively the RX\_Extended\_Min\_SAVE\_Config\_Time\_Capability attribute) defines the time  $T_{RX\_MIN\_SAVE\_CONFIG}$ . This time includes all implementation specific timings required to prepare for the reception of the next BURST after configuration during SAVE.
  - The attribute RX\_Extended\_Min\_SAVE\_Config\_Time\_Capability is optional, but should be implemented. If supported, then it takes precedence over RX\_Min\_SAVE\_Config\_Time\_Capability; and RX\_Min\_SAVE\_Config\_Time\_Capability shall be set to its maximum value defined in **Table 42**. Irrespective of the Capability in use, the time  $T_{RX\_MIN\_SAVE\_CONFIG}$  shall not exceed its maximum value defined in **Table 6**.
  - The TX\_Min\_SAVE\_Config\_Time\_Capability attribute (or alternatively the TX\_Extended\_Min\_SAVE\_Config\_Time\_Capability attribute) defines the time  $T_{TX\_MIN\_SAVE\_CONFIG}$ . This time shall not exceed its maximum value defined in **Table 6**. It includes all implementation specific timings required to prepare for the transmission of the next BURST after configuration during SAVE.
  - The attribute TX\_Extended\_Min\_SAVE\_Config\_Time\_Capability is optional, but should be implemented. If supported, then it takes precedence over TX\_Min\_SAVE\_Config\_Time\_Capability; and TX\_Min\_SAVE\_Config\_Time\_Capability shall be set to its maximum value defined in **Table 39**.
- After configuration, the protocol layer shall wait at least for the time  $T_{MIN\_SAVE\_CONFIG}$  before starting the BURST.

**Table 6  $T_{RX\_MIN\_SAVE\_CONFIG}$ ,  $T_{TX\_MIN\_SAVE\_CONFIG}$ , and  $T_{MIN\_SAVE\_CONFIG}$  Capabilities and Parameters**

| Attribute or Parameter      | Value                                                                                                                                                                                                                                              | Allowed Maximum | Units   |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------|
| $T_{RX\_MIN\_SAVE\_CONFIG}$ | IF (RX_Extended_Min_SAVE_Config_Time_Granularity_Capability = 1)<br>$T_{RX\_MIN\_SAVE\_CONFIG} = RX\_Extended\_Min\_SAVE\_Config\_Time\_Capability$<br>ELSE<br>$T_{RX\_MIN\_SAVE\_CONFIG} = RX\_Min\_SAVE\_Config\_Time\_Capability * 0.04$<br>END | 10 $\mu$ S      | $\mu$ S |
| $T_{TX\_MIN\_SAVE\_CONFIG}$ | IF (TX_Extended_Min_SAVE_Config_Time_Granularity_Capability = 1)<br>$T_{TX\_MIN\_SAVE\_CONFIG} = TX\_Extended\_Min\_SAVE\_Config\_Time\_Capability$<br>ELSE<br>$T_{TX\_MIN\_SAVE\_CONFIG} = TX\_Min\_SAVE\_Config\_Time\_Capability * 0.04$<br>END | 100 $\mu$ S     | $\mu$ S |
| $T_{MIN\_SAVE\_CONFIG}$     | IF (remote $T_{RX\_MIN\_SAVE\_CONFIG}$ is known)<br>$T_{MIN\_SAVE\_CONFIG} = MAX(local\ T_{TX\_MIN\_SAVE\_CONFIG}, remote\ T_{RX\_MIN\_SAVE\_CONFIG})$<br>ELSE<br>$T_{MIN\_SAVE\_CONFIG} = 100$<br>END                                             | 100 $\mu$ S     | $\mu$ S |

Each BURST starts from the SAVE state for that operating mode, with a transition from DIF-N to DIF-P. After a period of DIF-P called PREPARE, a sequence of 8b10b encoded symbols follows as specified in **Section 4.7.2.1**. After the last 8b10b SYMBOL of the BURST either a series of b0s or a series of b1s (TAIL-OF-BURST) is transmitted. A series of equal bits violate 8b10b code characteristics, and indicates whether the M-RX returns to the SAVE state of the current operating mode. In the case of PWM signaling, the last bit of the sequence is inverted to indicate the end of LINE activity.

Each BURST state contains a sub-state machine that specifies the sequence of events during a BURST, which is shown in **Figure 13**. There is much similarity between individual BURST states, but there are also distinct differences due to the exploited signaling schemes, which are explained in the following sections.

The following sections specify the details of the BURST sub-state machine.

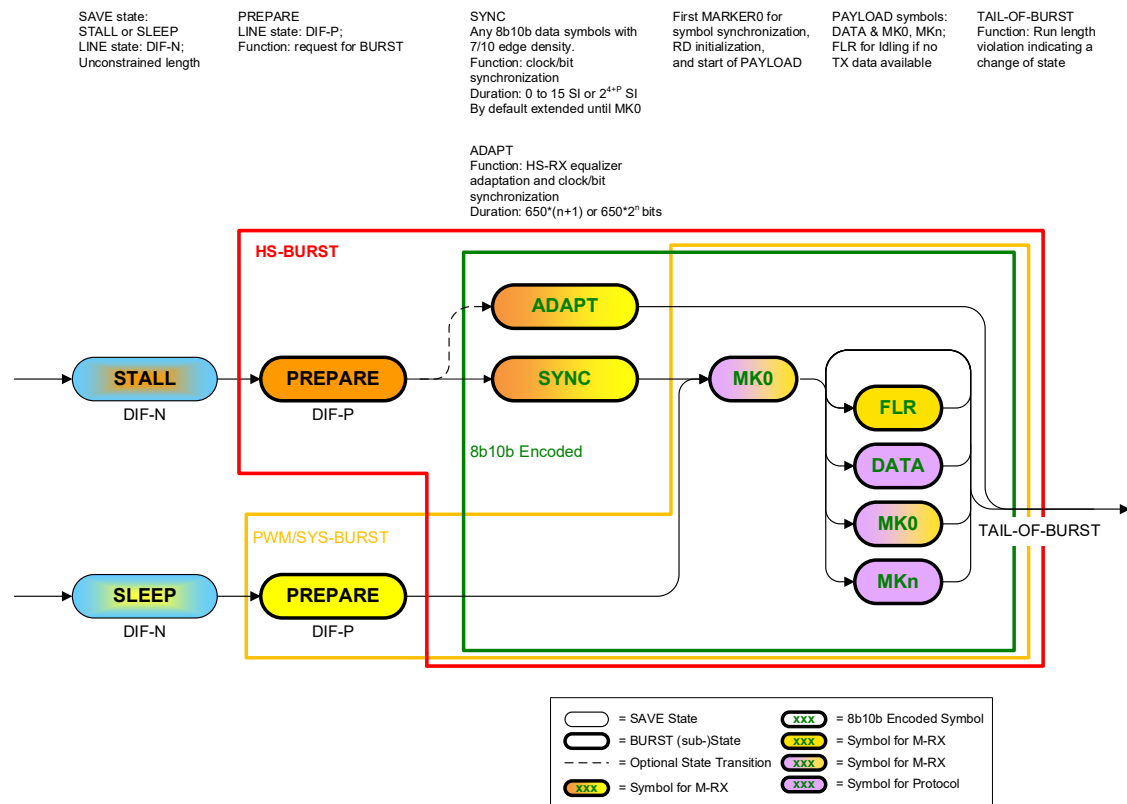


Figure 13 BURST SAVE: Detailed Sub FSM



#### 4.7.2.1 PREPARE for BURST

PREPARE is the initial sub-state of BURST which allows settling of LINE levels and transceiver settings before the bitstream is started. LINE state during PREPARE is DIF-P. If an M-RX is configured to terminate the LINE during the BURST, the termination shall be enabled during PREPARE. Signal integrity shall be maintained during any change of termination status. At the end of PREPARE, the LINE signals shall be settled. The length of PREPARE is configurable and specified in **Table 8**. The length of PREPARE in the local M-TX shall be greater than, or equal to the corresponding value of the remote M-RX parameters in the appropriate MODE, i.e.,  $T_{HS\_PREPARE}$  in HS-MODE,  $T_{PWM\_PREPARE}$  in PWM-MODE and  $T_{SYS\_PREPARE}$  in SYS-MODE.  $T_{PWM\_PREPARE}$  of the local M-TX shall not exceed the minimum value of  $T_{LINE-RESET-DETECT-LS}$ .

The PREPARE capabilities of the remote M-RX are used to configure the PREPARE duration of the local M-TX using the following equation:

$$M-TX\ LS\_PREPARE\_LENGTH \geq RX\_LS\_PREPARE\_LENGTH\_capability$$

#### 4.7.2.2 SYNC

For HS-MODE, the PREPARE sub-state period shall be followed by a SYNC sequence. The SYNC sequence is intended for bit synchronization of the M-RX to the embedded clock data stream. The SYNC sequence shall be a serialized subset of 8b10b data symbols with a high edge density for fast synchronization. Therefore, only symbols with at least seven transitions inside the symbol (out of nine possible transitions) shall be used for the SYNC sequence. Data symbols fulfilling this condition are listed in **Table 7**.

The SYNC sequence shall, by default, be generated by M-TX (TX\_SYNC\_Source = INTERNAL\_SYNC), but can be optionally configured to be provided by the protocol (TX\_SYNC\_Source = EXTERNAL\_SYNC). The default SYNC sequence shall be an alternating D10.5 and D26.5 pattern that may start with either of the two symbols. A SYNC pattern provided by the protocol shall only contain data symbols listed in **Table 7**. The SYNC sequence may start with RD of +1 or -1. However, for DC-balance, the SYNC sequence shall be encoded according to Running Disparity rules.

586

**Table 7 Valid Data Symbols for SYNC Sequence**

| Symbol Name | HGFEDCBA | RD = +1    | RD = -1    | Number of Transitions |
|-------------|----------|------------|------------|-----------------------|
|             |          | abcdeifghj | abcdeifghj |                       |
| D10.2       | 01001010 | 0101010101 | 0101010101 | 9                     |
| D21.5       | 10110101 | 1010101010 | 1010101010 | 9                     |
| D2.2        | 01000010 | 0100100101 | 1011010101 | 8                     |
| D4.2        | 01000100 | 0010100101 | 1101010101 | 8                     |
| D21.0       | 00010101 | 1010100100 | 1010101011 | 8                     |
| D21.4       | 10010101 | 1010100010 | 1010101101 | 8                     |
| D31.2       | 01011111 | 0101000101 | 1010110101 | 8                     |
| D5.2        | 01000101 | 1010010101 | 1010010101 | 8                     |
| D9.2        | 01001001 | 1001010101 | 1001010101 | 8                     |
| D10.5       | 10101010 | 0101011010 | 0101011010 | 8                     |
| D10.6       | 11001010 | 0101010110 | 0101010110 | 8                     |
| D21.1       | 00110101 | 1010101001 | 1010101001 | 8                     |
| D21.2       | 01010101 | 1010100101 | 1010100101 | 8                     |
| D22.5       | 10110110 | 0110101010 | 0110101010 | 8                     |
| D26.5       | 10111010 | 0101101010 | 0101101010 | 8                     |
| D1.2        | 01000001 | 1000100101 | 0111010101 | 7                     |
| D2.5        | 10100010 | 0100101010 | 1011011010 | 7                     |
| D2.6        | 11000010 | 0100100110 | 1011010110 | 7                     |
| D4.5        | 10100100 | 0010101010 | 1101011010 | 7                     |
| D4.6        | 11000100 | 0010100110 | 1101010110 | 7                     |
| D10.0       | 00001010 | 0101010100 | 0101011011 | 7                     |
| D10.4       | 10001010 | 0101010010 | 0101011101 | 7                     |
| D15.2       | 01001111 | 1010000101 | 0101110101 | 7                     |
| D16.2       | 01010000 | 1001000101 | 0110110101 | 7                     |
| D21.7       | 11110101 | 1010100001 | 1010101110 | 7                     |
| D22.0       | 00010110 | 0110100100 | 0110101011 | 7                     |
| D22.4       | 10010110 | 0110100010 | 0110101101 | 7                     |
| D23.5       | 10110111 | 0001011010 | 1110101010 | 7                     |
| D26.0       | 00011010 | 0101100100 | 0101101011 | 7                     |
| D26.4       | 10011010 | 0101100010 | 0101101101 | 7                     |
| D27.5       | 10111011 | 0010011010 | 1101101010 | 7                     |
| D29.5       | 10111101 | 0100011010 | 1011101010 | 7                     |
| D31.5       | 10111111 | 0101001010 | 1010111010 | 7                     |
| D31.6       | 11011111 | 0101000110 | 1010110110 | 7                     |

| Symbol Name | HGFEDCBA | RD = +1    | RD = -1    | Number of Transitions |
|-------------|----------|------------|------------|-----------------------|
|             |          | abcdeifghj | abcdeifghj |                       |
| D2.0        | 00000010 | 0100101011 | 1011010100 | 7                     |
| D2.4        | 10000010 | 0100101101 | 1011010010 | 7                     |
| D4.0        | 00000100 | 0010101011 | 1101010100 | 7                     |
| D4.4        | 10000100 | 0010101101 | 1101010010 | 7                     |
| D5.5        | 10100101 | 1010011010 | 1010011010 | 7                     |
| D5.6        | 11000101 | 1010010110 | 1010010110 | 7                     |
| D6.2        | 01000110 | 0110010101 | 0110010101 | 7                     |
| D9.5        | 10101001 | 1001011010 | 1001011010 | 7                     |
| D9.6        | 11001001 | 1001010110 | 1001010110 | 7                     |
| D10.1       | 00101010 | 0101011001 | 0101011001 | 7                     |
| D11.5       | 10101011 | 1101001010 | 1101001010 | 7                     |
| D12.2       | 01001100 | 0011010101 | 0011010101 | 7                     |
| D13.5       | 10101101 | 1011001010 | 1011001010 | 7                     |
| D18.2       | 01010010 | 0100110101 | 0100110101 | 7                     |
| D19.5       | 10110011 | 1100101010 | 1100101010 | 7                     |
| D20.2       | 01010100 | 0010110101 | 0010110101 | 7                     |
| D21.3       | 01110101 | 1010100011 | 1010101100 | 7                     |
| D21.6       | 11010101 | 1010100110 | 1010100110 | 7                     |
| D22.1       | 00110110 | 0110101001 | 0110101001 | 7                     |
| D22.2       | 01010110 | 0110100101 | 0110100101 | 7                     |
| D25.5       | 10111001 | 1001101010 | 1001101010 | 7                     |
| D26.1       | 00111010 | 0101101001 | 0101101001 | 7                     |
| D26.2       | 01011010 | 0101100101 | 0101100101 | 7                     |
| D31.0       | 00011111 | 0101001011 | 1010110100 | 7                     |
| D31.4       | 10011111 | 0101001101 | 1010110010 | 7                     |

587 The SYNC sequence has a minimum duration,  $T_{\text{SYNC}}$ , that is configurable in order to accommodate different  
588 application conditions as shown in **Table 8**.

589 The  $T_{\text{SYNC}}$  attributes of the remote M-RX are added to configure the SYNC duration of the local M-TX using  
590 the following equation:

591 M-TX SYNC\_range = M-RX SYNC\_range

592 M-TX SYNC\_length = M-RX SYNC\_length

593

**Table 8 PREPARE, SYNC, and ADAPT Attribute and Dependent Parameter Values**

| Attribute or Parameter | Value                                                                                                                                                                                                  | Units |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| HS_PREPARE_LENGTH      | 0 to 15                                                                                                                                                                                                | n/a   |
| $T_{HS\_PREPARE}$      | $HS\_PREPARE\_LENGTH * 2^{(GEAR - 1)}$                                                                                                                                                                 | SI    |
| LS_PREPARE_LENGTH      | 0 to 10                                                                                                                                                                                                | n/a   |
| $T_{PWM\_PREPARE}$     | $TPWM\_PREPARE\_calc = MAX(2^{(M-TX\_LS\_PREPARE\_LENGTH - 6)}, 1)$<br>$T_{PWM\_PREPARE} = MIN(T_{PWM\_PREPARE\_calc}, MIN(T_{LINE-RESET-DETECT-LS}))$                                                 | SI    |
| $T_{SYS\_PREPARE}$     | LS_PREPARE_LENGTH                                                                                                                                                                                      | SI    |
| SYNC_length            | 0 to 15                                                                                                                                                                                                | n/a   |
| SYNC_range             | 0 to 1                                                                                                                                                                                                 | n/a   |
| $T_{SYNC}$             | IF (SYNC_range = FINE)<br>$T_{SYNC} = SYNC\_length$<br>ELSE (IF SYNC_range = COARSE)<br>IF (M-RX)<br>$T_{SYNC} = MIN(2^{SYNC\_length}, 2^{14})$<br>ELSE<br>$T_{SYNC} = 2^{SYNC\_length}$<br>END<br>END | SI    |
| ADAPT_length           | 0 to 127                                                                                                                                                                                               | n/a   |
| ADAPT_range            | 0 to 1                                                                                                                                                                                                 | n/a   |
| $T_{ADAPT}$            | IF (ADAPT_range = FINE)<br>$T_{ADAPT} = 650 * (ADAPT\_length + 1)$<br>ELSE (IF ADAPT_range = COARSE)<br>$T_{ADAPT} = 650 * 2^{ADAPT\_length}$ ,<br>where ADAPT_length < 18<br>END                      | bit   |

594 In HS-BURST the SYNC sequence is followed by PAYLOAD that shall start with a MARKER0 (MK0). The  
595 Protocol Layer can request transmission of MARKER0 if 8b10b encoding is enabled. If transmission of  
596 MARKER0 is not requested before the configured SYNC length expires, and 8b10b encoding is enabled, the  
597 SYNC sequence shall be extended until the Protocol Layer requests transmission of MARKER0.  
598 SYS-BURST, and PWM-BURST do not include SYNC.

### 4.7.2.3 ADAPT

If an M-RX supports ADAPT, the PREPARE sub-state may be followed by the ADAPT sub-state for HS-G4 and HS-G5. The ADAPT sub-state is intended for the M-RX equalizer training to adapt to the channel characteristic, when receiving an HS data stream. The ADAPT state is executed individually and independently per HS-G4 and/or HS-G5 upon the initial HS-BURST entry for that HS-GEAR.

The ADAPT sequence shall start with an MK0 followed by an 8b10b encoded PRBS9 pattern completed by one b0 bit. Adding this bit enables complete encoding of the ADAPT sequence. The PRBS9 pattern is generated in the M-TX by a linear feedback shift register with feedback on the 5th and 9th taps as described by [ITU01]. The 8b10b encoded ADAPT sequence repeats every 650 bits. The ADAPT sub-state ends with the transmission of a TAIL-OF-BURST, upon which the M-RX and M-TX shall return to the STALL state.

The ADAPT sequence shall be generated by M-TX, and cannot be provided by the protocol.

The ADAPT sequence has a duration,  $T_{ADAPT}$ , that is configurable to accommodate different application conditions, as shown in **Table 8**. The ADAPT duration of an M-TX  $T_{ADAPT}$  is set according to the remote M-RX capabilities  $RX\_HS\_Gx\_ADAPT\_INITIAL\_Capability$  and  $RX\_HS\_Gx\_ADAPT\_REFRESH\_Capability$ . These capabilities describe the length for an Initial ADAPT after power-up and for a shorter Refresh ADAPT sequence during LINK operation, respectively. An M-RX shall store the equalizer settings in HIBERN8 and during LINE-RESET. A protocol can initiate ADAPT upon HIBERN8 exit or after LINE-RESET.

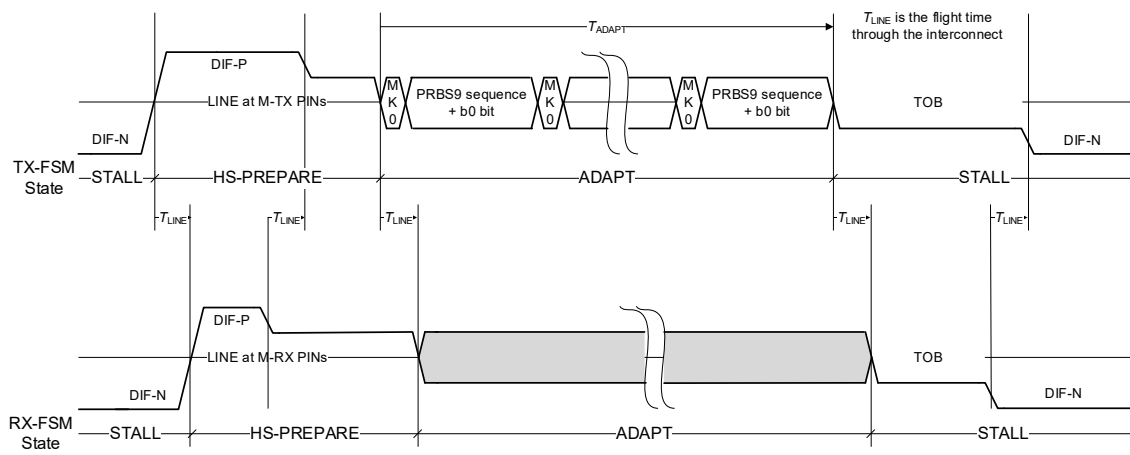
An M-RX shall complete its training in the ADAPT sub-state within an ADAPT duration of  $T_{ADAPT}$  signaled from the M-TX. An M-RX shall be able to detect the TOB at the end of ADAPT, but shall not interpret any part of the ADAPT sequence as TOB. The protocol shall set  $T_{ADAPT}$  of an M-TX to fulfil the remote M-RX requirements. A protocol should be able to set  $T_{ADAPT}$  of an M-TX also to larger values. The beginning of the M-TX and remote M-RX ADAPT sub-states can be misaligned. The M-TX and remote M-RX FSMs are aligned at STALL entry through TOB detection.

The ADAPT sub-state ends by returning to the STALL state. The remote M-RX can track RD but it shall not propagate errors during ADAPT. MODULEs can use  $T_{ADAPT}$  timers to time the end of ADAPT. When exiting the ADAPT sub-state to the STALL state, the TOB may be followed by an RCT.

#### Note:

*A configuration change affecting the signaling may need to be followed by another ADAPT sequence.*

**Figure 14** shows an example of an ADAPT operation. The duration of  $T_{ADAPT}$  is defined to start from the first MK0 up to the beginning of the TOB.



**Figure 14 ADAPT Operation**

During initial discovery, the local protocol requests and reads capabilities of MODULEs on both sides of the LINK. If HS-G4 and HS-G5 equalizer capability is detected on both sides, the protocol transfers remote M-RX ADAPT length capability into local M-TX ADAPT length configuration. The local protocol shall configure the following setting for an Initial ADAPT:

- $TX\_HS\_ADAPT\_Length \geq RX\_HS\_Gx\_ADAPT\_INITIAL\_Capability$

The local protocol shall configure the following setting for a Refresh ADAPT:

- $TX\_HS\_ADAPT\_Length \geq RX\_HS\_Gx\_ADAPT\_REFRESH\_Capability$

When a HS-G4 or HS-G5 BURST is initiated (through MLANE-AdaptStart.request) and ADAPT has been configured (through RX\_ADAPT\_Control), the M-TX transitions from PREPARE to the ADAPT sub-state instead of SYNC. The M-TX transitions from DIF-P to transmitting the ADAPT sequence. Both M-TX and M-RX remain in the ADAPT sub-state for the equalizer training for a duration of  $T_{ADAPT}$ , as configured in the local ADAPT length configuration (see **Table 8**). The M-RX signals exit from the ADAPT sub-state by flipping the ADAPT\_Control field of RX\_ADAPT\_Control from ADAPT to SYNC and returning to STALL.

The M-TX ADAPT length  $T_{ADAPT}$  can be calculated according to **Table 8** for configured TX\_HS\_ADAPT\_LENGTH.

The protocol can initiate subsequent retrainings through an M-LANE-AdaptStart.request service primitive, synchronously staging an RCT at both ends of the LINK, reconfiguring  $T_{ADAPT}$  at the M-TX, and writing the ADAPT\_control field of RX\_ADAPT\_Control from SYNC to ADAPT and setting the ADAPT\_type field of RX\_ADAPT\_Control to either INITIAL or REFRESH. During the execution of ADAPT, starting from M-LANE-AdaptStart to M-LANE-AdaptComplete, the protocol shall not issue an M-LANE-PREPARE.request.

#### 4.7.2.4 PAYLOAD of BURST

After SYNC or PREPARE period, PAYLOAD shall be transferred on request of the protocol. PAYLOAD starts with a MARKER0 and ends with the symbol before a TAIL-OF-BURST. Between the HEAD and TAIL symbols, any number of DATA0 to DATA255 or MARKER symbols can be transported in any order under protocol control via the PIF.

Note that the MARKER0 symbol has comma properties. This shall be utilized in the M-RX, to acquire, check and regain symbol alignment on any occurrence of MARKER0. If during a BURST at any time after the first MARKER0 the protocol does not provide the next symbol request on time, the M-TX will insert FILLER symbols (FLR) in order to prevent failure and corruption of the serial stream. The protocol layer may periodically provide MARKER0 for the purposes of self-healing or re-synchronization. For EMI reasons, protocols are strongly encouraged to limit consecutive repetitions of FILLER or any symbol.

#### 4.7.2.5 Closure of BURST

With the transmission of TAIL-OF-BURST, the BURST ends and the M-RX and M-TX shall return to the appropriate SAVE state; this constant bit sequence violates the 8b10b coding rules. The M-RX shall exit BURST mode on detection of the constant bit sequence.

#### 4.7.2.5.1 Closure and Return to SAVE

If the BURST closure condition for exit to SAVE state is transmitted (see **Table 9**), the M-RX shall become unterminated. The termination shall be disabled and the LINE state settled within the time defined by either  $RX\_Min\_STALL\_NoConfig\_Time\_Capability$  (exit to STALL) or  $RX\_Min\_SLEEP\_NoConfig\_Time\_Capability$  (exit to SLEEP). As shown in **Table 9**, the number and format of bits differ for different BURST states depending on the signaling scheme. In the table, N is an integer number of symbols represented by  $TX\_PWM\_BURST\_Closure\_Extension$  of the local M-TX, and shall be greater than, or equal to, the value of  $RX\_PWM\_Burst\_Closure\_Length\_Capability$  of the remote M-RX.

**Table 9 Summary of BURST Closure Conditions (TAIL-OF-BURST)**

| MODE | MODULE | Return to SAVE                                    |       |
|------|--------|---------------------------------------------------|-------|
|      |        | LINE Condition                                    | State |
| HS   | M-TX   | DIF-N for 20 $U_{IHS}$                            | STALL |
| HS   | M-RX   | DIF-N for 9 to 20 $U_{IHS}$                       | STALL |
| PWM  | M-TX   | $(9 + 10 \cdot N) \text{ PWM-b0} + \text{PWM-b1}$ | SLEEP |
| PWM  | M-RX   | $(\geq 9 \text{ PWM-b0}) + \text{PWM-b1}$         | SLEEP |
| SYS  | M-TX   | DIF-N for $\geq 10 \text{ } U_{ISys}$             | SLEEP |
| SYS  | M-RX   | DIF-N for 10 $U_{ISys}$                           | SLEEP |

#### 4.7.2.6 Example of an HS-BURST

A time domain illustration of HS-BURST operation is shown in **Figure 15**. In this example the M-RX is (default) configured to provide LINE termination during HS-BURST, which can be noticed by the signal level changes during PREPARE and (exit-to-)STALL.



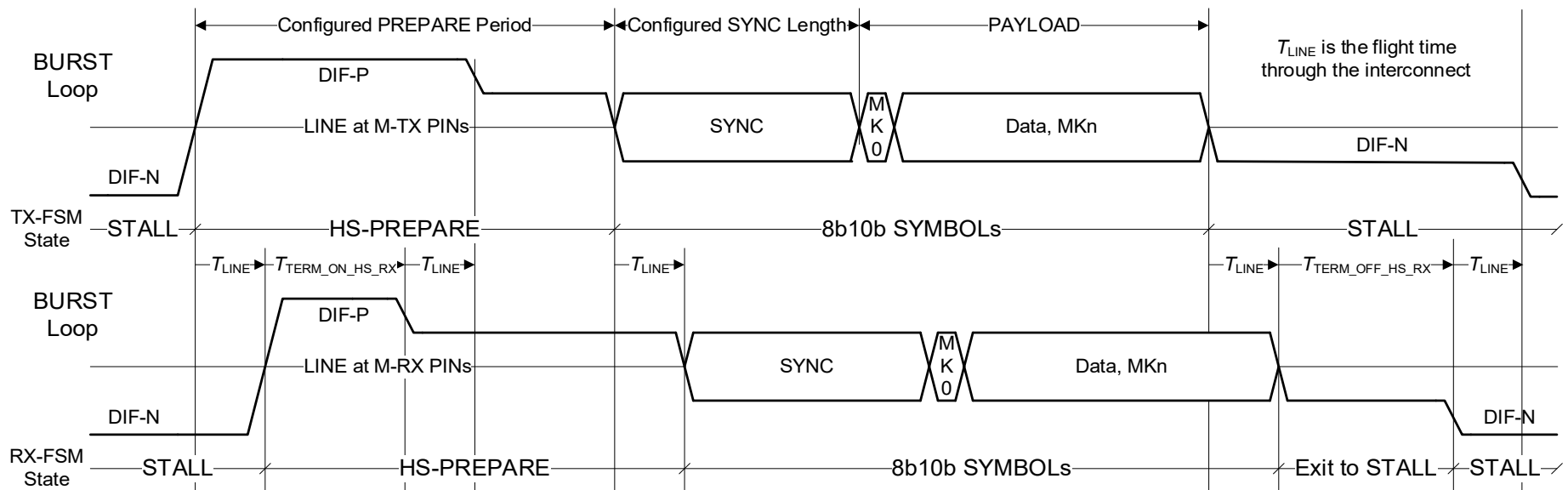


Figure 15 HS BURST Operation

### 4.7.3 BURST MODEs and GEARS

#### 4.7.3.1 HS-BURST

HS-BURST is the data transmission state of HS-MODE. HS-BURST starts from STALL on a transition to DIF-P. Data shall be 8b10b encoded in this mode and transmitted using NRZ signaling. After the last symbol of the BURST, a MODULE enters STALL state.

Eye monitor measurement can be executed during HS-BURST, see *Section 4.11* for details of Eye Monitor operation.

##### 4.7.3.1.1 HS-GEARS

A MODULE in HS-BURST shall only operate at the defined data rate,  $DR_{HS}$ . There are two RATE series, A and B, where each step in the series scales by a factor of two, while the speed rate difference between the two RATE series is about 15%, as listed in *Table 10*. If the data rates of the two RATE series are pair-wise coupled for closest rates (~15%), these individual couples are denoted as GEARS. A MODULE shall support both RATEs of a GEAR. A MODULE supporting HS-MODE shall support HS-G1. If a higher GEAR is supported all lower GEARS shall be supported as well.

**Table 10 HS BURST: RATE Series and GEARS**

| RATE A-series (Mbps) | RATE B-series <sup>1</sup> (Mbps) |         | High-Speed GEARS |
|----------------------|-----------------------------------|---------|------------------|
| 1248                 | 1459.2                            | 1456.0  | HS-G1 (A/B)      |
| 2496                 | 2918.4                            | 2912.0  | HS-G2 (A/B)      |
| 4992                 | 5836.8                            | 5824.0  | HS-G3 (A/B)      |
| 9984                 | 11673.6                           | 11648.0 | HS-G4 (A/B)      |
| 19968                | 23347.2                           | 23296.0 | HS-G5 (A/B)      |

**Note:**

1. The B series introduced are to allow integer multiples of reference frequencies 19.2 MHz or 26 MHz.

#### 4.7.3.2 PWM-BURST

PWM-BURST is the data transmission state of LS-MODE of Type-I LINKs. PWM-BURST starts from SLEEP on the transition to DIF-P. Data shall be 8b10b encoded in this mode and transmitted using PWM signaling. After the last symbol of the BURST, a sequence of same-value PWM bits is added, which creates an 8b10b run-length violation on the LINE. For a sequence of PWM-b0 with a trailing PWM-b1, both M-RX and M-TX shall return to SLEEP state. See **Table 9** for more details.

##### 4.7.3.2.1 PWM-GEARS

PWM-BURST has one PWM GEAR with a limited speed range. **Table 11** lists all the PWM-GEAR rate range. Only PWM-G1 is mandatory. PWM-G1 spans a speed range of a factor of three. This allows a continuum of possible rates. For PWM-G1 FIXED-RATIO signaling shall be applied.

**Table 11 PWM BURST GEAR**

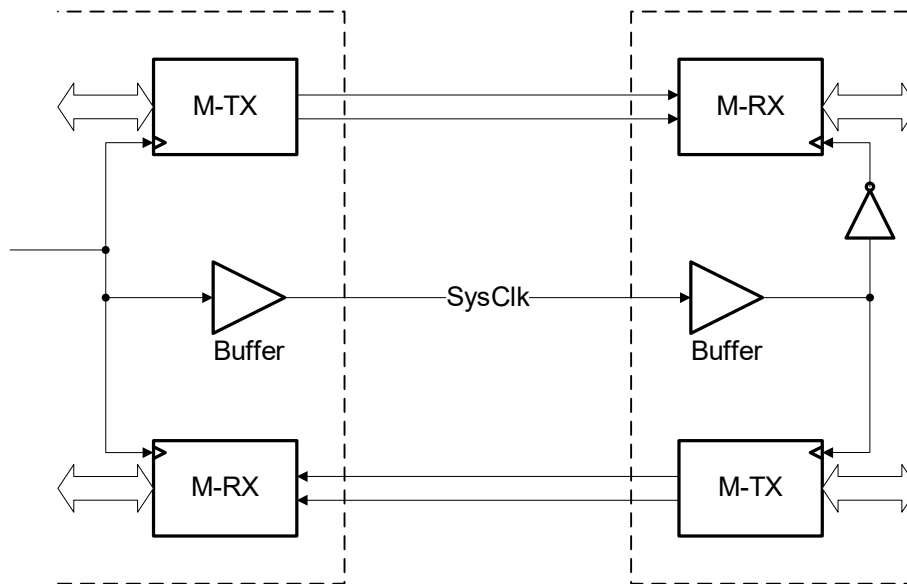
| PWM-GEARS | Min. (Mbps) | Max. (Mbps) |
|-----------|-------------|-------------|
| PWM-G1    | 3           | 9           |

### 4.7.3.3 System-clock Synchronous BURST (SYS-BURST)

SYS-BURST is the data transmission state of LS-MODE of Type-II LINKs. SYS-BURST starts from SLEEP on the transition to DIF-P. Data shall be 8b10b encoded in this mode and transmitted using reference-clock synchronous NRZ signaling. After the last symbol of the BURST, the LINE is driven to DIF-N state. The long DIF-N creates an 8b10b run-length violation which ends SYS-BURST and moves both M-RX and M-TX to SLEEP state.

In this mode, MODULEs depend on a shared reference clock for transmission. The transmission rate in this mode shall be an integer division of the shared reference clock frequency,  $f_{\text{SYS\_REF}}$ . The reference clock may originate from an independent system clock or from one of the two devices in the LINK. An example of the latter case is shown in **Figure 16**, where the device providing the clock is located on the left hand side of the figure.

This document only partially specifies this mode, as it also relies on the specifications of the reference clock, the timing relationship between the clock pin on the devices and the reference clock input of the MODULEs (PIF), and the timing between reference clock input of the MODULEs and the LINE signals. **Section 5** contains an informative guideline for timing between reference clock and LINE signals. The overall timing specifications for this signaling scheme shall be covered by the protocol specification utilizing this mode.



**Figure 16 Bidirectional SYS BURST Clocking Example**

23-Jun-2021

#### 4.7.4 BREAK States

BREAK states have special functions, which are entered by exceptional LINE sequences that do not occur during normal operating modes.

##### 4.7.4.1 LINE-RESET

This is the lowest level reset mechanism in order to reset the M-RX via the LINE during operation in case of malfunction. The duration of LINE-RESET depends on ResetMode (see [Section 7.3.7](#)). If ResetMode is LS\_MODE, then parameter  $T_{\text{LINE-RESET-LS}}$  applies. If ResetMode is HS\_MODE, then parameter  $T_{\text{LINE-RESET-HS}}$  applies. LINE-RESET can be initiated by the Protocol Layer on the M-TX side of a LINK using the M-CTRL-LINERESET.request primitive (see [Section 7.3.9](#)). A MODULE shall support LINE-RESET in all ACTIVATED states.

Before issuing M-CTRL-LINERESET.request with TActivateControl set to “ProtocolControlled”, the Protocol Layer issues M-LANE-BurstEnd.request and waits for  $T_{\text{ACTIVATE}}$  after the M-TX has generated M-LANE-SaveState.indication. This condition ensures the M-TX drives DIF-N for at least  $T_{\text{ACTIVATE}}$  so that an M-RX, which might be in HIBERN8, is ACTIVATED before the LINE-RESET condition is driven. For LINE-RESET, the M-TX shall drive DIF-P for  $T_{\text{LINE-RESET}}$ .

After the Protocol Layer issues M-CTRL-LINERESET.request with TActivateControl set to “PhyControlled”, the M-TX drives DIF-N for  $T_{\text{ACTIVATE}}$  before driving the LINE-RESET condition.  $T_{\text{ACTIVATE}}$  is calculated as described in [Table 5](#).

An M-RX shall be reset when DIF-P is observed on the LINE for  $T_{\text{LINE-RESET-DETECT-LS}}$  or  $T_{\text{LINE-RESET-DETECT-HS}}$ . The timing of the LINE-RESET detection window depends on ResetMode. If ResetMode is LS\_MODE, then parameter  $T_{\text{LINE-RESET-DETECT-LS}}$  applies. If ResetMode is HS\_MODE, then parameter  $T_{\text{LINE-RESET-DETECT-HS}}$  applies. The LINE-RESET timer shall not rely on correct protocol operation. LINE-RESET exits to SLEEP or STALL on a transition to DIF-N. LINE-RESET shall reset all configuration settings to their respective default values as specified in [Section 7.4](#).

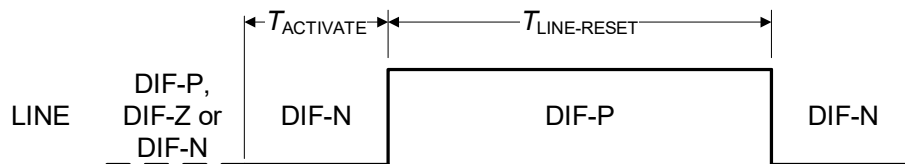


Figure 17 LINE-RESET Timing

Table 12 LINE-RESET and HIBERN8 Timer Values

| Parameter                         | Min. | Max. | Unit | Descriptions and Notes                                                           |
|-----------------------------------|------|------|------|----------------------------------------------------------------------------------|
| $T_{\text{LINE-RESET-LS}}$        | 3.1  | —    | ms   | A maximum value of 20 ms may be set at the option of the protocol. (Informative) |
| $T_{\text{LINE-RESET-HS}}$        | 0.31 | —    | ms   | A maximum value of 20 ms may be set at the option of the protocol. (Informative) |
| $T_{\text{LINE-RESET-DETECT-LS}}$ | 1    | 3    | ms   | —                                                                                |
| $T_{\text{LINE-RESET-DETECT-HS}}$ | 0.1  | 0.3  | ms   | —                                                                                |
| $T_{\text{HIBERN8\_ENTER\_TX}}$   | 50   | 1000 | ns   | —                                                                                |
| $T_{\text{HIBERN8\_ENTER\_RX}}$   | —    | 25   | ns   | —                                                                                |

#### 4.7.4.2 Re-Configuration Trigger (RCT)

737 A re-configuration trigger (RCT) is intended to provide a synchronous event upon which INLINE  
738 configuration attributes can be updated, ensuring MODULE attributes on both sides of a LANE remain  
739 consistent and interoperable. Both MODULEs shall detect an RCT.

740 The Protocol Layer shall provide sufficient time to the LANE following an RCT to complete re-configuration  
741 before requesting a new BURST as defined in *Section 7.4*.

742 All of the following conditions shall be fulfilled before a Re-Configuration Trigger is executed by a  
743 MODULE:

- 744 • A CFG-READY indication via the Protocol Interface (M-CTRL-CFGREADY.request)
- 745 • Entering or being in a SAVE state

## 4.8 Configuration

M-PHY provides significant flexibility advantages over other serial PHYs offering multiple optional MODEs and configurable Attributes. To support this level of flexibility, interoperability is managed in two ways, default parameter settings provide a minimum level of interoperation between MODULEs of the same type, while a robust configuration mechanism supports optimization of the PHY through the protocol for specific use-cases. Central to the configuration process is self-discovery where each MODULE contains a set of Attributes containing its capabilities. This information can be interrogated by the protocol using a CONFIG interface in the PIF. When coupled with the minimum dual-simplex LINK, this arrangement allows, through implementation in the protocol, a complete capability discovery and negotiation process avoiding the requirement of detailed knowledge of the LINK components at a system level.

### 4.8.1 Conceptual Configuration Process

Following an OFF state, or a LINE-RESET, a MODULE operates with default configuration settings. Under default operation the Protocol can retrieve MODULE capabilities, arbitrate more optimal configuration settings, then directly change OFFLINE settings, or make a change request for INLINE settings. This process consists of the following steps:

- DISCOVERY
  - Read MODULE capabilities, and determine desired, commonly supported configuration settings.
- PHY CONFIG
  - Request to change MODULE configuration settings.
- EFFECTUATE
  - Update the MODULE configuration settings.

To support the configuration process, the following four registries are anticipated:

- CAPABILITY registry; contains the capability information for a given MODULE.
- STATUS (also known as INLINE-SET) registry; contains effectuated INLINE configuration settings. These configuration settings are updated from the INLINE-CR registry by the MODULE upon an RCT.
- INLINE-CR registry, (shadow registry for the INLINE-SET registry); logs change requests for configuration settings of INLINE parameters, that is, settings that immediately impact actual signaling.
- OFFLINE-SET registry; contains configuration settings that do not directly impact signaling, and are therefore immediately effectuated with issuing of M-CTRL-CFGREADY.request. An RCT for effectuating these attributes is not required.

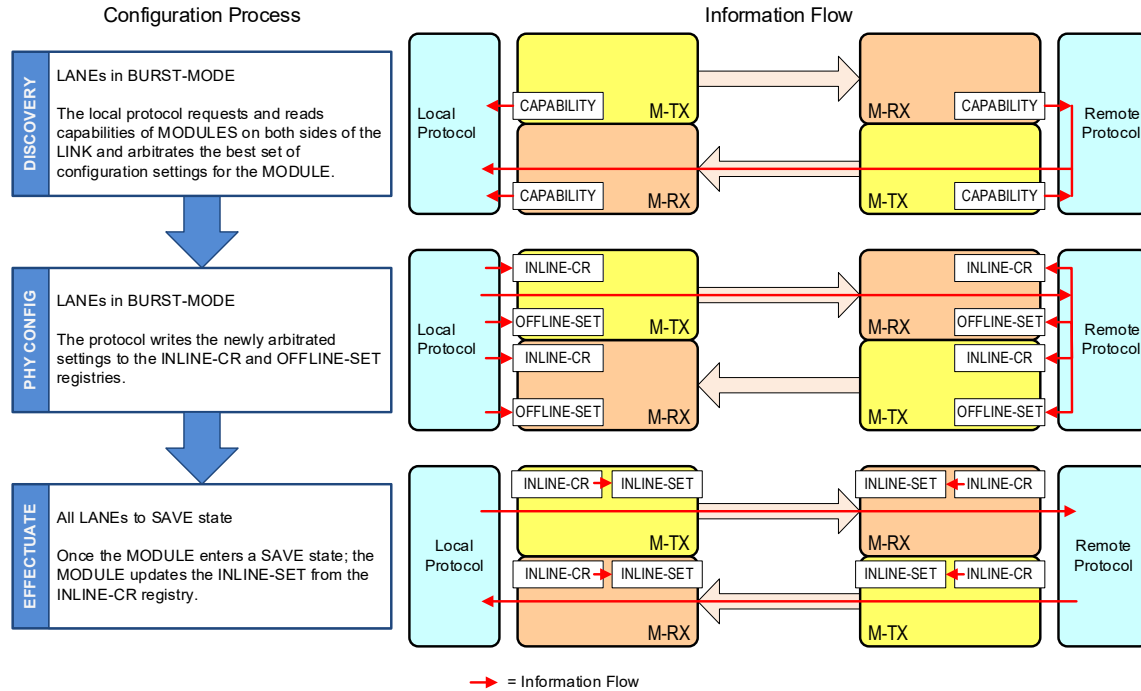
The nature of a configuration setting, whether it is INLINE or OFFLINE, depends on the mode of operation.

Each MODULE within a SUB-LINK may have its own set of registries, or an M-PORT implementation may combine common configuration settings, depending on the specific LINK composition. The implementation of the anticipated registries is outside the scope of this document.

The configuration process is detailed in the following sections.

#### 4.8.1.1 Configuration

**Figure 18** illustrates the information flow for a dual-simplex LINK, along with the steps of the configuration process for that LINK.



**Figure 18 Configuration Steps for LANE**

#### 4.8.2 Configuration Parameters

Configuration attributes for MODULEs are listed in *Section 7.4*.



## 4.9 Multiple LANE Provisions

This document governs individual LANES for a LINK. However, the LANE composition of a LINK is not specified by this document. This section specifies the provisions and constraints for multi-LANE SUB-LINK operation. This enables a multitude of possible LANE compositions for LINKs. The fine selection of allowed LANE combinations is left to the protocols on top of the Physical Layer.

There shall be no (tight) PHY-level requirements on timing alignment between SUB-LINKs.

The LANES of a SUB-LINK consisting of multiple LANES may each individually be in ACTIVATED, DISABLED or HIBERN8.

Independent activation of multiple LANES in a SUB-LINK is optional. If independent activation of multiple LANES in a SUB-LINK is supported, each MODULE shall satisfy all signaling requirements across multiple LANES used for a SUB-LINK. If independent activation of multiple LANES in a SUB-LINK is not supported, the number of LANES in BURST in a SUB-LINK shall only be changed while all LANES are either in a SAVE state or DISABLED.

Individual LANES of a SUB-LINK should only be DISABLED during initialization after power-up. When ACTIVATED, LANES of a SUB-LINK shall be in the same MODE at the same RATE. In HS-MODE, both SUB-LINKs shall use the same HS RATE series (A or B). SUB-LINKs may be operated in different GEARS. SUB-LINKs can be operated in different modes. SUB-LINKs may contain different numbers of LANES. Entry and exit of HIBERN8 of a SUB-LINK is correlated with the state of the SUB-LINK in the opposite direction. A SUB-LINK is considered to be in HIBERN8 when all its LANES are in HIBERN8 or DISABLED. LANE exit and entry of HIBERN8 are initiated from the M-TX side and controlled by the local protocols. The protocols on both sides control LANES such that both SUB-LINKs enter and exit HIBERN8 more, or less, simultaneously or shortly after each other. The detection means of the M-RX are just triggers for the further protocol action. At least one LANE of a TYPE-I SUB-LINK needs to remain enabled after power up.

When the requirement of both SUB-LINKs using the same HS RATE series is violated, proper operation of the LINK is not guaranteed and the behavior is undefined. To resume to normal operation, LINE-RESET on both SUB-LINKs or M-PHY reset at both peers shall be issued.

This document does not require functional symmetry of M-TXs and M-RXs for the SUB-LINKs of a LINK.

The allocation of PAYLOAD data over multiple LANES is left to the protocol specifications.

## 4.10 Test Modes

Test modes are special modes of operation which shall not happen during normal operation of a MODULE, which are intended to facilitate electrical, functional and protocol related tests. However, most tests can and should be executed using the normal operating modes. Protocols shall support generation of both burst and continuous mode signaling for the purposes of conformance testing. This may be accomplished via loopback, or dedicated test modes, depending on the application. See *Annex B* for further information.

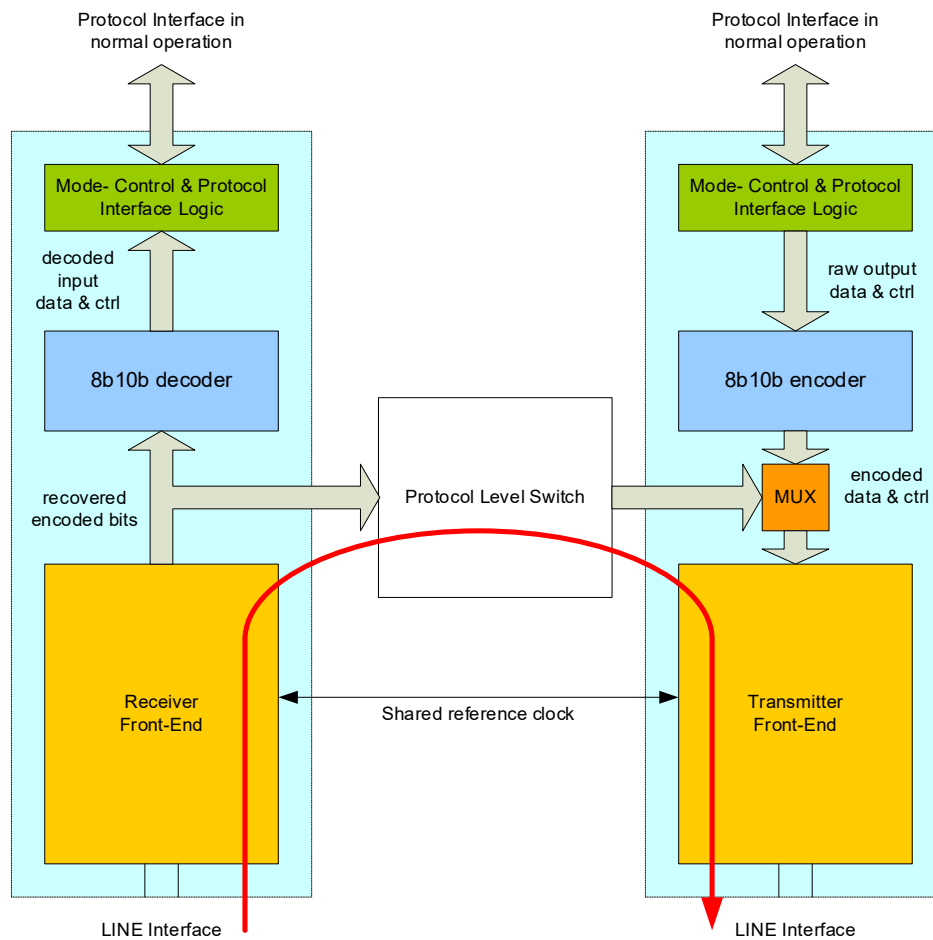
### 4.10.1 LOOPBACK Mode

LOOPBACK mode provides a transparent bit-by-bit path from an M-RX input to an M-TX output. This can be done only for commonly supported MODE and GEAR settings for the involved M-RX and M-TX. If multiple M-RXs or M-TXs are present in a complete LINK, the mapping of which M-RX is looped via which M-TX is either specified by the applicable protocol specification or is otherwise left to the implementor. The Physical Layer is set into LOOPBACK mode via configuration.

LOOPBACK retransmits via the M-TX the encoded LINE data as recovered by the M-RX without decoding (and re-encoding) the 8b10b symbols. The configured setup in the mode is illustrated in *Figure 19*. Bypassing the coders avoids bit error multiplication. For any mandatory test condition, the input data provided to the M-RX shall be 8b10b encoded. Furthermore, an implementation should use symbol streams with characteristics similar to what happens in the real application. LOOPBACK mode can for example be used for *BER* testing.

Although this mode allows a test setup to inject a non-8b10b encoded bit stream for experimental purposes, there shall not be mandatory requirements on the functionality or performance of the Physical Layer in this case. Because HS-MODE utilizes embedded-clock data recovery, it is essential that any input bit stream in HS-LOOPBACK contains sufficient edge density.

For LOOPBACK the RATES of M-RX and M-TX shall be identical, even though the MODULEs might be able to operate plesiochronously during normal operation. Note that this test mode is suitable to monitor the internal recovered bitstream of the M-RX on the outside via the M-TX, but not to characterize the M-TX performance.



835

**Figure 19 LOOPBACK Configuration**

## 4.11 Eye Monitor

The Eye Monitor function enables the system to get information about the receiver margin after the equalizer. This function is optional, and is intended to improve testability and debuggability for high speed transaction. The Eye Monitor function should be activated during HS-Burst operation, and shall be done after ADAPT. The Eye Monitor function path is non-destructive to the normal data path, and doesn't affect the main data even when the measurement is active.

Eye Margin is confirmed by whether there's an error between main data and Eye Monitor data sampled with voltage and/or timing offsets relative to the main data sampling point. During Eye Monitor measurement operation, the main data sampling point shall be taken properly, and assumed to be error free. When the receiver supports the Eye Monitor function, the EYEMON Capability Attributes in **Table 45** are set in advance, and the test offset configurations (i.e., EYEMON Configuration Attributes) in **Table 46** are set accordingly.

The Eye Monitor function starts by activating RX\_EYEMON\_Enable at RCT. M-RX shall not use the EYEMON configuration attribute while RX\_EYEMON\_Enable is disabled. All offset and count settings are sent before RX\_EYEMON\_Enable is asserted. The Eye Monitor measurement is performed while the error count remains below 63 and the test count remains below the configuration set by the system. Once the error count reaches the maximum, or if the test count reaches RX\_EYEMON\_Target\_Test\_Count as set by the system, then the receiver stops the measurement and clears RX\_EYEMON\_Start. The system should watch RX\_EYEMON\_Start by polling and reading the results after the measurement. Once the measurement is completed, RX\_EYEMON\_Enable is cleared by RCT and the Eye Monitor function is disabled. For more detail regarding the setting sequence, see **Annex G**.

The RX\_EYEMON Configuration Attributes are shown in **Table 46**.

### 4.11.1 RX\_EYEMON\_Start Control

RX\_EYEMON\_Start is activated at the same time that RX\_EYEMON\_Enable is activated, at RCT.

RX\_EYEMON\_Start is disabled in the following two cases:

- If RX\_EYEMON\_Test\_Count reaches or exceeds RX\_EYEMON\_Target\_Test\_Count. In this case, the PHY receiver should disable the RX\_EYEMON\_Start signal.
- If RX\_EYEMON\_Target\_Test\_Count = 0 (unlimited comparing case). In this case, RX\_EYEMON\_Start is kept high while RX\_EYEMON\_Enable = 1. When RX\_EYEMON\_Enable is set to 0 at RCT, RX\_EYEMON\_Start is cleared.

### 4.11.2 Clear Results

Eye Monitor results are cleared before starting Eye Monitor measurement, either at the main FSM LINE-RESET or by a system reset.

The Eye Monitor result data shall not be cleared by an unexpected state transition alone.

## 5 Electrical Characteristics

This section defines the electrical and low-level timing characteristics of M-TXs and M-RXs. The definitions of the common MODULE characteristics are followed by specific characteristics for HS-MODE, PWM-MODE, and SYS-MODE operation. Finally, this section specifies the general PIN characteristics for a MODULE.

The definitions within this section refer to a MODULE in certain MODEs, which are referred to as FUNCTIONs. The FUNCTIONs are listed with their abbreviations in *Table 13*.

**Table 13 FUNCTIONs and their Abbreviations**

| Abbreviation | FUNCTION         |
|--------------|------------------|
| HS-TX        | M-TX in HS-MODE  |
| PWM-TX       | M-TX in PWM-MODE |
| SYS-TX       | M-TX in SYS-MODE |
| HS-RX        | M-RX in HS-MODE  |
| PWM-RX       | M-RX in PWM-MODE |
| SYS-RX       | M-RX in SYS-MODE |
| SQ-RX        | M-RX in squelch  |

The names of the FUNCTIONs correspond with the operational states of the M-TXs and M-RXs as specified in *Section 4.6.3*. A MODULE does not need to support all FUNCTIONs, only those required for the intended application. FUNCTIONs required for an M-TX or an M-RX implementation are defined in *Section 4.4*, *Section 4.6*, *Section 4.7* and higher level protocol standards. Also, the high level timing of the FUNCTIONs and their operation are defined in *Section 4*.

The electrical and timing characteristics of the M-TX and the M-RX are defined at the PINs of an IC. Only MODULE characteristics that are observable at the PINs are subject to specification. These characteristics shall meet their specifications for any supported FUNCTION.

This specification is intended to be implementation agnostic. The section structure, which is based on FUNCTIONs, does not preclude integrated driver or receiver implementations. Although some figures in this section may suggest a certain driver or receiver implementation, they are used only for illustration purposes.

## 5.1 M-TX Characteristics

This document distinguishes three different operating modes and corresponding FUNCTIONS. Following the definition of the common M-TX electrical and timing characteristics, additional characteristics specific to HS-TX, PWM-TX, and SYS-TX are defined in this section.

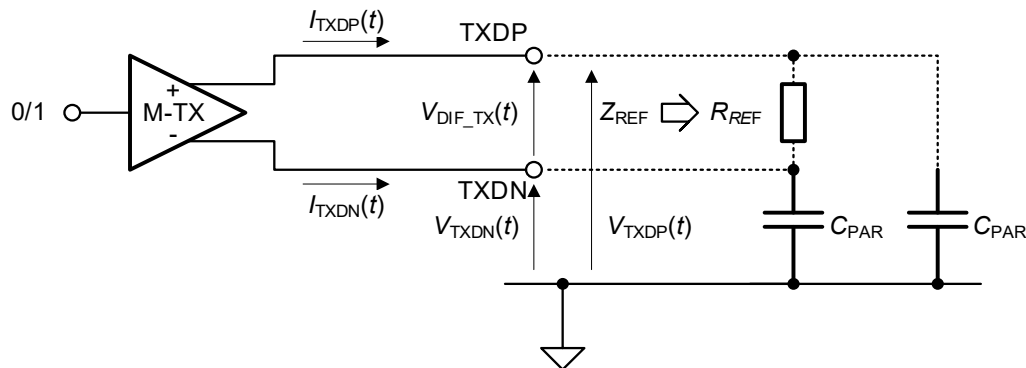
### 5.1.1 Common M-TX Characteristics

The common electrical and timing characteristics of an M-TX are defined in this section, which also contains the PIN and signal definitions. The common M-TX characteristics apply to the HS-TX, PWM-TX, and SYS-TX FUNCTIONS.

#### 5.1.1.1 PIN, Signal, and Reference Characteristic Definitions

An M-TX drives a low-voltage differential output signal at the PINs TXDP and TXDN either into a terminated, or an unterminated, load. TXDP and TXDN are defined as the positive and negative output PINs, respectively.

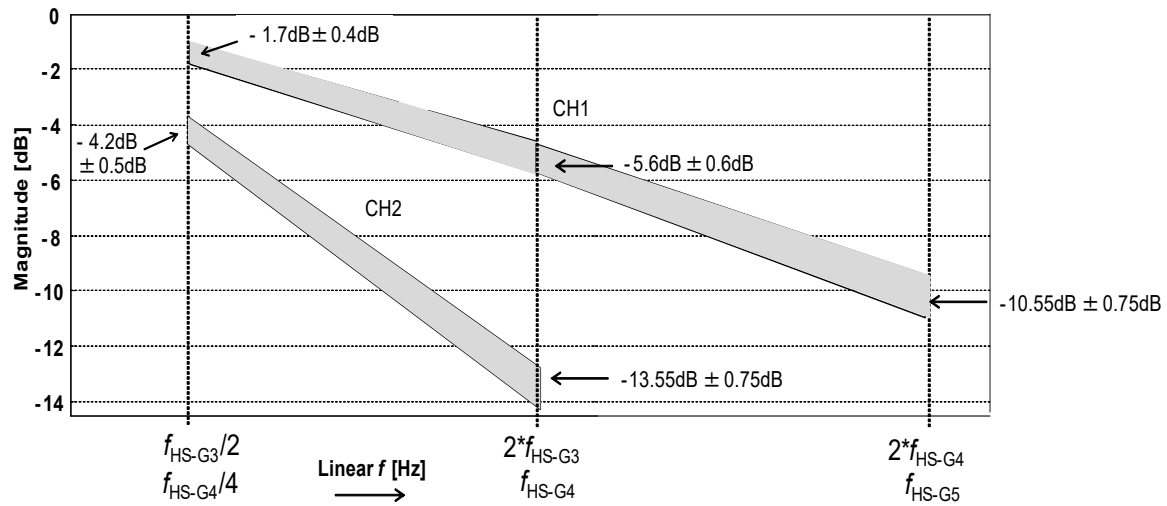
The PIN voltages and currents, as well as the reference load  $R_{REF}$  are shown in **Figure 20**.  $R_{REF\_RT}$  and  $R_{REF\_NT}$  are defined as reference loads for when the M-TX is terminated and not terminated, respectively.



**Figure 20 M TX PIN Voltages, PIN Currents, and Reference Loads**

Reference Channel CH1 is defined for operation in HS-G3, HS-G4 and HS-G5. Reference Channel CH2 is defined for operation in HS-G3 and HS-G4. The reference channels are defined by the channel insertion loss  $SDD_{IL\_REF\_CH}$ , the return loss  $SDD_{RL\_REF\_CH}$ , and the channel differential impedance  $R_{DIF\_REF\_CH}$  when terminated with  $R_{REF\_RT}$ . The maximum single-ended DC channel resistance is  $R_{DC\_REF\_CH}$ . The  $SDD_{IL\_REF\_CH}$  templates are shown in **Figure 21**. An M-PHY operating in HS-G3 and HS-G4 shall demonstrate TX eye opening conformance with Reference Channels CH1 or CH2. An M-PHY operating in HS-G5 shall demonstrate TX eye opening conformance only with Reference Channel CH1. The reference channels do not represent an actual LINK channel. The reference channel insertion loss template does not scale for HS-G4 and HS-G5. The reference channel definition is independent of HS-Gear.

A reference package model is defined for operation in HS-G4 and HS-G5 to represent an interconnect extension that affects the signal quality beyond the reference channels. The reference package model, with the addition of pad capacitance for HS-G4 or effective pad capacitance network model for HS-G5, shall conform to the HS-TX and HS-RX return loss defined in **Section 5.1.1.4** and **Section 5.2.1.5**.



**Figure 21 HS-G3, HS-G4, and HS-G5 Reference Channel Insertion Loss SDDIL\_REF\_CH Templates**

$V_{TXDP}(t)$  and  $V_{TXDN}(t)$  are defined as the signal voltages at TXDP and TXDN with respect to ground.  $V_{TXDP}$  and  $V_{TXDN}$  are defined as the voltage amplitudes of the  $V_{TXDP}(t)$  and  $V_{TXDN}(t)$  signals, respectively.

$I_{TXDP}(t)$  and  $I_{TXDN}(t)$  are defined as the output currents flowing out of TXDP and TXDN, respectively.  $I_{TXDP}$  and  $I_{TXDN}$  are defined as the current amplitudes of the  $I_{TXDP}(t)$  and  $I_{TXDN}(t)$  signals, respectively.

$Z_{REF}$  is the impedance of the reference load  $R_{REF}$ , which is bounded by the return loss  $SRL_{REF}$ .  $C_{PAR}$  illustrates parasitic capacitance that contributes to  $Z_{REF}$ ;  $C_{PAR}$  is not specified.  $Z_{REF\_RT}$  is defined as the complex impedance of  $R_{REF\_RT}$  and represents the AC reference load limit in the terminated state.  $SRL_{REF\_RT}$  is defined as the return loss of  $Z_{REF\_RT}$  and can be calculated using **Equation 1**.

$$SRL_{REF\_RT} = -20 \log \left| \frac{Z_{REF\_RT} + Z_R}{Z_{REF\_RT} - Z_R} \right| \quad (\text{Equation 1})$$

where  $Z_R$  is a defined reference impedance.  $SRL_{REF\_RT}$  is defined having a minimum value greater than  $SRL_{REF\_RT}[\text{MIN}]$  for all frequencies from 0 Hz up to  $f_{HS\_MAX}$  (see **Figure 22**).

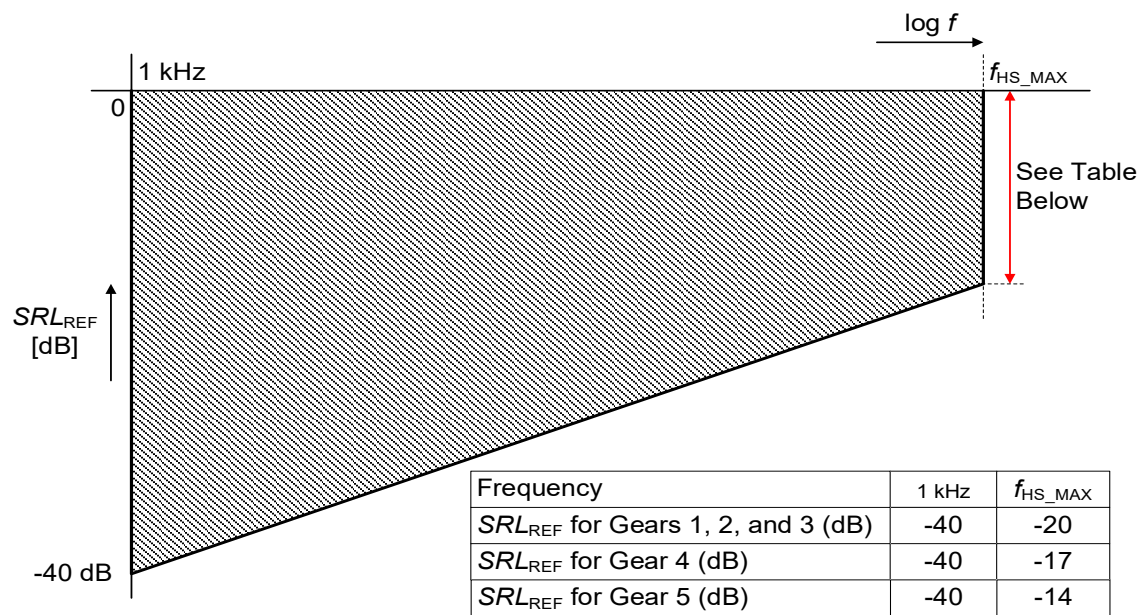


Figure 22 Template for Reference Return Loss

The HS frequency,  $f_{HS}$ , is half the HS data rate,  $DR_{HS}$ . Other characteristic frequencies during operation in HS-MODE are the maximum and minimum frequencies,  $f_{HS\_MAX}$  and  $f_{HS\_MIN}$ , respectively.  $f_{HS\_MAX}$  and  $f_{HS\_MIN}$  are used in the S-parameter templates. All these frequencies are defined as fractions of  $DR_{HS}$  as shown by the following equations:

$$f_{HS} = \frac{DR_{HS}}{2} \tag{Equation 2}$$

$$f_{HS\_MAX} = \frac{3 \times DR_{HS}}{4} \tag{Equation 3}$$

$$f_{HS\_MIN} = \frac{DR_{HS}}{10} \tag{Equation 4}$$



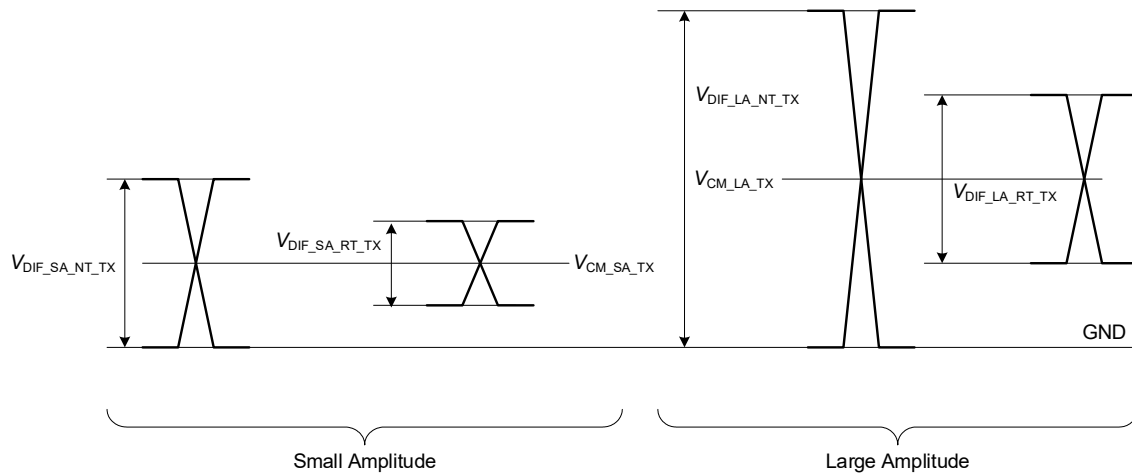
An M-TX drives a differential low-swing signal with either Large Amplitude or Small Amplitude. Small Amplitude should be used only for HS-G4 and below. The amplitude of the differential output signal is doubled when the M-TX drives an unterminated load compared to when it drives a terminated load. Differential output signals with large and small amplitudes for the terminated and unterminated states are shown in **Figure 23**. All single-ended voltage levels are relative to the ground voltage at the M-TX side.

The jitter of an HS-TX in HS-MODE is specified by means of a jitter transfer function with a corner frequency  $f_{C\_HS\_TX}$ . Jitter is integrated up to the upper transmitter cut-off frequency  $f_{U\_TX}$ . An additional lower cut-off frequency  $f_{STJ\_TX}$  is defined for the short term jitter of an HS-TX.

The jitter is defined for a BER of  $10^{-12}$  according to **[INC01]**. The mean ( $\mu$ ) of the distribution function is located at 0.

**Note:**

*The BER is changed from 10-10 to 10-12 in M-PHY v4.1. Other jitter definitions remain unchanged.  
The change in BER is tightening the M-TX jitter requirements.*



**Figure 23 M-TX Signal Levels**

943 The reference parameters for the M-TX are summarized in *Table 14*.

**Table 14 M-TX and HS-TX Reference Parameters**

| Symbol              | Values |                          |            | Unit       | Description                                                             |
|---------------------|--------|--------------------------|------------|------------|-------------------------------------------------------------------------|
|                     | Min.   | Nom.                     | Max.       |            |                                                                         |
| Reference Load      |        |                          |            |            |                                                                         |
| $R_{REF\_RT}$       | —      | 100                      |            | $\Omega$   | Reference load for when the M-TX is terminated.                         |
| $R_{REF\_NT}$       | 10     | —                        | —          | k $\Omega$ | Reference load for when the M-TX is not terminated.                     |
| $Z_R$               | —      | 100                      | —          | $\Omega$   | Reference impedance.                                                    |
| Reference Channel   |        |                          |            |            |                                                                         |
| $R_{DC\_REF\_CH}$   | —      | —                        | 4          | $\Omega$   | Reference channel single-ended DC resistance.                           |
| $SDD_{RL\_REF\_CH}$ | —      | —                        | -15        | dB         | Reference channel return loss limit from DC up to 12 GHz.               |
| $R_{DIF\_REF\_CH}$  | —      | 100                      | —          | $\Omega$   | Reference channel differential impedance.                               |
| Frequency           |        |                          |            |            |                                                                         |
| $f_{C\_HS\_G1\_TX}$ | —      | 2.0                      | —          | MHz        | Corner frequency of clock and data recovery transfer function in HS-G1. |
| $f_{C\_HS\_G2\_TX}$ | —      | 4.0                      | —          | MHz        | Corner frequency of clock and data recovery transfer function in HS-G2. |
| $f_{C\_HS\_G3\_TX}$ | —      | 8.0                      | —          | MHz        | Corner frequency of clock and data recovery transfer function in HS-G3. |
| $f_{C\_HS\_G4\_TX}$ | —      | 8.0                      | —          | MHz        | Corner frequency of clock and data recovery transfer function in HS-G4. |
| $f_{C\_HS\_G5\_TX}$ | —      | 16.0                     | —          | MHz        | Corner frequency of clock and data recovery transfer function in HS-G5. |
| $f_{U\_TX}$         | —      | $\frac{1}{2U_{I_{HS}}}$  | —          | Hz         | Upper frequency of jitter transfer function.                            |
| $f_{STJ\_TX}$       | —      | $\frac{1}{30U_{I_{HS}}}$ | —          | Hz         | Lower bound of short term jitter.                                       |
| $\zeta$             | —      | 0.707                    | —          | N.A.       | Damping ratio of jitter transfer function.                              |
| Limit for BER       |        |                          |            |            |                                                                         |
| $Q_{BER}$           | —      | 7.0345                   | —          | —          | Q-factor for a BER of $10^{-12}$                                        |
| BER                 | —      | —                        | $10^{-12}$ | —          | Target BER                                                              |

### 5.1.1.2 Differential and Common-mode Voltage

An M-TX drives a differential signal on the TXDP and TXDN PINs. The differential output voltage signal  $V_{\text{DIF\_TX}}(t)$  is defined as the difference of the voltage signals  $V_{\text{TXDP}}(t)$  and  $V_{\text{TXDN}}(t)$ .  $V_{\text{DIF\_TX}}$  is defined as the amplitude of  $V_{\text{DIF\_TX}}(t)$ .  $V_{\text{DIF\_TX}}(t)$  can be calculated from the following equation:

$$V_{\text{DIF\_TX}}(t) = V_{\text{TXDP}}(t) - V_{\text{TXDN}}(t) \quad (\text{Equation 5})$$

Separate AC and DC parameters are defined for  $V_{\text{DIF\_TX}}$ . The DC parameter  $V_{\text{DIF\_DC\_TX}}$  is defined for an M-TX which drives a steady DIF-N or a steady DIF-P LINE state into a reference load  $R_{\text{REF\_RT}}$  or  $R_{\text{REF\_NT}}$ . An M-TX shall drive a differential DC output voltage amplitude which meets the specified limits of  $V_{\text{DIF\_DC\_TX}}$ . When the differential DC output voltage amplitude remains within the specified limits of  $V_{\text{DIF\_DC\_TX}}$ , the LINE has settled.

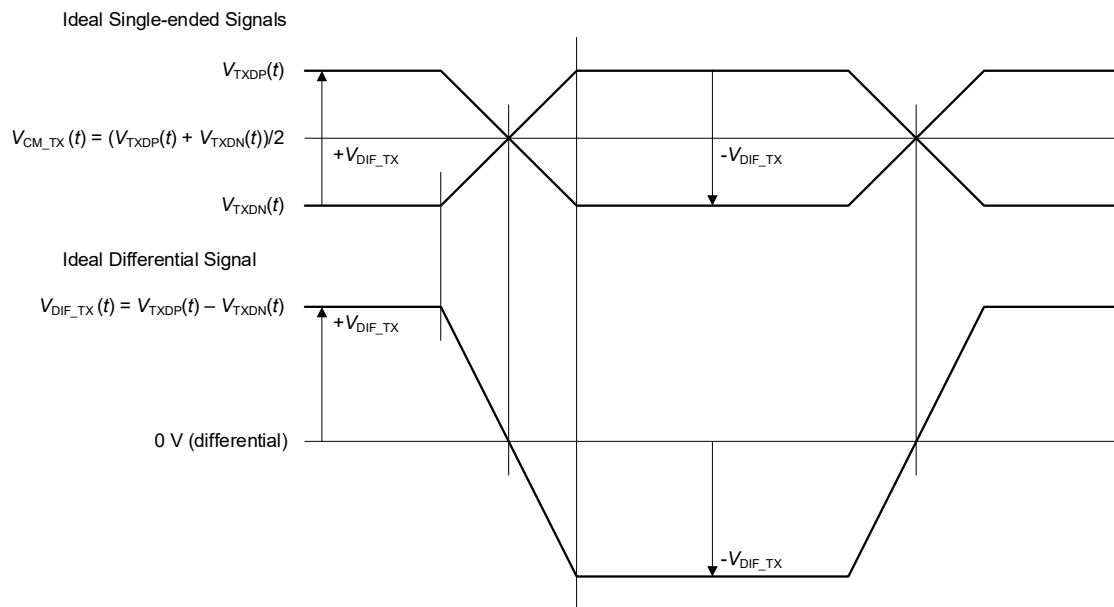
The AC parameter  $V_{\text{DIF\_AC\_TX}}$  is defined for an M-TX which drives a test pattern into a reference load  $R_{\text{REF\_RT}}$  or  $R_{\text{REF\_NT}}$ . For an HS-TX the lower limit of  $V_{\text{DIF\_AC\_TX}}$  is defined over the eye opening  $T_{\text{EYE\_TX}}$  as defined in **Section 5.1.2.9**. The upper limit of  $V_{\text{DIF\_AC\_TX}}$  is defined as the maximum differential output voltage, when the M-TX drives a test pattern into a reference load  $R_{\text{REF\_RT}}$  or  $R_{\text{REF\_NT}}$ . An M-TX shall drive a differential AC output voltage signal which meets the specified limits of  $V_{\text{DIF\_AC\_TX}}$ .

There is no definition for how long the lower limit of  $V_{\text{DIF\_AC\_TX}}$  has to be met for a PWM-TX or a SYS-TX.

The common-mode output voltage signal  $V_{\text{CM\_TX}}(t)$  is defined as the arithmetic mean value of the signal voltages  $V_{\text{TXDP}}(t)$  and  $V_{\text{TXDN}}(t)$  when the M-TX drives a test pattern into a reference load  $R_{\text{REF\_RT}}$  or  $R_{\text{REF\_NT}}$ .  $V_{\text{CM\_TX}}$  is defined as the amplitude of  $V_{\text{CM\_TX}}(t)$ .  $V_{\text{CM\_TX}}(t)$  can be calculated from the following equation:

$$V_{\text{CM\_TX}}(t) = \frac{V_{\text{TXDP}}(t) + V_{\text{TXDN}}(t)}{2} \quad (\text{Equation 6})$$

- 963 An M-TX shall drive a common-mode output voltage signal which meets the specified limits of  $V_{CM\_TX}$ .
- 964  $V_{DIF\_TX}(t)$  and  $V_{CM\_TX}(t)$  for ideal single-ended output signals  $V_{TXDP}(t)$  and  $V_{TXDN}(t)$  are shown in **Figure 24**.



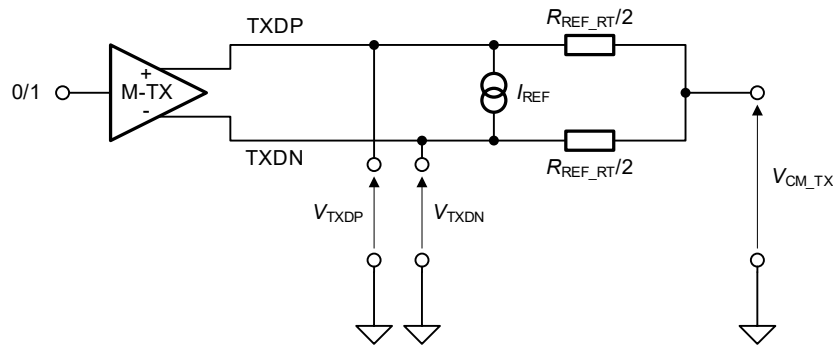
**Figure 24 Ideal Single-ended and Differential Signals**

### 5.1.1.3 Single-ended Output Resistance

The resistance  $R_{SE\_TX}$  is defined as the single-ended output resistance of an M-TX at both its TXDP and TXDN PINs.  $R_{SE\_TX}$  is defined for the case of a terminated M-TX that drives either a DIF-P or DIF-N LINE state with a reference load  $R_{REF\_RT}$  and a current source  $I$  connected between TXDP and TXDN as shown in **Figure 25**. A change of the current  $I$  results in a change of the PIN signal voltages  $V_{TXDP}$  and  $V_{TXDN}$ .

$I_{REF}$  is defined as the value of the current source  $I$  that causes a variation of  $V_{TXDP}$  and  $V_{TXDN}$  by  $\pm 25$  mV.

The single-ended output resistance shall conform with the specification limits of  $R_{SE\_TX}$  for both the DIF-N and DIF-P state. An implementation should keep the output resistance during state transitions close to the steady state output resistance.



**Figure 25 Measurement Setup for Single ended Output Resistance**

In **Equation 7** through **Equation 10**,  $\Delta V_{TXDP}$ ,  $\Delta V_{TXDN}$  and  $\Delta V_{CM\_TX}$  are defined as the voltages of the signals at the test points shown in **Figure 25** at two distinct times,  $t_1$  and  $t_2$ , where  $t_2 > t_1$ , such that  $\Delta V = V(t_1) - V(t_2)$ . The current source  $I$  sources  $I_{REF}$  and  $-I_{REF}$  at  $t_1$  and  $t_2$ , respectively.

The single-ended output resistance  $R_{SE\_TX}$  at TXDP can be calculated using the following equation:

$$R_{SE\_TX}(TXDP) = \frac{DV_{TXDP}}{-2I_{REF} - \frac{DV_{TXDP} - DV_{CM\_TX}}{R_{REF\_RT}/2}} \quad (\text{Equation 7})$$

Similarly, the single-ended output resistance  $R_{SE\_TX}$  at TXDN can be calculated using the following equation:

$$R_{SE\_TX}(TXDN) = \frac{DV_{TXDN}}{2I_{REF} - \frac{DV_{TXDN} - DV_{CM\_TX}}{R_{REF\_RT}/2}} \quad (\text{Equation 8})$$

$R_{SE\_PO\_TX}$  is defined as the single-ended output resistance of an M-TX in a STALL or SLEEP state at both the TXDP and TXDN PINs.  $R_{SE\_PO\_TX}$  is defined for a terminated M-TX, which drives either a DIF-N or a DIF-P LINE state, when a reference load  $R_{REF\_RT}$  is connected between TXDP and TXDN. If the optional  $R_{SE\_PO\_TX}$  is utilized, the single-ended output resistance of an M-TX in the STALL or SLEEP states shall conform with the specified limit of  $R_{SE\_PO\_TX}$ .

$V_{CM\_TX}$  and  $V_{DIF\_TX}$  shall stay in their specified limits during switching between  $R_{SE\_TX}$  and  $R_{SE\_PO\_TX}$ .  $R_{SE\_PO\_TX}$  is an optional feature of an M-TX, which is defined to allow for power optimization in the STALL and SLEEP states.

990  $R_{SE\_PO\_TX}$  is defined according to  $R_{SE\_TX}$ . Using the parameters of the  $R_{SE\_TX}$  definition, the single-ended  
 991 output resistance  $R_{SE\_PO\_TX}$  at TXDP can be calculated using the following equation:

$$R_{SE\_PO\_TX}(TXDP) = \frac{DV_{TXDP}}{-2I_{REF} - \frac{DV_{TXDP} - DV_{CM\_TX}}{R_{REF\_RT}/2}} \quad (Equation 9)$$

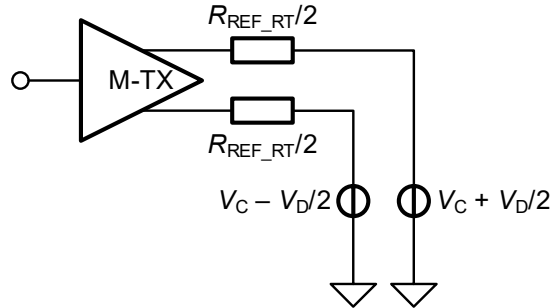
993 Similarly, the single-ended output resistance  $R_{SE\_PO\_TX}$  at TXDN can be calculated from the following  
 994 equation:

$$R_{SE\_PO\_TX}(TXDN) = \frac{DV_{TXDN}}{2I_{REF} - \frac{DV_{TXDN} - DV_{CM\_TX}}{R_{REF\_RT}/2}} \quad (Equation 10)$$

#### 5.1.1.4 Return Loss

The M-TX return loss parameters are based on a mixed-mode S-parameter matrix. The single ended S-parameters are characterized using the reference impedance  $R_{REF\_RT}/2$ .

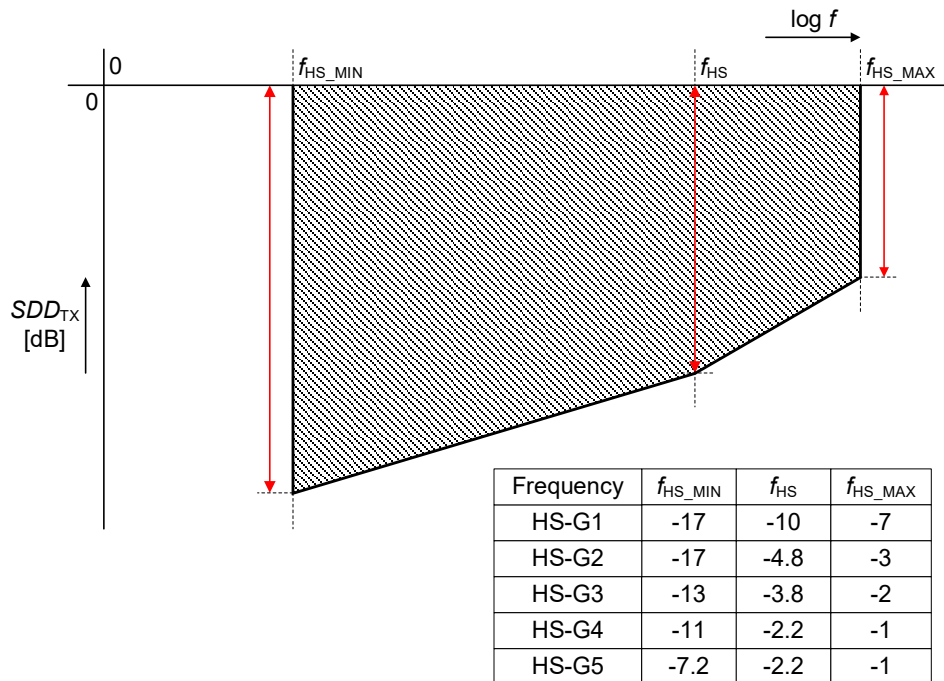
The characterization can be done with a setup as illustrated in **Figure 26**. In the figure,  $V_C$  denotes the common-mode voltage and  $V_D$  denotes the differential voltage.



**Figure 26 Measurement Setup for M-TX Return Loss**

The common-mode transmitter return loss,  $SCC_{TX}$ , and the differential transmitter return loss,  $SDD_{TX}$ , are defined for an M-TX transmitting a repetitive CRPAT into a reference load  $R_{REF\_RT}/4$  for  $SCC_{TX}$ , and  $R_{REF\_RT}$  for  $SDD_{TX}$ .  $SDD_{TX}$  and  $SCC_{TX}$  parameters are considered to be informative. When an M-TX supports Large Amplitude and Small Amplitude its  $SCC_{TX}$  and  $SDD_{TX}$  should conform with the specification limits for both amplitudes.  $SCC_{TX}$  and  $SDD_{TX}$  are defined at the PINs such that they include contributions from the on-chip circuitry as well as from the package.

The  $SDD_{TX}$  template is shown in **Figure 27** along with the return loss at corner frequencies  $f_{HS\_MIN}$ ,  $f_{HS}$  and  $f_{HS\_MAX}$ .  $SCC_{TX}$  is defined for frequencies up to  $f_{HS\_MAX}$ . An M-TX should fulfill both the common-mode transmitter return loss  $SCC_{TX}$  and the differential transmitter return loss  $SDD_{TX}$  specification limits.



**Figure 27 Template for Differential Transmitter Return Loss  $SDD_{TX}$**

**5.1.1.5 LINE Disturbance during M-TX Power-up**

1011 An M-TX in a Type-I LINK shall not cause a LINE condition upon the transition from the UNPOWERED  
1012 to a POWERED state which can be detected as a non-squelch state by the SQ-RX of the LANE. The allowed  
1013 LINE disturbance upon such a transition of an M-TX is hence restricted by the squelch pulse rejection as  
1014 defined in *Section 5.2.5*.

1015 Squelch detection is not used in a Type-II LINK. Hence, there is no restriction of the LINE disturbance caused  
1016 by an M-TX upon the transition from the UNPOWERED state to a POWERED state in such a LINK.



### 5.1.1.6 Common M-TX Parameters

1017 The common electrical and timing parameters of an M-TX are listed in *Table 15*.

**Table 15 Common M-TX Parameters**

| Symbol                                 | Values |      | Unit | Description                                                                                                                                                                                   |
|----------------------------------------|--------|------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | Min.   | Max. |      |                                                                                                                                                                                               |
| M-TX Electrical                        |        |      |      |                                                                                                                                                                                               |
| $V_{DIF\_DC\_LA\_RT\_TX}$              | 240    | 300  | mV   | Large Amplitude differential TX DC voltage when the M-TX is terminated <sup>8</sup> . Defined for $R_{REF\_RT}$ <sup>1</sup> and test pattern <sup>2</sup> . See <b>Section 5.1.1.2</b> .     |
| $V_{DIF\_AC\_LA\_RT\_TX}$ <sup>3</sup> | 220    | 310  | mV   | Large Amplitude differential TX AC voltage when the M-TX is terminated <sup>8</sup> . Defined for $R_{REF\_RT}$ <sup>1</sup> and CRPAT <sup>4</sup> . See <b>Section 5.1.1.2</b> .            |
| $V_{DIF\_DC\_LA\_NT\_TX}$              | 480    | 600  | mV   | Large Amplitude differential TX DC voltage when the M-TX is not terminated <sup>8</sup> . Defined for $R_{REF\_NT}$ <sup>5</sup> and test pattern <sup>2</sup> . See <b>Section 5.1.1.2</b> . |
| $V_{DIF\_AC\_LA\_NT\_TX}$ <sup>3</sup> | 440    | 620  | mV   | Large Amplitude differential TX AC voltage when the M-TX is not terminated <sup>8</sup> . Defined for $R_{REF\_NT}$ <sup>5</sup> and CRPAT <sup>4</sup> . See <b>Section 5.1.1.2</b> .        |
| $V_{DIF\_DC\_SA\_RT\_TX}$              | 100    | 190  | mV   | Small Amplitude differential TX DC voltage when the M-TX is terminated <sup>8</sup> . Defined for $R_{REF\_RT}$ <sup>1</sup> and test pattern <sup>2</sup> . See <b>Section 5.1.1.2</b> .     |
| $V_{DIF\_AC\_SA\_RT\_TX}$ <sup>3</sup> | 80     | 200  | mV   | Small Amplitude differential TX AC voltage when the M-TX is terminated <sup>8</sup> . Defined for $R_{REF\_RT}$ <sup>1</sup> and CRPAT <sup>4</sup> . See <b>Section 5.1.1.2</b> .            |
| $V_{DIF\_DC\_SA\_NT\_TX}$              | 200    | 380  | mV   | Small Amplitude differential TX DC voltage when the M-TX is not terminated <sup>8</sup> . Defined for $R_{REF\_NT}$ <sup>5</sup> and test pattern <sup>2</sup> . See <b>Section 5.1.1.2</b> . |
| $V_{DIF\_AC\_SA\_NT\_TX}$ <sup>3</sup> | 160    | 400  | mV   | Small Amplitude differential TX AC voltage when the M-TX is not terminated <sup>8</sup> . Defined for $R_{REF\_NT}$ <sup>5</sup> and CRPAT <sup>4</sup> . See <b>Section 5.1.1.2</b> .        |
| $V_{CM\_LA\_TX}$                       | 240    | 310  | mV   | Large Amplitude common-mode TX voltage <sup>8</sup> . Defined for $R_{REF\_RT}$ <sup>1</sup> or $R_{REF\_NT}$ <sup>5</sup> and CRPAT <sup>6</sup> . See <b>Section 5.1.1.2</b> .              |
| $V_{CM\_SA\_TX}$                       | 80     | 240  | mV   | Small Amplitude common-mode TX voltage <sup>8</sup> . Defined for $R_{REF\_RT}$ <sup>1</sup> or $R_{REF\_NT}$ <sup>5</sup> and CRPAT <sup>6</sup> . See <b>Section 5.1.1.2</b> .              |

| Symbol             | Values |      | Unit       | Description                                                                                                                                                            |
|--------------------|--------|------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | Min.   | Max. |            |                                                                                                                                                                        |
| M-TX Resistance    |        |      |            |                                                                                                                                                                        |
| $R_{SE\_TX}$       | 40     | 60   | $\Omega$   | Single-ended output resistance.<br>Defined for $R_{REF\_RT}$ <sup>1</sup> . See <b>Section 5.1.1.3</b> .                                                               |
| $R_{SE\_PO\_TX}$   |        | 10   | k $\Omega$ | Single-ended output resistance in STALL or SLEEP states.<br>Defined for $R_{REF\_RT}$ <sup>1</sup> . See <b>Section 5.1.1.3</b> .                                      |
| M-TX Return Loss   |        |      |            |                                                                                                                                                                        |
| $SCC_{HS\_G1\_TX}$ |        | -6.0 | dB         | Common-mode transmitter return loss.<br>Defined for $(R_{REF\_RT}/4)$ <sup>1</sup> up to $f_{HS\_MAX}$ and test pattern <sup>6</sup> .<br>See <b>Section 5.1.1.4</b> . |
| $SCC_{HS\_G2\_TX}$ |        | -4.0 | dB         | Common-mode transmitter return loss.<br>Defined for $(R_{REF\_RT}/4)$ <sup>1</sup> up to $f_{HS\_MAX}$ and test pattern <sup>6</sup> .<br>See <b>Section 5.1.1.4</b> . |
| $SCC_{HS\_G3\_TX}$ |        | -2.0 | dB         | Common-mode transmitter return loss.<br>Defined for $(R_{REF\_RT}/4)$ <sup>1</sup> up to $f_{HS\_MAX}$ and test pattern <sup>6</sup> .<br>See <b>Section 5.1.1.4</b> . |
| $SCC_{HS\_G4\_TX}$ |        | -2.0 | dB         | Common-mode transmitter return loss.<br>Defined for $(R_{REF\_RT}/4)$ <sup>1</sup> up to $f_{HS\_MAX}$ and test pattern <sup>6</sup> .<br>See <b>Section 5.1.1.4</b> . |
| $SCC_{HS\_G5\_TX}$ |        | -2.0 | dB         | Common-mode transmitter return loss.<br>Defined for $(R_{REF\_RT}/4)$ <sup>1</sup> up to $f_{HS\_MAX}$ and test pattern <sup>6</sup> .<br>See <b>Section 5.1.1.4</b> . |

**Note:**

1. External reference load  $R_{REF\_RT}$  and a reference impedance  $Z_{REF\_RT}$  that conform to  $SRL_{REF\_RT}$ .
2. Defined when driving both a DIF-N and a DIF-P LINE state.
3. The M-TX HS-G3, HS-G4 AC differential amplitude voltages are validated through eye-mask conformance at the end of a reference channel, CH1 or CH2, using a CRPAT test pattern. The M-TX HS-G5 AC differential amplitude voltages are validated through eye-mask conformance at the end of a reference channel, CH1, using a CRPAT test pattern. See **Section 5.1.2.9**.
4. Measurement based on accumulative eye diagram. Measurements are accomplished using the Compliant Random Pattern (CRPAT).
5. External reference load  $R_{REF\_NT}$  and capacitances at TXDP and at TXDN within the limit of  $C_{PIN\_RX}$ .
6. Defined for a repetitive CRPAT.
7. The listed parameters should be measured with de-emphasis turned off.
8.  $V_{CM\_TX}$  and  $V_{DIF\_TX}$  min values are increased for HS-G5 performance and are utilized for other HS-Gears to limit complexity of implementations.

### 5.1.2 HS-TX Characteristics

This section contains the electrical and timing characteristics specific to an HS-TX which are not covered by the common M-TX parameters in *Section 5.1.1*.

#### 5.1.2.1 Rise and Fall Times

The HS-TX rise and fall times,  $T_{R\_HS\_TX}$  and  $T_{F\_HS\_TX}$ , respectively, are defined as transition times between the 20% and 80% signal levels of the differential HS-TX output signal with an amplitude of  $V_{DIF\_DC\_TX}$ , when driving a pattern into a reference load,  $R_{REF\_RT}$ . The HS-TX rise and fall times are considered to be informative. The pattern used for measurement is a minimum of two  $UI_{HS}$  of DIF-P followed by a minimum of two  $UI_{HS}$  of DIF-N for  $T_{F\_HS\_TX}$ , and a minimum of two  $UI_{HS}$  of DIF-N, followed by a minimum of two  $UI_{HS}$  of DIF-P for  $T_{R\_HS\_TX}$ .

#### 5.1.2.2 Slew Rate

The slew rate  $SR_{DIF\_TX}$  is defined as the ratio  $\Delta V/\Delta T$ , where  $\Delta V$  is the absolute value of the voltage difference of the differential HS-TX output signal voltage measured at the 20% and 80% levels of  $V_{DIF\_DC\_SA\_RT\_TX}$  and  $\Delta T$  is the corresponding time difference when the HS-TX drives a reference load  $R_{REF\_RT}$  with Small Amplitude. The specification limits of  $SR_{DIF\_TX}$  shall be met by an HS-TX that supports slew rate control and which is operated in HS-G1.

The slew rate of the HS-TX should be controllable to allow for  $N$  different slew rate states.  $SR_{DIF\_TX}[1]$  and  $SR_{DIF\_TX}[N]$  denominate the slew rate for the fastest and for the slowest slew rate states, respectively. The number  $N$  is implementation-specific and is out of scope for this document. The slew rate states should cover a range defined by the maximum slew rate  $SR_{DIF\_TX}[MAX]$  and the minimum slew rate  $SR_{DIF\_TX}[MIN]$ . For at least one state the slew rate should be larger than  $SR_{DIF\_TX}[MAX]$ . For at least one state it should be smaller than  $SR_{DIF\_TX}[MIN]$ .

The slew rate shall be monotonically decreasing when stepping from faster to slower slew rate states, i.e.,  $SR_{DIF\_TX}[i]$  is larger than  $SR_{DIF\_TX}[i+1]$ , where  $i$  is in the range of 1 to  $N-1$ . It shall be monotonically increasing when stepping from slower to faster slew rate states. A given slew rate correspondence between setting and value is not intended to be specified, rather range and granularity are provided. The range of slew rate settings is intended to exceed the range of conformant slew rate values to allow control over common mode noise and EMI (see *Section 5.1.2.10.1*).

The resolution of the slew rate states  $\Delta SR_{DIF\_TX}$  is defined as the difference of the slew rates of two adjacent slew rate states divided by the slew rate of the slower state.

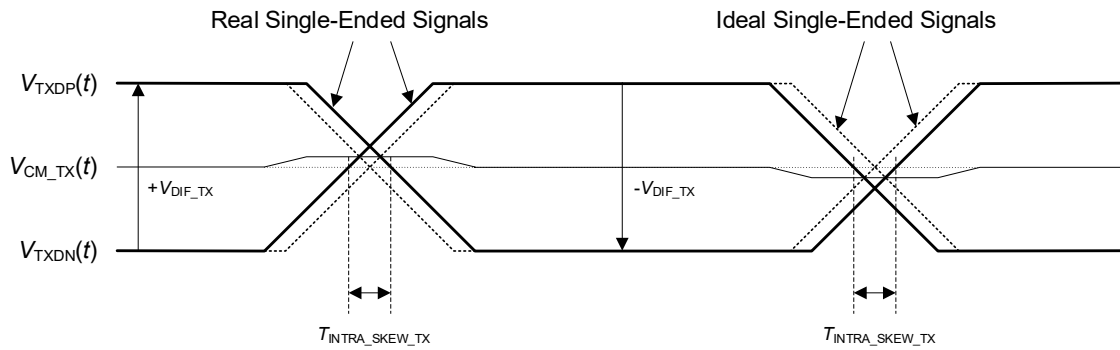
$\Delta SR_{DIF\_TX}$  can be calculated using the following equation:

$$DSR_{DIF\_TX} = \frac{SR_{DIF\_TX}[i] - SR_{DIF\_TX}[i+1]}{SR_{DIF\_TX}[i+1]} \quad (\text{Equation 11})$$

where  $SR_{DIF\_TX}[i+1]$  is the slew rate of the slower slew rate state and  $SR_{DIF\_TX}[i]$  is the slew rate of the adjacent faster slew rate state.  $\Delta SR_{DIF\_TX}$  shall be met between  $SR_{DIF\_TX}[1]$  and  $SR_{DIF\_TX}[N]$  (see *Table 16*).

### 5.1.2.3 Intra-LANE Output Skew

The transmitter intra-LANE output skew,  $T_{\text{INTRA\_SKEW\_TX}}$ , is defined as the time between the intersections of the single-ended output signals  $V_{\text{TXDP}}(t)$  and  $V_{\text{TXDN}}(t)$  with the averaged common-mode voltage  $V_{\text{CM\_TX}}$ , when the HS-TX drives a test pattern into a reference load  $R_{\text{REF\_RT}}$  or  $R_{\text{REF\_NT}}$ . The transmitter intra-lane output skew shall be in the specification limits of  $T_{\text{INTRA\_SKEW\_TX}}$ . A skew of the single-ended output signals results in a common-mode voltage ripple as illustrated in **Figure 28**.



**Figure 28 Impact of Signal Skew on Common Mode**

### 5.1.2.4 LANE-to-LANE Skew

The HS-TX LANE-to-LANE skew  $T_{\text{L2L\_SKEW\_HS\_TX}}$  is defined as the time between the zero crossings of the differential output signals  $V_{\text{DIF\_TX}}(t)$  of any two HS-TXs in one SUB-LINK, when both HS-TX drive a test pattern into identical reference load  $R_{\text{REF\_RT}}$ . The value of  $T_{\text{L2L\_SKEW\_HS\_TX}}$  is outside the scope of this document. If required, it shall be defined in the protocol specification.

### 5.1.2.5 Output Resistance Mismatch

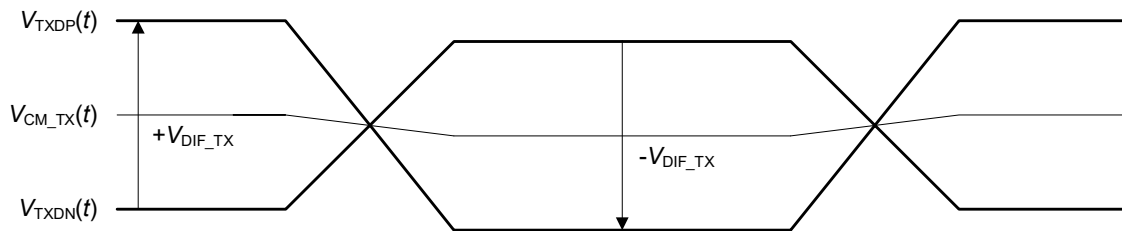
The HS-TX output resistance mismatch,  $\Delta R_{SE\_TX}$ , is defined as the difference of the single-ended output resistances,  $R_{SE\_TX}$ , at the TXDP and TXDN PINs, when the HS-TX drives CRPAT test pattern into a reference load,  $R_{REF\_RT}$ .  $R_{SE\_TX}$  is defined in **Section 5.1.1.3**.

$\Delta R_{SE\_TX}$  can be calculated from the following equation:

$$\Delta R_{SE\_TX} = R_{SE\_TX}(TXDP) - R_{SE\_TX}(TXDN) \quad (\text{Equation 12})$$

where  $R_{SE\_TX}(TXDP)$  is the output resistance driving either a DIF-N or a DIF-P and  $R_{SE\_TX}(TXDN)$  is the output resistance driving either a DIF-N or a DIF-P such that **Equation 12** has to be evaluated for four cases. The HS-TX output resistance mismatch shall be in the limits of  $\Delta R_{SE\_TX}$  for all four cases.

Transmitter output signal mismatch, as well as the transmitter output gain mismatch, originates from  $\Delta R_{SE\_TX}$ . The transmitter output gain mismatch definition is out of scope for this document. A transmitter output signal mismatch results in different signal transition times as well as in different differential DC output voltages  $V_{DIF\_DC\_TX}$  when driving a DIF-P or a DIF-N LINE state. Both effects cause a ripple of  $V_{CM\_TX}$ . An example of a  $V_{CM\_TX}$  ripple is illustrated in **Figure 29**.



**Figure 29 Impact of Output Signal Mismatch on Common Mode Voltage**

### 5.1.2.6 Transmitter Pulse Width

The transmitter pulse width  $T_{PULSE\_TX}$  of an HS-TX differential output signal is defined as the time between the zero crossings of a single bit of the differential output signal  $V_{DIF\_TX}(t)$  when driving a test pattern into a reference load  $R_{REF\_RT}$ . The transmitter pulse width of an HS-TX output signal should conform with the lower limit of  $T_{PULSE\_TX}$ .

### 5.1.2.7 Transmitter Jitter

To ensure interoperability among the components that comprise an end-to-end LANE, the jitter budget must be adhered to by the HS-TX and the HS-RX. The LINE characteristics are indirectly defined by the HS-TX jitter characteristics and by the HS-RX jitter tolerance. While the jitter budgets for reference clock and PLLs are not explicitly defined, the impact of these circuits is included in the jitter budget.

The transmitter total jitter  $TJ_{TX}$  is a convolution of the deterministic jitter  $DJ_{TX}$  and the random jitter  $RJ_{TX}$  of the differential output signal  $V_{DIF\_TX}(t)$  of the HS-TX.  $TJ_{TX}$  is the sum of the arithmetic sum of the deterministic jitter contributions  $DJ_{TX}[j]$ , where  $DJ_{TX}[j]$  are peak-to-peak values, and the square root of the sum of squared random jitter contributions  $RJ_{TX}[i]$  multiplied by two times the Q-factor  $Q_{BER}$ , which is a constant depending on the BER.

$TJ_{TX}$  can be calculated using following equation:

$$TJ_{TX} = \sum_j DJ_{TX}[j] + 2Q_{BER} \sqrt{\sum_i RJ_{TX}[i]^2} \quad (\text{Equation 13})$$

Using the dual-Dirac model,  $TJ_{TX}$  can be expressed by the following equation:

$$TJ_{TX} = DJ_{TX}(dd) + 2Q_{BER}s \quad (\text{Equation 14})$$

where  $DJ_{TX}(\delta\delta)$  is the time between two Dirac pulses and  $\sigma$  is the standard deviation of the Gaussian random jitter of the HS-TX.  $DJ_{TX}(\delta\delta)$  is the dual-Dirac model for the deterministic jitter of the HS-TX and  $\sigma$  is the model for the random jitter of the HS-TX. Further details of the dual-Dirac jitter model are described in [INC01].

This specification defines the  $TJ_{TX}$  and the  $DJ_{TX}(\delta\delta)$ . In addition, the short term total jitter,  $STTJ_{TX}$ , and the short term deterministic jitter,  $STDJ_{TX}(\delta\delta)$ , which limit the jitter within a  $^{30}U_{HS}$  signal sequence, are specified. The short term jitter corresponds to a high frequency jitter in the frequency domain.

The HS-TX jitter spectrum spans from very low frequencies up to high frequencies. In the low frequency range, HS-TX jitter can be significant down to a few kHz. An HS-RX at the other end of the LANE tracks the low frequency jitter components and behaves as a high pass jitter filter. Therefore the HS-TX jitter is filtered with a high-pass jitter transfer function  $H_{JTF}(s)$ , which is defined in the following equation:

$$H_{JTF}(s) = \frac{s^2}{s^2 + 2\zeta\omega_m s + \omega_m^2} \quad (\text{Equation 15})$$

where  $\omega_m = 2\pi f_m$  and  $f_m = \frac{f_{C \text{ HS TX}}}{\sqrt{1 + 2\zeta^2} + \sqrt{1 + (1 + 2\zeta^2)^2}}$ . The clock and data recovery transfer function can be expressed by the following equation:

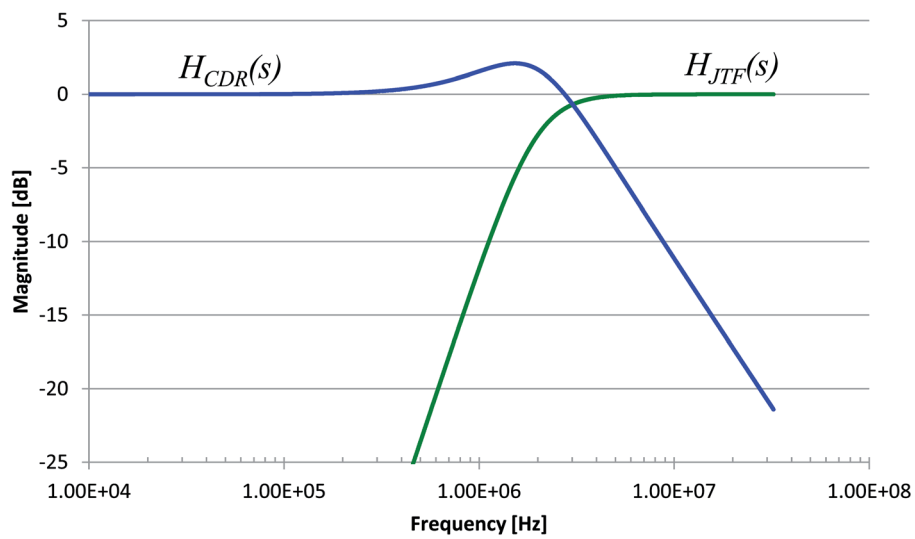
$$H_{CDR}(s) = \frac{2\zeta\omega_m s + \omega_m^2}{s^2 + 2\zeta\omega_m s + \omega_m^2} \quad (\text{Equation 16})$$

After  $H_{JTF}(s)$  is applied to the jitter,  $TJ_{TX}$  is determined by integrating over the frequency range from larger than 0 Hz up to  $f_{U\_TX}$ . **Figure 30** shows a plot of  $H_{JTF}(s)$  and  $H_{CDR}(s)$ .

A 1<sup>st</sup> order high-pass filter with the pole at  $f_{STJ\_TX}$  is applied to the transmit jitter to determine  $STTJ_{TX}$ .

The transmitter total jitter  $TJ_{TX}$  and deterministic jitter  $DJ_{TX}(\delta\delta)$  are defined for the differential output signal  $V_{DIF\_TX}(t)$  at the zero crossings when the HS-TX is driving a CRPAT test pattern into a reference load  $R_{REF\_RT}$  or  $R_{REF\_NT}$ . The transmitter total jitter and deterministic jitter of an HS-TX shall conform with the limits of  $TJ_{TX}$  and  $DJ_{TX}(\delta\delta)$ , respectively.

The transmitter short term total jitter  $STTJ_{TX}$  and short term deterministic jitter  $STDJ_{TX}(\delta\delta)$  are defined for the differential output signal  $V_{DIF\_TX}(t)$  at the zero crossings when the HS-TX is driving a CRPAT test pattern into a reference load  $R_{REF\_RT}$ . The transmitter short term total jitter and short term deterministic jitter of an HS-TX shall conform with the limits of  $STTJ_{TX}$  and  $STDJ_{TX}(\delta\delta)$ , respectively. Note that jitter in non-terminated mode cannot be practically measured.



1117

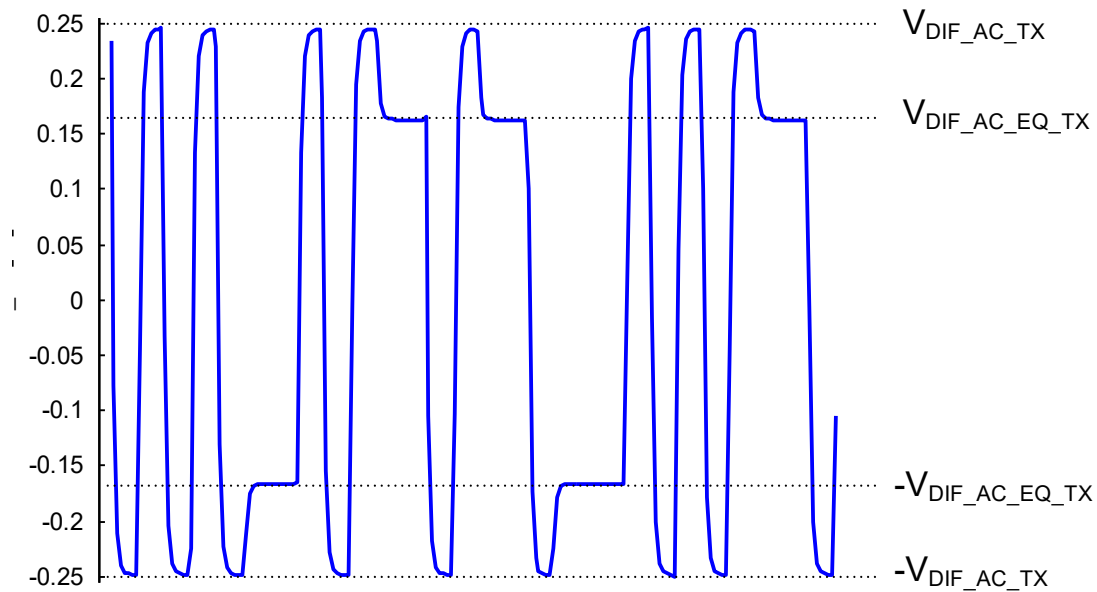
**Figure 30 Clock and Data Recovery Transfer Function and Jitter Transfer Function**

### 5.1.2.8 Transmitter De-emphasis

To mitigate additional channel-induced ISI at HS-G3 data rates or above, an HS-TX can utilize channel equalization in the form of de-emphasis. The transmitter de-emphasis has two taps, where the first tap is the cursor and the second tap is the first post-cursor. The taps are separated by  $UI_{HS}$  and the transmitter de-emphasis ratio  $EQ_{TX}$  determines the de-emphasis level. Two de-emphasis ratios are defined.

**Figure 31** shows an example transmit waveform with de-emphasis. After a logical bit transition, the amplitude of the differential output voltage signal  $V_{DIF\_TX}(t)$  conforms to the differential AC output voltage amplitude  $V_{DIF\_AC\_TX}$ . The next bit that retains the same logical state is reduced in amplitude. The differential AC output voltage amplitude with de-emphasis  $V_{DIF\_AC\_EQ\_TX}$  is defined as the reduced amplitude.  $EQ_{TX}$  is defined as the minus 20 log of the ratio of  $V_{DIF\_AC\_EQ\_TX}$  and  $V_{DIF\_AC\_TX}$  as shown in the following equation:

$$EQ_{TX} = -20\log\left(\frac{V_{DIF\_AC\_EQ\_TX}}{V_{DIF\_AC\_TX}}\right) \quad (\text{Equation 17})$$



**Figure 31 Example Transmit Waveform with De-emphasis**

An HS-TX with de-emphasis, when operated in HS-G3 or above, shall conform to the HS-G3, HS-G4 and HS-G5 transmitter eye diagram mask (see **Section 5.1.2.9**). The transmitter de-emphasis defined for HS-G3 and above may be implemented in HS-G1 and HS-G2 to facilitate longer channel lengths at lower data rates.



### 5.1.2.9 Transmitter Eye Opening

The transmitter eye opening,  $T_{\text{EYE\_TX}}$ , is defined as the duration in an eye diagram over which the absolute value of the differential HS-TX output signal has to be larger than the lower limit of  $V_{\text{DIF\_AC\_TX}}$  when the HS-TX transmits a test pattern into a reference load  $R_{\text{REF\_RT}}$ .  $T_{\text{EYE\_TX}}$  is defined for HS-G1 and HS-G2.

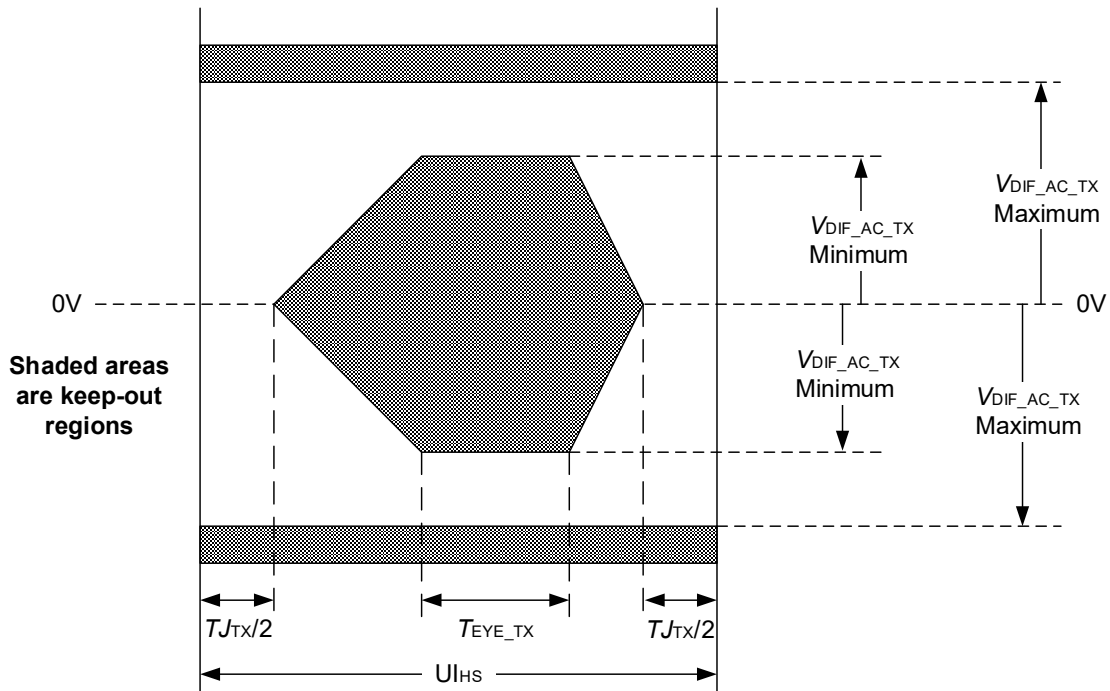
$T_{\text{EYE\_TX}}$ ,  $V_{\text{DIF\_AC\_TX}}$ , and  $TJ_{\text{TX}}/2$  define the eye mask for the accumulated M-TX signal in HS-G1 and HS-G2 as shown in **Figure 32**. The absolute value of the HS-TX differential output voltage signal shall be larger than the lower limit of  $V_{\text{DIF\_AC\_TX}}$  over  $T_{\text{EYE\_TX}}$ . The accumulated eye diagram shall conform to the eye diagram mask. For HS-G1 the position of  $T_{\text{EYE\_TX}}$  is limited by  $(UI_{\text{HS}} - T_{\text{EYE\_TX}})/2$  to the left and  $TJ_{\text{TX}}/2$  to the right. For HS-G2 the midpoint of  $T_{\text{EYE\_TX}}$  is located at  $UI_{\text{HS}}/2$ .

For HS-G3 the transmitter eye opening,  $T_{\text{EYE\_HS\_G3\_TX}}$  is defined as the duration over which no zero crossings of  $V_{\text{DIF\_TX}}(t)$  are allowed in the eye diagram when the HS-TX transmits a test pattern into a reference channel that is terminated with a reference load  $R_{\text{REF\_RT}}$ . Reference Channels CH1 and CH2 are defined in **Section 5.1.1.1**.

For HS-G4 and HS-G5 the transmitter eye openings,  $T_{\text{EYE\_HS\_G4\_TX}}$  and  $T_{\text{EYE\_HS\_G5\_TX}}$ , are defined as the duration over which no zero crossings of  $V_{\text{DIF\_TX}}(t)$  are allowed in the eye diagram when the HS-TX transmits a test pattern into a reference channel that is terminated with a reference load  $R_{\text{REF\_RT}}$  and after the reference package and reference channel equalizer. Reference Channels CH1 and CH2 and reference package are defined in **Section 5.1.1.1** and the receiver reference equalizers are defined in **Section 5.2.1.1**.

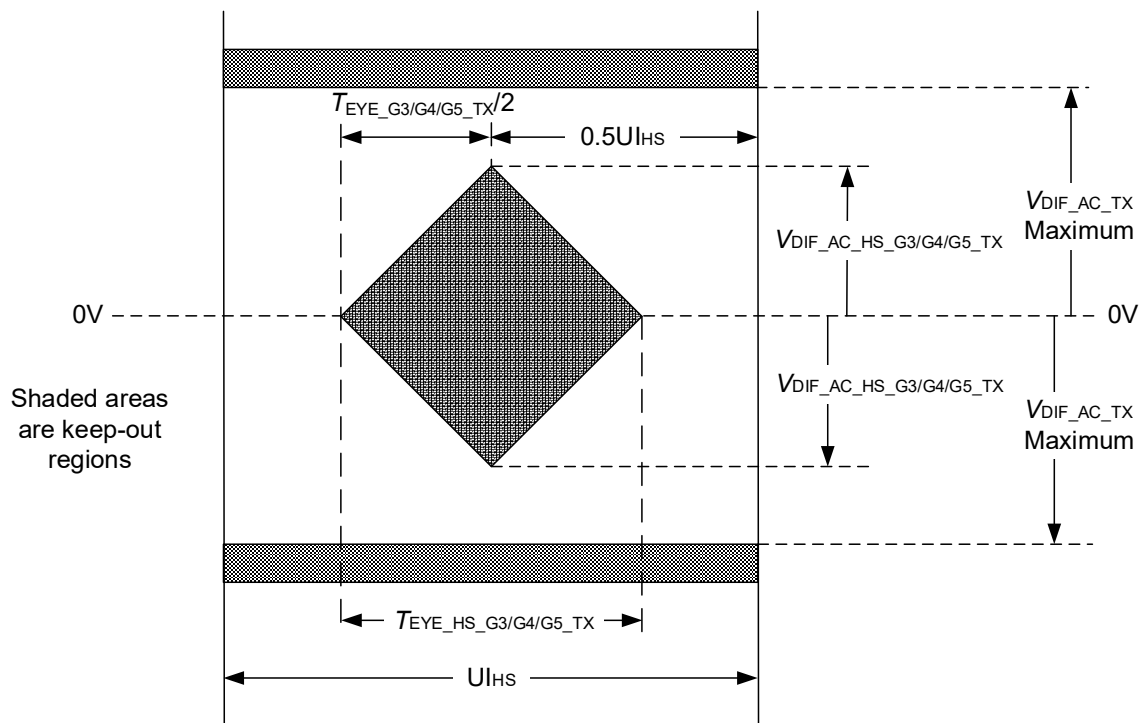
$T_{\text{EYE\_HS\_G3\_TX}}$  and  $V_{\text{DIF\_AC\_HS\_G3\_TX}}$  define the eye mask for the accumulated M-TX signal in HS-G3, as shown in **Figure 33**.  $T_{\text{EYE\_HS\_G4\_TX}}$ ,  $T_{\text{EYE\_HS\_G5\_TX}}$ , and  $V_{\text{DIF\_AC\_HS\_G4\_TX}}$ ,  $V_{\text{DIF\_AC\_HS\_G5\_TX}}$  define similarly the eye masks for the accumulated M-TX signal in HS-G4 and HS-G5. The accumulated M-TX signal in HS-G4 and HS-G5 is defined after the reference package and reference channel equalizer. The midpoints of  $T_{\text{EYE\_HS\_G3\_TX}}$ ,  $T_{\text{EYE\_HS\_G4\_TX}}$ , and  $T_{\text{EYE\_HS\_G5\_TX}}$  are located at  $UI_{\text{HS}}/2$ . The absolute value of the HS-TX differential output voltage signal shall be larger than  $V_{\text{DIF\_AC\_HS\_G3\_TX}}$  or  $V_{\text{DIF\_AC\_HS\_G4\_TX}}$  or  $V_{\text{DIF\_AC\_HS\_G5\_TX}}$  at the midpoint of  $T_{\text{EYE\_HS\_G3\_TX}}$  or  $T_{\text{EYE\_HS\_G4\_TX}}$  or  $T_{\text{EYE\_HS\_G5\_TX}}$ , respectively. The accumulated eye diagram of an HS-TX in HS-G3 or in HS-G4 or in HS-G5 shall conform to the HS-G3, HS-G4, and HS-G5 eye diagram mask, respectively.

The parameters shown in **Figure 32** and **Figure 33** are based on the accumulated eye for the target *BER*, where the total transmit jitter  $TJ_{\text{TX}}$  is defined around the mean of the zero crossings of the differential HS-TX output voltage signal.



1161

Figure 32 HS-G1 and HS-G2 Differential Transmit Eye Diagram



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Figure 33 HS-G3, HS-G4, and HS-G5 Differential Transmitter Eye Diagram

23-Jun-2021

**5.1.2.10 Power Spectral Magnitude Limit**

A power spectral magnitude limit is defined for the common-mode interference spectrum. A method of acquiring the common-mode interference spectrum of an HS-TX is also defined.

**5.1.2.10.1 Common-mode Power Spectral Magnitude Limit**

The common-mode interference spectrum of the HS-TX is impacted by the intra-lane timing skew and gain mismatches of the output signals at the TXDP and TXDN PINs. Slew rate control is an effective means of limiting electromagnetic interference (EMI) of an HS-TX. The level of interference, quantified by the power spectral density measured at TXDP and TXDN, can be controlled by the slew rate of the HS-TX signal waveform. Smaller slew rates result in a significant suppression of high frequency content of the HS-TX output power spectral density. The slew rate limit is application specific and interconnect dependent.

The common-mode power spectral magnitude limit is only valid and applicable to Type II and HS-G1.

A common-mode power spectral magnitude limit is defined along with a method of generating the spectra of an HS-TX.

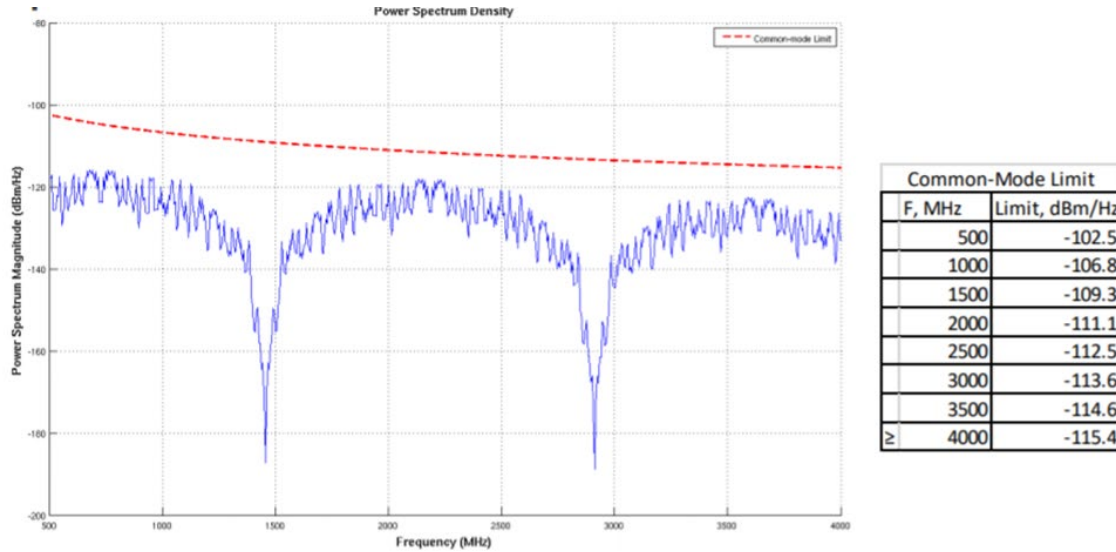
The common-mode power spectral magnitude limit  $PSML_{CM\_TX}$  for HS-G1 is shown in **Figure 34** and defined by:

$$PSML_{CM\_TX} = -180 - (14.3 \log_{10}(f \times 10^3) - 159) \quad (\text{Equation 18})$$

where  $f$  ranges between  $f_{HS\_MIN}$  to  $4 \cdot f_{HS\_MAX}$  in MHz.

$PSML_{CM\_TX}$  can be achieved by proper slew-rate control as well as by conforming to the limits of intra-lane timing skew, output resistance mismatch and output signal amplitude. Package isolation and victim-aggressor geometry should be considered in order to minimize interference. An M-TX should conform to  $PSML_{CM\_TX}$  when applicable. Spurs may violate  $PSML_{CM\_TX}$  outside the radio bands of interest. RATE Series A or B can be selected in order to minimize the interference.

For illustration purposes the common-mode power-spectral density of an 8b10b coded common-mode interference signal (gray curve) is also shown in **Figure 34**. This curve does not show the spurs at the fundamental frequency nor at the harmonics of the data signal.



**Figure 34 Common-Mode Power Spectral Magnitude Limit**

#### 5.1.2.10.2 Spectrum Generation Method

The method of acquiring the common-mode interference spectrum of an HS-TX can be applied both in simulation and measurement. The method is described in the following list:

- The operating condition of the HS-TX should be chosen such that it results in the maximum amplitude for the selected amplitude setting when the M-TX is terminated with a reference load RREF\_RT. In case the HS-TX is operated with Small Amplitude, the temperature, supply voltage, and process should be selected to result in a maximum HS-TX amplitude. This does not imply that the investigation has to be performed with Large Amplitude instead of Small Amplitude.
- A continuous CRPAT test pattern should be used, where regular signal sequences may be used.
- The HS-TX common-mode signal VCM\_TX(t) is calculated from the VTXDP(t) and VTXDN(t) signals.
- A load of 50 Ω is assumed for power calculation.
- The FFT of the common-mode signal with a Hamming window results in the interference spectrum, which has to be adjusted for the relevant bandwidth, is defined by:

$$PSD = 10\log_{10}\left(\frac{2 \times \text{abs}(FFT(x))^2}{f_s \times S}\right) + 10\log_{10}\left(\frac{1/R}{10^{-3}}\right) \quad (\text{Equation 19})$$

where

- x = signal to analyze
- f<sub>s</sub> = sampling frequency
- S = sum(Hamming window<sup>2</sup>), is the Hamming window correction factor
- R = 50 Ω (assumes an ideal probe antenna placed at the midpoint of the M-TX outputs)

#### 5.1.2.11 Transmitter Frequency Offset

1206 The transmitter frequency offset  $f_{\text{OFFSET\_TX}}$  is defined as the difference of the actual HS-TX frequency from  
1207 the nominal HS-TX frequency  $f_{\text{HS}}$ .  $f_{\text{OFFSET\_TX}}$  is defined at the zero crossings of the differential HS-TX output  
1208 signal when driving a test pattern into a reference load  $R_{\text{REF\_RT}}$ . The transmitter frequency offset of an HS-  
1209 TX shall conform with the limits of  $f_{\text{OFFSET\_TX}}$ . Modulation of the transmitter frequency (e.g., by spread  
1210 spectrum clocking) is not intended.

### 5.1.2.12 HS-TX Parameters

The electrical and timing parameters specific to an HS-TX are summarized in **Table 16**. Other than parameters  $V_{DIF\_AC\_HS\_G3\_TX}$ ,  $V_{DIF\_AC\_HS\_G4\_TX}$ ,  $V_{DIF\_AC\_HS\_G5\_TX}$  and  $T_{EYE\_HS\_G3\_TX}$ ,  $T_{EYE\_HS\_G4\_TX}$ ,  $T_{EYE\_HS\_G5\_TX}$  the HS-TX parameters are defined at the M-TX pins without de-emphasis. A channel with negligible loss or de-embedding method may be used to obtain the measurements of all other timing, slew-rate and jitter parameters.

**Table 16 HS-TX Parameters**

| Symbol                      | Values |      | Unit             | Description                                                                                                                                  |
|-----------------------------|--------|------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
|                             | Min.   | Max. |                  |                                                                                                                                              |
| HS-TX Electrical            |        |      |                  |                                                                                                                                              |
| $V_{DIF\_AC\_HS\_G3\_TX}$   | 40     | —    | mV               | Differential TX AC voltage in HS-G3. Defined for a reference channel, $R_{REF\_RT}$ and CRPAT. See <b>Section 5.1.2.9</b> .                  |
| $V_{DIF\_AC\_HS\_G4\_TX}$   | 40     | —    | mV               | Differential TX AC voltage in HS-G4. Defined for a reference channel, $R_{REF\_RT}$ and CRPAT. See <b>Section 5.1.2.9</b> .                  |
| $V_{DIF\_AC\_HS\_G5\_TX}^1$ | 30     | —    | mV               | Differential TX AC voltage in HS-G5. Defined for a reference channel, $R_{REF\_RT}$ and CRPAT. See <b>Section 5.1.2.9</b> .                  |
| HS-TX Timing                |        |      |                  |                                                                                                                                              |
| $f_{OFFSET\_TX}$            | -150   | 150  | ppm              | Transmitter frequency offset. Defined for RREF_ and CRPAT. See <b>Section 5.1.2.11</b> .                                                     |
| $T_{F\_HS\_TX}^2$           | 0.1    | —    | UI <sub>HS</sub> | Fall time. Defined for $R_{REF\_RT}^3$ and test pattern <sup>5</sup> . See <b>Section 5.1.2.1</b> .                                          |
| $T_{R\_HS\_TX}^2$           | 0.1    | —    | UI <sub>HS</sub> | Rise time. Defined for $R_{REF\_RT}^3$ and test pattern <sup>4</sup> . See <b>Section 5.1.2.1</b> .                                          |
| $SR_{DIF\_TX}[MAX]$         | 0.9    | —    | V/ns             | Maximum slew rate. Defined in HS-G1 for $V_{DIF\_DC\_SA\_RT\_TX}^6$ , $R_{REF\_RT}^3$ , and CRPAT. See <b>Section 5.1.2.2</b> .              |
| $SR_{DIF\_TX}[MIN]$         | —      | 0.35 | V/ns             | Minimum slew rate. Defined in HS-G1 for $V_{DIF\_DC\_SA\_RT\_TX}^5$ , $R_{REF\_RT}^3$ , and CRPAT. See <b>Section 5.1.2.2</b> .              |
| $\Delta SR_{DIF\_TX}$       | 1      | 30   | %                | Resolution of slew rate states. Defined in HS-G1 for $V_{DIF\_DC\_SA\_RT\_TX}^5$ , $R_{REF\_RT}^3$ , and CRPAT. See <b>Section 5.1.2.2</b> . |
| $T_{INTRA\_SKEW\_TX}$       | -0.06  | 0.06 | UI <sub>HS</sub> | Intra-lane output skew. Defined for $R_{REF\_RT}^3$ CRPAT. See <b>Section 5.1.2.3</b> .                                                      |
| $T_{PULSE\_TX}$             | 0.9    | —    | UI <sub>HS</sub> | Transmitter pulse width. Defined for $R_{REF\_RT}^3$ and CRPAT. See <b>Section 5.1.2.6</b> .                                                 |
| HS-TX Resistance            |        |      |                  |                                                                                                                                              |
| $\Delta R_{SE\_TX}$         | -6     | 6    | Ω                | Output resistance mismatch. Defined for $R_{REF\_RT}^3$ when driving DIF-N and DIF-P. See <b>Section 5.1.2.5</b> .                           |

| Symbol                           | Values |      | Unit             | Description                                                                                                                                                                                               |
|----------------------------------|--------|------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | Min.   | Max. |                  |                                                                                                                                                                                                           |
| HS-TX Jitter                     |        |      |                  |                                                                                                                                                                                                           |
| $T_{\text{EYE\_TX}}$             | 0.2    | —    | UI <sub>HS</sub> | Transmitter eye opening in HS-G1 and HS-G2 <sup>8</sup> . Defined for $R_{\text{REF\_RT}}$ <sup>3</sup> and CRPAT over a statistical confident record set <sup>9</sup> . See <b>Section 5.1.2.9</b> .     |
| $T_{\text{EYE\_HS\_G3\_TX}}$     | 0.55   | —    | UI <sub>HS</sub> | Transmitter eye opening in HS-G3. Defined for a Reference Channel, $R_{\text{REF\_RT}}$ <sup>3</sup> and CRPAT over a statistical confident record set <sup>8</sup> . See <b>Section 5.1.2.9</b> .        |
| $T_{\text{EYE\_HS\_G4\_TX}}$     | 0.5    | —    | UI <sub>HS</sub> | Transmitter eye opening in HS-G4. Defined for a Reference Channel, $R_{\text{REF\_RT}}$ <sup>3</sup> and CRPAT over a statistical confident record set <sup>8</sup> . See <b>Section 5.1.2.9</b> .        |
| $T_{\text{EYE\_HS\_G5\_TX}}$     | 0.3    | —    | UI <sub>HS</sub> | Transmitter eye opening in HS-G5. Defined for a Reference Channel, $R_{\text{REF\_RT}}$ <sup>3</sup> and CRPAT over a statistical confident record set <sup>8</sup> . See <b>Section 5.1.2.9</b> .        |
| $DJ_{\text{TX}}(\delta\delta)$   | —      | 0.15 | UI <sub>HS</sub> | Transmitter deterministic jitter <sup>10</sup> . Defined for $R_{\text{REF\_RT}}$ <sup>3</sup> and CRPAT for a statistical confident record set <sup>8,11</sup> . See <b>Section 5.1.2.7</b> .            |
| $TJ_{\text{TX}}$                 | —      | 0.32 | UI <sub>HS</sub> | Transmitter total jitter <sup>9</sup> . Defined for $R_{\text{REF\_RT}}$ <sup>3</sup> and CRPAT for a statistical confident record set <sup>8,10</sup> . See <b>Section 5.1.2.7</b> .                     |
| $STDJ_{\text{TX}}(\delta\delta)$ | —      | 0.10 | UI <sub>HS</sub> | Transmitter short term deterministic jitter <sup>9</sup> . Defined for $R_{\text{REF\_RT}}$ <sup>3</sup> and CRPAT for a statistical confident record set <sup>10,11</sup> . See <b>Section 5.1.2.7</b> . |
| $STTJ_{\text{TX}}$               | —      | 0.20 | UI <sub>HS</sub> | Transmitter short term total jitter <sup>9</sup> . Defined for $R_{\text{REF\_RT}}$ <sup>3</sup> and CRPAT for a statistical confident record set <sup>10,11</sup> . See <b>Section 5.1.2.7</b> .         |
| HS-TX De-emphasis                |        |      |                  |                                                                                                                                                                                                           |
| $EQ_{1\_TX}$                     | 0      | 4.5  | dB               | De-emphasis ratio defined for Small and Large Amplitude differential TX voltage. See <b>Section 5.1.2.8</b> .                                                                                             |
| $EQ_{2\_TX}$                     | 5.0    | 8.6  | dB               | De-emphasis ratio defined for Large Amplitude differential TX voltage. See <b>Section 5.1.2.8</b> .                                                                                                       |

**Note:**

1. Differential noise margin amplitude of 10mV is included into the Differential TX AC voltage for HS-G5 and above.
2.  $TR_{\text{HS\_TX}}$  and  $TF_{\text{HS\_TX}}$  are informative.
3. External reference load  $R_{\text{REF\_RT}}$  and a reference impedance  $Z_{\text{REF\_RT}}$  that conforms to  $SRL_{\text{REF\_RT}}$ .
4. Repetitive sequence of D.30.3 symbols to be used for test. Such a sequence is part of CJTPAT.
5. Values are specified for Small Amplitude. For Large Amplitude the slew rate is a factor of 1.85 larger.
6. External reference load  $R_{\text{REF\_RT}}$  and a reference impedance  $Z_{\text{REF\_RT}}$  that conforms to  $SRL_{\text{REF\_RT}}$ . The slew rate is only specified for when the M TX is terminated. When the M TX is not terminated, slew rate control is not strictly required due to smaller LINE power. However, slew rate control may also be used when the M TX is not terminated, but in this case how the slew rate control performs is not specified.
7. For slower slew rate settings the transmitter eye mask may be violated.
8. Filtered using a high-pass jitter transfer function  $H_{\text{JTF}}(s)$ .
9. Accumulated jitter as defined by the dual-Dirac model.
10. Measured for the target BER.
11. Filtered using a 1st order high-pass filter with a pole at  $f_{\text{STJ\_TX}}$ .

### 5.1.3 PWM-TX Characteristics

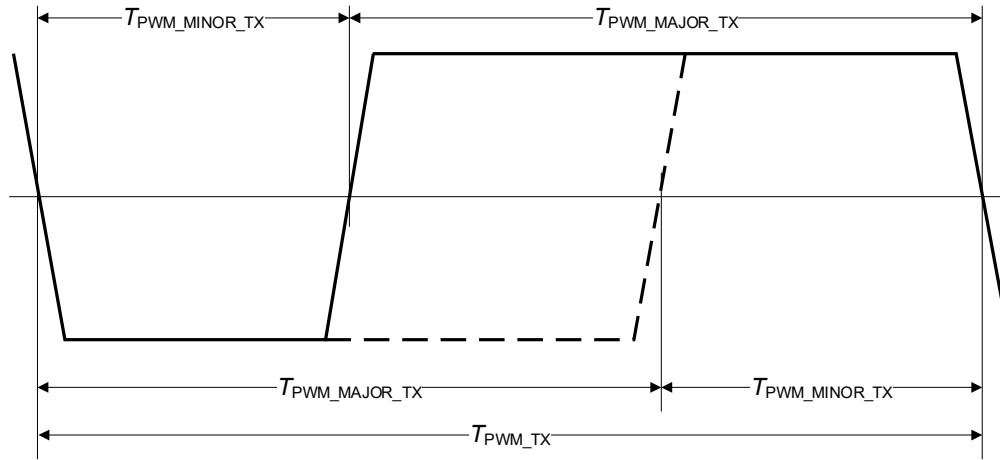
This section contains timing characteristics specific to a PWM-TX which are not covered by the common M-TX characteristics in **Section 5.1.1**. The PWM signaling scheme is defined in **Section 4.3.2**.

#### 5.1.3.1 PWM Bit Duration, Bit Duration Tolerance, and Ratio

A PWM bit consists out of a DIF-N LINE state followed by a DIF-P LINE state, which are either signaled for the minor duration  $T_{\text{PWM\_MINOR\_TX}}$  or for the major duration  $T_{\text{PWM\_MAJOR\_TX}}$ . The durations  $T_{\text{PWM\_MINOR\_TX}}$  and  $T_{\text{PWM\_MAJOR\_TX}}$  are defined as the time between the zero crossings of the differential output signal.

The PWM transmit bit duration  $T_{\text{PWM\_TX}}$  is defined as the duration between zero crossings of two consecutive falling edges of a differential signal at the PWM-TX output.  $T_{\text{PWM\_MINOR\_TX}}$ ,  $T_{\text{PWM\_MAJOR\_TX}}$ , and  $T_{\text{PWM\_TX}}$  are shown in **Figure 35**. The PWM transmit bit duration  $T_{\text{PWM\_TX}}$  is for all PWM GEARS the sum of its durations  $T_{\text{PWM\_MINOR\_TX}}$  and  $T_{\text{PWM\_MAJOR\_TX}}$ , as shown in the following equation:

$$T_{\text{PWM\_TX}} = T_{\text{PWM\_MINOR\_TX}} + T_{\text{PWM\_MAJOR\_TX}} \quad (\text{Equation 20})$$



**Figure 35 TX Minor and Major Duration in a PWM Signal**

$T_{\text{PWM\_MINOR\_TX}}$  and  $T_{\text{PWM\_MAJOR\_TX}}$  are determined by  $T_{\text{PWM\_TX}}$  and the PWM transmit ratio  $k_{\text{PWM\_TX}}$  for PWM-G1.  $k_{\text{PWM\_TX}}$  is defined as the ratio of  $T_{\text{PWM\_MAJOR\_TX}}$  and  $T_{\text{PWM\_MINOR\_TX}}$  of one PWM bit, as shown in the following equation:

$$k_{\text{PWM\_TX}} = \frac{T_{\text{PWM\_MAJOR\_TX}}}{T_{\text{PWM\_MINOR\_TX}}} \quad (\text{Equation 21})$$

The PWM transmit bit duration tolerance,  $TOL_{\text{PWM\_TX}}$ , is the allowed tolerance of an instantaneous PWM bit duration,  $T_{\text{PWM\_TX}}(i)$ , in PWM-MODE.  $TOL_{\text{PWM\_TX}}$  is defined as the ratio of  $T_{\text{PWM\_TX}}(i)$  and the average of  $N$  PWM transmit bit durations in PWM-MODE, as shown in the following equation:

$$TOL_{\text{PWM\_TX}} = \frac{T_{\text{PWM\_TX}}(i)}{\frac{1}{N} \sum_{j=1}^N T_{\text{PWM\_TX}}(j)} \quad (\text{Equation 22})$$

where  $N$  is a defined number of PWM bits, and  $i$  is in the range of 1 to  $N$ .



- 1236 While the  $T_{\text{PWM\_TX}}$  range is wide for a PWM GEAR,  $TOL_{\text{PWM\_TX}}$  limits the variation of  $T_{\text{PWM\_TX}}(i)$ .  
1237 A PWM-TX shall output a PWM signal with PWM transmit bit duration,  $T_{\text{PWM\_TX}}$ , in the specified range of  
1238 the operational PWM-GEAR during a PWM-BURST. For PWM-G1 the PWM transmit ratio  $k_{\text{PWM\_TX}}$  shall  
1239 be in the specified range for each PWM bit.  
1240 A PWM-TX shall output a PWM signal with PWM transmit bit duration tolerance in the limits of  $TOL_{\text{PWM\_TX}}$ .

### 5.1.3.2 Rise and Fall Time

- 1241 The PWM-TX rise and fall times,  $T_{\text{R\_PWM\_TX}}$  and  $T_{\text{F\_PWM\_TX}}$ , respectively, are defined as transition times  
1242 between the 20% and 80% signal levels of the differential PWM-TX output signal with an amplitude of  
1243  $V_{\text{DIF\_DC\_TX}}$ , when driving a reference load  $R_{\text{REF\_NT}}$ . The rise and fall times of a PWM-TX shall comply with  
1244 the limits of  $T_{\text{R\_PWM\_TX}}$  and  $T_{\text{F\_PWM\_TX}}$ .

### 5.1.3.3 LANE-to-LANE Skew

- 1245 The PWM-TX LANE-to-LANE skew  $T_{\text{L2L\_SKEW\_PWM\_TX}}$  is defined as the time between the zero crossings of  
1246 the falling edges of the differential output signals  $V_{\text{DIF\_TX}}(t)$  of any two PWM-TXs in one SUB-LINK, when  
1247 both PWM-TX drive a test pattern into identical reference loads  $R_{\text{REF\_RT}}$  or  $R_{\text{REF\_NT}}$ . The value of  
1248  $T_{\text{L2L\_SKEW\_PWM\_TX}}$  is outside the scope of this document. If required, it shall be defined in the protocol  
1249 specification.

### 5.1.3.4 PWM-TX Parameters

1250 The timing parameters specific to a PWM-TX are summarized in **Table 17**.

**Table 17 PWM-TX Parameters**

| Symbol                   | Values              |                     | Unit                 | Description                                                                                                                         |
|--------------------------|---------------------|---------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------|
|                          | Min.                | Max.                |                      |                                                                                                                                     |
| $T_{\text{PWM\_G1\_TX}}$ | $\frac{1}{9}$       | $\frac{1}{3}$       | $\mu\text{s}$        | PWM transmit bit duration in PWM-G1.<br>Defined for $R_{\text{REF\_NT}}^1$ and CRPAT. See <b>Section 5.1.3.1</b> .                  |
| $TOL_{\text{PWM\_TX}}$   | 0.90                | 1.10                | —                    | PWM transmit bit duration tolerance.<br>Defined for $R_{\text{REF\_NT}}^1$ and CRPAT in PWM-MODE. See <b>Section 5.1.3.1</b> .      |
| $N$                      | 50                  | 50                  | —                    | Number of PWM bits.<br>Sequence length for $TOL_{\text{PWM\_TX}}$ and $TOL_{\text{PWM\_G1\_LR\_TX}}$ . See <b>Section 5.1.3.1</b> . |
| $k_{\text{PWM\_TX}}$     | $\frac{0.63}{0.37}$ | $\frac{0.72}{0.28}$ | —                    | PWM transmit ratio for PWM-G1.<br>Defined for $R_{\text{REF\_NT}}^1$ and CRPAT. See <b>Section 5.1.3.1</b> .                        |
| $T_{\text{R\_PWM\_TX}}$  | —                   | 0.070               | $T_{\text{PWM\_TX}}$ | Rise time.<br>Defined for $R_{\text{REF\_NT}}^1$ and CRPAT. See <b>Section 5.1.3.2</b> .                                            |
| $T_{\text{F\_PWM\_TX}}$  | —                   | 0.070               | $T_{\text{PWM\_TX}}$ | Fall time.<br>Defined for $R_{\text{REF\_NT}}^1$ and CRPAT. See <b>Section 5.1.3.2</b> .                                            |

**Note:**

1. External reference load  $R_{\text{REF\_NT}}$  and capacitances at TXDP and at TXDN within the limit of  $C_{\text{PIN\_RX}}$ . If terminated state is supported external reference load  $R_{\text{REF\_RT}}$  and a reference impedance  $Z_{\text{REF\_RT}}$  which conforms to  $SRL_{\text{REF\_RT}}$  has to be verified additionally.

#### 5.1.4 SYS-TX Characteristics

1251 This section contains timing characteristics specific to a SYS-TX which are not covered by the common  
1252 M-TX characteristics in *Section 5.1.1*.

##### 5.1.4.1 Rise and Fall Times

1253 The SYS-TX rise and fall times,  $T_{R\_SYS\_TX}$  and  $T_{F\_SYS\_TX}$ , respectively, are defined as transition times between  
1254 the 20% and 80% signal levels of the differential SYS-TX output signal, with an amplitude of  $V_{DIF\_DC\_TX}$ ,  
1255 when driving a repetitive D.30.3 symbol sequence into reference load  $R_{REF\_NT}$ . The rise and fall times of a  
1256 SYS-TX shall comply with the limits of  $T_{R\_SYS\_TX}$  and  $T_{F\_SYS\_TX}$ .

##### 5.1.4.2 LANE-to-LANE Skew

1257 The SYS-TX LANE-to-LANE skew  $T_{L2L\_SKEW\_SYS\_TX}$  is defined as the time between the zero crossings of  
1258 the differential output signals  $V_{DIF\_TX}(t)$  of any two SYS-TXs in one SUB-LINK, when both SYS-TX drive  
1259 a test pattern into identical reference loads  $R_{REF\_RT}$  or  $R_{REF\_NT}$ . The value of  $T_{L2L\_SKEW\_SYS\_TX}$  is outside the  
1260 scope of this document. If required, it shall be defined in the protocol specification.

##### 5.1.4.3 Data-to-Clock Skew

1261 A system synchronous clocking scheme is used in the SYS-BURST mode, an example of which is shown in  
1262 *Figure 16*. Since the reference clock is not considered to be a part of an M-PHY, definition of the clock  
1263 characteristics is outside the scope of this specification. Parameters like the reference clock frequency, the  
1264 duty cycle distortion of the reference clock signal, or the rise and fall times of the reference clock signal have  
1265 to be covered in the protocol specification utilizing the M-PHY technology.

1266 The data-to-clock skew between the data signals of a SYS-TX and the reference clock signal has also to be  
1267 defined in the protocol specification, such that no unnecessary limitations for the clocking scheme or system  
1268 timing are put forth by this specification. This leaves maximum flexibility to the protocol specification, which  
1269 only has to adhere to the zero crossing of the SYS-TX output signal, when it is driving a reference load  
1270  $R_{REF\_RT}$  or  $R_{REF\_NT}$ , as reference timing point for such a definition. The data-to-clock skew has to be defined  
1271 for both SUB-LINKs. Interoperability in SYS-BURST mode thus has partly to be ensured by the protocol  
1272 specification.

1273 There might be applications for which a data-to-clock skew cannot be defined, e.g. in case of an external  
1274 reference clock signal. In such a case, the propagation delay between the external reference clock signal and  
1275 the SYS-TX data signals has to be defined in the protocol specification.

**5.1.4.4 SYS-TX Parameters**

1276 The timing parameters specific to a SYS-TX are summarized in **Table 18**.

**Table 18 SYS-TX Parameters**

| Symbol           | Values |      | Unit              | Description                                                                                                       |
|------------------|--------|------|-------------------|-------------------------------------------------------------------------------------------------------------------|
|                  | Min.   | Max. |                   |                                                                                                                   |
| $T_{R\_SYS\_TX}$ | —      | 0.20 | U <sub>ISYS</sub> | Rise time.<br>Defined for $R_{REF\_NT}$ <sup>1</sup> and test pattern <sup>2</sup> . See <b>Section 5.1.4.1</b> . |
| $T_{F\_SYS\_TX}$ | —      | 0.20 | U <sub>ISYS</sub> | Fall time.<br>Defined for $R_{REF\_NT}$ <sup>1</sup> and test pattern <sup>2</sup> . See <b>Section 5.1.4.1</b> . |

**Note:**

1. External reference load  $R_{REF\_NT}$  and capacitances at TXDP and at TXDN within the limit of  $C_{PIN\_RX}$ . If terminated state is supported external reference load  $R_{REF\_RT}$  and a reference impedance  $Z_{REF\_RT}$  which conforms to  $SRL_{REF\_RT}$  has to be verified additionally.
2. Repetitive sequence of D.30.3 symbols to be used for test. Such a sequence is part of CJTPAT.

## 5.2 M-RX Characteristics

This document distinguishes three different operating modes and corresponding FUNCTIONS. Following the definition of the common M-RX electrical and timing characteristics, which apply to HS-RX, PWM-RX, and SYS-RX, additional characteristics, which are specific to each receive FUNCTION, are defined in this section. The SQ-RX, which is an optional FUNCTION of an M-RX, is defined at the end of this section.

### 5.2.1 Common M-RX Characteristics

The common electrical and timing characteristics of an M-RX are defined in this section, which also contains the PIN and signal definitions. The common M-RX characteristics apply to the HS-RX, PWM-RX, and SYS-RX FUNCTIONS.

#### 5.2.1.1 PIN, Signal, and Reference Characteristic Definitions

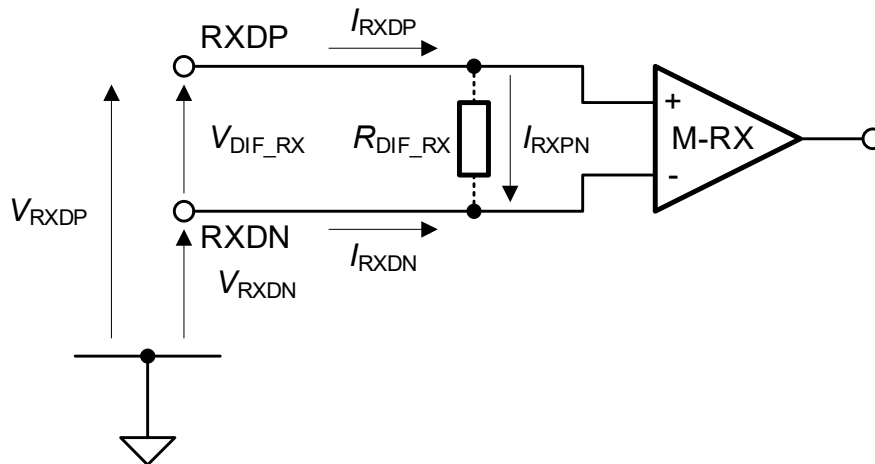
RXDP and RXDN are the input PINs of the M-RX. RXDP is defined as the positive input PIN and RXDN as the negative input PIN.

$V_{RXDP}(t)$  and  $V_{RXDN}(t)$  are defined as the voltage signals at these PINs with respect to ground.  $V_{RXDP}$  and  $V_{RXDN}$  are defined as the voltage amplitudes of the  $V_{RXDP}(t)$  and  $V_{RXDN}(t)$  signals, respectively.

$I_{RXDP}(t)$  and  $I_{RXDN}(t)$  are defined as the input currents flowing into RXDP and RXDN, respectively.  $I_{RXDP}$  and  $I_{RXDN}$  are defined as the current amplitudes of the  $I_{RXDP}(t)$  and  $I_{RXDN}(t)$  signals, respectively.

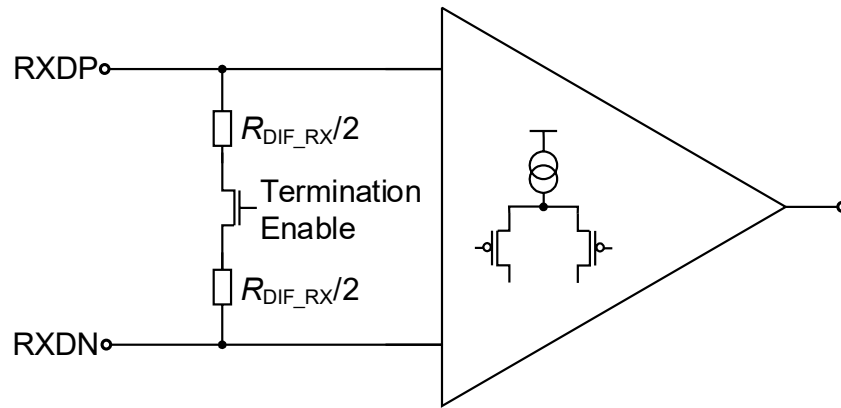
$I_{RXPN}(t)$  is defined as the current, which flows from RXDP to RXDN, in case the termination resistor is enabled.  $I_{RXPN}$  is defined as the current amplitude of  $I_{RXPN}(t)$ .

The PIN voltages and currents are shown in **Figure 36**.



**Figure 36 PIN Voltages and PIN Currents of an M-RX**

The M-RX contains a differential line receiver that supports the detection of M-TX signals having Large Amplitude as well as Small Amplitude. An M-RX has to support only FUNCTIONS required for the targeted application. An M-RX may contain a switchable differential termination resistor  $R_{DIF\_RX}$  between its input PINs RXDP and RXDN for improving the signal integrity. **Section 4.7.2** defines when  $R_{DIF\_RX}$  shall be enabled or disabled. When  $R_{DIF\_RX}$  is enabled, the M-RX is terminated, otherwise it is unterminated.



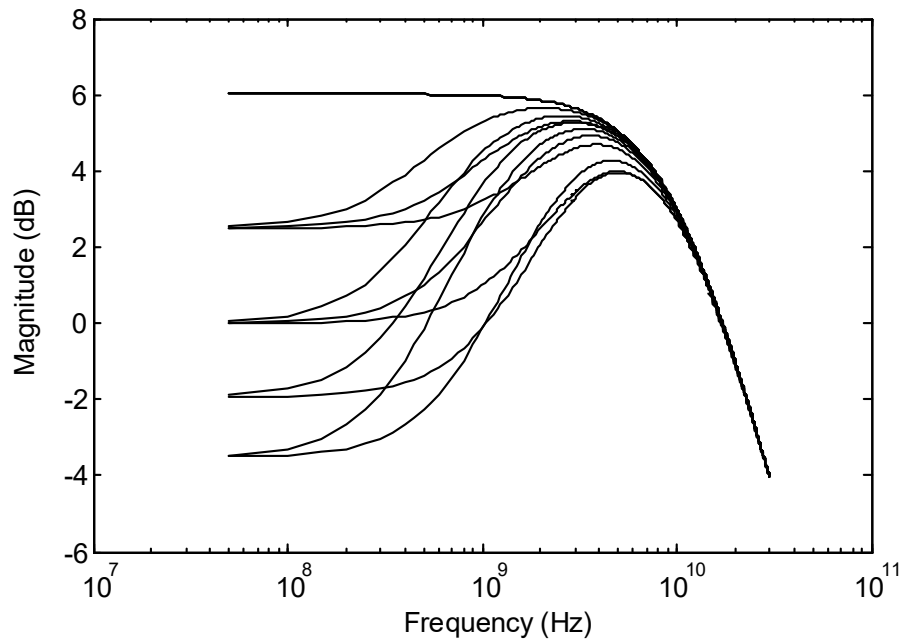
**Figure 37 M-RX Implementation Example**

A simplified diagram of an example implementation using a PMOS input stage is shown in **Figure 37**. The common-mode voltage of the LINE has to remain in the common-mode voltage limits upon switching of the termination resistor. This is achievable through an AC ground at the center tap of the termination resistor, for example, by use of a capacitor.

To mitigate additional channel induced ISI at HS-G4 and HS-G5 data rates, an HS-RX can utilize reference equalization in the form of a first-order continuous-time linear equalizer (CTLE) followed by a 1-tap decision feedback equalizer (DFE). The CTLE characteristics are defined by the following transfer function:

$$H(s) = \frac{A_{DC} w_{P1} w_{P2}}{w_z} \frac{s + w_z}{(s + w_{P1})(s + w_{P2})} \quad (\text{Equation 23})$$

where  $A_{DC}$  is the DC gain,  $\omega = 2\pi f$ ,  $f_z$ , is the zero,  $f_{P1}$  is the first pole, and  $f_{P2}$  is the second pole. **Figure 38** shows various valid CTLE transfer function curves. **Figure 38** shows examples of CTLE frequency responses, when the CTLE parameters are chosen within the limits defined in **Table 19**.

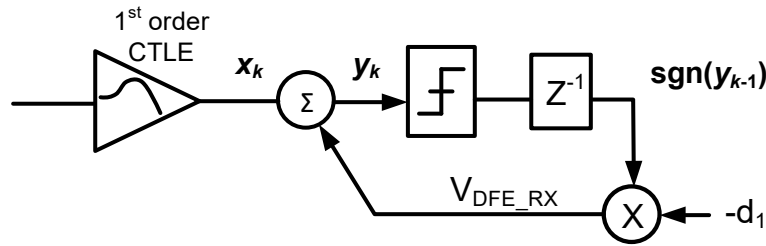


**Figure 38 Examples of CTLE Frequency Responses**

The DFE characteristics are defined by the following equation:

$$\begin{aligned} y_k &= x_k + V_{\text{DFE\_RX}} \\ y_k &= x_k - d_1 \text{sgn}(y_{k-1}) \end{aligned} \quad (\text{Equation 24})$$

where  $y_k$  is the output voltage signal of the DFE,  $x_k$  is the input voltage signal to the DFE,  $V_{\text{DFE\_RX}}$  is the DFE feedback voltage signal,  $k$  is the sample index of a data bit and  $d_1$  is the DFE feedback coefficient. **Figure 39** illustrates the Reference DFE diagram.



**Figure 39 Reference DFE Diagram**

The sinusoidal jitter tolerance of an M-RX in HS-MODE is specified by means of a sinusoidal jitter tolerance mask with a corner frequency  $f_{\text{C\_HS\_RX}}$ . Jitter is integrated up to the upper RX cut-off frequency  $f_{\text{U\_RX}}$ . The lower cut-off frequency  $f_{\text{SJ4\_RX}}$  is defined for the short term jitter of an M-RX.  $f_{\text{C\_HS\_RX}}$  is also the corner frequency of the receiver jitter tolerance.

Discrete test frequencies  $f_{\text{C\_HS\_RX}}$ ,  $f_{\text{SJ2\_RX}}$ ,  $f_{\text{SJ3\_RX}}$  and  $f_{\text{SJ4\_RX}}$  are defined for the sinusoidal jitter tolerance.  $f_{\text{SJ3\_RX}}$  is the system clock frequency of the chip, which in case of a Type-II M-PORT may be different than  $f_{\text{SYS\_REF}}$ .

The jitter is defined for a BER of  $10^{-12}$  according to [INC01]. The mean ( $\mu$ ) of the distribution function is located at 0.

**Note:**

*The BER is changed from  $10^{-10}$  to  $10^{-12}$  in M-PHY v4.1. Other jitter definitions remain unchanged.  
The change in BER is tightening the M-TX jitter requirements.*

The reference parameters for the M-RX are summarized in **Table 19**.

Table 19 M-RX Reference Parameters

| Symbol                              | Values |                          |            | Unit | Description                                                      |
|-------------------------------------|--------|--------------------------|------------|------|------------------------------------------------------------------|
|                                     | Min.   | Nom.                     | Max.       |      |                                                                  |
| Frequency                           |        |                          |            |      |                                                                  |
| $f_{U\_RX}$                         | —      | $\frac{1}{2U_{I_{HS}}}$  | —          | Hz   | Upper frequency for jitter tolerance.                            |
| $f_{C\_HS\_G1\_RX}$                 | —      | 2.0                      | —          | MHz  | Corner frequency of sinusoidal jitter tolerance mask in HS-G1.   |
| $f_{C\_HS\_G2\_RX}$                 | —      | 4.0                      | —          | MHz  | Corner frequency of sinusoidal jitter tolerance mask in HS-G2.   |
| $f_{C\_HS\_G3\_RX}$                 | —      | 8.0                      | —          | MHz  | Corner frequency of sinusoidal jitter tolerance mask in HS-G3.   |
| $f_{C\_HS\_G4\_RX}$                 | —      | 8.0                      | —          | MHz  | Corner frequency of sinusoidal jitter tolerance mask in HS-G4.   |
| $f_{C\_HS\_G5\_RX}$                 | —      | 16.0                     | —          | MHz  | Corner frequency of sinusoidal jitter tolerance mask in HS-G5.   |
| $f_{SJ0\_RX}$                       | —      | 10                       | —          | MHz  | Lower test frequency for sinusoidal tolerance.                   |
| $f_{SJ2\_RX}$                       | —      | 10                       | —          | MHz  | Test frequency for sinusoidal jitter.                            |
| $f_{SJ3\_RX}$                       | —      | $f_{SYSTEM}$             | —          | Hz   | Test frequency for sinusoidal jitter. Frequency of system clock. |
| $f_{SJ4\_RX}$                       | —      | $\frac{1}{30U_{I_{HS}}}$ | —          | Hz   | Test frequency for sinusoidal jitter.                            |
| Limit for $BER$                     |        |                          |            |      |                                                                  |
| $Q_{BER}$                           | —      | 7.0345                   | —          | —    | Q-factor for a $BER$ of $10^{-12}$                               |
| $BER$                               | —      | —                        | $10^{-12}$ | —    | Target $BER$                                                     |
| HS-RX Reference Equalizer for HS-G4 |        |                          |            |      |                                                                  |
| $A_{AC}$                            | —      | —                        | 6          | dB   | CTLE AC gain                                                     |
| $A_{DC}$                            | -4     | —                        | 6          | dB   | CTLE DC gain                                                     |
| $f_z^1$                             | 0.4    | —                        | 1.2        | GHz  | CTLE zero frequency                                              |
| $f_{P2}$                            | —      | —                        | 10         | GHz  | CTLE second pole frequency                                       |
| $V_{DFE\_RX}$                       | -60    | —                        | 60         | mV   | DFE feedback voltage signal                                      |
| HS-RX Reference Equalizer for HS-G5 |        |                          |            |      |                                                                  |
| $A_{AC}$                            | —      | —                        | 12         | dB   | CTLE AC gain                                                     |
| $A_{DC}$                            | -6     | —                        | 6          | dB   | CTLE DC gain                                                     |
| $f_z^1$                             | 0.4    | —                        | 2.5        | GHz  | CTLE zero frequency                                              |
| $f_{P2}$                            | —      | —                        | 20         | GHz  | CTLE second pole frequency                                       |
| $V_{DFE\_RX}$                       | -60    | —                        | 60         | mV   | DFE feedback voltage signal                                      |

**Note:**

- $f_{p1}$  is determined by other parameters:  $f_{p1} = f_z * 10^{((A_{AC} - A_{DC})/20)}$



### 5.2.1.2 Differential and Common-mode Voltage

The differential input voltage signal  $V_{\text{DIF\_RX}}(t)$  is defined as the difference of the voltage signals  $V_{\text{RXDP}}(t)$  and  $V_{\text{RXDN}}(t)$  at the M-RX PINs.  $V_{\text{DIF\_RX}}$  is defined as the amplitude of  $V_{\text{DIF\_RX}}(t)$ .  $V_{\text{DIF\_RX}}(t)$  can be calculated from the following equation:

$$V_{\text{DIF\_RX}}(t) = V_{\text{RXDP}}(t) - V_{\text{RXDN}}(t) \quad (\text{Equation 25})$$

The minimum value of  $V_{\text{DIF\_RX}}$  defines the minimum differential voltage amplitude of a test pattern an M-RX has to receive while the maximum value of  $V_{\text{DIF\_RX}}$  defines the maximum differential voltage amplitude of a test pattern an M-RX has to receive.

The receiver common-mode voltage signal  $V_{\text{CM\_RX}}(t)$  is defined as the arithmetic mean value of the voltage signals  $V_{\text{RXDP}}(t)$  and  $V_{\text{RXDN}}(t)$  when a test pattern is applied at the M-RX input PINs.  $V_{\text{CM\_RX}}$  is defined as the amplitude of  $V_{\text{CM\_RX}}(t)$ .  $V_{\text{CM\_RX}}(t)$  can be calculated from the following equation:

$$V_{\text{CM\_RX}}(t) = \frac{V_{\text{RXDP}}(t) + V_{\text{RXDN}}(t)}{2} \quad (\text{Equation 26})$$

The  $V_{\text{CM\_RX}}$  parameter values are defined such that they cover DC deviations, which can, e.g., be caused by a ground shift between an M-TX and an M-RX or by an output signal mismatch of the M-TX.

An M-RX shall detect a differential input signal at its RXDP and RXDN PINs with a differential voltage amplitude in the range of  $V_{\text{DIF\_RX}}$  and with common-mode voltage in the range of  $V_{\text{CM\_RX}}$ .

### 5.2.1.3 Termination Resistance

An M-RX may contain a switchable differential termination resistor  $R_{\text{DIF\_RX}}$ .  $R_{\text{DIF\_RX}}$  is defined by the ratio of the difference of the PIN voltage amplitudes  $V_{\text{RXDP}}$  and  $V_{\text{RXDN}}$  and the current amplitude  $I_{\text{RXPN}}$ , which flows from RXDP to RXDN, when the differential input voltage amplitude and the receiver common-mode voltage are both in the range of  $V_{\text{DIF\_RX}}$  and  $V_{\text{CM\_RX}}$ , respectively.

$R_{\text{DIF\_RX}}$  can be calculated from the following equation:

$$R_{\text{DIF\_RX}} = \frac{V_{\text{RXDP}} - V_{\text{RXDN}}}{I_{\text{RXPN}}} \quad (\text{Equation 27})$$

The termination resistance shall conform with the limits of  $R_{\text{DIF\_RX}}$ .

### 5.2.1.4 Differential Termination Switching Time

If an M-RX contains a differential termination resistor, it detects from the LINE state, when  $R_{\text{DIF\_RX}}$  has to be enabled or disabled, as defined in **Section 4.7.2**.

The differential termination enable time,  $T_{\text{TERM\_ON\_HS\_RX}}$ ,  $T_{\text{TERM\_ON\_PWM\_RX}}$  or  $T_{\text{TERM\_ON\_SYS\_RX}}$  (see **Figure 15** for HS example), is defined as the time from the zero crossing of the triggering DIF-N to DIF-P transition until the time when the differential input voltage reaches the evaluation level  $V_{\text{TERM\_ON\_EVAL}}$ , where  $V_{\text{TERM\_ON\_EVAL}}$  is defined as the 20% level of the voltage difference when the M-RX is not terminated and when the M-RX is terminated, as shown by the following equation:

$$V_{\text{TERM\_ON\_EVAL}} = V_{\text{DIF\_RT\_RX}} + 0.2(V_{\text{DIF\_NT\_RX}} - V_{\text{DIF\_RT\_RX}}) \quad (\text{Equation 28})$$

The differential termination enable time shall conform with the limit of the appropriate PREPARE time in **Table 8**.

$R_{\text{DIF\_RX}}$  is disabled through different triggering events for the HS-MODE, the PWM-MODE, and the SYS-MODE. This results in three different definitions of the differential termination disabled time. All termination disable times are defined using an evaluation level  $V_{\text{TERM\_OFF\_EVAL}}$ , which is defined as the 80% level of the voltage difference when the M-RX is not terminated and when the M-RX is terminated, as shown by the following equation:

$$V_{\text{TERM\_OFF\_EVAL}} = V_{\text{DIF\_RT\_RX}} + 0.8(V_{\text{DIF\_NT\_RX}} - V_{\text{DIF\_RT\_RX}}) \quad (\text{Equation 29})$$

In HS-MODE, the differential termination disable time,  $T_{\text{TERM\_OFF\_HS\_RX}}$ , is defined as the time starting after TOB until the time when the differential input voltage reaches  $V_{\text{TERM\_OFF\_EVAL}}$ . The differential termination disable time shall conform with the limit defined by  $\text{RX\_Min\_STALL\_NoConfig\_Time\_Capability}$  in HS-MODE.

In PWM-MODE, the differential termination disable time,  $T_{\text{TERM\_OFF\_PWM\_RX}}$ , is defined as the time starting after TOB until the time when the differential input voltage reaches  $V_{\text{TERM\_OFF\_EVAL}}$ . The differential termination disable time shall conform with the limit defined by  $\text{RX\_Min\_SLEEP\_NoConfig\_Time\_Capability}$  in PWM-MODE.

In SYS-MODE, the differential termination disable time,  $T_{\text{TERM\_OFF\_SYS\_RX}}$ , is defined as the time starting after TOB until the time when the differential input voltage reaches  $V_{\text{TERM\_OFF\_EVAL}}$ . The differential termination disable time shall conform with the limit defined by  $\text{RX\_Min\_SLEEP\_NoConfig\_Time\_Capability}$  in SYS-MODE.

5.2.1.5 Return Loss

The receiver return loss parameter is based on a mixed-mode S-parameter matrix. The single ended S-parameters are characterized using the reference impedance  $R_{REF\_RT}/2$ .

The characterization can be done with a setup, illustrated in **Figure 40**. VC denotes the common-mode voltage whereas VD denotes the differential voltage.

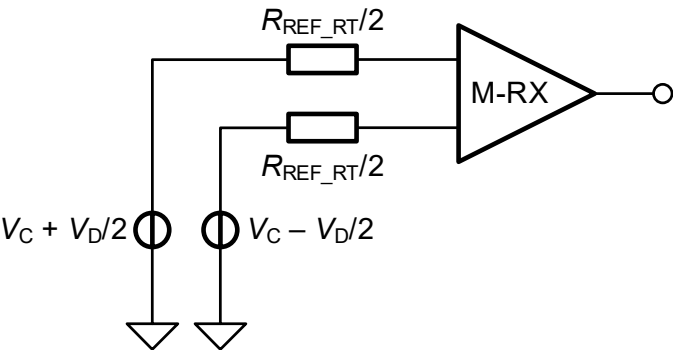


Figure 40 Measurement Setup for M-RX Return Loss

The differential receiver return loss,  $SDD_{RX}$ , is defined for an M-RX with the termination resistor enabled.  $SDD_{RX}$  is defined at the PINs such that it includes contributions from the on-chip circuitry as well as from the package. When the M-RX is not terminated, the PIN capacitance should be limited by  $C_{PIN\_RX}$ .

The  $SDD_{RX}$  is considered to be informative. The  $SDD_{RX}$  template is shown in **Figure 41** along with the return loss values at certain corner frequencies  $f_{HS\_MIN}$ ,  $f_{HS}$  and  $f_{HS\_MAX}$ , which are defined in **Section 5.1.1.1**. The differential receiver return loss of an M-RX should conform with the specification limits of  $SDD_{RX}$ .

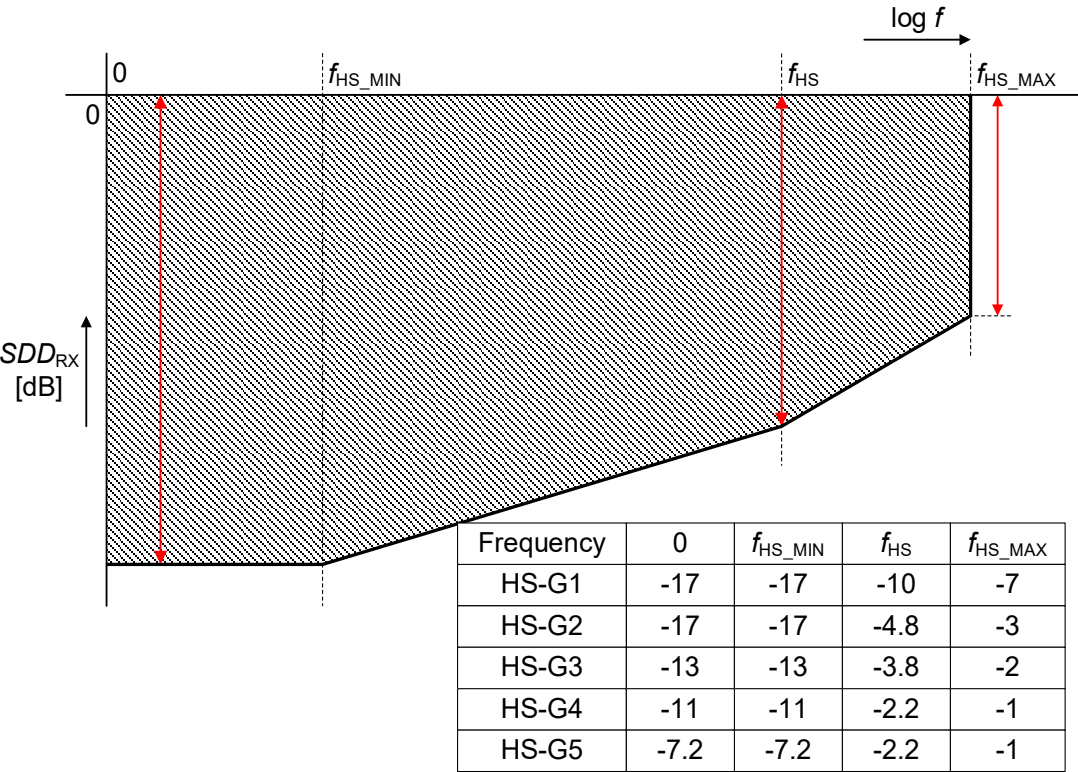


Figure 41 Template for Differential Receiver Return Loss  $SDD_{RX}$

### 5.2.1.6 Common M-RX Parameters

1394 The common electrical and timing parameters of an M-RX are summarized in *Table 20*.

**Table 20 Common M-RX Parameters**

| Symbol            | Values |      | Unit     | Description                                                                                                                       |
|-------------------|--------|------|----------|-----------------------------------------------------------------------------------------------------------------------------------|
|                   | Min.   | Max. |          |                                                                                                                                   |
| M-RX Electrical   |        |      |          |                                                                                                                                   |
| $V_{DIF\_RT\_RX}$ | 60     | 310  | mV       | Differential RX voltage amplitude in terminated state. Defined for CJTPAT <sup>1</sup> . See <b>Section 5.2.1.2</b> .             |
| $V_{DIF\_NT\_RX}$ | 120    | 620  | mV       | Differential RX voltage amplitude when the M-RX is not terminated. Defined for CJTPAT <sup>1</sup> . See <b>Section 5.2.1.2</b> . |
| $V_{CM\_RX}$      | 25     | 330  | mV       | RX common-mode voltage <sup>2</sup> . Defined for CJTPAT <sup>1</sup> . See <b>Section 5.2.1.2</b> .                              |
| M-RX Resistance   |        |      |          |                                                                                                                                   |
| $R_{DIF\_RX}$     | 80     | 110  | $\Omega$ | Differential input resistance <sup>3</sup> . Defined over $V_{DIF\_RX}$ range. See <b>Section 5.2.1.3</b> .                       |

**Note:**

1. Measurement based on accumulative eye diagram. Measurements are accomplished using the Compliant Jitter Tolerance Pattern (CJTPAT).
2. The values include a ground shift of  $\pm 50$  mV between the M-TX and M-RX.
3. The tolerance for the minimum and the maximum of  $R_{DIF\_RX}$  is different when a nominal resistance of 100  $\Omega$  is assumed. The reason for the 20  $\Omega$  decrease of the minimum is to cope with interconnect resistances below 50  $\Omega$ . However, for the maximum only an increase of 10  $\Omega$  is specified to limit the voltage drop over  $R_{DIF\_RX}$ .

## 5.2.2 HS-RX Characteristics

This section contains the electrical and timing characteristics specific to an HS-RX which are not covered by the common M-RX characteristics in **Section 5.2.1**.

### 5.2.2.1 LANE-to-LANE Skew

The HS-RX LANE-to-LANE skew  $T_{L2L\_SKEW\_HS\_RX}$  is defined as the time between the zero crossings of the differential input signal  $V_{DIF\_RX}(t)$  at any two HS-RXs in one SUB-LINK when test patterns are applied at both HS-RX PINs. The value of  $T_{L2L\_SKEW\_HS\_RX}$  is outside the scope of this document. If required, it shall be defined in the protocol specification.

### 5.2.2.2 Receiver Jitter Tolerance

The receiver total jitter tolerance,  $TJ_{RX}$ , is defined similarly to the transmitter total jitter,  $TJ_{TX}$ . The primary difference between  $TJ_{TX}$  and  $TJ_{RX}$  is that  $TJ_{TX}$  is a jitter characteristic and directly measured while  $TJ_{RX}$  is a jitter tolerance of HS-RX.  $TJ_{RX}$  is the sum of the receiver deterministic jitter tolerance,  $DJ_{RX}$ , and the receiver random jitter tolerance,  $RJ_{RX}$ , of the differential input signal  $V_{DIF\_RX}(t)$ .  $TJ_{RX}$  is defined for the required *BER* and is shown by the following equation:

$$TJ_{RX} = DJ_{RX} + RJ_{RX} \quad \text{(Equation 30)}$$

$TJ_{RX}$  and  $DJ_{RX}$  are defined relative to  $UI_{HS}$  over a frequency range from *DC* up to  $f_{U\_RX}$ , whereas  $RJ_{RX}$  is defined over the frequency from  $f_{C\_HS\_RX}$  to  $f_{U\_RX}$ .

The receiver short term total jitter tolerance,  $STTJ_{RX}$ , and the receiver short term deterministic jitter tolerance,  $STDJ_{RX}$ , are defined and limit the jitter within a  $30UI_{HS}$  signal sequence. The short term jitter corresponds to high frequency jitter in the frequency domain. In practice, short term deterministic jitter is dominated by data dependent jitter  $DDJ_{RX}$  and crosstalk originating from the transmitter and the channel.  $STTJ_{RX}$  is shown by the following equation:

$$STTJ_{RX} = STDJ_{RX} + STRJ_{RX} \quad \text{(Equation 31)}$$

where  $STRJ_{RX}$  is defined over the frequency range from  $f_{SJ4\_RX}$  up to  $f_{U\_RX}$ .  $STDJ_{RX}$  can be a combination of  $DDJ_{RX}$  and short term sinusoidal jitter  $STSJ_{RX}$ . The frequency of  $STSJ_{RX}$  is greater than  $f_{SJ4\_RX}$ .

The receiver total short term jitter tolerance,  $STTJ_{RX}$  is defined for the differential input signal  $V_{DIF\_RX}(t)$  at the zero crossings and conforms to the accumulated differential input voltage amplitude  $V_{DIF\_ACC\_RX}$  when a CJTPAT test pattern is applied at an HS-RX.

The receiver total jitter tolerance  $TJ_{RX}$  is defined for the differential input signal  $V_{DIF\_RX}(t)$  at the zero crossings and conforms to the accumulated differential input voltage amplitude  $V_{DIF\_ACC\_RX}$  when a CJTPAT test pattern is applied at an HS-RX.

The deterministic jitter tolerance,  $DJ_{RX}$ , can be expressed by the following equation:

$$DJ_{RX} = STDJ_{RX} + SJ_{RX}(f) \quad \text{(Equation 32)}$$

A sinusoidal jitter tolerance mask is given in **Figure 42**, which defines the sinusoidal jitter tolerance amplitude depending on frequency. The sinusoidal jitter is characterized by its oscillation frequency  $f_{SJ\_RX}$  and its peak-to-peak amplitude, which is identical to the receiver sinusoidal jitter tolerance  $SJ_{RX}(f)$ . In practice sinusoidal jitter may or may not be present in an M-PHY LINK. However, sinusoidal jitter tolerance is a convenient method for quantifying the HS-RX behavior and its effective minimum jitter tracking bandwidth.

The sinusoidal jitter tolerance  $SJ_{RX}(f)$  is defined in two frequency regions that range from  $f_{SJ0\_RX}$  up to  $f_{C\_HS\_RX}$  and from  $f_{C\_HS\_RX}$  up to  $f_{SJ4\_RX}$ . The low frequency sinusoidal jitter  $f_{SJ0\_RX}$  is defined by the following equation:

$$f_{SJ0\_RX} = \frac{f_{C\_HS\_RX}}{10} \quad (\text{Equation 33})$$

The sinusoidal jitter tolerance mask for shared reference clock topologies continues with -20dB/dec slope below the  $f_{C\_HS\_RX}$  and does not exceed  $SJ_{RX}(f)$  at  $f_{SJ0\_RX}$  for frequencies below  $f_{SJ0\_RX}$  as shown in **Figure 42**. **Table 19** contains a list of frequencies for conformance testing.  $f_{SJ3\_RX}$  is included in this list if it is within the range set by  $f_{SJ0\_RX}$  and  $f_{SJ4\_RX}$ . The jitter amplitudes are given in **Table 21**. The reference clock jitter and M-RX PLL jitter are not explicitly defined. However, reference clock and M-RX PLL jitter characteristics are included in the receiver jitter tolerance.

An HS-RX shall tolerate a CJTPAT test pattern with a deterministic jitter tolerance  $DJ_{RX}$  onto which random jitter tolerance  $RJ_{RX}$  is superpositioned, where the value of  $RJ_{RX}$  is indirectly specified through **Equation 30**.

An HS-RX shall tolerate a CJTPAT test pattern with short term deterministic jitter tolerance  $STDJ_{RX}$  onto which short term random jitter tolerance  $STRJ_{RX}$  is superpositioned, where the value of  $STRJ_{RX}$  is indirectly specified through **Equation 31**.

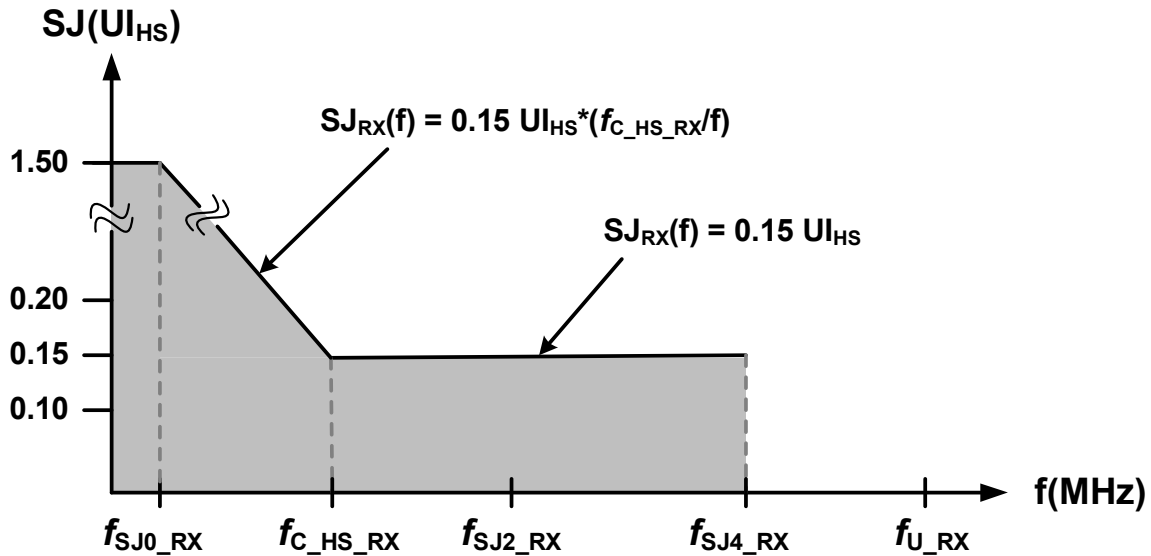


Figure 42 Sinusoidal Jitter Tolerance Mask

### 5.2.2.3 Receiver Eye Opening and Accumulated Differential Receiver Input Voltage

The minimum value of  $V_{\text{DIF\_RX}}$ , as described in **Section 5.2.1.2**, defines the minimum instantaneous differential input voltage amplitude at the M-RX PINs. In addition, the accumulated differential receiver input voltage  $V_{\text{DIF\_ACC\_RX}}$  is defined as the minimum differential voltage amplitude within an accumulated eye diagram generated from a CJTPAT test pattern.  $V_{\text{DIF\_ACC\_HS\_G1\_RX}}$ ,  $V_{\text{DIF\_ACC\_HS\_G2\_RX}}$ ,  $V_{\text{DIF\_ACC\_HS\_G3\_RX}}$ ,  $V_{\text{DIF\_ACC\_HS\_G4\_RX}}$  and  $V_{\text{DIF\_ACC\_HS\_G5\_RX}}$  are the accumulated differential receiver input voltages for an HS-RX operated in HS-G1, HS-G2, HS-G3, HS-G4, and HS-G5 respectively.  $V_{\text{DIF\_ACC\_HS\_G3\_RX}}$ ,  $V_{\text{DIF\_ACC\_HS\_G4\_RX}}$  and  $V_{\text{DIF\_ACC\_HS\_G5\_RX}}$  are defined at the midpoint of the eye,  $UI_{\text{HS}}/2$ .

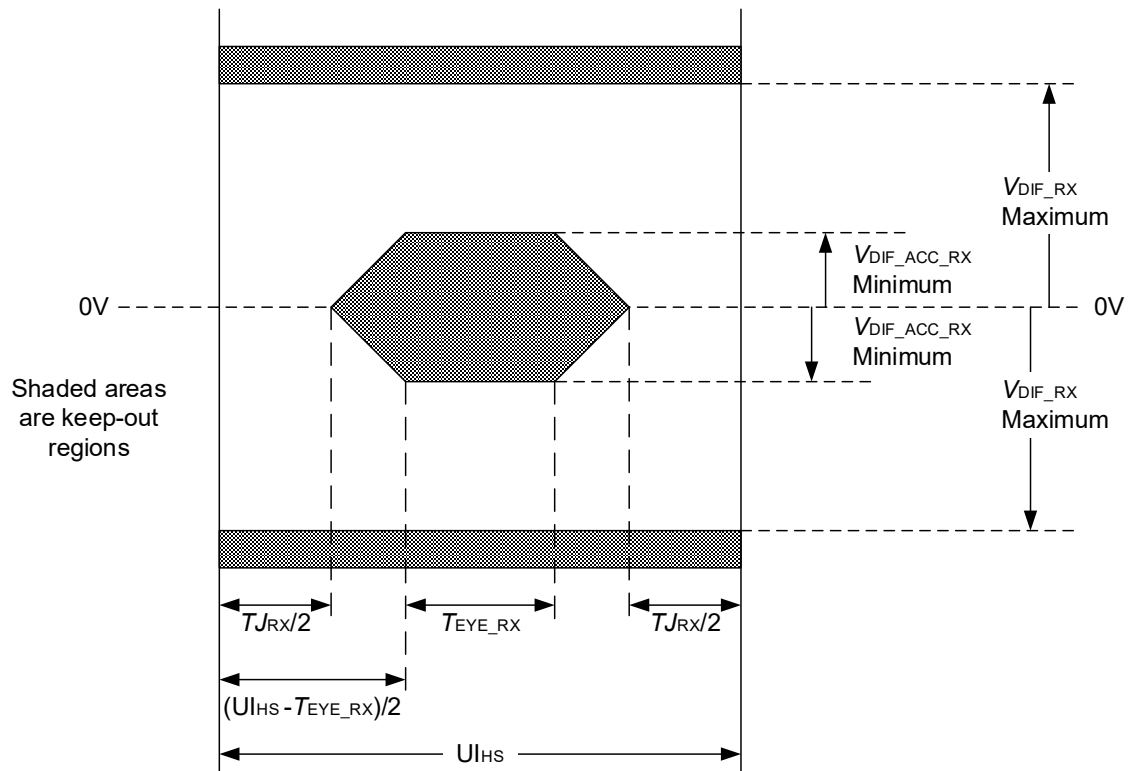
For HS-G1 and HS-G2, the receiver eye opening,  $T_{\text{EYE\_RX}}$ , is defined as the duration over which the differential voltage amplitude has to be larger than  $V_{\text{DIF\_ACC\_RX}}$  in the accumulated eye diagram generated from a CJTPAT test pattern. For HS-G3, HS-G4 and HS-G5 the receiver eye openings,  $T_{\text{EYE\_HS\_G3\_RX}}$ ,  $T_{\text{EYE\_HS\_G4\_RX}}$  and  $T_{\text{EYE\_HS\_G5\_RX}}$  are defined as the duration over which no zero crossings of  $V_{\text{DIF\_RX}}(t)$  are allowed in the eye diagram generated from a CJTPAT test pattern. The total receiver jitter tolerance  $TJ_{\text{RX}}$  is defined as the duration between the earliest and latest zero crossing at one crossing point in the accumulated eye diagram as defined in **Section 5.2.1.2**.

$V_{\text{DIF\_ACC\_RX}}$ ,  $T_{\text{EYE\_RX}}$ , and  $TJ_{\text{RX}}/2$  define the HS-G1 and HS-G2 eye mask for the accumulated M-RX signal as shown in **Figure 43**. The absolute value of the HS-RX differential input voltage signal shall be larger than the lower limit of  $V_{\text{DIF\_ACC\_RX}}$  over the receiver eye opening  $T_{\text{EYE\_RX}}$  and the accumulated eye diagram shall conform with the eye diagram mask. The position of  $T_{\text{EYE\_RX}}$  is centred in the middle of the eye.

$V_{\text{DIF\_ACC\_HS\_G3\_RX}}$ ,  $T_{\text{EYE\_HS\_G3\_RX}}$  and  $TJ_{\text{RX}}/2$  define the HS-G3 eye mask for the accumulated M-RX signal as shown in **Figure 44**. Similarly,  $V_{\text{DIF\_ACC\_HS\_G4\_RX}}$ ,  $T_{\text{EYE\_HS\_G4\_RX}}$ ,  $V_{\text{DIF\_ACC\_HS\_G5\_RX}}$ ,  $T_{\text{EYE\_HS\_G5\_RX}}$  and  $TJ_{\text{RX}}/2$  define the HS-G4 and HS-G5 eye masks. The absolute value of the HS-RX differential input voltage signal shall be larger than the lower limit of  $V_{\text{DIF\_ACC\_HS\_G3\_RX}}$  or  $V_{\text{DIF\_ACC\_HS\_G4\_RX}}$  or  $V_{\text{DIF\_ACC\_HS\_G5\_RX}}$ . Additionally, the accumulated eye diagram shall conform to the eye diagram mask. The position of  $T_{\text{EYE\_HS\_G3\_RX}}$ ,  $T_{\text{EYE\_HS\_G4\_RX}}$  and  $T_{\text{EYE\_HS\_G5\_RX}}$  is centered in the middle of the eye.

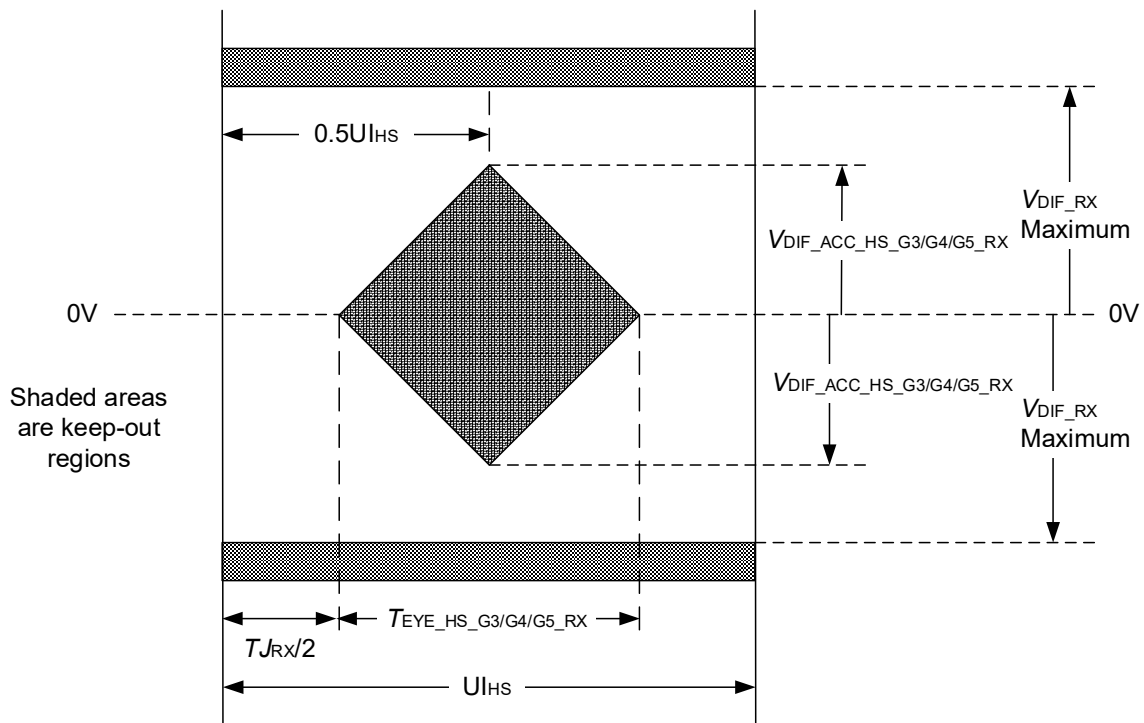
An HS-RX shall receive an input signal at the RXDP and RXDN PINs which conforms to the limits of  $V_{\text{DIF\_ACC\_RX}}$ ,  $T_{\text{EYE\_RX}}$ , and  $TJ_{\text{RX}}$  for the respective HS-Gear. The accumulated eye diagram in HS-G4 and HS-G5 shall be post-processed with the reference package and receiver reference equalizer. Definitions visualized in **Figure 43** and **Figure 44** are based on the accumulated eye for the target BER.

For conformance test, the accumulated eye diagram should closely meet the keep-out region of the eye mask but the accumulated eye diagram is not required to touch the keep-out region of the eye mask in all points.



1475

Figure 43 HS-G1 and HS-G2 Differential Receiver Eye Diagram



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Figure 44 HS-G3, HS-G4, HS-G5 Differential Receiver Eye Diagram



#### 5.2.2.4 Receiver Pulse Width

The receiver pulse width,  $T_{\text{PULSE\_RX}}$ , is defined as the minimum time between the zero crossings of the differential input signal  $V_{\text{DIF\_RX}}(t)$  when a test pattern is applied at the RXDP and RXDN PINs of an HS-RX.  $T_{\text{PULSE\_RX}}$  is shown in **Figure 45** for a DIF-P pulse. Each symbol should conform with the receiver pulse width in **Figure 45** to ensure reliable reception.

An HS-RX shall detect an input signal with a receiver pulse width that conforms with the limit of  $T_{\text{PULSE\_RX}}$ .

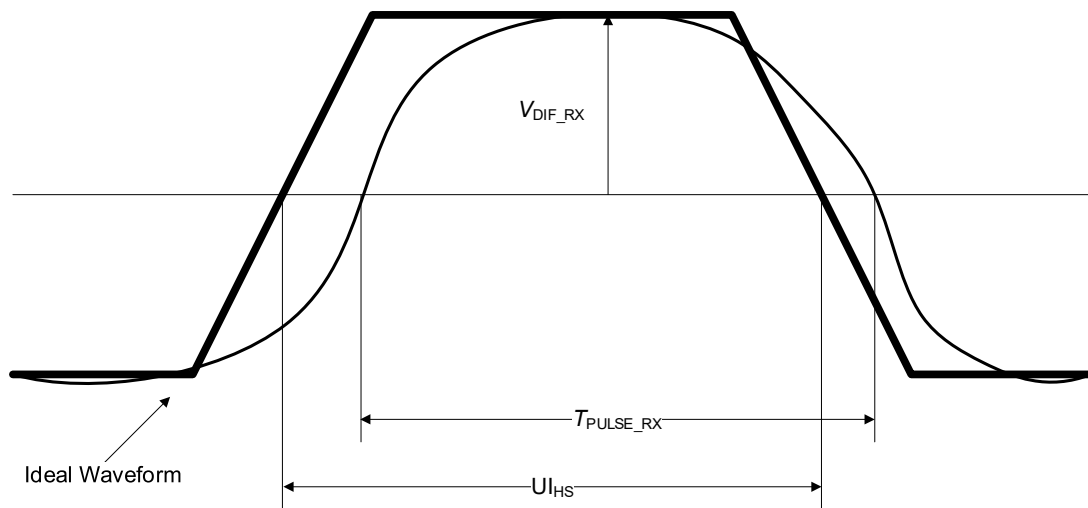


Figure 45 Receiver Pulse Width

### 5.2.2.5 Eye Monitor

The Eye Monitor function enables confirmation of the status of the eye after the receiver equalizer. The Eye Monitor test operation shall be done after ADAPT.

The system sets a Timing Offset and a Voltage Offset, both relative to the data sampling point. Mismatched data are counted as errors by comparing the sampled data with the data from the selected offset point. The system can use these result comparison errors to confirm the eye margin. Setting resolutions and offsets are detailed in subsections below.

The Eye Monitor function is optional. When a PHY receiver supports the Eye Monitor function, EYEMON Capability Attribute RX\_EYEMON\_Capability (see **Table 45**) is set to 1.

See also: *Annex G, Eye Monitor Sequence Examples*.

#### 5.2.2.5.1 Voltage and Timing Setting Resolutions

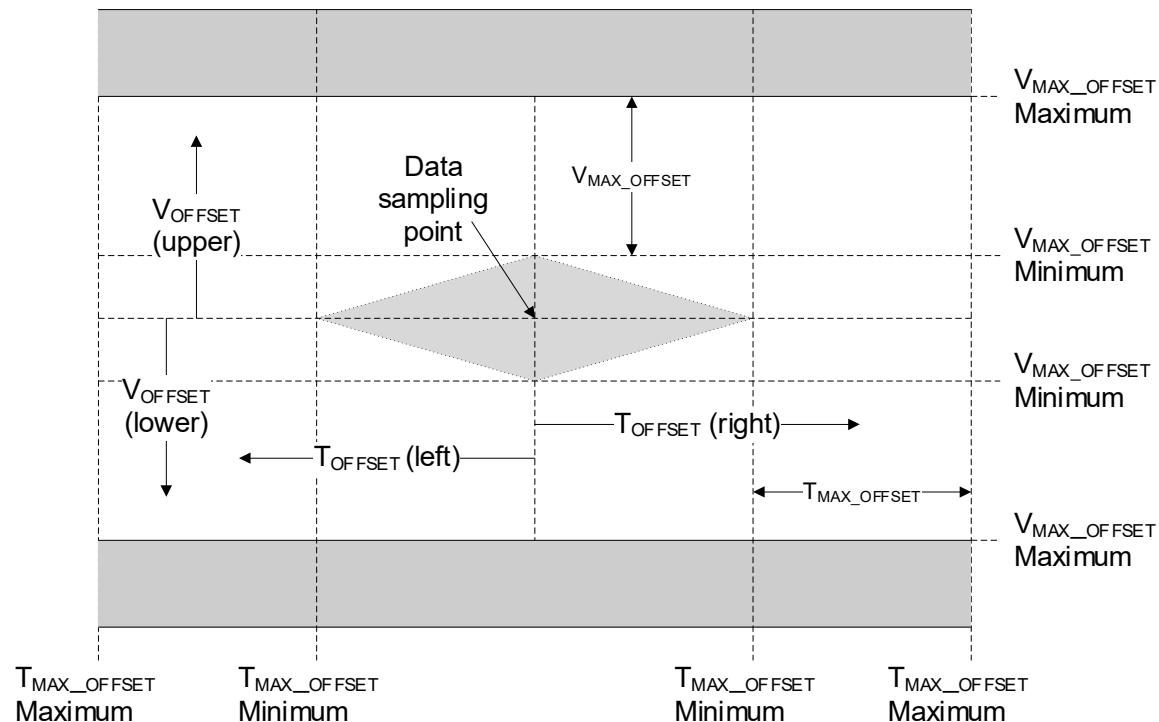
The Voltage and Timing setting resolutions are given by the following equations, based on HS-RX Parameters defined in **Table 21**:

$$RES_{TIMING} = \frac{T_{MAX\_OFFSET}}{N_{MAX\_TIMING\_STEPS}} \quad (\text{Equation 34})$$

$$RES_{VOLTAGE} = \frac{V_{MAX\_OFFSET}}{N_{MAX\_VOLTAGE\_STEPS}} \quad (\text{Equation 35})$$

#### 5.2.2.5.2 Voltage and Timing Offsets

**Figure 46** illustrates the Eye Monitor offset range, in the voltage dimension (vertical axis) and the timing dimension (horizontal axis). Voltage and timing test measurement offsets relative to the data sampling point are set using the EYEMON Configuration Attributes (see **Table 46**).



**Figure 46 Eye Monitor Offset Range**

**Offset Values:** Voltage and timing offset values are discrete and are based on the voltage resolution and the timing resolution, respectively. Offset values are given by the following equations (which are based on EYEMON Configuration Attributes specified in **Table 46**, and on the setting resolutions given by **Equation 34** and **Equation 35** above):

$$T_{\text{OFFSET}} = \text{RX\_EYEMON\_Timing\_Steps} \times \text{RES}_{\text{TIMING}} \quad (\text{Equation 36})$$

$$V_{\text{OFFSET}} = \text{RX\_EYEMON\_Voltage\_Steps} \times \text{RES}_{\text{VOLTAGE}} \quad (\text{Equation 37})$$

**Offset Directions:** Voltage and timing offset directions (i.e., positive vs. negative relative to the data sampling point) are indicated via EYEMON Configuration Attributes specified in **Table 46**:

- Timing Offsets: Bit 6 of RX\_EYEMON\_Timing\_Steps indicates right side (0) vs. left side (1)
- Voltage Offsets: Bit 6 of RX\_EYEMON\_Voltage\_Steps indicates upper side (0) vs. lower side (1)

**Maximum Offsets:** The maximum Timing and Voltage Offsets are set via HS-RX Parameters  $T_{\text{MAX\_OFFSET}}$  and  $V_{\text{MAX\_OFFSET}}$ .

- The Eye Monitor offset configuration value is set within these maximum offset values.
- If a receiver has the maximum valid offset, then the maximum offset value applies to both sides:
  - Timing Offsets: The maximum offset value is applied to both the left side and the right side
  - Voltage Offsets: The maximum offset value is applied to both the upper side and the lower side
- A receiver can support both the Timing Offset and the Voltage Offset, or only one of them.
  - If only the Voltage Offset is supported, then the Capability Attribute  $T_{\text{MAX\_OFFSET}}$  shall be set to 0.
  - If only the Timing Offset is supported, then the Capability Attribute  $V_{\text{MAX\_OFFSET}}$  shall be set to 0.

**Offset Steps:** The Timing and Voltage Offset measurements are stepped (quantized). The number of offset steps for each side is set using HS-RX Parameters defined in **Table 21**:

- Parameter  $N_{\text{MAX\_TIMING\_STEPS}}$  is the timing step value at HS-RX Parameter  $T_{\text{MAX\_OFFSET}}$ . The maximum value for this parameter is set using Capability Attribute RX\_EYEMON\_Timing\_MAX\_Steps\_Capability (see **Table 45**).
- Parameter  $N_{\text{MAX\_VOLTAGE\_STEPS}}$  is the voltage step value at HS-RX Parameter  $V_{\text{MAX\_OFFSET}}$ . The maximum value for this parameter is set using Capability Attribute RX\_EYEMON\_Voltage\_MAX\_Steps\_Capability (see **Table 45**).
- Symmetry in the vertical vs. horizontal measurements is achieved when parameters  $N_{\text{MAX\_TIMING\_STEPS}}$  and  $N_{\text{MAX\_VOLTAGE\_STEPS}}$  are set to the same value.
- To determine the device's offset value per step, read Capability Attributes RX\_EYEMON\_Timing\_MAX\_Steps\_Capability and RX\_EYEMON\_Voltage\_MAX\_Steps\_Capability (see **Table 45**).

**5.2.2.6 HS-RX Parameters**

1532 The electrical and timing parameters of the HS-RX are summarized in *Table 21*.

**Table 21 HS-RX Parameters**

| Symbol                     | Values        |      | Unit             | Description                                                                                                                                        |
|----------------------------|---------------|------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | Min.          | Max. |                  |                                                                                                                                                    |
| HS-RX Electrical           |               |      |                  |                                                                                                                                                    |
| $V_{DIF\_ACC\_RX}$         | 40            | —    | mV               | Accumulated differential receiver input voltage for HS-G1 <sup>1</sup> . Defined for CJTPAT <sup>2</sup> . See <b>Section 5.2.2.3</b> .            |
| $V_{DIF\_ACC\_HS\_G2\_RX}$ | 40            | —    | mV               | Accumulated differential receiver input voltage for HS-G2 <sup>1</sup> . Defined for CJTPAT <sup>2</sup> . See <b>Section 5.2.2.3</b> .            |
| $V_{DIF\_ACC\_HS\_G3\_RX}$ | 40            | —    | mV               | Accumulated differential receiver input voltage for HS-G3 <sup>1</sup> . Defined for CJTPAT <sup>2</sup> . See <b>Section 5.2.2.3</b> .            |
| $V_{DIF\_ACC\_HS\_G4\_RX}$ | 40            | —    | mV               | Accumulated differential receiver input voltage for HS-G4 <sup>1</sup> . Defined for CJTPAT <sup>2</sup> . See <b>Section 5.2.2.3</b> .            |
| $V_{DIF\_ACC\_HS\_G5\_RX}$ | 30            | —    | mV               | Accumulated differential receiver input voltage for HS-G5 <sup>1</sup> . Defined for CJTPAT <sup>2</sup> . See <b>Section 5.2.2.3</b> .            |
| HS-RX Timing               |               |      |                  |                                                                                                                                                    |
| $T_{EYE\_RX}$              | 0.20          | —    | UI <sub>HS</sub> | Receiver eye opening. Defined for CJTPAT <sup>2</sup> over a statistical confident record set <sup>4</sup> . See <b>Section 5.2.2.3</b> .          |
| $T_{EYE\_HS\_G3\_RX}$      | 1 - $T_{JRX}$ | —    | UI <sub>HS</sub> | Receiver eye opening in HS-G3. Defined for CJTPAT <sup>2</sup> over a statistical confident record set <sup>6</sup> . See <b>Section 5.2.2.3</b> . |
| $T_{EYE\_HS\_G4\_RX}$      | 1 - $T_{JRX}$ | —    | UI <sub>HS</sub> | Receiver eye opening in HS-G4. Defined for CJTPAT <sup>2</sup> over a statistical confident record set <sup>6</sup> . See <b>Section 5.2.2.3</b> . |
| $T_{EYE\_HS\_G5\_RX}$      | 1 - $T_{JRX}$ | —    | UI <sub>HS</sub> | Receiver eye opening in HS-G5. Defined for CJTPAT <sup>2</sup> over a statistical confident record set <sup>6</sup> . See <b>Section 5.2.2.3</b> . |
| $T_{PULSE\_RX}$            | 0.80          | —    | UI <sub>HS</sub> | Receiver pulse width. Defined for CJTPAT <sup>2</sup> . See <b>Section 5.2.2.4</b> .                                                               |

| Symbol                           | Values |                                       | Unit             | Description                                                                                                                                                                                                                                    |
|----------------------------------|--------|---------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | Min.   | Max.                                  |                  |                                                                                                                                                                                                                                                |
| HS-RX Jitter for HS-G4 and below |        |                                       |                  |                                                                                                                                                                                                                                                |
| $DJ_{RX}$                        | —      | 0.35                                  | UI <sub>HS</sub> | Receiver deterministic jitter <sup>3</sup> over the frequency range from $f_{C\_HS\_RX}$ to $f_{U\_RX}$ which includes $STDJ_{RX}$ . Defined for CJTPAT for a statistical confident record set <sup>2,4,5</sup> . See <b>Section 5.2.2.2</b> . |
| $SJ_{RX}(f)$                     | —      | 0.15                                  | UI <sub>HS</sub> | Receiver sinusoidal jitter tolerance <sup>3</sup> . Defined for CJTPAT and frequency range from $f_{C\_HS\_RX}$ to $f_{SJ4\_RX}$ . See <b>Section 5.2.2.2</b> .                                                                                |
|                                  | —      | $\frac{0.15 \times f_{C\_HS\_RX}}{f}$ | UI <sub>HS</sub> | Receiver sinusoidal jitter tolerance <sup>3</sup> . Defined for CJTPAT and frequency range from $f_{SJ0\_RX}$ to $f_{C\_HS\_RX}$ . See <b>Section 5.2.2.2</b> .                                                                                |
| $STDJ_{RX}$                      | —      | 0.20                                  | UI <sub>HS</sub> | Receiver deterministic jitter over the frequency range from $f_{SJ4\_RX}$ to $f_{U\_RX}$ . Defined for CJTPAT for a statistical confident record set <sup>2,5,6</sup> . See <b>Section 5.2.2.2</b> .                                           |
| $TJ_{RX}$                        | —      | 0.52                                  | UI <sub>HS</sub> | Receiver total jitter tolerance <sup>3</sup> for galvanic interconnect. Defined for CJTPAT and a statistical confident record set <sup>2,4,5</sup> . See <b>Section 5.2.2.2</b> .                                                              |
| $STTJ_{RX}$                      | —      | 0.30                                  | UI <sub>HS</sub> | Receiver total jitter over the frequency range from $f_{SJ4\_RX}$ to $f_{U\_RX}$ . Defined for CJTPAT and a statistical confident record set <sup>2,5, 6</sup> . See <b>Section 5.2.2.2</b> .                                                  |

| Symbol                 | Values |                                       | Unit             | Description                                                                                                                                                                                                                                    |
|------------------------|--------|---------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | Min.   | Max.                                  |                  |                                                                                                                                                                                                                                                |
| HS-RX Jitter for HS-G5 |        |                                       |                  |                                                                                                                                                                                                                                                |
| $DJ_{RX}$              | —      | 0.53                                  | UI <sub>HS</sub> | Receiver deterministic jitter <sup>3</sup> over the frequency range from $f_{C\_HS\_RX}$ to $f_{U\_RX}$ which includes $STDJ_{RX}$ . Defined for CJTPAT for a statistical confident record set <sup>2,4,5</sup> . See <b>Section 5.2.2.2</b> . |
| $SJ_{RX}(f)$           | —      | 0.15                                  | UI <sub>HS</sub> | Receiver sinusoidal jitter tolerance <sup>3</sup> . Defined for CJTPAT and frequency range from $f_{C\_HS\_RX}$ to $f_{SJ4\_RX}$ . See <b>Section 5.2.2.2</b> .                                                                                |
|                        | —      | $\frac{0.15 \times f_{C\_HS\_RX}}{f}$ | UI <sub>HS</sub> | Receiver sinusoidal jitter tolerance <sup>3</sup> . Defined for CJTPAT and frequency range from $f_{SJ0\_RX}$ to $f_{C\_HS\_RX}$ . See <b>Section 5.2.2.2</b> .                                                                                |
| $STDJ_{RX}$            | —      | 0.38                                  | UI <sub>HS</sub> | Receiver deterministic jitter over the frequency range from $f_{SJ4\_RX}$ to $f_{U\_RX}$ . Defined for CJTPAT for a statistical confident record set <sup>2,5,6</sup> . See <b>Section 5.2.2.2</b> .                                           |
| $TJ_{RX}$              | —      | 0.7                                   | UI <sub>HS</sub> | Receiver total jitter tolerance <sup>3</sup> for galvanic interconnect. Defined for CJTPAT and a statistical confident record set <sup>2,4,5</sup> . See <b>Section 5.2.2.2</b> .                                                              |
| $STTJ_{RX}$            | —      | 0.48                                  | UI <sub>HS</sub> | Receiver total jitter over the frequency range from $f_{SJ4\_RX}$ to $f_{U\_RX}$ . Defined for CJTPAT and a statistical confident record set <sup>2,5, 6</sup> . See <b>Section 5.2.2.2</b> .                                                  |

| Symbol                         | Values |      | Unit                    | Description                                                                                                          |
|--------------------------------|--------|------|-------------------------|----------------------------------------------------------------------------------------------------------------------|
|                                | Min.   | Max. |                         |                                                                                                                      |
| HS-RX Eye Monitor              |        |      |                         |                                                                                                                      |
| T <sub>MAX_OFFSET</sub>        | 0.15   | 0.50 | UI <sub>HS</sub>        | Maximum timing offset value. Same offset is applied to both right side and left side. See <b>Section 5.2.2.5</b> .   |
| V <sub>MAX_OFFSET</sub>        | 30     | 310  | mV                      | Maximum voltage offset value. Same offset is applied to both upper side and lower side. See <b>Section 5.2.2.5</b> . |
| N <sub>MAX_TIMING_STEPS</sub>  | 3      | 63   | —                       | Offset step value at T <sub>MAX_OFFSET</sub> . See <b>Section 5.2.2.5</b> .                                          |
| N <sub>MAX_VOLTAGE_STEPS</sub> | 6      | 63   | —                       | Offset step value at V <sub>MAX_OFFSET</sub> . See <b>Section 5.2.2.5</b> .                                          |
| RES <sub>TIMING</sub>          | 2.3    | 167  | mUI <sub>HS</sub> /step | Resolution of eye monitor timing step. See <b>Equation 36</b> .                                                      |
| RES <sub>VOLTAGE</sub>         | 0.476  | 51.7 | mV/step                 | Resolution of eye monitor voltage step. See <b>Equation 37</b> .                                                     |

**Note:**

1. Measurement based on accumulative eye diagram.
2. The test has to be performed at the maximum data rate of the applicable HS-GEAR.
3. Accumulated jitter as defined by the jitter model in **Section 5.2.2.2**.
4. Filtered using a high-pass jitter transfer function  $H_{JTF}(s)$ .
5. Measured for the target BER.
6. Filtered using a 1<sup>st</sup> order high-pass filter with a pole at  $f_{SJ4\_RX}$ .

### 5.2.3 PWM-RX Characteristics

This section contains the timing characteristics specific to a PWM-RX which are not covered by the common M-RX characteristics in *Section 5.2.1*. The PWM signaling scheme is defined in *Section 4.3.2*.

#### 5.2.3.1 Differential Receiver Input Voltage

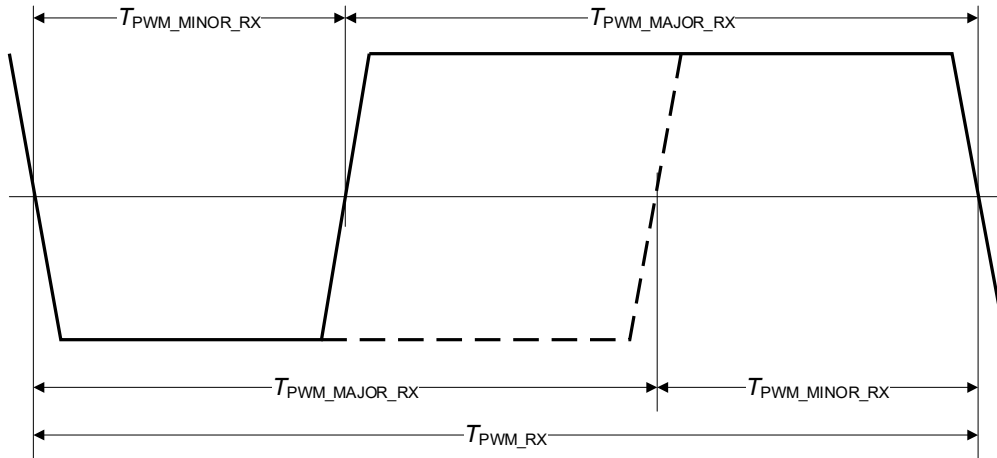
The minimum value of  $V_{\text{DIF\_RX}}$ , as described in *Section 5.2.1.2*, defines the minimum instantaneous differential input voltage amplitude at the M-RX PINs.

#### 5.2.3.2 PWM Bit Duration, Bit Duration Tolerance, and Ratio

The PWM receive bit duration  $T_{\text{PWM\_RX}}$  is defined as the duration between zero crossings of two consecutive falling edges of a differential signal at the PWM-RX input.  $T_{\text{PWM\_MINOR\_RX}}$ ,  $T_{\text{PWM\_MAJOR\_RX}}$ , and  $T_{\text{PWM\_RX}}$  are shown in *Figure 47*. The PWM receive bit duration  $T_{\text{PWM\_RX}}$  is, for all PWM GEARS, the sum of its durations  $T_{\text{PWM\_MINOR\_RX}}$  and  $T_{\text{PWM\_MAJOR\_RX}}$ , as shown in the following equation:

$$T_{\text{PWM\_RX}} = T_{\text{PWM\_MINOR\_RX}} + T_{\text{PWM\_MAJOR\_RX}} \quad (\text{Equation 38})$$

The limits of  $T_{\text{PWM\_RX}}$  are, for all PWM GEARS, identical to the limits of  $T_{\text{PWM\_TX}}$ .



**Figure 47 RX Minor and Major Duration in a PWM Signal**

$T_{\text{PWM\_MINOR\_RX}}$  and  $T_{\text{PWM\_MAJOR\_RX}}$  are determined by  $T_{\text{PWM\_RX}}$  and the PWM receive ratio  $k_{\text{PWM\_RX}}$  for PWM-G1.  $k_{\text{PWM\_RX}}$  is defined as the ratio of  $T_{\text{PWM\_MAJOR\_RX}}$  and  $T_{\text{PWM\_MINOR\_RX}}$  of one PWM bit, as shown in following equation:

$$k_{\text{PWM\_RX}} = \frac{T_{\text{PWM\_MAJOR\_RX}}}{T_{\text{PWM\_MINOR\_RX}}} \quad (\text{Equation 40})$$



1548 The PWM receive bit duration tolerance,  $TOL_{PWM\_RX}$ , is the allowed tolerance of an instantaneous PWM bit  
1549 duration,  $T_{PWM\_RX}(i)$ , in PWM-MODE.  $TOL_{PWM\_RX}$  is defined as the ratio of  $T_{PWM\_RX}(i)$  and the average of  $N$   
1550 PWM receive bit durations in PWM-MODE, as shown in the following equation:

$$TOL_{PWM\_RX} = \frac{T_{PWM\_RX}(i)}{\frac{1}{N} \sum_{j=1}^N T_{PWM\_RX}(j)}$$

(Equation 41)

1552 where  $N$  is a defined number of PWM bits, and  $i$  is in the range of 1 to  $N$ .

1553 While the  $T_{PWM\_RX}$  range is wide for a PWM GEAR,  $TOL_{PWM\_RX}$  limits the variation of  $T_{PWM\_RX}(i)$  during  
1554 PWM-MODE.

1555 A PWM-RX shall detect a PWM input signal with a PWM receive bit duration,  $T_{PWM\_RX}$ , in the specified  
1556 range of the operational PWM GEAR during a PWM-BURST. For PWM-G1, the PWM receive ratio  $k_{PWM\_RX}$   
1557 shall be in the specified range for each PWM bit.

1558 A PWM-RX shall detect a PWM input signal with PWM receive bit duration tolerance in the limits of  
1559  $TOL_{PWM\_RX}$ .

### 5.2.3.3 Rise and Fall Time

1560 The PWM-RX rise and fall times,  $T_{R\_PWM\_RX}$  and  $T_{F\_PWM\_RX}$ , respectively, are defined as transition times  
1561 between the 20% and 80% signal levels of the differential PWM-RX input signal with an amplitude of  
1562  $V_{DIF\_RX}$ .

1563 A PWM-RX shall detect a PWM input signal with rise and fall times that comply with the limits of  $T_{R\_PWM\_RX}$   
1564 and  $T_{F\_PWM\_RX}$ .

### 5.2.3.4 LANE-to-LANE Skew

1565 The PWM-RX LANE-to-LANE skew  $T_{L2L\_SKEW\_PWM\_RX}$  is defined as the time between the zero crossings of  
1566 the falling edges of the differential input signal  $V_{DIF\_RX}(t)$  at any two PWM-RXs in one SUB-LINK when test  
1567 patterns are applied at both PWM-RX PINs. The value of  $T_{L2L\_SKEW\_PWM\_RX}$  is outside the scope of this  
1568 document. If required, it shall be defined in the protocol specification.

### 5.2.3.5 PWM-RX Parameters

1569 The timing parameters of the PWM-RX are shown in *Table 22*.

**Table 22 PWM-RX Parameters**

| Symbol                   | Values              |                     | Unit                 | Description                                                                                                                      |
|--------------------------|---------------------|---------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------|
|                          | Min.                | Max.                |                      |                                                                                                                                  |
| PWM-RX Timing            |                     |                     |                      |                                                                                                                                  |
| $T_{\text{PWM\_G1\_RX}}$ | $\frac{1}{9}$       | $\frac{1}{3}$       | $\mu\text{s}$        | PWM receive bit duration in PWM-G1. Defined for CJTPAT. See <b>Section 5.2.3.2</b> .                                             |
| $TOL_{\text{PWM\_RX}}$   | 0.82                | 1.18                | —                    | PWM receive bit duration tolerance. Defined for CJTPAT in PWM-MODE. See <b>Section 5.2.3.2</b> .                                 |
| $N$                      | 50                  | 50                  | —                    | Number of PWM bits. Sequence length for $TOL_{\text{PWM\_RX}}$ and $TOL_{\text{PWM\_G1\_LR\_RX}}$ . See <b>Section 5.2.3.2</b> . |
| $k_{\text{PWM\_RX}}$     | $\frac{0.60}{0.40}$ | $\frac{0.75}{0.25}$ | —                    | PWM receive ratio for PWM-G1. Defined for CJTPAT. See <b>Section 5.2.3.2</b> .                                                   |
| $T_{\text{R\_PWM\_RX}}$  | —                   | 0.14                | $T_{\text{PWM\_RX}}$ | Rise time defined for CJTPAT.                                                                                                    |
| $T_{\text{F\_PWM\_RX}}$  | —                   | 0.14                | $T_{\text{PWM\_RX}}$ | Fall time defined for CJTPAT.                                                                                                    |

**Note:**

1. Measurements based on accumulative eye diagram.

### 5.2.4 SYS-RX Characteristics

1570 This section contains the timing characteristics specific to a SYS-RX which are not covered by the common  
1571 M-RX characteristics in **Section 5.2.1**.

#### 5.2.4.1 LANE-to-LANE Skew

1572 The SYS-RX LANE-to-LANE skew  $T_{\text{L2L\_SKEW\_SYS\_RX}}$  is defined as the time between the zero crossings of  
1573 the differential input signal  $V_{\text{DIF\_RX}}(t)$  at any two SYS-RXs in one SUB-LINK when test patterns are applied  
1574 at both SYS-RX pins. The value of  $T_{\text{L2L\_SKEW\_SYS\_RX}}$  is outside the scope of this document. If required, it shall  
1575 be defined in the protocol specification.

#### 5.2.4.2 Setup and Hold Times

1576 Some parameters of the SYS-BURST mode have to be defined by the protocol specification, as described in  
1577 **Section 5.1.4.3**. The setup and hold times of the data signal at the SYS-RX input with respect to the reference  
1578 clock signal belong to the parameters which are defined in the protocol specification. The zero crossing of  
1579 the differential signal at the SYS-RX input is used as reference timing point for such a definition. Thus,  
1580 Interoperability in SYS-BURST mode is partly ensured by the protocol specification.

## 5.2.5 SQ-RX Characteristics

This section contains the electrical and timing characteristics specific to a SQ-RX which are not covered by the common M-RX characteristics in **Section 5.2.1**. The SQ-RX drives a DIF-Z LINE state in certain states. Additionally, the SQ-RX can monitor the LINE state to detect a non-squelch state. The operation of the SQ-RX is described in **Section 4.6**.

### 5.2.5.1 Squelch Common-mode Voltage and Squelch Differential Voltage

The squelch common-mode voltage signal  $V_{CM\_SQ}(t)$  is defined as the arithmetic mean value of the voltage signals  $V_{RXDP}(t)$  and  $V_{RXDN}(t)$  when the SQ-RX drives a DIF-Z at RXDP and RXDN while the LINE is not driven from the M-TX.  $V_{CM\_SQ}$  is defined as the amplitude of  $V_{CM\_SQ}(t)$ .  $V_{CM\_SQ}(t)$  can be calculated from following equation:

$$V_{CM\_SQ}(t) = \frac{V_{RXDP}(t) + V_{RXDN}(t)}{2} \quad (\text{Equation 41})$$

A SQ-RX shall keep the squelch common-mode voltage at the M-RX PINs within the limits of  $V_{CM\_SQ}$ , while driving a DIF-Z and the LINE is not driven from the M-TX.

The squelch differential voltage signal  $V_{DIF\_SQ}(t)$  is defined as the difference of the signal voltages  $V_{RXDP}(t)$  and  $V_{RXDN}(t)$  at the M-RX PINs when the SQ-RX drives a DIF-Z at RXDP and RXDN while the LINE is not driven from the M-TX.  $V_{DIF\_SQ}$  is defined as the amplitude of  $V_{DIF\_SQ}(t)$ .  $V_{DIF\_SQ}(t)$  can be calculated from following equation:

$$V_{DIF\_SQ}(t) = V_{RXDP}(t) - V_{RXDN}(t) \quad (\text{Equation 42})$$

The SQ-RX shall control the signal voltages at the M-RX PINs such that the squelch differential voltage is below the limit of  $V_{DIF\_SQ}$ , while the SQ-RX drives a DIF-Z and the LINE is not driven from the M-TX.

The limits of  $V_{CM\_SQ}$  and of  $V_{DIF\_SQ}$  can be achieved by use of a differential resistor or two single-ended resistors, for instance.  $V_{CM\_SQ}$  and of  $V_{DIF\_SQ}$  impose limits on the M-RX input resistances at RXDP and RXDN. The lower value of the M-RX input resistances at RXDP and RXDN has to be such that an M-TX with Small Amplitude can drive the LINE from the squelch state to the non-squelch state while the SQ-RX is driving DIF-Z. The upper value of the M-RX input resistances is limited by the PIN leakage currents of the M-TX, the PIN leakage currents of the M-RX, and the mismatch of the M-TX PIN leakage currents. The M-RX input resistances has to be such that the limits of  $V_{CM\_SQ}$  and of  $V_{DIF\_SQ}$  are met for the specified M-RX and M-TX PIN leakage currents while the SQ-RX is driving DIF-Z.

### 5.2.5.2 Squelch Exit Voltage

1607 The squelch exit voltage  $V_{SQ}$  is the threshold voltage of the SQ-RX, which shall operate when the  
 1608 common-mode voltage is in the  $V_{CM\_SQ}$  range. When enabled the SQ-RX shall indicate a squelch state of the  
 1609 LINE, as long as the voltage difference of  $V_{RXDN}$  and  $V_{RXDP}$  is smaller than the minimum squelch exit voltage  
 1610  $V_{SQ}$ , i.e., squelch shall be indicated when the following relation holds:

$$1611 \quad V_{RXDN}(t) - V_{RXDP}(t) < V_{SQ, MIN} \quad (\text{Equation 43})$$

1612 When enabled the SQ-RX shall indicate a non-squelch state of the LINE, as long as the voltage difference of  
 1613  $V_{RXDN}$  and  $V_{RXDP}$  is larger than the maximum squelch exit voltage  $V_{SQ}$ , i.e., non-squelch shall be indicated  
 1614 when the following relation holds:

$$1615 \quad V_{RXDN}(t) - V_{RXDP}(t) > V_{SQ, MAX} \quad (\text{Equation 44})$$

1616 The SQ-RX does not need to detect if  $V_{RXDP}$  is by more than  $V_{SQ}$  larger than  $V_{RXDN}$ , because it is only required  
 1617 to detect the transition of the LINE state from DIF-Z to DIF-N.

### 5.2.5.3 Squelch Exit Time

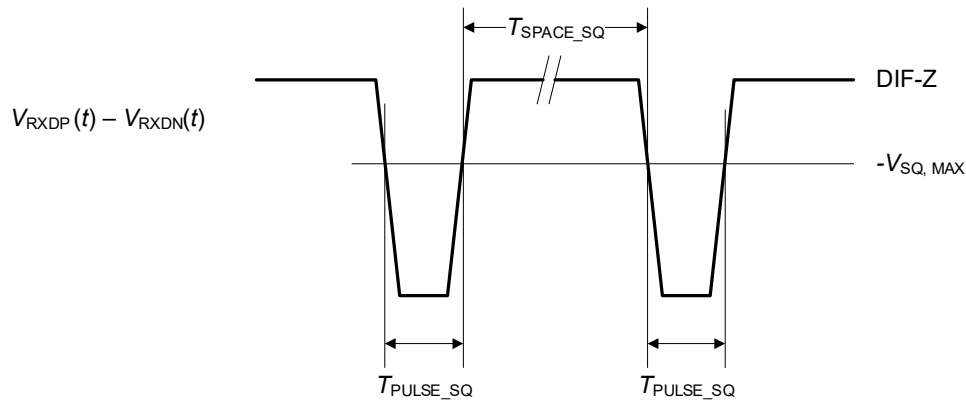
1618 The squelch exit time  $T_{SQ}$  is the duration from non-squelch detection until the M-RX enters the SLEEP state.  
 1619  $T_{SQ}$  is defined from the crossing of the differential signal  $V_{RXDN} - V_{RXDP}$  with  $V_{SQ, MAX}$  until the M-RX enters  
 1620 the SLEEP state. No value is defined for  $T_{SQ}$ , which is an M-RX internal characteristic. However the DIF-N,  
 1621 which is signaled by the M-TX upon exit of the HIBERN8 state for the period  $T_{ACTIVATE}$ , is an upper bound  
 1622 for  $T_{SQ}$ . A lower bound is the pulse width of a DIF-N pulse, which is detected as a non-squelch state by the  
 1623 SQ-RX. This pulse width is not specified, but bounded by the squelch pulse rejection.

#### 5.2.5.4 Squelch Pulse and RF Rejection

The squelch noise pulse width  $T_{\text{PULSE\_SQ}}$  is defined as the time the M-RX input signal  $V_{\text{RXDN}}(t) - V_{\text{RXDP}}(t)$  is larger than  $V_{\text{SQ\_MAX}}$ .  $T_{\text{PULSE\_SQ}}$  is shown in **Figure 48**. The SQ-RX shall reject single input noise pulses with an amplitude beyond  $V_{\text{SQ\_MAX}}$  and shorter than the squelch noise pulse width  $T_{\text{PULSE\_SQ}}$ , where the pulse is of a rectangular shape.

The noise pulse spacing  $T_{\text{SPACE\_SQ}}$  is defined as the time between the crossings of two adjacent pulse edges of two different, but adjacent, noise pulses with  $V_{\text{SQ\_MAX}}$ . Multiple pulses shall be rejected by the SQ-RX when the duration between adjacent pulses is larger than  $T_{\text{SPACE\_SQ}}$ . An example is shown in **Figure 48**.

Furthermore, the SQ-RX has to be tolerant to superimposed RF interferences onto the  $V_{\text{RXDP}}(t)$  and  $V_{\text{RXDN}}(t)$  signals. This implies an input signal filter. The RF interference is modelled by a sinusoidal signal with a peak interference amplitude  $V_{\text{INT\_SQ}}$  and an interference frequency  $f_{\text{INT\_SQ}}$ . The RF interference is superimposed on the M-RX input signal. The SQ-RX shall meet all specifications in presence of RF interferences with a peak interference amplitude  $V_{\text{INT\_SQ}}$  and frequencies higher than the interference frequency  $f_{\text{INT\_SQ}}$ . The interference shall not cause glitches or incorrect operation during signal transitions.



**Figure 48 Pulse Rejection and Non squelch State Detection**

**5.2.5.5 SQ-RX Parameters**

1638 The electrical and timing parameters of the SQ-RX are summarized in *Table 23*.

**Table 23 SQ-RX Parameters**

| Symbol           | Values |      | Unit | Description                                                                                       |
|------------------|--------|------|------|---------------------------------------------------------------------------------------------------|
|                  | Min.   | Max. |      |                                                                                                   |
| SQ-RX Electrical |        |      |      |                                                                                                   |
| $V_{SQ}$         | 50     | 140  | mV   | Squelch exit voltage.<br>Defined for test pattern. See <b>Section 5.2.5.2</b> .                   |
| $V_{DIF\_SQ}$    | –      | 20   | mV   | Squelch differential voltage amplitude.<br>Defined for test pattern. See <b>Section 5.2.5.1</b> . |
| $V_{CM\_SQ}$     | 0      | 330  | mV   | Squelch common-mode voltage.<br>Defined for test pattern. See <b>Section 5.2.5.1</b> .            |
| $V_{INT\_SQ}$    | –      | 200  | mV   | Peak interference amplitude.<br>Defined for test pattern. See <b>Section 5.2.5.4</b> .            |
| $f_{INT\_SQ}$    | 500    | –    | MHz  | Interference frequency.<br>Defined for test pattern. See <b>Section 5.2.5.4</b> .                 |
| SQ-RX Timing     |        |      |      |                                                                                                   |
| $T_{PULSE\_SQ}$  | –      | 20   | ns   | Noise pulse width.<br>Defined for test pattern. See <b>Section 5.2.5.4</b> .                      |
| $T_{SPACE\_SQ}$  | 500    | –    | ns   | Noise pulse spacing.<br>Defined for test pattern. See <b>Section 5.2.5.4</b> .                    |

### 5.3 PIN Characteristics

1639 The PIN characteristics of an M-TX and of an M-RX are defined in this section.

#### 5.3.1 PIN Capacitance

1640 The single-ended PIN capacitance  $C_{\text{PIN\_RX}}$  of the M-RX is defined as the capacitance between the RXDP and  
1641 RXDN PINs to ground.  $C_{\text{PIN\_RX}}$  is the lump sum of all single-ended capacitance at an M-RX PIN. The  
1642 single-ended PIN capacitance should conform to the limit of  $C_{\text{PIN\_RX}}$ .

1643 The PIN capacitance mismatch  $\Delta C_{\text{PIN\_RX}}$  of an M-RX is defined as the difference of the PIN capacitances at  
1644 RXDP and RXDN. The PIN capacitance mismatch shall conform to the limits of  $\Delta C_{\text{PIN\_RX}}$ .  $\Delta C_{\text{PIN\_RX}}$  limits  
1645 the timing skew between the single-ended signals.

#### 5.3.2 PIN Signal Voltage Range

1646 The PIN signal voltage  $V_{\text{PIN}}$  is defined as the single-ended signal voltage of an M-RX or M-TX PIN to ground.  
1647 No structure within an M-RX or M-TX shall be damaged when a DC voltage that is within the limits of  $V_{\text{PIN}}$   
1648 is applied at a PIN for an indefinite period of time. The single-ended output signals of an M-TX shall conform  
1649 with the limits of  $V_{\text{PIN}}$ .

#### 5.3.3 PIN Leakage Current

1650 The PIN leakage current,  $I_{\text{LEAK}}$ , is defined as the PIN current flowing in, or out, of a MODULE when a  
1651 single-ended voltage in the PIN signal voltage range,  $V_{\text{PIN}}$ , is applied at a MODULE PIN while the differential  
1652 voltage at the MODULE PINs is in the squelch common-mode voltage range,  $V_{\text{CM\_SQ}}$ .  $I_{\text{LEAK}}$  is defined for a  
1653 MODULE that does not drive the LINE and, in the case of an M-RX, with its termination resistor disabled.  
1654 The PIN leakage current of a MODULE PIN shall conform with the limits of  $I_{\text{LEAK}}$ .

1655 The PIN leakage current mismatch  $\Delta I_{\text{LEAK\_TX}}$  is defined as the difference of the PIN leakage currents at the  
1656 TXDP and TXDN PINs of an M-TX, when signal voltages are applied which conform with the  $V_{\text{CM\_SQ}}$  and  
1657  $V_{\text{DIF\_SQ}}$  ranges. The PIN leakage current mismatch shall stay in the limits of  $\Delta I_{\text{LEAK\_TX}}$ .

#### 5.3.4 Ground Shift

1658 The ground shift  $V_{\text{GND SH}}$  is defined as the ground potential difference of an M-TX and M-RX within a LANE.  
1659 The ground shift of MODULEs within a LANE shall conform with the limits of  $V_{\text{GND SH}}$ .

1660 The ground shift is taken into account in the definition of signal voltage parameters. It does not need to be  
1661 added on top of any signal parameter.

### 5.3.5 PIN Parameters

1662 The common PIN characteristics of an M-RX and M-TX are summarized in **Table 24**.

**Table 24 PIN Parameters**

| Symbol                       | Values |      | Unit | Description                                                                          |
|------------------------------|--------|------|------|--------------------------------------------------------------------------------------|
|                              | Min.   | Max. |      |                                                                                      |
| $C_{\text{PIN\_RX}}$         | —      | 1.5  | pF   | PIN capacitance. Recommended PIN capacitance to ground at an M-RX PIN <sup>1</sup> . |
| $\Delta C_{\text{PIN\_RX}}$  | -0.15  | 0.15 | pF   | Mismatch of PIN capacitance. Mismatch of M-RX PIN capacitances <sup>2</sup> .        |
| $V_{\text{PIN}}$             | -100   | 600  | mV   | PIN signal voltage range. Signal voltage range.                                      |
| $I_{\text{LEAK}}$            | -10    | 10   | μA   | PIN leakage current.                                                                 |
| $\Delta I_{\text{LEAK\_TX}}$ | -5     | 5    | μA   | M-TX PIN leakage current mismatch.                                                   |
| $V_{\text{GND SH}}$          | -50    | 50   | mV   | Ground shift. Ground shift between M-TX and M-RX.                                    |

**Note:**

1. Includes package capacitance.
2. For recommended PIN capacitance only.



## 6 Electrical Interconnect (Informative)

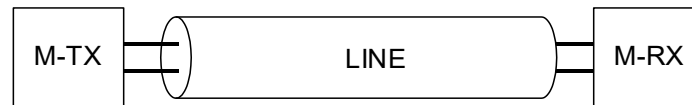
This section provides an implementation guideline for the interconnect simulation environment. A methodology is also provided that allows the evaluation of interconnect performance.

A LINE is defined as an interconnect between an M-TX and an M-RX that conducts the LANE signals. These signals include differential signaling for both high speed and low speed data transfer. Thus, a LINE should be implemented by means of balanced, differential, point-to-point transmission lines referenced to ground.

A LINE might consist of several cascaded transmission lines, such as printed circuit boards, flex-foils, or cable connections that might also include vias and connectors.

An M-PHY LANE is a unidirectional connection between an M-TX and an M-RX using a LINE as an interconnect. Overall LANE performance is determined by the combination of these three elements. **Figure 49** shows a simple point-to-point interconnect.

The interconnect under test should reflect the application such that M-PHY specifications are met at the M-RX inputs. Therefore, the interconnect model needs to include noise source and target components when applicable, e.g. when simulating several LANEs, a supply distribution network, side-band signals, or sensitive or noisy elements such as radio antennas.



**Figure 49 Point-to-Point Interconnect**

### 6.1 Line Characteristics

The LINE delay,  $T_{\text{LINE}}$ , is defined as the time during which a signal is transmitted from the M-TX to M-RX through the LINE. The  $T_{\text{LINE}}$  parameters and test conditions are summarized in **Table 25**.

**Table 25 Interconnect Parameters**

| Parameter  | Symbol            | Values |      | Unit | Note / Test Condition                                                                                                                 |
|------------|-------------------|--------|------|------|---------------------------------------------------------------------------------------------------------------------------------------|
|            |                   | Min.   | Max. |      |                                                                                                                                       |
| LINE delay | $T_{\text{LINE}}$ | —      | 7    | ns   | Measured between zero crossings at test points <sup>1</sup> with conformance test signal source <sup>2</sup> and pattern <sup>3</sup> |

**Note:**

1. Measured between LINE input port and LINE output port, which is terminated by a reference resistor,  $R_{\text{REF}}$ , and reference capacitors,  $C_{\text{PIN\_RX}}$ , at both pins.
2. External signal source connected to LINE input port.
3. Test pattern CJTPAT at maximum data rate.

## 6.2 Methodology

The method described here imports a LANE's S-parameters into a simulation environment that includes worst case models for M-TX and M-RX as well as stress patterns. The resulting time domain simulation, from which voltage and timing can be obtained, is compared against those defined for the M-RX in **Section 5**.

The interconnect characteristics are completely defined by its mixed-mode S-parameter models, i.e. insertion loss, return loss, and coupling effects. These parameters are sufficient to completely characterize all interconnect-induced parasitic effects including impedance mismatch and discontinuities, insertion loss, crosstalk, jitter amplification, jitter attenuation and insertion. A long interconnect tends to be dominated by insertion loss and crosstalk, while a short interconnect tends to be dominated by impedance discontinuities. Since both types of LINE are possible, it is necessary to provide a means of characterizing the interconnect that comprehends all possible LANE characteristics.

It is also necessary to take into account the LANE's S-parameters with a worst case M-TX behavioral model and stress patterns, e.g. CJTPAT. The time domain results can be compared against the parameters defined in **Section 5**.

## 6.3 Methodology Guidance for Validating a LANE

An interconnect can be characterized using the following methodology:

- Extract S-parameters (insertion loss, return loss and, if applicable, coupling) for the interconnect under test.
- Import S-parameters into an interface simulation environment based on M-TX and M-RX models.
- Tune the models to produce electrical characteristics as defined in Section 5.
- Worst case M-TX model feeds stressed test patterns, e.g. CJTPAT, into interconnect S-parameters model and then into M-RX model.
- Compare simulation results to specifications that should be respected at the input of the M-RX.

### 6.3.1 Interconnect S-parameters Extraction

Interconnect S-parameters should be taken over the entire signaling spectral range, which extends from the Low Speed minimal bit rate to more than five times  $f_{U\_RX}$  of the highest supported GEAR. Both near and far end return loss, as well as forward and reverse insertion loss, should be measured since most simulation tools require a complete mixed-mode S-parameter representation.

### 6.3.2 Simulation Environment Setup

A simple end-to-end simulation environment is illustrated in **Figure 50**. The environment includes an M-TX model, LINE model using interconnect S-parameters, an M-RX model, and stress pattern. A single LANE environment suffices for a topology with minimal crosstalk. Otherwise, the simulation environment needs to include noise source and noise target components as shown in **Figure 51**.

At a minimum, the M-TX behavioral model should have an ideal source and consider all transmitter specification parameters listed in **Section 5.1** to accommodate worst case simulations over the whole parameter range. Moreover, a realistic M-TX model should include package parasitic elements. Some parameters need to be simulated over their minimum to maximum range in order to guarantee worst case voltage and time margins for a particular interconnect. For example,  $V_{DIF\_TX}$  produces worst case eye margins when set to a minimum and at the same time produces low crosstalk.

The test pattern needs to provide worst case results while conforming to 8b10b coding rules. The simulation environment should use CJTPAT since most M-PHY electrical parameters are derived from it. Other test patterns that create different stress conditions should also be considered. When the simulation environment is composed of several LANEs, either the same time-shifted pattern, or different patterns, can be used for individual LANEs.

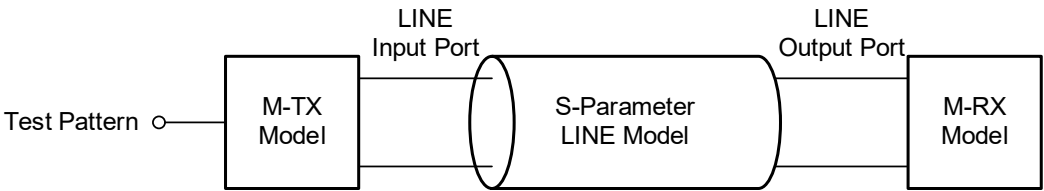


Figure 50 Single LANE Simulation Environment

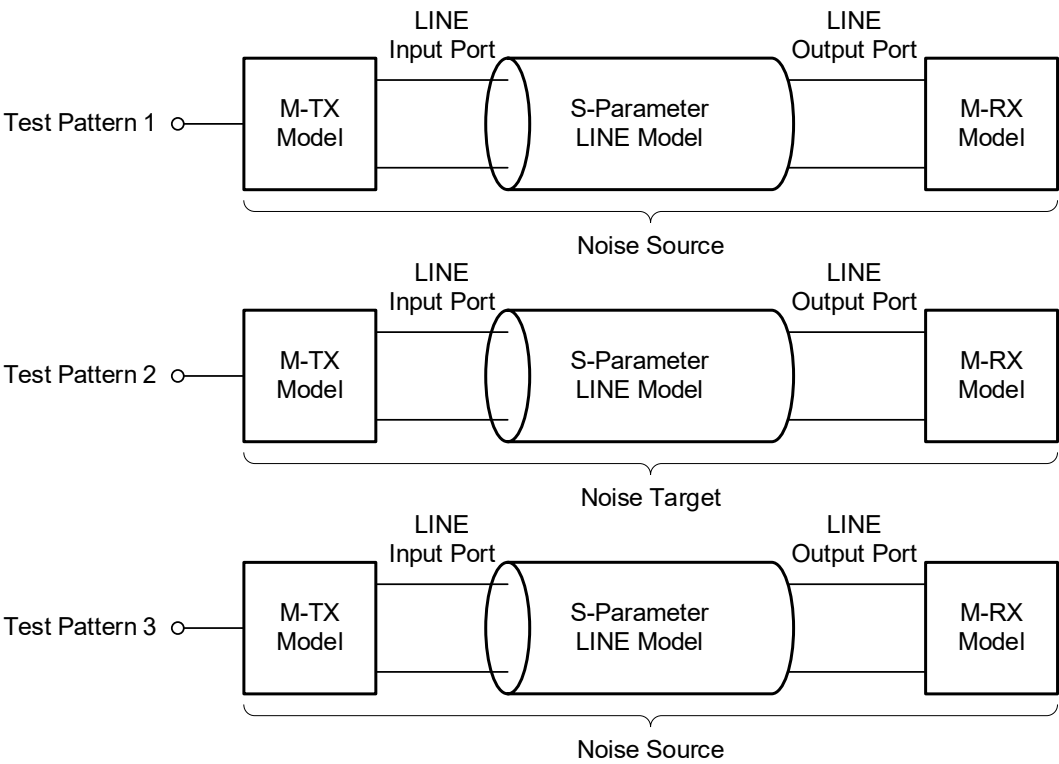
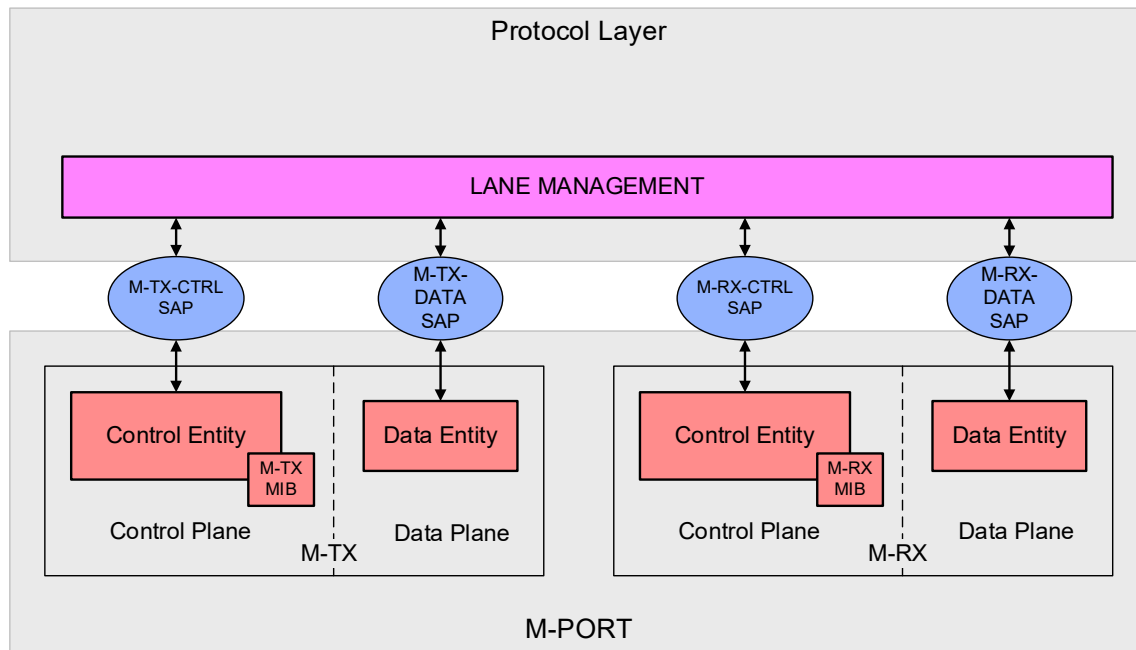


Figure 51 Multiple LANE Simulation Environment

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## 7 The Protocol Interface

This section defines the Protocol Interface of M-PORTs. This interface connects an M-PORT with the Protocol Layer that utilize M-PHY for the Physical Layer. Protocols applying M-PHY technology include UniPro<sup>SM</sup> and DigRF<sup>SM</sup> v4. The M-PORT Protocol Interface is represented in **Figure 52**.



**Figure 52 M-PORT Protocol Interface**

The normative interface specification is based on service access points (SAPs) and service primitives. M-TX-DATA SAP (M-TX Data Service Access Point) and M-RX-DATA SAP (M-RX Data Service Access Point) provide access to the data services of an M-TX and an M-RX, respectively. M-TX-CTRL SAP (M-TX Control Service Access Point) and M-RX-CTRL SAP (M-RX Control Service Access Point) provide access to configuration and reset services of an M-TX and M-RX, respectively.

All data transported across LANEs goes through, and is controlled by, the M-TX-DATA and M-RX-DATA SAPs, while the M-TX and M-RX local RESET, LINE-RESET, mode and parameter settings (configuration) are controlled through the M-TX-CTRL and M-RX-CTRL SAPs.

An M-PORT may consist of one or more M-TXs and one or more M-RXs. All individual M-TXs and M-RXs in an M-PORT are independent from the Protocol Interface perspective and each MODULE has its own DATA and CTRL SAP. Constraints on supported MODULE functionality of multi-LANE SUB-LINKS are specified in **Section 4.9**. LINK composition and usage of LANEs shall be defined by protocols that utilize M-PHY technology for the Physical Layer.

## 7.1 Service Primitive Naming Convention

This document uses an OSI-conforming naming convention for service primitives. Service primitive names are structured as follows:

```

<service-primitive> ::= <name-of-service-primitive> ( {<parameter>, }*)
<name-of-service-primitive> ::= <layer-identifier> - <service-primitive-name> . <primitive>
<parameter> ::= <service control information> | <service user data>
<layer-identifier> ::= M (or M-LANE or M-CTRL)
<service-primitive-name> ::= e.g. SYMBOL | PREPARE | CFGGET | CFGSET | ...
<primitive> ::= request | indication | response | confirm

```

Services are specified by describing the service primitives and parameters that characterize them. A service may have one or more related primitives that constitute the activity that is related to that particular service. Each service primitive may have zero or more parameters that convey the information required to provide the service.

A primitive can be one of four generic types:

- Request: The request primitive is passed from the Protocol Layer to a MODULE to request that a service is initiated by the MODULE.
- Indication: The indication primitive is passed from a MODULE to the Protocol Layer to indicate an event that is significant to the Protocol Layer. This event may be logically related to a remote service request, or it may be caused by a LANE event.
- Response: The response primitive is passed from Protocol Layer to a MODULE to complete a procedure previously invoked by an indication primitive.
- Confirm: The confirm primitive is passed from a MODULE to the Protocol Layer to convey the results of one or more associated previous service requests.

23-Jun-2021

## 7.2 M-TX-DATA and M-RX-DATA SAP

The M-TX-DATA SAP and M-RX-DATA SAP contain service primitives for data transfer between the Protocol Layer and the MODULEs of an M-PORT. More specifically, M-TX-DATA SAP provides service primitives for sending data, FILLER symbols, changing the LINE state between BURST-SAVE loop, and sending programmable synchronization pattern during SYNC period of HS-BURST. M-RX-DATA SAP provides service primitives to transfer received data, indicate LINE state change between BURST-SAVE loop and reception of FILLER symbols to the Protocol Layer. Each MODULE (M-TX or M-RX) shall have its own SAP (M-TX-DATA SAP or M-RX-DATA SAP, respectively). **Table 26** and

Table 27 give an overview of the service primitives provided by the M-TX-DATA SAP and the M-RX-DATA SAP, respectively, and displays the respective section numbers.

**Table 26 M-TX-DATA SAP Service Primitives**

| Name                 | Request       | Indication    | Response | Confirm       |
|----------------------|---------------|---------------|----------|---------------|
| M-LANE-SYMBOL        | <b>7.2.1</b>  | n/a           | n/a      | <b>7.2.3</b>  |
| M-LANE-PREPARE       | <b>7.2.4</b>  | n/a           | n/a      | <b>7.2.6</b>  |
| M-LANE-SYNC          | <b>7.2.7</b>  | n/a           | n/a      | <b>7.2.8</b>  |
| M-LANE-BurstEnd      | <b>7.2.9</b>  | n/a           | n/a      | <b>7.2.11</b> |
| M-LANE-SaveState     | n/a           | <b>7.2.13</b> | n/a      | n/a           |
| M-LANE-AdaptStart    | <b>7.2.15</b> | n/a           | n/a      | <b>7.2.16</b> |
| M-LANE-AdaptComplete | n/a           | <b>7.2.17</b> | n/a      | n/a           |

**Table 27 M-RX-DATA SAP Service Primitives**

| Name                 | Request | Indication    | Response | Confirm |
|----------------------|---------|---------------|----------|---------|
| M-LANE-SYMBOL        | n/a     | <b>7.2.2</b>  | n/a      | n/a     |
| M-LANE-PREPARE       | n/a     | <b>7.2.5</b>  | n/a      | n/a     |
| M-LANE-BurstEnd      | n/a     | <b>7.2.10</b> | n/a      | n/a     |
| M-LANE-HIBERN8Exit   | n/a     | <b>7.2.12</b> | n/a      | n/a     |
| M-LANE-MRXSaveState  | n/a     | <b>7.2.14</b> | n/a      | n/a     |
| M-LANE-AdaptComplete | n/a     | <b>7.2.17</b> | n/a      | n/a     |

1774 There are parameters associated with some of these primitives. **Table 28** defines the names, types and valid  
 1775 ranges of these parameters.

1776 **Table 28 Parameters of M-TX-DATA SAP and M-RX-DATA Service Primitives**

| Name           | Type    | Valid Range      | Description                                                                        |
|----------------|---------|------------------|------------------------------------------------------------------------------------|
| DataN_Ctrl     | Boolean | FALSE, TRUE      | Data symbol or control symbol selector                                             |
| DataValue      | Integer | 0 to 1023        | Normal PAYLOAD data.<br>When 8b10b coding is enabled, the valid range is 0 to 255. |
| MarkerN_Filler | Boolean | FALSE, TRUE      | MARKER or FILLER control symbol selection                                          |
| MarkerNumber   | Integer | 0 to 6           | Type of MARKER symbol selector                                                     |
| 3b4b_Error     | Boolean | FALSE, TRUE      | 3b4b Sub-block coding error                                                        |
| 5b6b_Error     | Boolean | FALSE, TRUE      | 5b6b Sub-block coding error                                                        |
| Res_Error      | Boolean | FALSE, TRUE      | Reserved symbol error                                                              |
| RD_Error       | Boolean | FALSE, TRUE      | Running Disparity error                                                            |
| State          | Enum    | SLEEP, STALL     | Entering SAVE state                                                                |
| Status         | Boolean | ACCEPTED, BUSY   | Indicates acceptance of symbol                                                     |
| SyncData       | Integer | 0 to 1023        | Data for programmable synchronization sequence                                     |
| WaitType       | Enum    | NoConfig, Config | Indicates minimum waiting time in SAVE state                                       |

1777 The following sections define the meaning of M-TX-DATA SAP and M-RX-DATA SAP service primitives  
 1778 and their associated parameters.



## 7.2.1 M-LANE-SYMBOL.request

This primitive requests the transmission of either a PAYLOAD data symbol or a control symbol from the Protocol Layer to an M-TX. The control symbol can be either a MARKER symbol or a FILLER symbol. See *Section 4.5.2* and *Section 4.7.2* for constraints on MARKER usage by the Protocol.

### 7.2.1.1 Semantics of the Service Primitive

The semantics of the M-LANE-SYMBOL.request are as follows:

```
M-LANE-SYMBOL.request (
    DataN_Ctrl,
    DataValue,
    MarkerN_Filler,
    MarkerNumber
)
```

*Table 29* specifies the parameters for the M-LANE-SYMBOL request primitive.

**Table 29 Parameters for the M-LANE-SYMBOL.request Primitive**

| Name           | Type    | Valid Range            | Description                                                                                                                                                                                                                                                 |
|----------------|---------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DataN_Ctrl     | Boolean | FALSE = 0,<br>TRUE = 1 | DataN_Ctrl set to "FALSE" selects value associated with the DataValue parameter for transmission; DataN_Ctrl set to "TRUE" chooses either a MARKER or a FILLER control symbol based on the value of MarkerN_Filler parameter for transmission.              |
| DataValue      | Integer | 0 to 1023              | Normal PAYLOAD data.<br>When 8b10b coding is enabled, the valid range is 0 to 255.<br>This parameter shall be ignored when DataN_Ctrl set to "TRUE".                                                                                                        |
| MarkerN_Filler | Boolean | FALSE = 0,<br>TRUE = 1 | MarkerN_Filler set to "FALSE" selects MARKER symbol based on the value of MarkerNumber parameter for transmission; MarkerN_Filler set to "TRUE" selects the FILLER symbol for transmission; This parameter shall be ignored when DataN_Ctrl set to "FALSE". |
| MarkerNumber   | Integer | 0 to 6                 | MarkerNumber set to n selects MKn for transmission, where n is any number in the valid range. For example, if MarkerNumber = 0, MK0 is selected.<br>This parameter shall be ignored when DataN_Ctrl is set to "FALSE" or MarkerN_Filler is set to "TRUE".   |

### 7.2.1.2 When Generated

This primitive shall be generated by the Protocol Layer in order to transmit a data symbol, a MARKER symbol or a FILLER symbol over the LINE. This primitive, with details as M-LANE-SYMBOL.request (FALSE, DataValue, X, X), where DataValue takes valid range as defined in *Table 29* and X means ignore that parameter, shall be generated by the Protocol Layer in order to transmit a symbol over the LINE.

Since MARKER0 symbol is needed to achieve symbol boundary synchronization at M-RX, before actual PAYLOAD data transmission starts, the Protocol Layer shall generate this primitive with MARKER0 symbol

details, i.e., M-LANE-SYMBOL.request (TRUE, X, FALSE, 0), where X means ignore that parameter, at the very beginning of a data transmission BURST. In other words, the Protocol Layer shall generate this primitive with MARKER0 symbol details after issuing an M-LANE-PREPARE.request, but before issuing this primitive with details other than MARKER0 symbol.

This primitive with MKn symbol details, i.e., M-LANE-SYMBOL.request (TRUE, X, FALSE, n), where X means ignore a parameter and n is the MarkerNumber, is used by the Protocol Layer to request a MARKERn during a BURST.

Protocol Layer may request transmission of a FILLER symbol, explicitly, by using this primitive with FILLER symbol details, i.e., M-LANE-SYMBOL.request (TRUE, X, TRUE, X), where X means ignore that parameter. Note that M-TX will insert FILLER symbols autonomously in a BURST state as described in **Section 4.7.4.2**.

The Protocol Layer shall not exceed the valid range of any parameter. The Protocol Layer shall not request this primitive with DataN\_Ctrl set to “TRUE” when 8b10b coding is disabled. If this primitive is requested with DataN\_Ctrl set to “TRUE” when 8b10b coding is disabled, the MODULE might not behave properly. A MODULE shall not verify the validity of any parameter value. Out of range values might lead to malfunction of a MODULE.

### 7.2.1.3 Effect on Receipt

When this primitive is requested with DataN\_Ctrl set to “FALSE” and 8b10b encoding is enabled, the M-TX shall encode the DataValue byte into an 8b10b Data symbol and then transfer the symbol over the LINE.

When this primitive is requested with DataN\_Ctrl set to “FALSE” and 8b10b coding is disabled, the M-TX shall transfer the symbol in DataValue unchanged over the LINE.

When this primitive is requested with DataN\_Ctrl set to “TRUE”, MarkerN\_Filler set to “FALSE”, MarkerNumber set to a valid MARKER symbol number, and 8b10b encoding is enabled, the M-TX shall transmit an 8b10b control symbol corresponding to the requested MARKER symbol over the LINE.

When this primitive is requested with DataN\_Ctrl set to “TRUE” and MarkerN\_Filler set to “TRUE”, and 8b10b coding is enabled, the M-TX shall transmit the 8b10b control symbol corresponding to the FILLER symbol over the LINE.

Refer to **Section 4.5** for encoding and serialization process.

## 7.2.2 M-LANE-SYMBOL.indication

This primitive reports the reception of a data PAYLOAD byte or a MARKER or a FILLER symbol over the LINE.

### 7.2.2.1 Semantics of the Service Primitive

The semantics of the M-LANE-SYMBOL.indication primitive are as follows:

```
M-LANE-SYMBOL.indication(
    DataN_Ctrl,
    DataValue,
    MarkerN_Filler,
    MarkerNumber,
    3b4b_Error,
    5b6b_Error,
    RD_Error,
    Res_Error
)
```

**Table 30** specifies the parameters for the M-LANE-SYMBOL.indication primitive.

1838

**Table 30 Parameters for the M-LANE-SYMBOL.indication Primitive**

| Name           | Type    | Valid Range            | Description                                                                                                                                                                                                                                                                                                                                              |
|----------------|---------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DataN_Ctrl     | Boolean | FALSE = 0,<br>TRUE = 1 | When DataN_Ctrl set to "FALSE", the value associated with the DataValue parameter shall be considered as a received PAYLOAD symbol;<br>When DataN_Ctrl is set to "TRUE", the value of MarkerN_Filler shall be used to identify the type of control symbol received                                                                                       |
| DataValue      | Integer | 0 to 1023              | Indicates normal PAYLOAD data, one symbol in length. When 8b10b decoding is enabled, the valid range is 0 to 255. This parameter shall be ignored when DataN_Ctrl set to "TRUE"                                                                                                                                                                          |
| MarkerN_Filler | Boolean | FALSE = 0,<br>TRUE = 1 | If the value set to MarkerN_Filler is "FALSE", then the value associated with a MarkerNumber parameter shall be used in identifying the type of MARKER symbol received;<br>When MarkerN_Filler is set to "TRUE", it shall be considered as reception of a FILLER symbol over the LINE;<br>This parameter shall be ignored when DataN_Ctrl set to "FALSE" |
| MarkerNumber   | Integer | 0 to 6                 | MarkerNumber set to n indicates MKn is received, where n is any number in the valid range. For example, if MarkerNumber = 0, MK0 is received. This parameter shall be ignored when DataN_Ctrl is set to "FALSE" or MarkerN_Filler is set to "TRUE".                                                                                                      |
| 3b4b_Error     | Boolean | FALSE = 0,<br>TRUE = 1 | 3b4b Sub-block coding error;<br>FALSE: No error detected<br>TRUE: Error detected                                                                                                                                                                                                                                                                         |
| 5b6b_Error     | Boolean | FALSE = 0,<br>TRUE = 1 | 5b6b Sub-block coding error;<br>FALSE: No error detected<br>TRUE: Error detected                                                                                                                                                                                                                                                                         |
| RD_Error       | Boolean | FALSE = 0,<br>TRUE = 1 | Running Disparity error;<br>FALSE: No error detected<br>TRUE: Error detected                                                                                                                                                                                                                                                                             |
| Res_Error      | Boolean | FALSE = 0,<br>TRUE = 1 | Reserved symbol error;<br>FALSE: No error detected<br>TRUE: Error detected                                                                                                                                                                                                                                                                               |

#### 7.2.2.2 When Generated

- 1839 This primitive shall be generated by the M-RX when an 8b10b data symbol or a control symbol corresponding  
1840 to any valid MARKER symbol, or FILLER is received over the LINE.
- 1841 When 8b10b decoding is disabled, DataN\_Ctrl shall be set to "FALSE", DataValue shall carry the symbol as  
1842 received and all other fields shall be ignored.
- 1843 When 8b10b decoding is enabled, if the received 8b10b symbol is a valid data symbol, DataValue shall carry  
1844 the decoded PAYLOAD byte. In this case, 3b4b\_Error, 5b6b\_Error, Res\_Error and DataN\_Ctrl shall be set  
1845 to "FALSE", and all other parameter values, except DataValue, shall be ignored

When 8b10b decoding is enabled, if the received 8b10b symbol is a MARKER symbol, then the M-RX shall set DataN\_Ctrl to “TRUE”, MarkerN\_Filler to “FALSE”, and MarkerNumber to a number corresponding to the received MARKER symbol. DataValue may be set to “0”. The Protocol Layer shall ignore DataValue. All error parameters shall be set to “FALSE”.

When 8b10b decoding is enabled, if the received 8b10b symbol is a FILLER symbol, then the M-RX shall set DataN\_Ctrl to “TRUE” and MarkerN\_Filler to “TRUE”. DataValue and MarkerNumber may be set to “0”. The Protocol Layer shall DataValue and MarkerNumber. All error parameters shall be set to “FALSE”.

When 8b10b decoding is enabled, if the received 8b10b symbol is an invalid symbol, DataValue shall carry the remapped PAYLOAD byte, with potentially incorrect bits for the invalid sub-block, but correct bits of the valid sub-block. In this case, 3b4b\_Error or 5b6b\_Error shall be set to “TRUE”, depending on which of the sub-blocks was in error.

When 8b10b decoding is enabled, if the received 8b10b symbol is a valid, but reserved symbol (i.e. not equal to a data symbol, a MARKER symbol or FILLER), DataValue shall carry the remapped PAYLOAD byte. In this case, Res\_Error shall be set to “TRUE”.

When 8b10b decoding is enabled, if the Running Disparity (RD) in the M-RX (See **Section 4.5.3**) computes an RD error for the currently received 8b10b symbol, the RD\_Error parameter shall be set to “TRUE”. This setting shall not depend on the other error parameters described above.

### 7.2.2.3 Effect on Receipt

On receipt of the M-LANE-SYMBOL.indication primitive, the Protocol Layer is notified of the availability of inbound data byte, or the reception of a MARKER symbol or FILLER symbol, by the M-RX and generating a corresponding MARKER number or FILLER indication, and error information at M-RX. The Protocol Layer shall consume the data byte or a MARKER number along with error information, and may carry out appropriate Protocol action.

The Protocol Layer shall ignore MarkerN\_Filler and MarkerNumber when DataN\_Ctrl is set to “FALSE”, and it shall ignore DataValue when DataN\_Ctrl is set to “TRUE”. The Protocol Layer shall ignore MarkerNumber when MarkerN\_Filler is set to “TRUE”.

### 7.2.3 M-LANE-SYMBOL.confirm

This primitive informs the Protocol Layer that the M-TX has completed the previously issued M-LANE-SYMBOL.request.

#### 7.2.3.1 Semantics of the Service Primitive

The semantics of M-LANE-SYMBOL.confirm primitive are as follows

```
M-LANE-SYMBOL.confirm(
    Status
)
```

**Table 31** specifies the parameters for the M-LANE-SYMBOL.confirm primitive.

**Table 31 Parameters for the M-LANE-SYMBOL.confirm Primitive**

| Name   | Type    | Valid Range               | Description                                                                                                                                                                                                                                                            |
|--------|---------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Status | Boolean | ACCEPTED = 0,<br>BUSY = 1 | Status = ACCEPTED means that M-TX has accepted the previously requested symbol for transmission and ready for new request to be served.<br>Status = BUSY means that M-TX has rejected the previously requested symbol; the Protocol Layer may issue the request again. |

#### **7.2.3.2 When Generated**

1879 This primitive shall be generated when the M-TX has either accepted or rejected the previously issued  
1880 M-LANE-SYMBOL.request primitive. It also confirms that the M-TX may accept another request to transfer  
1881 a PAYLOAD byte or a MARKER or a FILLER symbol from the Protocol Layer.

#### **7.2.3.3 Effect on Receipt**

1882 Following the issuing of an M-LANE-SYMBOL.request and prior to the reception of an  
1883 M-LANE-SYMBOL.confirm primitive, the Protocol Layer shall not trigger a new  
1884 M-LANE-SYMBOL.request primitive. Upon receiving this primitive the Protocol Layer may issue a new  
1885 data or MARKER symbol, or configuration request or retry the previously rejected symbol request.

#### 7.2.4 M-LANE-PREPARE.request

1886 This primitive requests the M-TX to enter into a BURST state, either HS-BURST, PWM-BURST or  
 1887 SYS-BURST depending upon the mode of operation, from the power saving state. See *Section 4* for more  
 1888 details on BURST state, power saving state and operating modes.

##### 7.2.4.1 Semantics of the Service Primitive

1889 The semantics of the M-LANE-PREPARE.request primitive are as follows:

```
1890 M-LANE-PREPARE.request (
1891 )
```

1892 This primitive has no parameter.

##### 7.2.4.2 When Generated

1893 The Protocol Layer shall issue this primitive to request the M-TX to enter from power saving state to BURST  
 1894 state corresponding to the M-TX mode of operation. This primitive shall only be issued when the M-TX is  
 1895 in power saving state.

##### 7.2.4.3 Effect on Receipt

1896 The M-TX shall enter into the BURST state following the sequence of operation as described in *Section 4.7.2*.

#### 7.2.5 M-LANE-PREPARE.indication

1897 This primitive informs the Protocol Layer that the M-RX is coming out of power saving state and entering  
 1898 into a BURST state (HS-BURST, PWM-BURST, or SYS-BURST) or ADAPT state depending on the M-RX  
 1899 mode of configuration. See *Section 4* for more details on BURST state, power saving state and operating  
 1900 modes.

##### 7.2.5.1 Semantics of the Service Primitive

1901 The semantics of the M-LANE-PREPARE.indication primitive are as follows:

```
1902 741 M-LANE-PREPARE.indication (
1903 742 )
```

1904 This primitive has no parameter.

##### 7.2.5.2 When Generated

1905 The M-RX shall issue this primitive to the Protocol Layer when M-RX detects the start of the PREPARE  
 1906 sub-state period while it is in power saving state.

##### 7.2.5.3 Effect on Receipt

1907 The Protocol Layer shall accept M-RX entering to the BURST state corresponding to the M-RX mode of  
 1908 operation and shall be prepared to receive data.

## 7.2.6 M-LANE-PREPARE.confirm

1909 This primitive informs the Protocol Layer that the M-TX has started entering into BURST state following  
1910 the reception of M-LANE-PREPARE.request.

### 7.2.6.1 Semantics of the Service Primitive

1911 The semantics of M-LANE-PREPARE.confirm primitive are as follows

```
1912 748 M-LANE-PREPARE.confirm (
1913 749 )
```

1914 This primitive has no parameter.

### 7.2.6.2 When Generated

1915 This primitive shall be generated by the M-TX when it enters into a PREPARE period upon the reception of  
1916 an M-LANE-PREPARE.request primitive.

### 7.2.6.3 Effect on Receipt

1917 Upon receiving this primitive the Protocol Layer may issue a programmable synchronization sequence  
1918 through M-LANE-SYNC.request primitive when the M-TX is configured to receive external synchronization  
1919 pattern from the protocol. Otherwise, the Protocol Layer may issue MARKER0 symbol request at any time  
1920 during the SYNC period.

## 7.2.7 M-LANE-SYNC.request

1921 This primitive requests the transmission of a programmable sync pattern byte wise over the LINE. For more  
1922 details on SYNC sequences see *Section 4.7.2.2*.

### 7.2.7.1 Semantics of the Service Primitive

1923 The semantics of the M-LANE-SYNC.request primitive are as follows:

```
1924 M-LANE-SYNC.request (
1925     SyncData
1926 )
```

1927 *Table 32* specifies the parameters for the M-LANE-SYNC.request primitive

1928 **Table 32 Parameters for M-LANE-SYNC.request Primitive**

| Name     | Type    | Valid Range | Description                                        |
|----------|---------|-------------|----------------------------------------------------|
| SyncData | Integer | 0 to 1023   | A byte of data from the programmable sync sequence |

### 7.2.7.2 When Generated

1929 This primitive shall be generated by the Protocol Layer to request the transmission of a synchronization  
1930 pattern to be provided by the protocol. This primitive only has effect if the M-TX is configured to send a  
1931 programmable synchronization sequence (i.e., TX\_SYNC\_Source is EXTERNAL\_SYNC and the  
1932 configured GEAR requires a synchronization pattern). The synchronization sequence shall be issued to the  
1933 M-TX one byte at a time using this primitive. The Protocol Layer shall wait for the M-LANE-SYNC.confirm  
1934 primitive before issuing this primitive again. The first issue of this primitive shall only take place after the  
1935 Protocol Layer receives M-LANE-PREPARE.confirm primitive from the M-TX to the previously issued  
1936 M-LANE-PREPARE.request primitive and before SYNC period starts.

**7.2.7.3 Effect on Receipt**

1937 When 8b10b coding is enabled, the M-TX shall encode SyncData byte as an 8b10b symbol and then transfer  
 1938 the symbol over the LINE. When 8b10b coding is disabled, the M-TX shall transfer the symbol in SyncData  
 1939 unchanged over the LINE. Upon transmission of SyncData byte the M-TX shall issue an  
 1940 M-LANE-SYNC.confirm primitive to the Protocol Layer.

**7.2.8 M-LANE-SYNC.confirm**

1941 This primitive informs the Protocol Layer that the M-TX has completed the previously issued service request  
 1942 M-LANE-SYNC.request.

**7.2.8.1 Semantics of the Service Primitive**

1943 The semantics of M-LANE-SYNC.confirm primitive are as follows

1944 M-LANE-SYNC.confirm (   
 1945 )

1946 This primitive has no parameter.

**7.2.8.2 When Generated**

1947 This primitive shall be generated when the M-TX has completed serving the previously issued  
 1948 M-LANE-SYNC.request primitive and is ready to accept another synchronization sequence symbol from the  
 1949 Protocol Layer to transfer.

**7.2.8.3 Effect on Receipt**

1950 The Protocol Layer may issue a new synchronization symbol or MARKER symbol request upon receiving  
 1951 this primitive. The Protocol Layer shall not issue a new M-LANE-SYNC.request until a previously issued  
 1952 M-LANE-SYNC.request has been responded with this primitive.

**7.2.9 M-LANE-BurstEnd.request**

1953 This primitive requests the M-TX to send TAIL-OF-BURST sequence.

**7.2.9.1 Semantics of the Service Primitive**

1954 The semantics of the M-LANE-BurstEnd.request primitive are as follows:

1955 M-LANE-BurstEnd.request (   
 1956 )

1957 This primitive has no parameter.

**7.2.9.2 When Generated**

1958 The Protocol Layer shall issue this primitive to request the M-TX end BURST state and enter a SAVE state.

**7.2.9.3 Effect on Receipt**

1959 The M-TX shall end the BURST state following the sequence of operation as described in *Section 4.7.2.5*.



### 7.2.10 M-LANE-BurstEnd.indication

1960 This primitive reports the reception of a BURST CLOSURE condition to the Protocol as described in  
1961 *Section 4.7.2.5*.

#### 7.2.10.1 Semantics of the Service Primitive

1962 The semantics of the M-LANE-BurstEnd.indication primitive are as follows:

1963 M-LANE-BurstEnd.indication (   
1964 )

1965 This primitive has no parameter.

#### 7.2.10.2 When Generated

1966 This primitive shall be generated by the M-RX to the Protocol Layer when M-RX detects a sequence of b0  
1967 or b1 on LINE over the period defined in *Section 4.7.2.5* while it is in BURST state.

#### 7.2.10.3 Effect on Receipt

1968 Protocol Layer shall accept end of the BURST state and shall consider that the M-RX is entering either into  
1969 SAVE state when sequence of b0 received over a period as described in *Section 4.7.2.5.1*.

### 7.2.11 M-LANE-BurstEnd.confirm

1970 This primitive informs the Protocol Layer that the M-TX has started sending a TAIL-OF-BURST sequence  
1971 following the reception of M-LANE-BurstEnd.request.

#### 7.2.11.1 Semantics of the Service Primitive

1972 The semantics of M-LANE-BurstEnd.confirm primitive are as follows:

1973 M-LANE-BurstEnd.confirm( WaitType )

1974 *Table 33* specifies the parameters for the M-LANE-BurstEnd.confirm primitive

1975 **Table 33 Parameters for M-LANE-BurstEnd.confirm Primitive**

| Name     | Type | Valid Range                 | Description                                               |
|----------|------|-----------------------------|-----------------------------------------------------------|
| WaitType | Enum | NoConfig = 0,<br>Config = 1 | Indicates minimum wait time before new BURST is requested |

#### 7.2.11.2 When Generated

1976 The M-TX shall generate this primitive once it starts sending a TAIL-OF-BURST sequence after the reception  
1977 of an M-LANE-BurstEnd.request primitive.

#### 7.2.11.3 Effect on Receipt

1978 Upon receiving this primitive, the Protocol Layer shall wait before asserting M-LANE-PREPARE.request  
1979 for:

- 1980 • At least  $T_{MIN\_SAVE\_CONFIG}$  if any configuration request is made (i.e., if WaitType is “Config”), or  
1981 • At least TX\_Min\_SLEEP\_NoConfig\_Time\_Capability if in SLEEP state, or  
1982 • TX\_Min\_STALL\_NoConfig\_Time\_Capability if in STALL state when no configuration request is made  
1983 (i.e., if WaitType is “NoConfig”) after the M-TX issues the M-LANE-SaveState.indication primitive.

**7.2.12 M-LANE-HIBERN8Exit.indication**

1984 This primitive reports the exit of HIBERN8 state to the Protocol as described in *Section 4.7.1.3*.

**7.2.12.1 Semantics of the Service Primitive**

1985 The semantics of the M-LANE-HIBERN8Exit.indication primitive are as follows:

1986 M-LANE-HIBERN8Exit.indication (   
1987 )

1988 This primitive has no parameter.

**7.2.12.2 When Generated**

1989 This primitive shall be generated by the M-RX to the Protocol Layer when M-RX detects exit of HIBERN8  
1990 state as defined in *Section 4.7.1.3* while M-RX is in HIBERN8 state.

**7.2.12.3 Effect on Receipt**

1991 Protocol Layer shall accept exit of HIBERN8 state and shall consider that the M-RX is entering either to  
1992 SLEEP or to STALL state based on the value of RX\_MODE attribute. The Protocol Layer may use this  
1993 primitive to get out of hibernation.

**7.2.13 M-LANE-SaveState.indication**

1994 This primitive reports entry into a SAVE state to the Protocol.

**7.2.13.1 Semantics of the Service Primitive**

1995 The semantics of the M-LANE-SaveState.indication primitive are as follows:

1996 M-LANE-SaveState.indication( State )

1997 *Table 34* specifies the parameters for the M-LANE-SaveState.indication primitive

1998 **Table 34 Parameters for M-LANE-SaveState.indication Primitive**

| Name  | Type | Valid Range             | Description |
|-------|------|-------------------------|-------------|
| State | Enum | SLEEP = 0,<br>STALL = 1 | SAVE state  |

**7.2.13.2 When Generated**

1999 This primitive shall be generated by an M-TX when the M-TX enters either SLEEP state or STALL state.

**7.2.13.3 Effect on Receipt**

2000 Protocol Layer shall accept entry of the M-TX into either SLEEP or STALL state.

#### 7.2.14 M-LANE-MRXSaveState.indication

2001 This primitive reports M-RX entry into a SAVE state to the Protocol.

##### 7.2.14.1 Semantics of the Service Primitive

2002 The semantics of the M-LANE-MRXSaveState.indication primitive are as follows:

2003 `M-LANE-MRXSaveState.indication( State )`

2004 *Table 35* specifies the parameters for the M-LANE-MRXSaveState.indication primitive

2005 **Table 35 Parameters for M-LANE-MRXSaveState.indication Primitive**

| Name  | Type | Valid Range             | Description |
|-------|------|-------------------------|-------------|
| State | Enum | SLEEP = 0,<br>STALL = 1 | SAVE state  |

##### 7.2.14.2 When Generated

2006 This primitive shall be generated by an M-RX when the M-RX enters either SLEEP state or STALL state.

##### 7.2.14.3 Effect on Receipt

2007 Protocol Layer shall accept entry of the M-RX into either SLEEP or STALL state.

#### 7.2.15 M-LANE-AdaptStart.request

2008 This primitive requests the transmission of the ADAPT sequence as described in *Section 4.7.2.3*.

##### 7.2.15.1 Semantics of the Service Primitive

2009 The semantics of the M-LANE-AdaptStart.request primitive are as follows:

2010 `M-LANE-AdaptStart.request (`  
2011 `)`

2012 This primitive has no parameter.

##### 7.2.15.2 When Generated

2013 This primitive shall be generated by the Protocol Layer to request the ADAPT sub-state upon completion of  
2014 the PREPARE sub-state. The Protocol Layer shall issue this primitive when M-TX is configured in HS-  
2015 MODE and is in STALL state. The Protocol Layer may control the duration of the ADAPT sequence period  
2016 by setting the appropriate value of TX\_HS\_ADAPT\_LENGTH prior to issuing this primitive.

##### 7.2.15.3 Effect on Receipt

2017 M-TX shall prepare to transition from the STALL state to the PREPARE sub-state upon receiving this  
2018 primitive. After completion of the PREPARE period, the M-TX shall transmit the ADAPT sequence for  
2019  $T_{ADAPT}$  as described in *Section 4.7.2.3*.

### 7.2.16 M-LANE-AdaptStart.confirm

2020 This primitive confirms the Start of the ADAPT sequence initiated due to the previously issued  
2021 M-LANE-AdaptStart.request primitive.

#### 7.2.16.1 Semantics of the Service Primitive

2022 The semantics of the M-LANE-AdaptStart.confirm primitive are as follows:

2023 M-LANE-AdaptStart.confirm (  
2024 )

2025 This primitive has no parameter.

#### 7.2.16.2 When Generated

2026 This primitive shall be generated by M-TX upon entering the PREPARE sub-state due to reception of the M-  
2027 LANE-AdaptStart.request primitive.

#### 7.2.16.3 Effect on Receipt

2028 The Protocol Layer shall wait for completion of the ADAPT sequence indicated by M-TX through generation  
2029 of the M-LANE-AdaptComplete.indication service primitive and shall not request a new burst in between.

### 7.2.17 M-LANE-AdaptComplete.indication

2030 This primitive indicates the completion of the ADAPT sequence.

#### 7.2.17.1 Semantics of the Service Primitive

2031 The semantics of the M-LANE-AdaptComplete.indication primitive are as follows:

2032 M-LANE-AdaptComplete.indication (  
2033 )

2034 This primitive has no parameter.

#### 7.2.17.2 When Generated

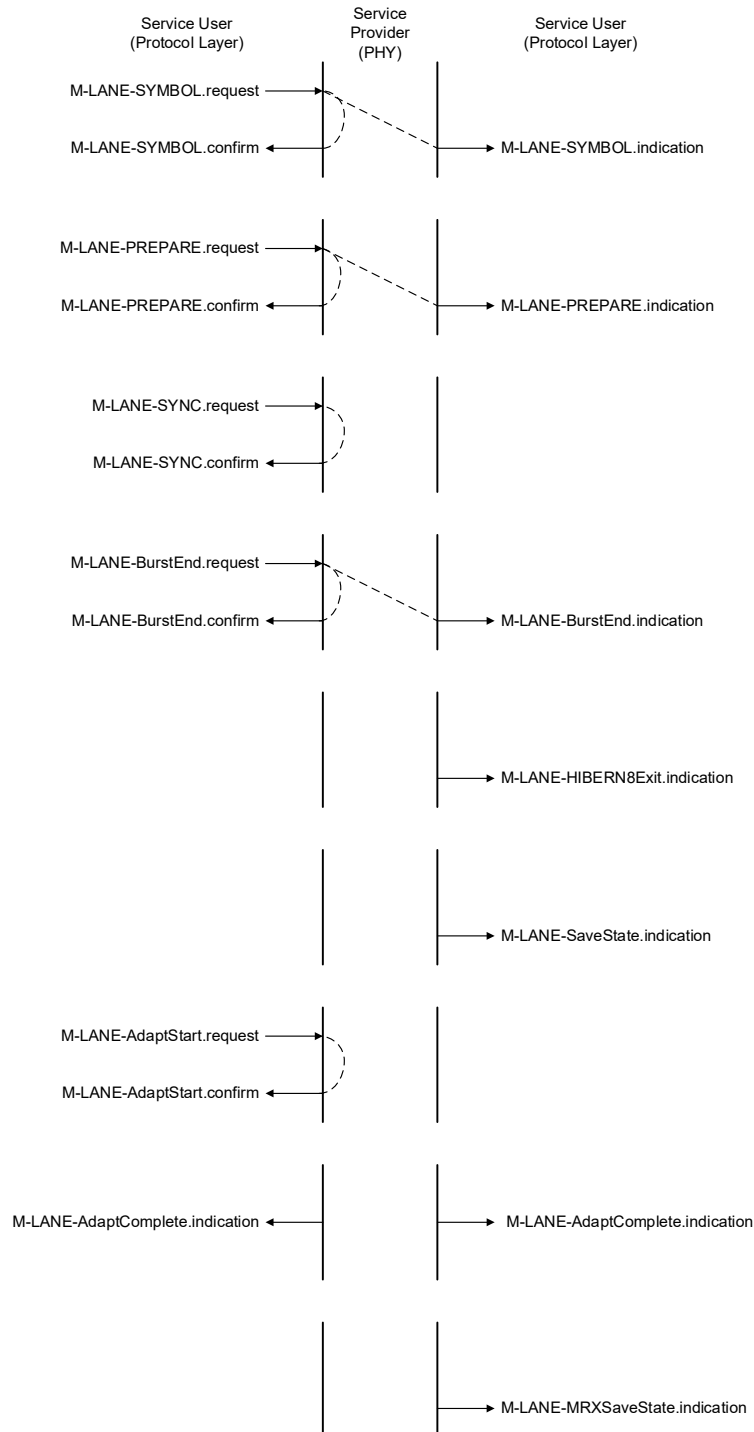
2035 This primitive shall be generated by the M-TX and by the M-RX upon completion of the ADAPT sequence  
2036 as described in *Section 4.7.2.3*. After the ADAPT sequence, MODULE shall return to the STALL state,.

#### 7.2.17.3 Effect on Receipt

2037 The Protocol Layer shall consider completion of the ADAPT sequence when it receives this primitive. The  
2038 Protocol Layer may start a new burst.

## 7.2.18 Sequence of Service Primitives

The possible relationships among primitives at M-TX-DATA SAP and M-RX-DATA SAP are illustrated by the given time sequence diagrams shown in **Figure 53**. They also indicate a possible logical relationship in terms of time. Primitives that occur earlier in time and connected by dotted lines are logical predecessors of subsequent primitives.



**Figure 53 Sequence of Primitives at M-TX-DATA SAP and M-RX-DATA SAP**

### 7.3 M-TX-CTRL SAP and M-RX-CTRL SAP

2044 M-TX-CTRL SAP and M-RX-CTRL SAP contain service primitives for configuring M-TX and M-RX,  
2045 respectively, and obtaining capability and status information from these MODULES. **Table 36** and **Table 37**  
2046 give an overview of the service primitives provided by M-TX-CTRL SAP and M-RX-CTRL SAP,  
2047 respectively, and display the respective section numbers. There are parameters associated with these  
2048 primitives. **Section 7.4** defines the name, type and valid range of these parameters.

2049

**Table 36 M-TX-CTRL SAP Service Primitives**

| Name             | Request                  | Indication | Response | Confirm                  |
|------------------|--------------------------|------------|----------|--------------------------|
| M-CTRL-CFGGET    | M-CTRL-CFGGET.request    | n/a        | n/a      | M-CTRL-CFGGET.confirm    |
| M-CTRL-CFGSET    | M-CTRL-CFGSET.request    | n/a        | n/a      | M-CTRL-CFGSET.confirm    |
| M-CTRL-CFGREADY  | M-CTRL-CFGREADY.request  | n/a        | n/a      | M-CTRL-CFGREADY.confirm  |
| M-CTRL-RESET     | M-CTRL-RESET.request     | n/a        | n/a      | M-CTRL-RESET.confirm     |
| M-CTRL-LINERESET | M-CTRL-LINERESET.request | n/a        | n/a      | M-CTRL-LINERESET.confirm |

2050

**Table 37 M-RX-CTRL SAP Service Primitives**

| Name             | Request                 | Indication                  | Response | Confirm                 |
|------------------|-------------------------|-----------------------------|----------|-------------------------|
| M-CTRL-CFGGET    | M-CTRL-CFGGET.request   | n/a                         | n/a      | M-CTRL-CFGGET.confirm   |
| M-CTRL-CFGSET    | M-CTRL-CFGSET.request   | n/a                         | n/a      | M-CTRL-CFGSET.confirm   |
| M-CTRL-CFGREADY  | M-CTRL-CFGREADY.request | n/a                         | n/a      | M-CTRL-CFGREADY.confirm |
| M-CTRL-RESET     | M-CTRL-RESET.request    | n/a                         | n/a      | M-CTRL-RESET.confirm    |
| M-CTRL-LINERESET | n/a                     | M-CTRL-LINERESET.indication | n/a      | n/a                     |

2051 The parameters associated with these primitives are defined in **Table 38** with the name, type and valid range.

2052

**Table 38 Parameters of M-TX-CTRL SAP and M-RX-CTRL SAP Service Primitives**

| Name             | Type                 | Valid Range                                                     | Description                                                                                                                                                      |
|------------------|----------------------|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MIBattribute     | Attribute name       | Any AttributeID as defined in M-TX and M-RX Attributes          | The name of the MIB attribute                                                                                                                                    |
| MIBvalue         | Depends on attribute | Depends on the attribute as defined in M-TX and M-RX Attributes | The value of the MIB attribute                                                                                                                                   |
| TActivateControl | Enum                 | ProtocolControlled = 0,<br>PhyControlled = 1                    | Indicates which Layer controls $T_{\text{ACTIVATE}}$ time and driving DIF-N. TActivateControl is an optional parameter. The default value is ProtocolControlled. |

2053 The following sections define the meaning of M-TX-CTRL SAP and M-RX-CTRL SAP service primitives  
 2054 and their associated parameters.

### 7.3.1 M-CTRL-CFGGET.request

2055 This primitive requests information about a MIB attribute, which are defined in M-TX and M-RX Attributes.

#### 7.3.1.1 Semantics of the Service Primitive

2056 The semantics of the M-CTRL-CFGGET.request primitive are as follows:

```
2057 M-CTRL-CFGGET.request (
2058     MIBattribute
2059 )
```

2060 The primitive parameter is defined in *Table 38*.

#### 7.3.1.2 When Generated

2061 This primitive is generated by the Protocol Layer to obtain information of an MIBattribute from a  
 2062 MODULE's MIB. The Protocol Layer shall ensure that the requested MIBattribute exists. The MODULE  
 2063 may not check the validity of an MIBattribute. Undefined attribute names may result in malfunctioning of a  
 2064 MODULE. After issuing an M-CTRL-CFGGET.request primitive, the Protocol Layer shall wait for the  
 2065 M-CTRL-CFGGET.confirm primitive reception before issuing a new configuration service request.

#### 7.3.1.3 Effect on Receipt

2066 The MODULE retrieves value of the requested attribute from its MIB and responds with  
 2067 M-CTRL-CFGGET.confirm that gives the result.



### 7.3.2 M-CTRL-CFGGET.confirm

2068 This primitive reports the result of a service request on MIBAttribute.

#### 7.3.2.1 Semantics of the Service Primitive

2069 The semantics of the M-CTRL-CFGGET.confirm primitive are as follows:

```
2070 M-CTRL-CFGGET.confirm  (  
2071                         MIBvalue  
2072                         )
```

2073 The primitive parameters are defined in *Table 38*.

#### 7.3.2.2 When Generated

2074 This primitive shall be generated by a MODULE in response to the most recent M-CTRL-CFGGET.request  
2075 by the Protocol Layer. The MIBvalue parameter shall contain the value of the requested MIBAttribute.

#### 7.3.2.3 Effect on Receipt

2076 The Protocol Layer shall accept this primitive in order to receive the value of the requested MIBAttribute.  
2077 The MIBvalue parameter will carry this value.

### 7.3.3 M-CTRL-CFGSET.request

2078 This primitive requests to set an MIB attribute indicated by the parameter MIBAttribute to the value hold by  
2079 the parameter MIBvalue.

#### 7.3.3.1 Semantics of the Service Primitive

2080 The semantics of the M-CTRL-CFGSET.request primitive are as follows:

```
2081 M-CTRL-CFGSET.request  (  
2082                         MIBAttribute,  
2083                         MIBvalue  
2084                         )
```

2085 The primitive parameters are defined in *Table 38*.

#### 7.3.3.2 When Generated

2086 The Protocol Layer shall generate this primitive to set an MIB attribute indicated by MIBAttribute parameter  
2087 with the value of MIBvalue parameter. The Protocol Layer shall ensure that the requested MIBAttribute exists  
2088 and the MIBvalue is in valid range of the requested MIBAttribute. A MODULE may not check the validity of  
2089 MIBAttribute and MIBvalue. Undefined attribute names or out of range attribute values may result in  
2090 malfunctioning of the MODULE. After issuing an M-CTRL-CFGSET.request primitive, the Protocol Layer  
2091 shall wait for the M-CTRL-CFGSET.confirm primitive reception before issuing a new configuration service  
2092 request.

#### 7.3.3.3 Effect on Receipt

2093 The MODULE shall set the specified MIBAttribute with the value carried by MIBvalue in its MIB registry.  
2094 If setting the value of an MIBAttribute implies a specific action, then this action shall not be performed until  
2095 the M-CTRL-CFGREADY.request primitive is received. The MODULE shall respond with  
2096 M-CTRL-CFGSET.confirm after registering the MIBvalue for the requested attribute.

**7.3.4 M-CTRL-CFGSET.confirm**

2097 This primitive confirms registering the attribute value based on the last issued request to set the value of an  
 2098 attribute in the MIB.

**7.3.4.1 Semantics of the Service Primitive**

2099 The semantics of the M-CTRL-CFGSET.confirm primitive are as follows:

2100 M-CTRL-CFGSET.confirm (   
 2101 )

2102 This primitive has no parameter.

**7.3.4.2 When Generated**

2103 This primitive shall be generated by a MODULE in response to the most recent M-CTRL-CFGSET.request  
 2104 by the Protocol Layer after setting the value of the requested MIBattribute.

**7.3.4.3 Effect on Receipt**

2105 The Protocol Layer is informed about serving the M-CTRL-CFGSET.request issued previously. The Protocol  
 2106 Layer may issue another service request upon receiving this primitive.

**7.3.5 M-CTRL-CFGREADY.request**

2107 This primitive requests a MODULE to update the operation settings of MIB attribute(s) with the  
 2108 corresponding MIB values that are issued through previous M-CTRL-CFGSET.request.

**7.3.5.1 Semantics of the Service Primitive**

2109 The semantics of the M-CTRL-CFGREADY.request primitive are as follows:

2110 M-CTRL-CFGREADY.request (   
 2111 )

2112 This primitive has no parameter.

**7.3.5.2 When Generated**

2113 The Protocol Layer shall issue this primitive after sending all setting requests to MIB attributes that compose  
 2114 a consistent new configuration parameter set. Issuing this primitive enables the MODULE to perform specific  
 2115 actions based on the MIB attributes set and the values assigned to these attributes. If a MODULE is in BURST  
 2116 state when this primitive is issued, then the Protocol Layer shall bring the MODULE into power saving state  
 2117 before specific actions can be taken and the new setting become effective.

**7.3.5.3 Effect on Receipt**

2118 The MODULE shall perform specific actions, if any, required upon receiving this primitive, based on the  
 2119 configuration set requests received before. These actions shall be performed, if needed, when the MODULE  
 2120 is entering into or in power saving state.

### 7.3.6 M-CTRL-CFGREADY.confirm

2121 This primitive reports the reception of M-CTRL-CFGREADY.request to update the operation settings to the  
2122 configured MIB attribute(s).

#### 7.3.6.1 Semantics of the Service Primitive

2123 The semantics of the M-CTRL-CFGREADY.confirm primitive are as follows:

2124 M-CTRL-CFGREADY.confirm (  
2125 )

2126 This primitive has no parameter.

#### 7.3.6.2 When Generated

2127 This primitive shall be generated by the MODULE in response to the reception of  
2128 M-CTRL-CFGREADY.request by the Protocol Layer.

#### 7.3.6.3 Effect on Receipt

2129 The Protocol Layer is informed about registering the M-CTRL-CFGREADY.request issued previously. Upon  
2130 receiving this primitive, if the MODULE is in BURST state, then the Protocol Layer shall request the  
2131 MODULE enter into power saving state.

### 7.3.7 M-CTRL-RESET.request

2132 This primitive requests the MODULE reset to its Power-on Reset state. All previous configuration settings  
2133 are lost.

#### 7.3.7.1 Semantics of the Service Primitive

2134 The semantics of the M-CTRL-RESET.request primitive are as follows:

2135 M-CTRL-RESET.request (ResetMode  
2136 )

2137 This primitive has one parameter:

| Name      | Type | Valid Range            | Description                                   |
|-----------|------|------------------------|-----------------------------------------------|
| ResetMode | Enum | LS_MODE=0<br>HS_MODE=1 | This parameter defines default operating mode |

#### 7.3.7.2 When Generated

2138 The Protocol Layer issues this request when it is desired to reset the MODULE to its default state and settings.

#### 7.3.7.3 Effect on Receipt

2139 When the Protocol Layer issues this request, the MODULE shall enter into DISABLED state specified in  
2140 *Section 4.7.1.4*.

2141 The ResetMode parameter chooses the default operating mode for the MODULE. When ResetMode is set to  
2142 1, the default operating mode is HS\_MODE. When set to 0, the default operating mode is LS\_MODE.

### 7.3.8 M-CTRL-RESET.confirm

2143 This primitive shall only be utilized for modeling purposes of Protocol Layer.

2144 This primitive informs the Protocol Layer that the MODULE has completed previously requested RESET  
2145 action and ready to service any request.

#### 7.3.8.1 Semantics of the Service Primitive

2146 The semantics of the M-CTRL-RESET.confirm primitive are as follows

2147 M-CTRL-RESET.confirm (   
2148 )

2149 This primitive has no parameter.

#### 7.3.8.2 When Generated

2150 After a request from the Protocol Layer to reset the MODULE, the MODULE shall generate this primitive  
2151 upon completion of initialization and ready to receive a service request.

#### 7.3.8.3 Effect on Receipt

2152 Upon receiving this primitive the Protocol Layer should aware that the MODULE has completed  
2153 initialization, reset all configuration settings to default values and entered HIBERN8 state.

### 7.3.9 M-CTRL-LINERESET.request

2154 This primitive requests an M-TX perform a LINE-RESET action. All configuration settings (rates,  
2155 amplitudes, etc.) are lost and reset to default values. The M-TX also asserts a signal on the LINE so that the  
2156 remote M-RX recognizes the LINE-RESET state and acts as defined in *Section 4.7.4.1*.

#### 7.3.9.1 Semantics of the Service Primitive

2157 The semantics of the M-CTRL-LINERESET.request primitive are as follows:

2158 M-CTRL-LINERESET.request (   
2159 TActivateControl   
2160 )

2161 The primitive parameter is defined in *Table 38*.

#### 7.3.9.2 When Generated

2162 The Protocol Layer shall issue M-LANE-BurstEnd.request and wait for  $T_{ACTIVATE}$  after the M-TX has  
2163 generated M-LANE-SaveState.indication before issuing M-CTRL-LINERESET.request with  
2164 TActivateControl set to “ProtocolControlled”.

2165 If M-CTRL-LINERESET.request with TActivateControl set to “PhyControlled” is issued when the M-TX is  
2166 in BURST state, the Protocol Layer shall be aware the PAYLOAD of an ongoing BURST is interrupted  
2167 immediately, and the M-TX might not trigger the proper BURST closure condition.

#### 7.3.9.3 Effect on Receipt

2168 Upon receiving this request with TActivateControl set to “ProtocolControlled”, the M-TX shall immediately  
2169 drive the LINE-RESET condition as described in *Section 4.7.4.1*.

2170 If this request is received with TActivateControl set to “PhyControlled”, the M-TX shall immediately drive  
2171 DIF-N on the LINE for  $T_{ACTIVATE}$  before driving the LINE-RESET condition.

### 7.3.10 M-CTRL-LINERESET.indication

2172 This primitive reports to the Protocol Layer that the M-RX has been reset by a LINE-RESET

#### 7.3.10.1 Semantics of the Service Primitive

2173 The semantics of the M-CTRL-LINERESET.indication primitive are as follows:

2174 M-CTRL-LINERESET.indication (   
2175 )

2176 This primitive has no parameter.

#### 7.3.10.2 When Generated

2177 When M-RX detects LINE-RESET as described in *Section 4.7.4.1*, it shall indicate the same to the Protocol  
2178 Layer using this primitive.

#### 7.3.10.3 Effect on Receipt

2179 When the Protocol Layer receives this primitive, it should be aware that the LANE is reset by a LINE-RESET  
2180 and both M-TX and M-RX on this LANE will be in default state with default attribute values.

### 7.3.11 M-CTRL-LINERESET.confirm

2181 This primitive informs the Protocol Layer that the MODULE has completed a previously requested  
2182 LINE-RESET action.

#### 7.3.11.1 Semantics of the Service Primitive

2183 The semantics of the M-CTRL-LINERESET.confirm primitive are as follows:

2184 M-CTRL-LINERESET.confirm (   
2185 )

2186 This primitive has no parameter.

#### 7.3.11.2 When Generated

2187 After a request from the Protocol Layer to an M-TX to reset the LANE by a LINE-RESET, the M-TX shall  
2188 issue this primitive upon completion of the LINE-RESET operation as described in *Section 4.7.4.1*.

2189 After exiting LINE-RESET, the Protocol Layer shall keep the M-TX for a given LANE in SAVE state for at  
2190 least T<sub>MIN\_SAVE\_CONFIG</sub>.

#### 7.3.11.3 Effect on Receipt

2191 Upon receiving this primitive the Protocol Layer should aware that the M-TX has completed LINE-RESET  
2192 activity and reset all configuration settings to default values while entering into SLEEP or STALL state.

7.3.12 Sequence of Service Primitives

The possible relationships among primitives at M-TX-CTRL SAP and M-RX-CTRL SAP are illustrated by the given time sequence diagrams shown in *Figure 54*. They also indicate a possible logical relationship in terms of time. Primitives that occur earlier in time and connected by dotted lines are logical predecessors of subsequent primitives.

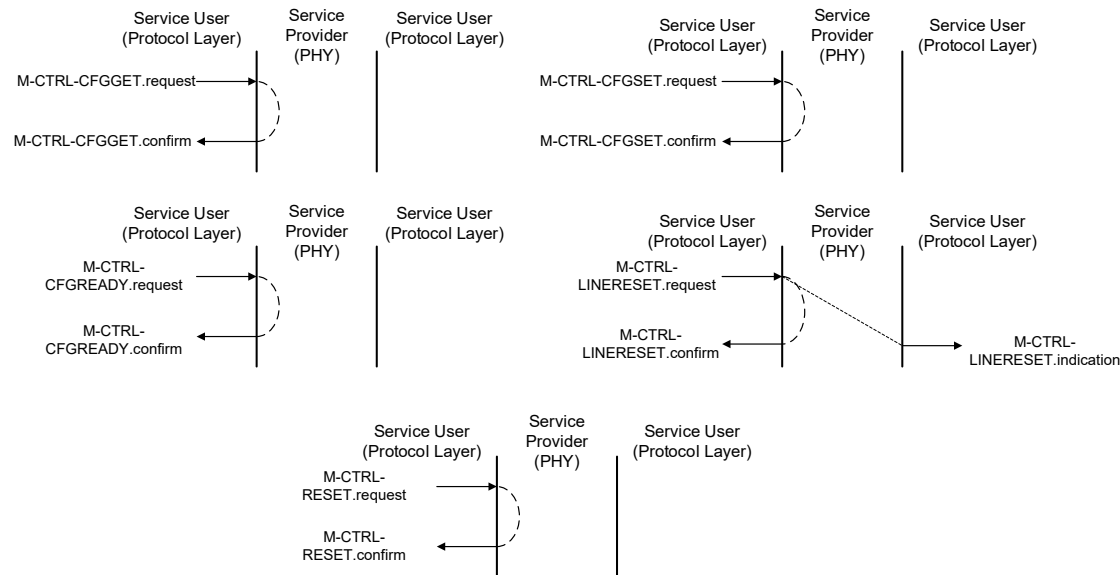


Figure 54 Sequence of Service Primitives at M-TX-CTRL SAP and M-RX-CTRL SAP

## 7.4 M-TX and M-RX Attributes

Capability, configuration and status attributes for an M-TX are listed in *Table 39*, *Table 40*, and *Table 41*, respectively, and for an M-RX these attributes are listed in *Table 42*, *Table 43*, and *Table 44*. Capability attributes describe the capabilities of an implementation and shall be read-only. Currently, only one status attribute is defined for a MODULE to provide the current operating state of the MODULE. No request, such as M-CTRL-CFGSET.request, shall be made by Protocol to write any value to any capability or status attribute. Any write request, such as M-CTRL-CFGSET.request, to a capability or status attribute shall be ignored and shall not be responded by a MODULE.

Configuration attributes are used for configuring a MODULE based on applicable capabilities, if there are any, to control its behavior. Configuration attributes shall be readable and writable. A write request, such as M-CTRL-CFGSET, to a configuration attribute shall hold a valid AttributeID and attribute value corresponding to that AttributeID. The attribute value shall not violate range of values of applicable capabilities, if any, for that attribute. Validity check of AttributeID and its corresponding value for a write request may not be performed in a MODULE. A read request, such as M-CTRL-CFGGET, to a configuration or capability attribute shall hold a valid AttributeID. Validity check of AttributeID for a read request may not be performed in a MODULE.

No request, such as M-CTRL-CFGGET.request, shall be made by the Protocol Layer to read a value from a write-only attribute. Any read request, such as M-CTRL-CFGGET.request, to a write-only attribute shall be ignored and shall not be responded by a MODULE.

The “Attribute Name” column in the tables specifies a symbolic name in a human readable form for an attribute.

The “AttributeID” column contains a hexadecimal code for an attribute which shall be used in read or write request made to an attribute. The parameter MIBattribute of M-CTRL-CFGGET.request and M-CTRL-CFGSET.request service primitives shall contain AttributeID of an attribute.

The “Description” column of an attribute provides a brief description of the attribute and four optional fields.

- The “Existence depends on” field of an attribute contains capability attributes that are applicable for its existence. An attribute becomes an Existence-dependant attribute if the “Description” contains an “Existence Depends on” field. An Existence-dependent attribute exists if all attributes listed in its “Existence Depends on” field are “TRUE”. Before making any read or write access to an existence dependant attribute, the Protocol shall ensure that all the applicable attributes for its existence are realizable to logical “TRUE” condition. If any of the attributes listed in the “Existence Depends on” field of an Existence-dependant attribute results in a logical “FALSE” condition then no access shall be made to that Existence-dependant attribute. For example, before accessing TX\_Advanced\_Hibern8Time\_Capability attribute, TX\_Advanced\_Granularity\_Capability attribute’s value is verified because the latter attribute is listed in the former attribute’s “Existence Depends on” field (see *Table 39*). The TX\_Advanced\_Hibern8Time\_Capability attribute is accessed if and only if TX\_Advanced\_Granularity\_Capability attribute’s value is “TRUE”.
- The “Value depends on” field of an attribute contains capability attributes that are applicable for defining its value. While writing to an attribute that has a “Value Depends on” field, the value being written to the attribute shall not exceed the worst-case value limits defined for those capability attributes that are listed in its “Value Depends on” field. For example, to set the TX\_HSGEAR attribute’s value, the TX\_HSGEAR\_Capability attribute value must be read, as this attribute is listed in the former attribute’s “Value Depends on” field. For example, if the value of the TX\_HSGEAR\_Capability attribute is 3, then the value of the TX\_HSGEAR attribute must be in the range [1, 3] (worst-case value limit).
- The “Req’d Values” field is applicable only to configuration attributes. If a configuration attribute is supported by a MODULE, then the MODULE shall support all values or range of values specified in the The “Req’d Values” field of that configuration attribute.

- 2246 • The “Reset Value” field is applicable to configuration attributes only and specifies the default value of  
2247 an attribute. A configuration attribute shall hold this default value after exiting the DISABLED state.
- 2248 The “FSM” column of an attribute contains those FSM types that this attribute shall be applicable. So, this  
2249 column specifies the validity of an attribute to be used in either TYPE-I or TYPE-II or both (TYPE-I and  
2250 TYPE-II).
- 2251 The “Type” column of an attribute specifies the type of data (as used in most common programming  
2252 languages) it holds.
- 2253 The “Bits” column of an attribute either recommends or mandates which bits to use for representing the  
2254 possible values listed inside an attribute’s value range.
- 2255 The “Range” column of an attribute specifies permissible limits of range of values that an attribute can take.  
2256 Supported value range for an attribute shall not exceed the range of values specified in the “Range” column  
2257 of that attribute.



2258

**Table 39 M-TX Capability Attributes**

| Attribute Name                                  | AttributeID | Description                                                                                                                                                                                 | FSM    | Type | Bits                | Range                                                                                                |
|-------------------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|---------------------|------------------------------------------------------------------------------------------------------|
| TX_HSMODE_Capability                            | 0x01        | Specifies support for HS-MODE.                                                                                                                                                              | Both   | Bool | B[0] <sup>1</sup>   | TRUE = 1                                                                                             |
| TX_HSGEAR_Capability                            | 0x02        | Specifies supported HS-GEARs.                                                                                                                                                               | Both   | Enum | B[2:0] <sup>1</sup> | HS_G1_ONLY = 1,<br>HS_G1_TO_G2 = 2,<br>HS_G1_TO_G3 = 3,<br>HS_G1_TO_G4 = 4,<br>HS_G1_TO_G5 = 5       |
| TX_PWMG0_Capability                             | 0x03        | Specifies support for PWM-G0.                                                                                                                                                               | TYPE-I | Bool | B[0] <sup>1</sup>   | NO = 0 <sup>3</sup>                                                                                  |
| TX_PWMGEAR_Capability                           | 0x04        | Specifies support for PWM-GEAR.                                                                                                                                                             | TYPE-I | Enum | B[2:0] <sup>1</sup> | PWM_G1_ONLY = 1,<br>Reserved: 2 to 7 <sup>3</sup>                                                    |
| TX_Amplitude_Capability                         | 0x05        | Specifies supported signal amplitude levels.                                                                                                                                                | Both   | Enum | B[1:0] <sup>1</sup> | SMALL_AMPLITUDE_ONLY = 1,<br>LARGE_AMPLITUDE_ONLY = 2,<br>LARGE_AND_SMALL_AMPLITUDE = 3 <sup>3</sup> |
| TX_ExternalSYNC_Capability                      | 0x06        | Specifies support for external SYNC pattern.                                                                                                                                                | Both   | Bool | B[0]                | FALSE = 0,<br>TRUE = 1                                                                               |
| TX_HS_Unterminated_LINE_Drive_Capability        | 0x07        | Specifies whether or not M-TX supports driving an unterminated LINE in HS-MODE.                                                                                                             | Both   | Bool | B[0] <sup>1</sup>   | NO = 0,<br>YES = 1                                                                                   |
| TX_LS_Terminated_LINE_Drive_Capability          | 0x08        | Specifies whether or not M-TX supports driving a terminated LINE in LS-MODE.                                                                                                                | Both   | Bool | B[0] <sup>1</sup>   | NO = 0,<br>YES = 1                                                                                   |
| TX_Min_SLEEP_NoConfig_Time_Capability           | 0x09        | Specifies minimum time (in SI) in SLEEP state needed when inline configuration was not performed.                                                                                           | Both   | Int  | B[3:0] <sup>1</sup> | 1 to 15                                                                                              |
| TX_Min_STALL_NoConfig_Time_Capability           | 0x0A        | Specifies minimum time (in SI) in STALL state needed when inline configuration was not performed.                                                                                           | Both   | Int  | B[7:0] <sup>1</sup> | 1 to 255                                                                                             |
| TX_Min_SAVE_Config_Time_Capability <sup>2</sup> | 0x0B        | Specifies minimum reconfiguration time (in 40 ns steps). This applies only to SLEEP and STALL states.<br>Is set to 250 (10000 ns), if TX_Extended_Min_SAVE_Config_Time_c capability exists. | Both   | Int  | B[7:0]              | 1 to 250 (10000 ns)                                                                                  |

| Attribute Name                       | AttributeID | Description                                                                                                                                                                                                  | FSM    | Type | Bits              | Range                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TX_REF_CLOCK_SHARED_Capability       | 0x0C        | Specifies support for a shared reference clock.                                                                                                                                                              | TYPE-I | Bool | B[0] <sup>1</sup> | NO = 0,<br>YES = 1                                                                                                                                                                                                                                                                                                                                                                                  |
| TX_PHY_MajorMinor_Release_Capability | 0x0D        | Specifies the major and minor numbers of the M-PHY version supported by the M-TX.                                                                                                                            | Both   | Int  | B[7:4]            | Major version number, 0 to 9                                                                                                                                                                                                                                                                                                                                                                        |
|                                      |             |                                                                                                                                                                                                              |        |      | B[3:0]            | Minor version number, 0 to 9                                                                                                                                                                                                                                                                                                                                                                        |
| TX_PHY_Editorial_Release_Capability  | 0x0E        | Specifies the sequence number of the M-PHY version supported by the M-TX.                                                                                                                                    | Both   | Int  | B[7:0]            | 0 to 99                                                                                                                                                                                                                                                                                                                                                                                             |
| TX_Hibern8Time_Capability            | 0x0F        | Specifies minimum time (in 100 $\mu$ s steps) in HIBERN8 state.                                                                                                                                              | Both   | Int  | B[7:0]            | 1 to 128 (100 $\mu$ s to 12.8 ms)                                                                                                                                                                                                                                                                                                                                                                   |
| TX_Advanced_Granularity_Capability   | 0x10        | Support and degree of fine granularity steps for a reduced time in HIBERN8 state. If a finer granularity is specified, all coarser granularities shall be supported.                                         | Both   | Int  | B[2:1]            | step size<br>b00 = 4 $\mu$ s<br>b01 = 8 $\mu$ s<br>b10 = 16 $\mu$ s<br>b11 = 32 $\mu$ s                                                                                                                                                                                                                                                                                                             |
|                                      |             |                                                                                                                                                                                                              |        |      | B[0]              | supports fine granularity steps:<br>No = 0 (100 $\mu$ s step)<br>Yes = 1                                                                                                                                                                                                                                                                                                                            |
| TX_Advanced_Hibern8Time_Capability   | 0x11        | Specifies minimum time in HIBERN8 state when advanced granularity is supported in steps defined by TX_Advanced_Granularity_Capability.<br><i>Existence depends on:</i><br>TX_Advanced_Granularity_Capability | Both   | Int  | B[7:0]            | 1 to 128                                                                                                                                                                                                                                                                                                                                                                                            |
| TX_HS_Equalizer_Setting_Capability   | 0x12        | Support for transmit path de-emphasis for HS-MODE                                                                                                                                                            | Both   | Int  | B[6:0]            | TX equalization capabilities <sup>4</sup> :<br>B[0]: De-emphasis of 0.8 dB supported<br>B[1]: De-emphasis of 1.6 dB supported<br>B[2]: De-emphasis of 2.5 dB supported<br>B[3]: De-emphasis of 3.5 dB supported<br>B[4]: De-emphasis of 4.7 dB supported<br>B[5]: De-emphasis of 6.0 dB supported<br>B[6]: De-emphasis of 7.6 dB supported<br><br>For all bits:<br>1: Supported<br>0: Not supported |

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| Attribute Name                                          | AttributeID | Description                                                                                                                                                                                           | FSM  | Type | Bits   | Range                           |
|---------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|--------|---------------------------------|
| TX_Extended_Min_SAVE_Config_Time_Granularity_Capability | 0x13        | Support for TX_Extended_Min_SAVE_Config_Time_Capability.                                                                                                                                              | Both | Int  | B[0]   | 0: Not supports<br>1: Supported |
| TX_Extended_Min_SAVE_Config_Time_Capability             | 0x14        | Specifies minimum reconfiguration time (in 1 $\mu$ s steps). This applies only to SLEEP and STALL states.<br><i>Existence depends on:</i><br>TX_Extended_Min_SAVE_Config_Time_Granularity_Capability. | Both | Int  | B[6:0] | 1 to 100 (100 $\mu$ S)          |

**Note:**

1. Recommended bit assignment.
2. There is a potential timing mismatch between the setting of this capability, implementation width of RMMI when the prior gear is PWM-G1. As neither timing nor width is defined at the RMMI interface, if such a hazard exists for an M-PHY implementation, then the M-PHY IP has to advertise this in its data sheet. This allows the implementer to ensure the protocol and the selected M-PHY IP are in a good match and allows sufficient time for attributes to be updated. Since the sensitivity is only in PWM-G1, the protocol has to take the current GEAR into account while computing the minimum time between BURSTs after a configuration change.
3. PWM-G0 and PWM-G2 to PWM-G7 are no longer supported from M-PHY v5.0. Associated attributes are obsolete.
4. The TX equalization values are intended to cover a certain range. The attribute settings provide some possible nominal values. Actual values are implementation specific. Also tolerances are not specified.

Table 40 M-TX Configuration Attributes

| Attribute Name   | AttributeID | Description                                                                                                                                                                        | FSM    | Type | Bits                | Range                                                             |
|------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|---------------------|-------------------------------------------------------------------|
| TX_MODE          | 0x21        | M-TX operating mode.<br><i>Req'd Value:</i> LS_MODE and HS_MODE<br><i>Reset Value:</i> Depends on ResetMode<br>(see <b>Section 7.3.7.1</b> )                                       | Both   | Enum | B[1:0] <sup>1</sup> | LS_MODE = 1,<br>HS_MODE = 2                                       |
| TX_HSRATE_Series | 0x22        | HS mode RATE series value of M-TX.<br><i>Req'd Value:</i> A and B<br><i>Reset Value:</i> A                                                                                         | Both   | Enum | B[1:0] <sup>1</sup> | A = 1,<br>B = 2                                                   |
| TX_HSGEAR        | 0x23        | HS-GEAR value of M-TX.<br><i>Value depends on:</i> TX_HSGEAR_Capability<br><i>Req'd Value:</i> HS_G1<br><i>Reset Value:</i> HS_G1                                                  | Both   | Enum | B[2:0] <sup>1</sup> | HS_G1 = 1,<br>HS_G2 = 2,<br>HS_G3 = 3,<br>HS_G4 = 4,<br>HS_G5 = 5 |
| TX_PWMGEAR       | 0x24        | PWM-GEAR value of M-TX.<br><i>Value depends on:</i> TX_PWMGEAR_Capability<br><i>Req'd Value:</i> PWM_G1<br><i>Reset Value:</i> PWM_G1                                              | TYPE-I | Enum | B[2:0] <sup>1</sup> | Reserved: 0,<br>PWM_G1 = 1,<br>Reserved: 2 to 7 <sup>11</sup>     |
| TX_Amplitude     | 0x25        | Type of drive strength on PINs at M-TX.<br><i>Value depends on:</i><br>TX_Amplitude_Capability<br>TX_HSGEAR_Capability <sup>13</sup><br><i>Reset Value:</i> LARGE_AMPLITUDE        | Both   | Enum | B[1:0] <sup>1</sup> | SMALL_AMPLITUDE = 1,<br>LARGE_AMPLITUDE = 2                       |
| TX_HS_SlewRate   | 0x26        | Slew Rate control of M-TX output driver.<br><i>Reset Value:</i> see <sup>2</sup>                                                                                                   | Both   | Int  | B[7:0] <sup>3</sup> | 0 to 255 <sup>4</sup>                                             |
| TX_SYNC_Source   | 0x27        | Source of synchronization pattern at M-TX.<br><i>Existence depends on:</i><br>TX_ExternalSync_Capability<br><i>Req'd Value:</i> INTERNAL_SYNC<br><i>Reset Value:</i> INTERNAL_SYNC | Both   | Enum | B[0] <sup>1</sup>   | INTERNAL_SYNC = 0,<br>EXTERNAL_SYNC = 1                           |

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| Attribute Name                    | AttributeID | Description                                                                                                                                                                                                                         | FSM                            | Type | Bits                | Range                                                               |
|-----------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|------|---------------------|---------------------------------------------------------------------|
| TX_HS_SYNC_LENGTH                 | 0x28        | High Speed Synchronization pattern length of M-TX in SI.<br><i>Req'd Values:</i> FINE, COARSE, 0 to 15<br><i>Reset Values:</i><br>IF (Tx_ResetMode = HS_MODE)<br>COARSE, 12<br>ELSE<br>COARSE, 15                                   | Both                           | Int  | B[7:6]              | SYNC_range<br>FINE = 0,<br>COARSE = 1                               |
|                                   |             |                                                                                                                                                                                                                                     |                                |      | B[5:0]              | SYNC_length <sup>6</sup><br>1 to 15 for FINE,<br>0 to 15 for COARSE |
| TX_HS_PREPARE_LENGTH <sup>7</sup> | 0x29        | HS PREPARE length multiplier for M-TX.<br><i>Req'd Values:</i> 0 to 15 <sup>6</sup><br><i>Reset Value:</i> 15 <sup>6</sup>                                                                                                          | Both                           | Int  | B[3:0] <sup>1</sup> | 0 to 15                                                             |
| TX_LS_PREPARE_LENGTH <sup>7</sup> | 0x2A        | PWM-BURST or SYS-BURST PREPARE length multiplier for M-TX.<br><i>Req'd Values:</i> 0 to 10 <sup>7</sup><br><i>Reset Value:</i> 10 <sup>7</sup>                                                                                      | Both                           | Int  | B[3:0] <sup>1</sup> | 0 to 10                                                             |
| TX_HIBERN8_Control                | 0x2B        | M-TX HIBERN8 state control.<br><i>Req'd Values:</i> ENTER, EXIT<br><i>Reset Value:</i><br>ENTER for Local RESET<br>EXIT for LINE-RESET                                                                                              | TYPE-I<br>TYPE-II <sup>8</sup> | Bool | B[0] <sup>1</sup>   | EXIT = 0,<br>ENTER = 1                                              |
| TX_LCC_Enable                     | 0x2C        | LCCs support by the M-TX.<br>This attribute setting is immediately effective upon receipt of M-CTRL-CFGREADY.request (see <b>Section 4.7.4.2</b> ).<br><i>Req'd Values:</i> NO <sup>12</sup><br><i>Reset Value:</i> NO <sup>9</sup> | TYPE-I                         | Bool | B[0] <sup>1</sup>   | NO = 0 <sup>12</sup>                                                |
| TX_PWM_BURST_Closure_Extension    | 0x2D        | BURST CLOSURE sequence duration in SI. The value shall be greater than, or equal to, the value of RX_PWM_Burst_Closure_Length_Capability.<br><i>Req'd Values:</i> 0 to 255<br><i>Reset Value:</i> 32                                | TYPE-I                         | Int  | B[7:0] <sup>1</sup> | 0 to 255                                                            |
| TX_BYPASS_8B10B_Enable            | 0x2E        | Bypass 8b10b encoding operation at M-TX.<br><i>Req'd Value:</i> FALSE<br><i>Reset Value:</i> FALSE                                                                                                                                  | Both                           | Bool | B[0] <sup>1</sup>   | FALSE = 0,<br>TRUE = 1                                              |

| Attribute Name                       | AttributeID | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | FSM    | Type | Bits              | Range                                     |
|--------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|-------------------|-------------------------------------------|
| TX_DRIVER_POLARITY                   | 0x2F        | M-TX output driver polarity.<br><i>Req'd Values:</i> NORMAL, INVERTED<br><i>Reset Value:</i> NORMAL for local RESET.<br>LINE-RESET shall not reset the value of this attribute.                                                                                                                                                                                                                                                                                                             | Both   | Enum | B[0] <sup>1</sup> | NORMAL = 0,<br>INVERTED = 1               |
| TX_HS_Unterminated_LINE_Drive_Enable | 0x30        | Enable M-TX to drive unterminated LINE in HS-MODE.<br><i>Existence depends on:</i><br>TX_HS_Unterminated_LINE_Drive_Capability<br><i>Req'd Values:</i> NO<br><i>Reset Value:</i> NO                                                                                                                                                                                                                                                                                                         | Both   | Bool | B[0] <sup>1</sup> | NO = 0                                    |
| TX_LS_Terminated_LINE_Drive_Enable   | 0x31        | Enable M-TX to drive terminated LINE in LS-MODE.<br><i>Existence depends on:</i><br>TX_LS_Terminated_LINE_Drive_Capability<br><i>Req'd Values:</i> NO, YES<br><i>Reset Value:</i> NO                                                                                                                                                                                                                                                                                                        | Both   | Bool | B[0] <sup>1</sup> | NO = 0,<br>YES=1                          |
| TX_LCC_Sequencer                     | 0x32        | To set bits for carrying out multiple LCC-READ or LCC-WRITE operations.<br>To perform an LCC operation the corresponding bit for this attribute shall be set.<br>TX_LCC_Sequencer bits are cleared by M-TX when it issues the associated LCC request.<br>This attribute setting is immediately effective upon receipt of M-CTRL-CFGREADY.request (see <b>Section 4.7.4.2</b> ).<br><i>Req'd Values:</i> 0 <sup>12</sup><br><i>Reset Values:</i><br>0 (no READ or WRITE operation requested) | TYPE-I | Enum | B[7:0]            | B[7:0] Reserved, set to 0 <sup>12</sup> . |

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| Attribute Name               | AttributeID | Description                                                                                                                                                                                                                                                                                          | FSM    | Type | Bits   | Range                                                                                   |
|------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|--------|-----------------------------------------------------------------------------------------|
| TX_Min_ActivateTime          | 0x33        | Specifies minimum activate time needed in 100 $\mu$ s steps.<br>This value must be greater than or equal to the smaller of RX_Min_ActivateTime_Capability or RX_Advanced_Min_ActivateTime_Capability from the remote M-RX.<br><i>Reset Values:</i><br>IF (Tx_ResetMode = HS_MODE)<br>4<br>ELSE<br>15 | Both   | Int  | B[3:0] | 1 to 15                                                                                 |
| TX_PWM_G6_G7_SYNC_LENGTH     | 0x34        | Synchronization pattern length of M-TX, in SI, for PWM-G6 and PWM-G7 in LS-MODE.<br><i>Existence depends on:</i><br>TX_PWMGEAR_Capability<br><i>Req'd Values:</i> COARSE, 15 <sup>11</sup><br><i>Reset Values:</i> COARSE, 15                                                                        | TYPE-I | Int  | B[7:6] | SYNC_range<br>FINE = 0,<br>COARSE = 1                                                   |
|                              |             |                                                                                                                                                                                                                                                                                                      |        |      | B[5:0] | SYNC_length <sup>5</sup><br>0 to 15                                                     |
| TX_Advanced_Granularity_Step | 0x35        | Support and degree of fine granularity steps for T <sub>ACTIVATE</sub><br><i>Reset Value:</i> 0                                                                                                                                                                                                      | Both   | Int  | B[2:1] | step size<br>b00 = 4 $\mu$ s<br>b01 = 8 $\mu$ s<br>b10 = 16 $\mu$ s<br>b11 = 32 $\mu$ s |
|                              |             |                                                                                                                                                                                                                                                                                                      |        |      | B[0]   | Supports advanced granularity<br>No = 0<br>Yes = 1                                      |
| TX_Advanced_Granularity      | 0x36        | Specifies minimum activate time when advanced granularity is supported in steps defined by TX_Advanced_Granularity_Step. The configured value has to meet the requirements of T <sub>ACTIVATE</sub> in <i>Table 5</i> .<br><i>Reset Value:</i> 15                                                    | Both   | Int  | B[3:0] | 1 to 15                                                                                 |

| Attribute Name             | AttributeID | Description                                                                                                                                                                                                                                                                                                                                                          | FSM  | Type | Bits   | Range                                                                                                                                                                                                                                                                                                                        |
|----------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TX_HS_Equalizer_Setting    | 0x37        | HS Transmit path de-emphasis value selection.<br><i>Existence depends on:</i><br>TX_HS_Equalizer_Setting_Capability<br><i>Value depends on:</i><br>TX_HS_Equalizer_Setting_Capability<br><i>Required Values: Support for de-emphasis from 0 dB up to 8.6 dB</i><br><i>Reset Value: 0 for local RESET.</i><br>LINE-RESET shall not reset the value of this attribute. | Both | Int  | B[2:0] | b000: No de-emphasis selected,<br>b001: De-emphasis of 0.8 dB selected,<br>b010: De-emphasis of 1.6 dB selected,<br>b011: De-emphasis of 2.5 dB selected,<br>b100: De-emphasis of 3.5 dB selected,<br>b101: De-emphasis of 4.7 dB selected,<br>b110: De-emphasis of 6.0 dB selected,<br>b111: De-emphasis of 7.6 dB selected |
| TX_Min_SLEEP_NoConfig_Time | 0x38        | This amount of time (in SI) ensures M-RX has transitioned to termination disable entering SLEEP. Greater than or equal to the larger of TX_Min_SLEEP_NoConfig_Time_Capability and RX_Min_SLEEP_NoConfig_Time_Capability.<br><i>Reset Value: 15.</i>                                                                                                                  | Both | Int  | B[3:0] | 1 to 15                                                                                                                                                                                                                                                                                                                      |
| TX_Min_STALL_NoConfig_Time | 0x39        | This amount of time (in SI) ensures M-RX has transitioned to termination disable entering STALL. Greater than or equal to the larger of TX_Min_STALL_NoConfig_Time_Capability and RX_Min_STALL_NoConfig_Time_Capability.<br><i>Reset Value: 255.</i>                                                                                                                 | Both | Int  | B[7:0] | 1 to 255                                                                                                                                                                                                                                                                                                                     |
| TX_HS_ADAPT_LENGTH         | 0x3A        | High Speed adaptation pattern length of M-TX.<br><i>Required Values: FINE, COARSE, 0 to 127</i><br><i>Reset Values: COARSE, 17</i>                                                                                                                                                                                                                                   | Both | Int  | B[7]   | ADAPT_range<br>FINE = 0,<br>COARSE = 1                                                                                                                                                                                                                                                                                       |
|                            |             |                                                                                                                                                                                                                                                                                                                                                                      |      |      | B[6:0] | ADAPT_length <sup>10</sup><br>0 to 127 for FINE<br>0 to 17 for COARSE                                                                                                                                                                                                                                                        |

**Note:**

1. Recommended bit assignment.
2. Implementation should ensure that the TX\_HS\_SlewRate value does not violate other parameter specifications.
3. 256 steps monotonically decreasing.
4. "0" represents the fastest slew rate value and "255" represents the slowest slew rate value. Maximum number of possible steps are 256 (0 to 255). An implementation may support less than 256 steps but be able to interpret the 8-bit range.
5. Actual SYNC length is calculated using the formula for TSYNC in **Table 7**.
6. Actual HS PREPARE length is calculated using the formula for THS\_PREPARE in **Table 7** with TX\_HSGEAR.



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7. *Actual PWM PREPARE length is calculated using the formula for TPWM\_PREPARE in **Table 7** with TX\_PWMGEAR.*
8. *TYPE-II with embedded HIBERN8 exit control. See Section 4.7.1.3.*
9. *The Reset Value was changed from YES to NO from M-PHY version v4.1 onward.*
10. *Actual ADAPT length is calculated using the formula for TADAPT in **Table 7**.*
11. *PWM-G0 and PWM-G2 to PWM-G7 are no longer supported in M-PHY v5.0. Associated attributes are obsolete.*
12. *LINE-CFG is no longer supported in M-PHY v5.0. Associated attributes are obsolete.*
13. *Small amplitude is not set in HS-G5 and above.*

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**Table 41 M-TX Status Attributes**

| Attribute Name | AttributeID | Description                           | FSM  | Type | Bits                | Range                                                                                                                                        |
|----------------|-------------|---------------------------------------|------|------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| TX_FSM_State   | 0x41        | To read out the current state of M-TX | Both | Enum | B[3:0] <sup>1</sup> | DISABLED = 0,<br>HIBERN8 = 1,<br>SLEEP = 2,<br>STALL = 3,<br>LS-BURST = 4,<br>HS-BURST = 5,<br>Reserved = 6 <sup>2</sup> ,<br>LINE-RESET = 7 |

**Note:**

1. Recommended bit assignment.
2. Value “6” was used for LINE-CFG up to M-PHY v4.1. This functionality is no longer supported in M-PHY v5.0.

**Table 42 M-RX Capability Attributes**

| Attribute Name                                  | AttributeID | Description                                                                                                                                                                              | FSM    | Type | Bits                | Range                                                                                          |
|-------------------------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|---------------------|------------------------------------------------------------------------------------------------|
| RX_HSMODE_Capability                            | 0x81        | Specifies support for HS-MODE.                                                                                                                                                           | Both   | Bool | B[0] <sup>1</sup>   | TRUE = 1                                                                                       |
| RX_HSGEAR_Capability                            | 0x82        | Specifies supported HS-GEARS.                                                                                                                                                            | Both   | Enum | B[2:0] <sup>1</sup> | HS_G1_ONLY = 1,<br>HS_G1_TO_G2 = 2,<br>HS_G1_TO_G3 = 3,<br>HS_G1_TO_G4 = 4,<br>HS_G1_TO_G5 = 5 |
| RX_PWMG0_Capability                             | 0x83        | Specifies support for PWM-G0.                                                                                                                                                            | TYPE-I | Bool | B[0] <sup>1</sup>   | NO = 0 <sup>7</sup>                                                                            |
| RX_PWMGEAR_Capability                           | 0x84        | Specifies supported PWM-GEAR.                                                                                                                                                            | TYPE-I | Enum | B[2:0] <sup>1</sup> | PWM_G1_ONLY = 1,<br>Reserved: 2 to 7 <sup>7</sup>                                              |
| RX_HS_Unterminated_Capability                   | 0x85        | Specifies support for disconnection of resistive termination in HS-MODE.                                                                                                                 | Both   | Bool | B[0] <sup>1</sup>   | NO = 0                                                                                         |
| RX_LS_Terminated_Capability                     | 0x86        | Specifies support for enabling resistive termination in LS-MODE.                                                                                                                         | TYPE-I | Bool | B[0] <sup>1</sup>   | NO = 0,<br>YES = 1                                                                             |
| RX_Min_SLEEP_NoConfig_Time_Capability           | 0x87        | Specifies minimum time (in SI) in SLEEP state needed when inline configuration was not performed.                                                                                        | Both   | Int  | B[3:0] <sup>1</sup> | 1 to 15                                                                                        |
| RX_Min_STALL_NoConfig_Time_Capability           | 0x88        | Specifies minimum time (in SI) in STALL state needed when inline configuration was not performed.                                                                                        | Both   | Int  | B[7:0] <sup>1</sup> | 1 to 255                                                                                       |
| RX_Min_SAVE_Config_Time_Capability <sup>2</sup> | 0x89        | Specifies minimum reconfiguration time (in 40 ns steps). This applies only to SLEEP and STALL states.<br>Is set to 250 (10000 ns) if RX_Extended_Min_SAVE_Config_Time_Capability exists. | Both   | Int  | B[7:0]              | 1 to 250 (10000 ns)                                                                            |
| RX_REF_CLOCK_SHARED_Capability                  | 0x8A        | Specifies support for a shared reference clock.                                                                                                                                          | TYPE-I | Bool | B[0] <sup>1</sup>   | YES = 1                                                                                        |
| RX_HS_G1_SYNC_LENGTH_Capability                 | 0x8B        | High Speed GEAR 1 Synchronization pattern length in SI.                                                                                                                                  | Both   | Int  | B[7:6]              | SYNC_range<br>FINE = 0,<br>COARSE = 1                                                          |
|                                                 |             |                                                                                                                                                                                          |        |      | B[5:0]              | SYNC_length <sup>3</sup><br>1 to 15 for FINE,<br>0 to 12 for COARSE                            |
| RX_HS_G1_PREPARE_LENGTH_Capability <sup>6</sup> | 0x8C        | HS-G1 PREPARE length multiplier for M-RX.                                                                                                                                                | Both   | Int  | B[3:0] <sup>1</sup> | 0 to 15                                                                                        |

| Attribute Name                                  | AttributeID | Description                                                                                                                                            | FSM    | Type | Bits                | Range                                                               |
|-------------------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|---------------------|---------------------------------------------------------------------|
| RX_LS_PREPARE_LENGTH_Capability <sup>5</sup>    | 0x8D        | PWM-BURST or SYS-BURST PREPARE length multiplier for M-RX.                                                                                             | Both   | Int  | B[3:0] <sup>1</sup> | 0 to 10                                                             |
| RX_PWM_Burst_Closure_Length_Capability          | 0x8E        | Specifies minimum burst closure time (in SI) necessary to guarantee complete data processing inside M-RX                                               | TYPE-I | Int  | B[4:0]              | 0 to 31                                                             |
| RX_Min_ActivateTime_Capability                  | 0x8F        | Specifies minimum activate time needed in 100 $\mu$ s steps.                                                                                           | Both   | Int  | B[3:0]              | 1 to 4                                                              |
| RX_PHY_MajorMinor_Release_Capability            | 0x90        | Specifies the major and minor numbers of the M-PHY version supported by the M-RX.                                                                      | Both   | Int  | B[7:4]              | Major version number, 0 to 9                                        |
|                                                 |             |                                                                                                                                                        |        |      | B[3:0]              | Minor version number, 0 to 9                                        |
| RX_PHY_Editorial_Release_Capability             | 0x91        | Specifies the sequence number of the M-PHY version supported by the M-RX.                                                                              | Both   | Int  | B[7:0]              | 0 to 99                                                             |
| RX_Hibern8Time_Capability                       | 0x92        | Specifies minimum time (in 100 $\mu$ s steps) in HIBERN8 state.                                                                                        | Both   | Int  | B[7:0]              | 1 to 128<br>(100 $\mu$ s to 12.8 ms)                                |
| RX_PWM_G6_G7_SYNC_LENGTH_Capability             | 0x93        | Synchronization pattern length, in SI, for PWM-G6 and PWM-G7 in LS-MODE <sup>7</sup> .<br><i>Existence depends on:</i><br><i>RX_PWMGEAR_Capability</i> | TYPE-I | Int  | B[7:0]              | Reserved                                                            |
| RX_HS_G2_SYNC_LENGTH_Capability                 | 0x94        | High Speed GEAR 2 Synchronization pattern length in SI.<br><i>Existence depends on:</i><br><i>RX_HSGEAR_Capability</i>                                 | Both   | Int  | B[7:6]              | SYNC_range<br>FINE = 0,<br>COARSE = 1                               |
|                                                 |             |                                                                                                                                                        |        |      | B[5:0]              | SYNC_length <sup>3</sup><br>1 to 15 for FINE,<br>0 to 15 for COARSE |
| RX_HS_G3_SYNC_LENGTH_Capability                 | 0x95        | High Speed GEAR 3 Synchronization pattern length in SI.<br><i>Existence depends on:</i><br><i>RX_HSGEAR_Capability</i>                                 | Both   | Int  | B[7:6]              | SYNC_range<br>FINE = 0,<br>COARSE = 1                               |
|                                                 |             |                                                                                                                                                        |        |      | B[5:0]              | SYNC_length <sup>3</sup><br>1 to 15 for FINE,<br>0 to 15 for COARSE |
| RX_HS_G2_PREPARE_LENGTH_Capability <sup>4</sup> | 0x96        | HS-G2 PREPARE length multiplier for M-RX.<br><i>Existence depends on:</i><br><i>RX_HSGEAR_Capability</i>                                               | Both   | Int  | B[3:0] <sup>1</sup> | 0 to 15                                                             |

| Attribute Name                                  | AttributeID | Description                                                                                                                                                                                                                 | FSM  | Type | Bits                | Range                                                                                                           |
|-------------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|---------------------|-----------------------------------------------------------------------------------------------------------------|
| RX_HS_G3_PREPARE_LENGTH_Capability <sup>4</sup> | 0x97        | HS-G3 PREPARE length multiplier for M-RX.<br><i>Existence depends on:</i><br><i>RX_HSGEAR_Capability</i>                                                                                                                    | Both | Int  | B[3:0] <sup>1</sup> | 0 to 15                                                                                                         |
| RX_Advanced_Granularity_Capability              | 0x98        | Support and degree of fine granularity steps for $T_{\text{HIBERN8}}$ and $T_{\text{ACTIVATE}}$ .                                                                                                                           | Both | Int  | B[2:1]              | step size<br>b00 = 4 $\mu\text{s}$<br>b01 = 8 $\mu\text{s}$<br>b10 = 16 $\mu\text{s}$<br>b11 = 32 $\mu\text{s}$ |
|                                                 |             |                                                                                                                                                                                                                             |      |      | B[0]                | supports fine granularity steps<br>No = 0 (100 $\mu\text{s}$ step)<br>Yes = 1                                   |
| RX_Advanced_Hibern8Time_Capability              | 0x99        | Specifies minimum time in HIBERN8 state when advanced granularity is supported in steps defined by <i>RX_Advanced_Granularity_Capability</i> .<br><i>Existence depends on:</i><br><i>RX_Advanced_Granularity_Capability</i> | Both | Int  | B[7:0]              | 1 to 128                                                                                                        |
| RX_Advanced_Min_ActivateTime_Capability         | 0x9A        | Specifies minimum activate time when advanced granularity is supported in steps defined by <i>RX_Advanced_Granularity_Capability</i> .<br><i>Existence depends on:</i><br><i>RX_Advanced_Granularity_Capability</i>         | Both | Int  | b[3:0]              | 1 to 14                                                                                                         |
| RX_HS_G4_SYNC_LENGTH_Capability                 | 0x9B        | High Speed GEAR 4 Synchronization pattern length in SI.<br><i>Existence depends on:</i><br><i>RX_HSGEAR_Capability</i>                                                                                                      | Both | Int  | B[7:6]              | SYNC_range<br>FINE = 0,<br>COARSE = 1                                                                           |
|                                                 |             |                                                                                                                                                                                                                             |      |      | B[5:0]              | SYNC_length <sup>3</sup><br>1 to 15 for FINE,<br>0 to 15 for COARSE                                             |
| RX_HS_G4_PREPARE_LENGTH_Capability <sup>4</sup> | 0x9C        | HS-G4 PREPARE length multiplier for M-RX.<br><i>Existence depends on:</i><br><i>RX_HSGEAR_Capability</i>                                                                                                                    | Both | Int  | B[3:0] <sup>1</sup> | 0 to 15                                                                                                         |
| RX_HS_Equalizer_Setting_Capability              | 0x9D        | Support for receiver path channel equalization for HS-MODE.                                                                                                                                                                 | Both | Int  | B[0]                | B0 = 0: No equalization<br>B0 = 1: Yes equalization                                                             |
| RX_HS_G4_ADAPT_REFRESH_Capability               | 0x9E        | REFRESH is used for RX adaptation refresh.<br>High Speed GEAR 4 adaptation pattern length of M-RX.                                                                                                                          | Both | Int  | B[7]                | ADAPT_range<br>FINE = 0,<br>COARSE = 1                                                                          |

| Attribute Name                                          | AttributeID | Description                                                                                                                                                                     | FSM  | Type | Bits                | Range                                                                 |
|---------------------------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|---------------------|-----------------------------------------------------------------------|
|                                                         |             | <i>Existence depends on:</i><br><i>RX_HS_Equalizer_Setting_Capability</i>                                                                                                       |      |      | B[6:0]              | ADAPT_length <sup>6</sup><br>0 to 127 for FINE,<br>0 to 17 for COARSE |
| RX_HS_G4_ADAPT_INITIAL_Capability                       | 0x9F        | INITIAL is used for initial RX adaptation.<br>High Speed GEAR 4 adaptation pattern length of M-RX.<br><i>Existence depends on:</i><br><i>RX_HS_Equalizer_Setting_Capability</i> | Both | Int  | B[7]                | ADAPT_range<br>FINE = 0,<br>COARSE = 1                                |
|                                                         |             |                                                                                                                                                                                 |      |      | B[6:0]              | ADAPT_length <sup>6</sup><br>0 to 127 for FINE,<br>0 to 17 for COARSE |
| RX_HS_G5_SYNC_LENGTH_Capability                         | 0x71        | High Speed GEAR 5 Synchronization pattern length in SI.<br><i>Existence depends on:</i><br><i>RX_HSGEAR_Capability</i>                                                          | Both | Int  | B[7:6]              | SYNC_range<br>FINE = 0,<br>COARSE = 1                                 |
|                                                         |             |                                                                                                                                                                                 |      |      | B[5:0]              | SYNC_length <sup>3</sup><br>1 to 15 for FINE,<br>0 to 15 for COARSE   |
| RX_HS_G5_PREPARE_LENGTH_Capability <sup>4</sup>         | 0x72        | HS-G5 PREPARE length multiplier for M-RX.<br><i>Existence depends on:</i><br><i>RX_HSGEAR_Capability</i>                                                                        | Both | Int  | B[3:0] <sup>1</sup> | 0 to 15                                                               |
| RX_HS_G5_ADAPT_REFRESH_Capability                       | 0x73        | REFRESH is used for RX adaptation refresh.<br>High Speed GEAR 5 adaptation pattern length of M-RX.<br><i>Existence depends on:</i><br><i>RX_HS_Equalizer_Setting_Capability</i> | Both | Int  | B[7]                | ADAPT_range<br>FINE = 0,<br>COARSE = 1                                |
|                                                         |             |                                                                                                                                                                                 |      |      | B[6:0]              | ADAPT_length <sup>6</sup><br>0 to 127 for FINE,<br>0 to 17 for COARSE |
| RX_HS_G5_ADAPT_INITIAL_Capability                       | 0x74        | INITIAL is used for initial RX adaptation.<br>High Speed GEAR 5 adaptation pattern length of M-RX.<br><i>Existence depends on:</i><br><i>RX_HS_Equalizer_Setting_Capability</i> | Both | Int  | B[7]                | ADAPT_range<br>FINE = 0,<br>COARSE = 1                                |
|                                                         |             |                                                                                                                                                                                 |      |      | B[6:0]              | ADAPT_length <sup>6</sup><br>0 to 127 for FINE,<br>0 to 17 for COARSE |
| RX_Extended_Min_SAVE_Config_Time_Granularity_Capability | 0x75        | Support for<br>RX_Extended_Min_SAVE_Config_Time_Capability.                                                                                                                     | Both | Int  | B[0]                | 0: Not Supported<br>1: Supported                                      |

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| Attribute Name                              | AttributeID | Description                                                                                                                                                                                               | FSM  | Type | Bits   | Range                |
|---------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|--------|----------------------|
| RX_Extended_Min_SAVE_Config_Time_Capability | 0x76        | Specifies minimum reconfiguration time (in 1 $\mu$ s steps). This applies only to SLEEP and STALL states.<br><i>Existence depends on:</i><br>RX_Extended_Min_SAVE_Config_Time_Gr<br>anularity_Capability. | Both | Int  | B[6:0] | 1 to 10 (10 $\mu$ s) |

**Note:**

1. Recommended bit assignment.
2. There is a potential timing mismatch between the setting of this capability, implementation width of RMMI when the prior gear is PWM-G1. As neither timing nor width is defined at the RMMI interface, if such a hazard exists for an M-PHY implementation, then the M-PHY IP has to advertise this in its data sheet. This allows the implementer to ensure the protocol and the selected M-PHY IP are in a good match and allows sufficient time for attributes to be updated. Since the sensitivity is only in PWM-G1, the protocol has to take the current GEAR into account while computing the minimum time between BURSTS after a configuration change.
3. Actual SYNC length is calculated using the formula for  $T_{\text{SYNC}}$  in **Table 8**.
4. Actual HS PREPARE length is calculated using the formula for  $T_{\text{HS\_PREPARE}}$  in **Table 8** with RX\_HXGEAR.
5. Actual PWM PREPARE length is calculated using the formula for  $T_{\text{PWM\_PREPARE}}$  in **Table 8** with RX\_PWMGEAR.
6. Actual ADAPT length is calculated using the formula for  $T_{\text{ADAPT}}$  in **Table 8**.
7. PWM-G0 and PWM-G2 to PWM-G7 are no longer supported in M-PHY v5.0. Associated attributes are obsolete.

Table 43 M-RX Configuration Attributes

| Attribute Name            | AttributeID | Description                                                                                                                                                                                     | FSM    | Type | Bits                | Range                                                             |
|---------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|---------------------|-------------------------------------------------------------------|
| RX_MODE                   | 0xA1        | Operating mode.<br><i>Req'd Value:</i> LS_MODE and HS_MODE<br><i>Reset Value:</i> Depends on ResetMode<br>(see <b>Section 7.3.7.1</b> )                                                         | Both   | Enum | B[1:0] <sup>1</sup> | LS_MODE = 1,<br>HS_MODE = 2                                       |
| RX_HSRATE_Series          | 0xA2        | HS mode RATE series value.<br><i>Req'd Values:</i> A and B<br><i>Reset Value:</i> A                                                                                                             | Both   | Enum | B[1:0] <sup>1</sup> | A = 1,<br>B = 2                                                   |
| RX_HSGEAR                 | 0xA3        | Current HS-GEAR.<br><i>Value depends on:</i><br><i>RX_HSGEAR_Capability</i><br><i>Req'd Value:</i> HS_G1<br><i>Reset Value:</i> HS_G1                                                           | Both   | Enum | B[2:0] <sup>1</sup> | HS_G1 = 1,<br>HS_G2 = 2,<br>HS_G3 = 3,<br>HS_G4 = 4,<br>HS_G5 = 5 |
| RX_PWMGEAR                | 0xA4        | Current PWM-GEAR.<br><i>Req'd Value:</i> PWM_G1<br><i>Reset Value:</i> PWM_G1                                                                                                                   | TYPE-I | Enum | B[2:0] <sup>1</sup> | Reserved: 0,<br>PWM_G1 = 1,<br>Reserved: 2 to 7 <sup>4</sup>      |
| RX_LS_Terminated_Enable   | 0xA5        | Enable resistive termination of M-RX in LS-MODE.<br><i>Existence depends on:</i><br><i>RX_LS_Terminated_Capability</i><br><i>Req'd Value:</i> OFF<br><i>Reset Value:</i> OFF                    | TYPE-I | Bool | B[0] <sup>1</sup>   | OFF = 0,<br>ON = 1                                                |
| RX_HS_Unterminated_Enable | 0xA6        | Enable disconnection of resistive termination of M-RX in HS-MODE.<br><i>Existence depends on:</i><br><i>RX_HS_Unterminated_Capability</i><br><i>Req'd Value:</i> OFF<br><i>Reset Value:</i> OFF | Both   | Bool | B[0] <sup>1</sup>   | OFF = 0                                                           |



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| Attribute Name                | AttributeID | Description                                                                                                                                                                      | FSM                            | Type | Bits              | Range                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RX_Enter_HIBERN8              | 0xA7        | M-RX entry to HIBERN8 state control.<br><i>Req'd Values:</i> YES, NO<br><i>Reset Value:</i><br>YES for Local RESET,<br>NO for LINE-RESET                                         | TYPE-I<br>TYPE-II <sup>2</sup> | Bool | B[0] <sup>1</sup> | NO = 0: Protocol Layer shall not set the value of this attribute to "NO".<br>When the M-RX is in HIBERN8 state, upon squelch detection the M-RX exits HIBERN8 state (to SLEEP or STALL state) and resets this attribute value to NO.<br>YES = 1: Can be set by the Protocol.<br>The M-RX enters from SLEEP or STALL state to HIBERN8 state, if it is not already in HIBERN8 state. |
| RX_BYPASS_8B10B_Enable        | 0xA8        | Bypass 8b10b Decoding at the M-RX.<br><i>Req'd Value:</i> FALSE<br><i>Reset Value:</i> FALSE                                                                                     | Both                           | Bool | B[0] <sup>1</sup> | FALSE = 0,<br>TRUE = 1                                                                                                                                                                                                                                                                                                                                                             |
| RX_Termination_Force_Enable   | 0xA9        | Force connection of differential termination resistance, $R_{DIF\_RX}$ , to enabled state, for RX S-Parameter test purposes.<br><i>Req'd Value:</i> NO<br><i>Reset Value:</i> NO | Both                           | Bool | B[0] <sup>1</sup> | NO = 0,<br>YES = 1                                                                                                                                                                                                                                                                                                                                                                 |
| RX_ADAPT_Control <sup>3</sup> | 0xAA        | M-RX ADAPT sub-state control and ADAPT_type.<br><i>Req'd Values:</i><br>ADAPT Control: ADAPT, SYNC<br>ADAPT Type: REFRESH, INITIAL<br><i>Reset Value:</i> INITIAL, SYNC          | Both                           | Int  | B[1]              | ADAPT_type<br>INITIAL = 0,<br>REFRESH = 1                                                                                                                                                                                                                                                                                                                                          |
|                               |             |                                                                                                                                                                                  |                                |      | B[0]              | ADAPT_control<br>ADAPT = 0,<br>SYNC = 1                                                                                                                                                                                                                                                                                                                                            |
| RX_RECEIVER_POLARITY          | 0xAB        | M-RX receiver polarity.<br><i>Req'd Values:</i> NORMAL, INVERTED<br><i>Reset Value:</i> NORMAL for local RESET.<br>LINE-RESET shall not reset the value of this attribute.       | Both                           | Enum | B[0]              | NORMAL = 0,<br>INVERTED = 1                                                                                                                                                                                                                                                                                                                                                        |

**Note:**

1. Recommended bit assignment.
2. TYPE-II with embedded HIBERN8 exit control. See **Section 4.7.1.3**.
3. The local module sets to SYNC after completion of the ADAPT sub-state.
4. PWM-G0 and PWM-G2 to PWM-G7 are no longer supported in M-PHY v5.0. Associated attributes are obsolete.

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**Table 44 M-RX Status Attributes**

| Attribute Name | AttributeID | Description                           | FSM  | Type | Bits                | Range                                                                                                                                        |
|----------------|-------------|---------------------------------------|------|------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| RX_FSM_State   | 0xC1        | To read out the current state of M-RX | Both | Enum | B[3:0] <sup>1</sup> | DISABLED = 0,<br>HIBERN8 = 1,<br>SLEEP = 2,<br>STALL = 3,<br>LS-BURST = 4,<br>HS-BURST = 5,<br>Reserved = 6 <sup>2</sup> ,<br>LINE-RESET = 7 |

**Note:**

1. Recommended bit assignment.
2. Value “6” was used for LINE-CFG up to M-PHY v4.1. This functionality is no longer supported in M-PHY v5.0.

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**Table 45 EYEMON Capability Attributes**

| Attribute Name                                       | AttributeID | Description                                                                                                                                                                         | FSM  | Type | Bits   | Range                                                 |
|------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|--------|-------------------------------------------------------|
| RX_EYEMON_Capability                                 | 0xF1        | Specifies support for eye monitor function.                                                                                                                                         | Both | Int  | B[0]   | FALSE=0,<br>TRUE=1                                    |
| RX_EYEMON_Timing_MAX_Steps_Capability <sup>1</sup>   | 0xF2        | Timing step maximum from center sampling point.<br>Monotonicity and same step size toward both positive and negative side.<br><i>Existence depends on:</i><br>RX_EYEMON_Capability  | Both | Int  | B[5:0] | 3 to 63                                               |
| RX_EYEMON_Timing_MAX_Offset_Capability <sup>2</sup>  | 0xF3        | Timing MAX offset.<br>Minimum $\pm 0.15$ UI, value 15 to<br>Maximum $\pm 0.50$ UI, value 50.<br><i>Existence depends on:</i><br>RX_EYEMON_Capability                                | Both | Int  | B[5:0] | 0 (No support) or<br>15 to 50<br>(0.15 UI to 0.50 UI) |
| RX_EYEMON_Voltage_MAX_Steps_Capability <sup>3</sup>  | 0xF4        | Voltage step maximum from center sampling point.<br>Monotonicity and same step size toward both positive and negative side.<br><i>Existence depends on:</i><br>RX_EYEMON_Capability | Both | Int  | B[5:0] | 6 to 63                                               |
| RX_EYEMON_Voltage_MAX_Offset_Capability <sup>4</sup> | 0xF5        | Voltage MAX offset.<br>Minimum $\pm 30$ mV, value 3 to<br>Maximum $\pm 310$ mV, value 31.<br><i>Existence depends on:</i><br>RX_EYEMON_Capability                                   | Both | Int  | B[4:0] | 0 (No support) or<br>3 to 31<br>(30 mV to 310 mV)     |

**Note:**

1. Corresponding to  $N_{MAX\_TIMING\_STEPS}$  in **Table 21**.
2. Corresponding to  $T_{MAX\_OFFSET}$  in **Table 21**.
3. Corresponding to  $N_{MAX\_VOLTAGE\_STEPS}$  in **Table 21**.
4. Corresponding to  $V_{MAX\_OFFSET}$  in **Table 21**.

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**Table 46 EYEMON Configuration Attributes**

| Attribute Name              | AttributeID | Description                                                                                                                                                                                                                                                                                                       | FSM  | Type | Bits   | Range                                  |
|-----------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|--------|----------------------------------------|
| RX_EYEMON_Enable            | 0xF6        | Eye Monitor Test control register.<br><i>Existence depends on:</i><br>RX_EYEMON_Capability<br><i>Req'd Value:</i> DISABLE, ENABLE<br><i>Reset Value:</i> DISABLE                                                                                                                                                  | Both | Bool | B[0]   | DISABLE=0,<br>ENABLE=1                 |
| RX_EYEMON_Timing_Steps      | 0xF7        | Timing offset step setting.<br><i>Existence depends on:</i><br>RX_EYEMON_Capability<br><i>Value depends on:</i><br>RX_EYEMON_Timing_MAX_Steps_Capability<br>RX_EYEMON_Timing_MAX_Offset_Capability<br><i>Req'd Value:</i> 0 to RX_EYEMON_Timing_MAX_Steps_Capability<br><i>Reset Value:</i> Right Direction, 0    | Both | Int  | B[6]   | Right Direction=0,<br>Left Direction=1 |
|                             |             |                                                                                                                                                                                                                                                                                                                   |      |      | B[5:0] | 0 to 63                                |
| RX_EYEMON_Voltage_Steps     | 0xF8        | Voltage offset step setting.<br><i>Existence depends on:</i><br>RX_EYEMON_Capability<br><i>Value depends on:</i><br>RX_EYEMON_Voltage_MAX_Steps_Capability<br>RX_EYEMON_Voltage_MAX_Offset_Capability<br><i>Req'd Value:</i> 0 to RX_EYEMON_Voltage_MAX_Steps_Capability<br><i>Reset Value:</i> Plus Direction, 0 | Both | Int  | B[6]   | Plus Direction=0,<br>Minus Direction=1 |
|                             |             |                                                                                                                                                                                                                                                                                                                   |      |      | B[5:0] | 0 to 63                                |
| RX_EYEMON_Target_Test_Count | 0xF9        | Target test count: Value = $3 \cdot \log_2$ [Actual Eye Monitor Test Bit Count]                                                                                                                                                                                                                                   | Both | Int  | B[6:0] | 0 to 127                               |

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**Table 47 EYEMON Status Attributes**

| Attribute Name                       | AttributeID | Description                                                                                                                                                                                                                                                                           | FSM  | Type | Bits   | Range               |
|--------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|--------|---------------------|
| RX_EYEMON_Testing_Count <sup>1</sup> | 0xFA        | To read out the tested bit count of Eye Monitor test.<br>Value= $3 \times \log_2$ [Actual Eye Monitor Test Bit Count]<br>Maximum 127 = $2^{42.333...} = 5.54 \times 10^{12}$ bit<br>Stopped by RX_EYEMON_Start=0 (STOP), or by reaching or exceeding the RX_EYEMON_Target_Test_Count. | Both | Int  | B[6:0] | 0 to 127            |
| RX_EYEMON_Error_Count <sup>1</sup>   | 0xFB        | To read out the current error bit count during Eye Monitor test.                                                                                                                                                                                                                      | Both | Int  | B[5:0] | 0 to 63             |
| RX_EYEMON_Start <sup>2</sup>         | 0xFC        | Eye Monitor Test start/activation flag.                                                                                                                                                                                                                                               | Both | Bool | B[0]   | STOP=0,<br>ACTIVE=1 |

**Note:**

1. Cleared at the start of a new EYEMON measurement, or LINE-RESET of main FSM, or system reset (see **Section 4.11.2**). If the value is saturated during measurement, then it is stopped at the maximum value.
2. Cleared by receiver upon error count saturation, or when testing count reaches target value, or when RX\_EYEMON\_Enable is disabled (see **Section 4.11.2**).

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## Annex A Signaling Interface Description (normative)

The signaling interface described in this annex, the Reference M-PHY MODULE Interface (RMMI), is optional. However, if a MODULE includes the RMMI it shall implement it as described in this annex.

The RMMI signaling interface for a MODULE (M-TX or M-RX) consists of two independent interfaces for control service primitives (M-TX-CTRL SAP and M-RX-CTRL SAP) and for data transfer service primitives (M-TX-DATA SAP and M-RX-DATA SAP). An M-PORT with multiple M-TXs or M-RXs uses a set of signals defined for M-TX or M-RX for each MODULE. To keep the same structure used for SAP definitions, the signaling interface of a MODULE is divided into DATA and CTRL signaling interfaces.

A shadow memory bank inside the MODULE implements the INLINE-CR registry as defined in Section 4.8.1, and a separate effective configuration bank implements the INLINE-SET and OFFLINE-SET registries. Both the shadow memory and the effective configuration banks are written sequentially. However, the entire contents of the shadow memory bank can be uploaded to the effective configuration bank in a single, Protocol Layer-requested operation.

Due to the high data rates supported in M-PHY implementations, the width of the data buses conveying data to and from the Physical Layer can be increased, and different parallelization options are provided.

Finally, testability extensions to the CTRL signal interface are also included in this specification. However, the definition of the internal M-RX and M-TX structures controlled by these extensions is out of scope for this document.

**Section A.2** and **Section A.3** define the signals used in the signaling interface of an M-TX, and M-RX, respectively. While the CTRL signaling interfaces for M-TXs and M-RXs cannot be identical, this annex provides a common signal definition for M-TX-CTRL SAP and M-RX-CTRL SAP to the furthest extent possible. M-TX-DATA SAP and M-RX-DATA SAP signaling interfaces are, by their nature, substantially different.

### A.1 One Hot Coding of Control Symbols

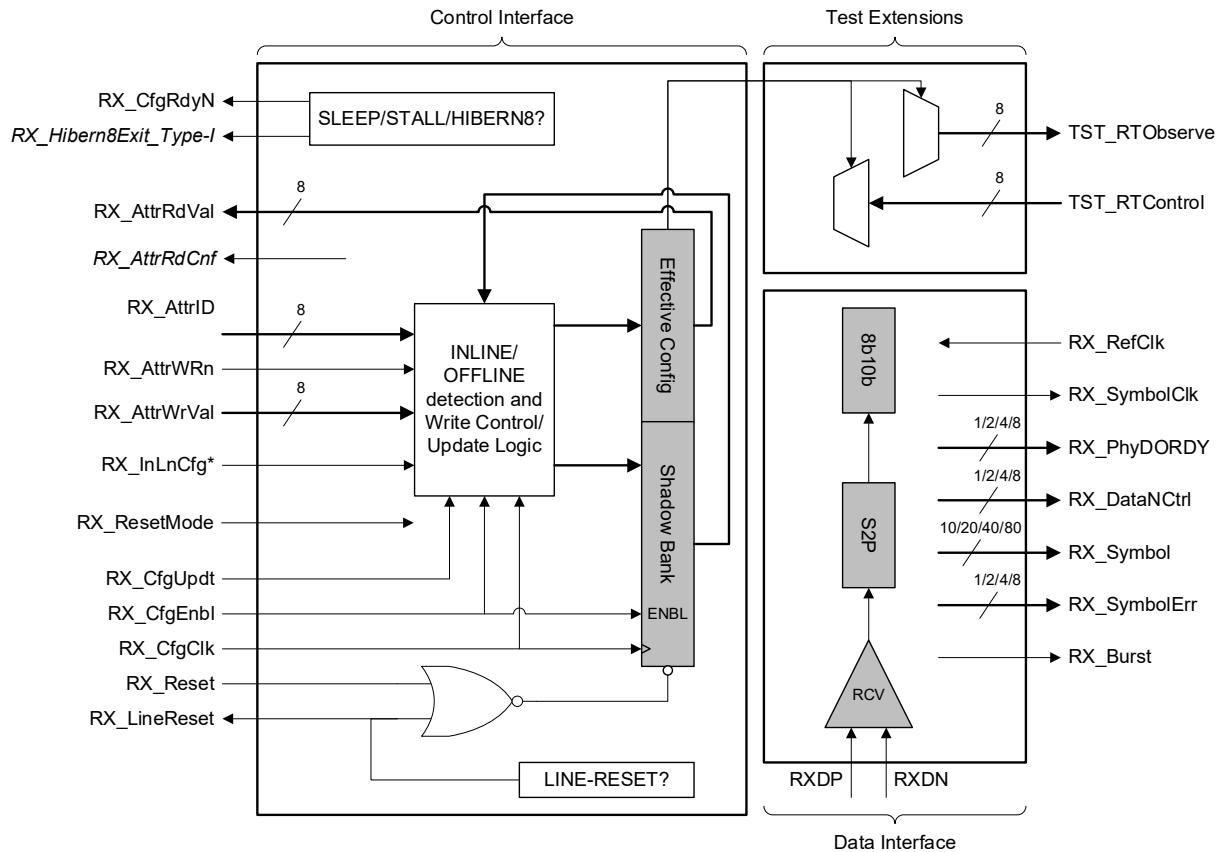
**Table 48** defines the One-Hot coding of control symbols.

**Table 48 One-Hot Coding of Control Symbols**

| One-Hot Code | Type of Control Symbol at TX | Type of Control Symbol at RX |
|--------------|------------------------------|------------------------------|
| 0000 0000    | Reserved                     | Reserved                     |
| 0000 0001    | MARKER0                      | MARKER0                      |
| 0000 0010    | MARKER1                      | MARKER1                      |
| 0000 0100    | MARKER2                      | MARKER2                      |
| 0000 1000    | MARKER3                      | MARKER3                      |
| 0001 0000    | MARKER4                      | MARKER4                      |
| 0010 0000    | MARKER5                      | MARKER5                      |
| 0100 0000    | MARKER6                      | MARKER6                      |
| 1000 0000    | FILLER                       | FILLER                       |

## A.2 The M-RX Signaling Interface

A schematic overview of the M-RX signaling interface is shown in **Figure 55**.



RX\_InLnCfg\* - This signal is optional. Implementations may choose appropriate fixed value for backward compatibility.

**Figure 55 M-RX Signal Interfaces Diagram**

### A.2.1 M-RX Signal Description

In **Table 49** through **Table 51**, entries in the “Direction” column specifies the direction of each signal from the perspective of the M-RX. An input signal (abbreviated as “I”) is driven by the Protocol Layer. An output signal (abbreviated as “O”) is driven by the M-RX.

The “Detection Type” column indicates the relevant condition for a given signal. A Detection Type of “Level” means the relevant information is either a high or low level on the signal. A Detection Type of “Transition” means a change from high-to-low or low-to-high causes the described action. A Detection Type of “Clock” indicates the signal is used to synchronize other signals on the interface. A Detection Type of “Asynch” means the signal changes state asynchronously to the relevant clock signal.



**Table 49 M-RX-CTRL Interface Signals**

| Signal Name  | Direction | Detection Type | Width | Signal Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------|-----------|----------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RX_CfgClk    | I         | Clock          | 1     | Control Interface Clock.<br>All M-RX-CTRL interface signals, with the exception of RX_Reset and <i>RX_Hibern8Exit_Type-I</i> , are synchronous with this signal.<br>The exact frequency of RX_CfgClk is implementation specific. Choice of frequency should consider squelch detection, adequate measurement of $T_{\text{LINE-RESET}}$ and minimizing interface access latencies.<br>RX_CfgClk may change frequency depending on state, but is expected to be available in all M-RX states except DISABLED and UNPOWERED. |
| RX_Reset     | I         | Asynch         | 1     | RX_Reset is the active-high asynchronous reset for all logic inside the M-RX. RX_Reset implements the local RESET function as defined in <b>Section 4.7</b> .<br>The Protocol Layer, or other source, shall set RX_Reset to “1” for at least 100 ns.                                                                                                                                                                                                                                                                       |
| RX_ResetMode | I         | Level          | 1     | RX_ResetMode selects default operating mode value.<br>0: LS_MODE<br>1: HS_MODE<br>This signal controls RX_MODE attribute reset value (includes LINE-RESET)<br>RX_ResetMode implements the local RESET parameter ResetMode as defined in <b>Section 4.7</b> .<br>RX_ResetMode shall be changed only when M-RX is in DISABLED state.<br>RX_ResetMode shall have the same value as TX_ResetMode.                                                                                                                              |
| RX_LineReset | O         | Transition     | 1     | RX_LineReset indicates the status of the LINE-RESET action in the M-RX.<br>M-RX shall set RX_LineReset to “1” when LINE-RESET is detected.<br>M-RX shall set RX_LineReset to “0” once it has transitioned to the LINE-RESET exit state (see <b>Section 4.7.4.1</b> ).                                                                                                                                                                                                                                                      |
| RX_AttrID    | I         | Level          | 8     | RX_AttrID carries the AttributeID of M-RX Configuration attributes for read or write operations, or M-RX Capability attribute for read operation.                                                                                                                                                                                                                                                                                                                                                                          |
| RX_AttrRdVal | O         | Level          | 8     | RX_AttrRdVal carries the attribute value read from an M-RX-MIB attribute specified by RX_AttrID.<br>The M-RX-MIB attribute value should be held on this bus until a subsequent read command is issued by the protocol.<br>When RX_AttrRdCnf is not implemented, the M-RX shall provide the specified attribute value within one-half of the RX_CfgClk period.                                                                                                                                                              |

| Signal Name  | Direction | Detection Type | Width | Signal Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------|-----------|----------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RX_AttrRdCnf | O         | Level          | 1     | RX_AttrRdCnf confirms that RX_AttrRdVal carries a new value.<br>M-RX shall set RX_AttrRdCnf to "1" for a single RX_CfgClk cycle to inform the Protocol Layer that RX_AttrRdVal carries a recently requested attribute read value.<br>RX_AttrRdCnf is an optional signal.                                                                                                                                                                                                                                                                                                                           |
| RX_AttrWrVal | I         | Level          | 8     | RX_AttrWrVal carries the attribute value to write to an M-RX-MIB attribute specified by RX_AttrID.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| RX_AttrWRn   | I         | Level          | 1     | RX_AttrWRn specifies the operation, read or write, to perform on an M-RX-MIB attribute.<br>The Protocol Layer shall set RX_AttrWRn to "0" to indicate a read operation.<br>The Protocol Layer shall set RX_AttrWRn to "1" to indicate a write operation.                                                                                                                                                                                                                                                                                                                                           |
| RX_CfgEnbl   | I         | Level          | 1     | Config Enable<br>The Protocol Layer shall set RX_CfgEnbl to "1" for a single RX_CfgClk cycle to perform an attribute read, or write, operation.<br>The Protocol Layer shall set RX_CfgEnbl, RX_AttrID, RX_AttrWRn and RX_AttrWrVal in the same RX_CfgClk cycle.                                                                                                                                                                                                                                                                                                                                    |
| RX_CfgUpdt   | I         | Transition     | 1     | RX_CfgUpdt transfers the contents of the INLINE-CR registry to the effective configuration bank during a SAVE state. RX_CfgUpdt indicates the completion of the required configuration settings to the MODULE for effectuating configuration change requests atomically.<br>The Protocol Layer shall set RX_CfgUpdt to "1" for a single RX_CfgClk cycle to trigger the upload of the entire M-RX shadow memory contents to the effective configuration bank. The Protocol Layer shall move the MODULE into a SAVE state, if not already in a SAVE state, before the new settings become effective. |

| Signal Name                  | Direction | Detection Type     | Width | Signal Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------|-----------|--------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RX_CfgRdyN                   | O         | Level              | 1     | <p>RX_CfgRdyN indicates the M-RX cannot process a register write command to its effective configuration bank.</p> <p>The M-RX shall set this signal to “1” in the same RX_CfgClk cycle that triggers its internal FSM exit from SLEEP, STALL, or HIBERN8 state to any other state.</p> <p>The M-RX may also set this signal to “1” while it is processing a Protocol-issued change to its effective configuration bank.</p> <p>The M-RX shall set this signal to “0” when its internal FSM is in SLEEP, STALL, or HIBERN8 state and the MODULE is ready to accept a register write command to any register of its effective configuration bank.</p> <p>For a RX_Reset (local RESET command), the M-RX shall set RX_CfgRdyN to “1” asynchronously.</p> <p>If the Protocol Layer issues write commands to the M-RX effective configuration bank (including RX_CfgUpdt) while RX_CfgRdyN is set to “0”, the M-RX shall process those commands immediately. If the Protocol Layer issues write commands to the M-RX effective configuration bank while RX_CfgRdyN is set to “1”, the specific M-RX behavior is dependent on the command itself addressing an OFFLINE-SET or INLINE-SET attribute. The M-RX shall execute a write command to an OFFLINE-SET Attribute in the effective configuration bank. The M-RX shall redirect a write command to an INLINE-SET attribute in the effective configuration bank to the associated shadow register.</p> <p>The M-RX shall not ignore a write command or Rx_CfgUpdt request from the Protocol Layer except in UNPOWERED and DISABLED states, or when local RESET is asserted.</p> <p>The M-RX shall respond to read commands from the Protocol Layer regardless of the value of RX_CfgRdyN.</p> <p>The M-RX shall process register write commands to its shadow memory bank regardless of the value of RX_CfgRdyN.</p> |
| <i>RX_Hibern8Exit_Type-I</i> | O         | Transition, Asynch | 1     | <p><i>RX_Hibern8Exit_Type-I</i> indicates the M-RX is exiting HIBERN8. The M-RX sets <i>RX_Hibern8Exit_Type-I</i> to “1” when it detects a HIBERN8 exit condition on the LINE (see <b>Section 4.7.1.3</b> and <b>Figure 11</b>). The M-RX sets <i>RX_Hibern8Exit_Type-I</i> to “0” when the M-RX is in either HIBERN8 or DISABLED state. A Type-I implementation shall include this signal.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

2300

**Table 50 M-RX-DATA Interface Signals**

| Signal Name  | Direction | Detection Type | Width            | Signal Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------|-----------|----------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RX_RefClk    | I         | Clock          | 1                | Reference Clock.<br>RX_RefClk may not be accessible in the M-RX-DATA interface for an M-PHY implementation that comprises an integrated clock multiplier.<br>RX_RefClk shall have no specific phase relationship requirement to any signal in the M-RX-DATA interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| RX_SymbolClk | O         | Clock          | 1                | Symbol Clock.<br>All M-RX-DATA interface signals are synchronous with this signal.<br>The M-RX may disable RX_SymbolClk generation when the M-RX is not in PWM-BURST, SYS-BURST, or HS-BURST states.<br>The M-RX shall provide the minimum number of cycles to transfer all M-RX data to the Protocol Layer. At the end of BURST, the M-RX shall provide a minimum of two additional clock cycles beyond the de-assertion of RX_Burst.<br>In HS-MODE and SYS-MODE, RX_SymbolClk shall have a period of 10 UI for a 10-bit RX_Symbol bus, 20 UI for a 20-bit RX_Symbol bus, 40 UI for a 40-bit RX_Symbol bus, or 80 UI for an 80-bit RX_Symbol bus.<br>In PWM-MODE, RX_SymbolClk shall have a period of 10 $T_{PWM-RX}$ for a 10-bit RX_Symbol bus, 20 $T_{PWM-RX}$ for a 20-bit RX_Symbol bus, 40 $T_{PWM-RX}$ for a 40-bit RX_Symbol bus or 80 $T_{PWM-RX}$ for an 80-bit RX_Symbol bus.<br>The behavior of RX_SymbolClk must be glitch-free even when this signal is being enabled or disabled. The M-RX shall not provide a RX_SymbolClk “1” or “0” pulse with a duration less than one-quarter of the nominal RX_SymbolClk period. |
| RX_Symbol    | O         | Level          | 10, 20, 40 or 80 | RX_Symbol is used for BURST data transfer to the Protocol Layer. The contents of this bus depend on the interface width (10, 20, 40 or 80 bits, corresponding to 1, 2, 4 or 8 parallel symbols, respectively), and also on whether or not the 10b8b decoding function is bypassed.<br>When the 10b8b decoding function is disabled, RX_Symbol carries the raw data as received on the LINES, parallelized according to the implemented width. The LSb of RX_Symbol shall correspond to the earliest received bit.<br>When the 10b8b decoding function is enabled, only the 8, 16, 32 or 64 LSbs of RX_Symbol are used to carry the decoded DATA or control symbol. The M-RX shall set the remaining MSbs to “0”.<br>Control symbols shall be decoded as listed in <b>Table 48</b> .                                                                                                                                                                                                                                                                                                                                                    |

| Signal Name     | Direction                      | Detection Type | Width        | Signal Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                 |                                |   |                       |   |                          |   |                           |   |                           |   |                           |   |                           |   |                           |   |                           |
|-----------------|--------------------------------|----------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------------------------------|---|-----------------------|---|--------------------------|---|---------------------------|---|---------------------------|---|---------------------------|---|---------------------------|---|---------------------------|---|---------------------------|
| RX_PhyDORDY     | O                              | Level          | 1, 2, 4 or 8 | <p>PHY Data Output Ready.</p> <p>RX_PhyDORDY indicates data is available in the corresponding RX_Symbol bus range. The width of RX_PhyDORDY is one, two, four or eight bits depending on the RX_Symbol bus width of 10, 20, 40 or 80 bits, respectively.</p> <p>Each bit in RX_PhyDORDY corresponds to a 10b8b symbol in RX_Symbol bus.</p> <table><tr><td>RX_PhyDORDY bit</td><td>RX_Symbol bits (10b8b enabled)</td></tr><tr><td>0</td><td>bits[9:0] (bits[7:0])</td></tr><tr><td>1</td><td>bits[19:10] (bits[15:8])</td></tr><tr><td>2</td><td>bits[29:20] (bits[23:16])</td></tr><tr><td>3</td><td>bits[39:30] (bits[31:24])</td></tr><tr><td>4</td><td>bits[49:40] (bits[39:32])</td></tr><tr><td>5</td><td>bits[59:50] (bits[47:40])</td></tr><tr><td>6</td><td>bits[69:60] (bits[55:48])</td></tr><tr><td>7</td><td>bits[79:70] (bits[63:56])</td></tr></table> <p>The M-RX shall set each bit of RX_PhyDORDY to “1” for every RX_SymbolClk cycle that the corresponding RX_Symbol bus range contains new data.</p> <p>The M-RX shall set each bit of RX_PhyDORDY bit to “0” for every RX_SymbolClk cycle that the corresponding RX_Symbol bus range does not contain new data.</p> <p>The Protocol Layer shall always be ready to consume the data from the M-RX.</p> <p>During ADAPT, the RX_Phy_DORDY signal shall be set to zero.</p> | RX_PhyDORDY bit | RX_Symbol bits (10b8b enabled) | 0 | bits[9:0] (bits[7:0]) | 1 | bits[19:10] (bits[15:8]) | 2 | bits[29:20] (bits[23:16]) | 3 | bits[39:30] (bits[31:24]) | 4 | bits[49:40] (bits[39:32]) | 5 | bits[59:50] (bits[47:40]) | 6 | bits[69:60] (bits[55:48]) | 7 | bits[79:70] (bits[63:56]) |
| RX_PhyDORDY bit | RX_Symbol bits (10b8b enabled) |                |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                 |                                |   |                       |   |                          |   |                           |   |                           |   |                           |   |                           |   |                           |   |                           |
| 0               | bits[9:0] (bits[7:0])          |                |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                 |                                |   |                       |   |                          |   |                           |   |                           |   |                           |   |                           |   |                           |   |                           |
| 1               | bits[19:10] (bits[15:8])       |                |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                 |                                |   |                       |   |                          |   |                           |   |                           |   |                           |   |                           |   |                           |   |                           |
| 2               | bits[29:20] (bits[23:16])      |                |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                 |                                |   |                       |   |                          |   |                           |   |                           |   |                           |   |                           |   |                           |   |                           |
| 3               | bits[39:30] (bits[31:24])      |                |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                 |                                |   |                       |   |                          |   |                           |   |                           |   |                           |   |                           |   |                           |   |                           |
| 4               | bits[49:40] (bits[39:32])      |                |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                 |                                |   |                       |   |                          |   |                           |   |                           |   |                           |   |                           |   |                           |   |                           |
| 5               | bits[59:50] (bits[47:40])      |                |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                 |                                |   |                       |   |                          |   |                           |   |                           |   |                           |   |                           |   |                           |   |                           |
| 6               | bits[69:60] (bits[55:48])      |                |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                 |                                |   |                       |   |                          |   |                           |   |                           |   |                           |   |                           |   |                           |   |                           |
| 7               | bits[79:70] (bits[63:56])      |                |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                 |                                |   |                       |   |                          |   |                           |   |                           |   |                           |   |                           |   |                           |   |                           |
| RX_DataNCtrl    | O                              | Level          | 1, 2, 4 or 8 | <p>RX_DataNCtrl indicates the type of symbol on the indicated range of RX_Symbol.</p> <p>The width of RX_DataNCtrl is one, two, four or eight bits depending on the RX_Symbol bus width of 10, 20, 40 or 80 bits, respectively.</p> <p>RX_DataNCtrl are mapped the same as RX_PhyDORDY.</p> <p>The M-RX shall set the corresponding bit of RX_DataNCtrl to “0” when the related RX_Symbol bus range carries a data symbol.</p> <p>The M-RX shall set the corresponding bit of RX_DataNCtrl to “1” when the related RX_Symbol bus range carries a control symbol or a reserved symbol.</p> <p>The M-RX shall set all bits of RX_DataNCtrl to “0” when 10b8b decoding is bypassed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                 |                                |   |                       |   |                          |   |                           |   |                           |   |                           |   |                           |   |                           |   |                           |

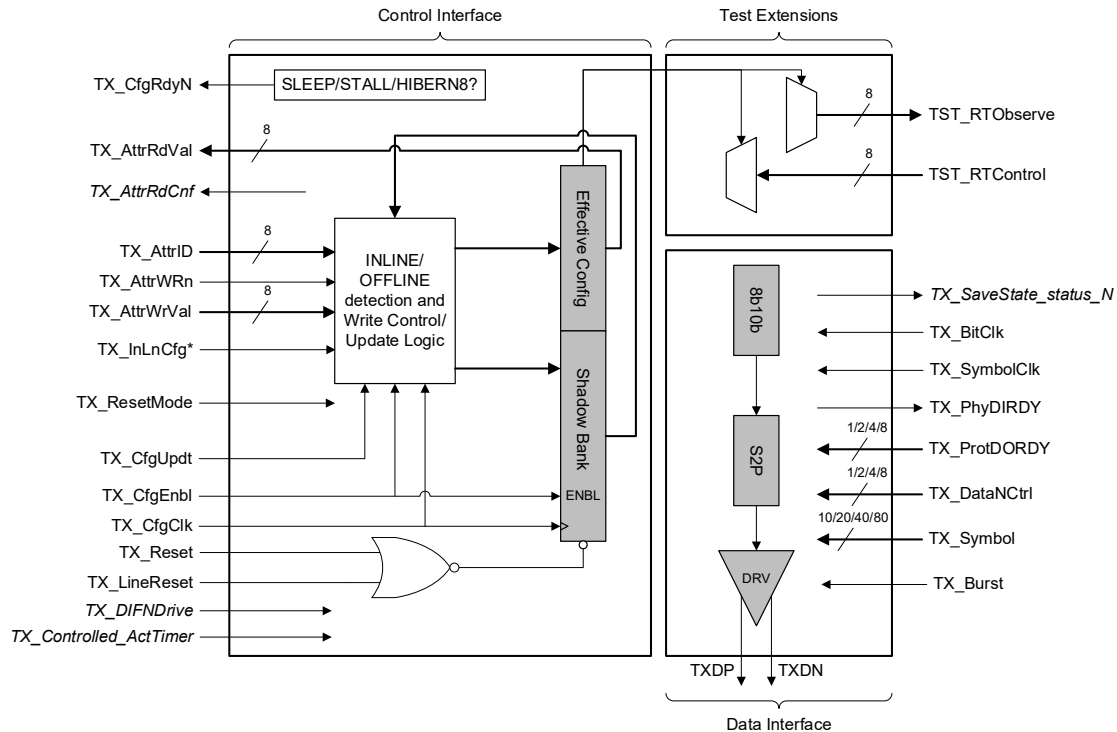
| Signal Name  | Direction | Detection Type | Width        | Signal Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------|-----------|----------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RX_SymbolErr | O         | Level          | 1, 2, 4 or 8 | <p>The width of RX_SymbolErr is one, two, four or eight bits depending on the RX_Symbol bus width of 10, 20, 40 or 80 bits, respectively.</p> <p>The M-RX shall set each bit of RX_SymbolErr to “1” for one RX_SymbolClk cycle when any of the following conditions on the corresponding RX_Symbol bus range are “TRUE”:</p> <ul style="list-style-type: none"> <li>• The 3b4b sub-block is in error while decoding the related 8b10b symbol received over the LINE</li> <li>• The 5b6b sub-block is in error while decoding the related 8b10b symbol received over the LINE</li> <li>• The Running Disparity algorithm computes an RD error for the related 8b10b symbol received over the LINE</li> <li>• The related 8b10b symbol received over the LINE is a reserved symbol</li> </ul> <p>The M-RX shall set all bits of RX_SymbolErr to “0” for all other conditions.</p> <p>RX_Symbol shall carry, in the corresponding bus range, the remapped PAYLOAD byte.</p> <p>The M-RX shall set all bits of RX_SymbolErr to “0” when 10b8b decoding is bypassed.</p> |
| RX_Burst     | O         | Transition     | 1            | <p>RX_Burst provides a framing window to the Protocol Layer for received BURSTS.</p> <p>The M-RX shall set RX_Burst to “1” when it detects the start of a PREPARE period.</p> <p>The M-RX shall set RX_Burst to “0” when it detects any of the BURST exit conditions and all 8b10b PAYLOAD data has been sent to the Protocol Layer via RX_Symbol (see <b>Section 4.7.2.5</b>). The M-RX shall set RX_Burst to “0” when it detects the ADAPT sub-state exit condition (see <b>Section 4.7.2.3</b>).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

**Table 51 M-RX Test Extensions**

| Signal Name   | Direction | Detection Type | Width | Signal Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------|-----------|----------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TST_RTObserve | O         | Asynch         | 8     | TST_RTObserve makes internal M-RX real-time signals observable, e.g. through DMA, by the Protocol Layer, or external test equipment. These signals are asynchronous to any clock on the M-RX-DATA or M-RX-CTRL interfaces.<br>Signals are selected by programming implementation-specific M-RX registers using the M-RX-CTRL interface.<br>The M-RX implementation shall not require TST_RTObserve for normal operation.                                                              |
| TST_RTControl | I         | Asynch         | 8     | TST_RTControl carries real-time signals to control implementation-specific signals, e.g. test features, inside the M-RX. These signals are asynchronous to any clock on the M-RX-DATA or M-RX-CTRL interfaces.<br>Internal multiplexers with signals on this bus are selected by programming implementation-specific M-RX registers using the M-RX-CTRL interface.<br>The M-RX implementation shall not require any specific behavior or value on TST_RTControl for normal operation. |

### A.3 The M-TX Signaling Interface

2301 A schematic overview of the M-TX signaling interface is shown in **Figure 56**.



TX\_InLnCfg\* - This signal is optional. Implementations may choose appropriate fixed value for backward compatibility.

**Figure 56 M-TX Signal Interfaces Diagram**

#### A.3.1 M-TX Signal Description

2303 In **Table 52** through **Table 54**, entries in the “Direction” column specifies the direction of each signal from  
 2304 the perspective of the M-TX. An input signal (abbreviated as “I”) is driven by the Protocol Layer. An output  
 2305 signal (abbreviated as “O”) is driven by the M-TX.

2306 The “Detection Type” column indicates the relevant condition for a given signal. A Detection Type of “Level”  
 2307 means the relevant information is either a high or low level on the signal. A Detection Type of “Transition”  
 2308 means a change from high-to-low or low-to-high causes the described action. A Detection Type of “Clock”  
 2309 indicates the signal is used to synchronize other signals on the interface. A Detection Type of “Asynch” means  
 2310 the signal changes state asynchronously to the relevant clock signal.



**Table 52 M-TX-CTRL Interface Signals**

| Signal Name  | Direction | Detection Type | Width | Signal Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------|-----------|----------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TX_CfgClk    | I         | Clock          | 1     | Identical behavior as RX_CfgClk                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| TX_Reset     | I         | Asynch         | 1     | Identical behavior as RX_Reset                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| TX_ResetMode | I         | Level          | 1     | Identical behavior as RX_ResetMode                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| TX_AttrID    | I         | Level          | 8     | Identical behavior as RX_AttrID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| TX_AttrRdVal | O         | Level          | 8     | Identical behavior as RX_AttrRdVal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| TX_AttrRdCnf | O         | Level          | 1     | Identical behavior as RX_AttrRdCnf.<br>TX_AttrRdCnf is an optional signal.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| TX_AttrWrVal | I         | Level          | 8     | Identical behavior as RX_AttrWrVal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| TX_AttrWRn   | I         | Level          | 1     | Identical behavior as RX_AttrWRn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| TX_CfgEnbl   | I         | Level          | 1     | Identical behavior as RX_CfgEnbl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| TX_CfgUpdt   | I         | Transition     | 1     | Identical behavior as RX_CfgUpdt. Protocol shall not issue TX_CfgUpdt between the protocol setting TX_Burst to "1" and the M-TX setting TX_SaveState_Status_N to "1".                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| TX_CfgRdyN   | O         | Level          | 1     | Identical behavior as RX_CfgRdyN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| TX_LineReset | I         | Transition     | 1     | TX_LineReset triggers the M-TX to issue a LINE-RESET condition.<br>When TX_Controlled_ActTimer is set to "0", or TX_Controlled_ActTimer is not implemented, the Protocol Layer shall set TX_Burst to "0" and wait for $T_{ACTIVATE}$ after the M-TX sets TX_SaveState_Status_N to "0" before it sets TX_LineReset to "1" for one TX_CfgClk cycle. After the Protocol Layer sets TX_LineReset to "1" for one TX_CfgClk cycle, the M-TX shall immediately drive the LINE-RESET condition (see <b>Section 4.7.4.1</b> ).<br>When TX_Controlled_ActTimer is set to "1", after the Protocol Layer sets TX_LineReset to "1" for one TX_CfgClk cycle, the M-TX shall immediately drive DIF-N for $T_{ACTIVATE}$ , irrespective of its current state, before it drives the LINE-RESET condition. |

| Signal Name                   | Direction | Detection Type | Width | Signal Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------|-----------|----------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>TX_Controlled_ActTimer</i> | I         | Level          | 1     | <p><i>TX_Controlled_ActTimer</i> informs the M-TX which Layer controls the <math>T_{\text{ACTIVATE}}</math> time. <i>TX_Controlled_ActTimer</i> is an optional signal.</p> <p>The Protocol Layer shall set <i>TX_Controlled_ActTimer</i> to "1" to inform the M-TX to drive DIF-N for at least <math>T_{\text{ACTIVATE}}</math>, irrespective of the current M-TX state, before driving DIF-P for <math>T_{\text{LINE-RESET}}</math> after TX_LineReset is asserted. The M-TX shall also control the <math>T_{\text{ACTIVATE}}</math> time upon HIBERN8 exit.</p> <p>When <i>TX_Controlled_ActTimer</i> is set to "0", the Protocol Layer shall wait <math>T_{\text{ACTIVATE}}</math> after M-TX sets TX_SaveState_Status_N to "0" before asserting TX_LineReset. If <i>TX_Controlled_ActTimer</i> is set to "0", when TX_LineReset is asserted, the M-TX shall immediately drive the LINE-RESET condition. The Protocol Layer shall also control the <math>T_{\text{ACTIVATE}}</math> time upon HIBERN8 exit.</p> |
| <i>TX_DIFNDrive</i>           | I         | Level          | 1     | <p>Required only if <i>TX_Controlled_ActTimer</i> is 0 and M-TX supports RSE_PO_TX.</p> <p>When set to 1, supports the protocol control of M-TX output resistance to RSE_TX.</p> <p>If <i>TX_Controlled_ActTimer</i> is 0, the protocol shall set this signal to 1 for at least <math>T_{\text{ACTIVATE}}</math> before issuing a pulse on TX_LineReset or on exiting HIBERN8.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

**Table 53 M-TX-DATA Interface Signals**

| Signal Name                  | Direction | Detection Type | Width | Signal Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------|-----------|----------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>TX_SaveState_Status_N</i> | O         | Level          | 1     | <i>TX_SaveState_Status_N</i> indicates the M-TX is entering or exiting a SAVE state. The Protocol Layer can use this signal to understand when the M-TX is not transmitting PREPARE, SYNC, HOB, PAYLOAD, TOB, or BURST Extension. The M-TX sets <i>TX_SaveState_Status_N</i> to “0” when it enters into a SAVE state. The M-TX sets <i>TX_SaveState_Status_N</i> to “1” when it exits a SAVE state.<br>A Type-I implementation shall include this signal. Though not required, a Type-II implementation should include this signal. There may be a delay between the LINE state change and the indication of SAVE state entry.                                                                                                                                                                                                                                                                                                                                                          |
| TX_BitClk                    | I         | Clock          | 1     | Bit Clock<br>TX_BitClk is used to transmit data bits over the LINES.<br>TX_BitClk may not be accessible in the M-TX-DATA interface for M-PHY implementations that comprise an integrated clock multiplier.<br>TX_BitClk shall have no specific phase relationship requirement to any signal in the M-TX-DATA interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| TX_SymbolClk                 | I         | Clock          | 1     | Symbol Clock<br>All M-TX-DATA interface signals are synchronous with this signal.<br>The Protocol Layer may disable TX_SymbolClk generation when the M-TX is not in PWM-BURST, SYS-BURST, or HS-BURST states. For this purpose, the Protocol Layer shall read the M-TX FSM state attribute.<br>In HS-MODE and SYS-MODE, TX_SymbolClk shall have a period of 10 UI for a 10-bit TX_Symbol bus, 20 UI for a 20-bit TX_Symbol bus, 40 UI for a 40-bit TX_Symbol bus, or 80 UI for an 80-bit TX_Symbol bus.<br>In PWM-MODE, TX_SymbolClk shall have a period of 10 $T_{PWM\_TX}$ for a 10-bit TX_Symbol bus, 20 $T_{PWM\_TX}$ for a 20-bit RX_Symbol bus, 40 $T_{PWM\_TX}$ for a 40-bit TX_Symbol bus, or 80 $T_{PWM\_TX}$ for an 80-bit TX_Symbol bus.<br>The behavior of TX_SymbolClk must be glitch-free even when this signal is being enabled or disabled.<br>TX_SymbolClk shall not have a “1” or “0” pulse with a duration less than one-quarter of the nominal TX_SymbolClk period. |
| TX_PhyDIRDY                  | O         | Level          | 1     | PHY Data Input Ready<br>TX_PhyDIRDY indicates the M-TX is ready to accept new data on the TX_Symbol bus.<br>The M-TX shall set TX_PhyDIRDY to “1” when the M-TX is ready to consume data.<br>The M-TX shall set TX_PhyDIRDY to “0” when the M-TX is busy.<br>The Protocol Layer should not update TX_Symbol while TX_PhyDIRDY is “0” and TX_Burst is “1”.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Signal Name       | Direction                      | Detection Type | Width            | Signal Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                   |                                |   |                       |   |                          |   |                           |   |                           |   |                           |   |                           |   |                           |   |                           |
|-------------------|--------------------------------|----------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------------------------|---|-----------------------|---|--------------------------|---|---------------------------|---|---------------------------|---|---------------------------|---|---------------------------|---|---------------------------|---|---------------------------|
| TX_Symbol         | I                              | Level          | 10, 20, 40 or 80 | <p>TX_Symbol is used for BURST data transfer to the M-TX. The contents of this bus depend on the interface width (10, 20, 40 or 80 bits, corresponding to 1, 2, 4 or 8 parallel symbols, respectively), and also on whether the 8b10b encoding function in the M-TX is bypassed.</p> <p>When the M-TX 8b10b encoding function is bypassed, TX_Symbol carries the raw data to send on the LINES, parallelized according to the implemented width. The LSb of TX_Symbol shall correspond to the earliest transmitted bit.</p> <p>When the M-TX 8b10b encoding function is enabled, only the 8, 16, 32 or 64 LSbs of TX_Symbol are used to carry the unencoded DATA or control symbol. The M-TX shall ignore the unused MSbs of TX_Symbol. The Protocol Layer should set the unused MSbs to “0”.</p> <p>Control symbols shall be encoded as listed in <b>Table 48</b>.</p> <p>TX_Symbol is accepted by the M-TX on every TX_SymbolClk cycle in which TX_ProtDORDY, TX_PhyDIRDY and TX_Burst are “1”.</p>                                                                                                                                                                                                                                                                                                                                                                                                                             |                   |                                |   |                       |   |                          |   |                           |   |                           |   |                           |   |                           |   |                           |   |                           |
| TX_ProtDORDY      | I                              | Level          | 1, 2, 4 or 8     | <p>Protocol Data Output Ready</p> <p>TX_ProtDORDY indicates data is available in the corresponding TX_Symbol bus range. The width of TX_ProtDORDY is one, two, four or eight bits depending on the TX_Symbol bus width of 10, 20, 40, or 80 bits, respectively.</p> <p>Each bit in TX_ProtDORDY corresponds to a range of bits in the TX_Symbol bus.</p> <table><thead><tr><th>TX_ProtDORDY Bits</th><th>TX_Symbol bits (8b10b enabled)</th></tr></thead><tbody><tr><td>0</td><td>bits[9:0] (bits[7:0])</td></tr><tr><td>1</td><td>bits[19:10] (bits[15:8])</td></tr><tr><td>2</td><td>bits[29:20] (bits[23:16])</td></tr><tr><td>3</td><td>bits[39:30] (bits[31:24])</td></tr><tr><td>4</td><td>bits[49:40] (bits[39:32])</td></tr><tr><td>5</td><td>bits[59:50] (bits[47:40])</td></tr><tr><td>6</td><td>bits[69:60] (bits[55:48])</td></tr><tr><td>7</td><td>bits[79:70] (bits[63:56])</td></tr></tbody></table> <p>When a TX_Symbol bus range contains new data, the Protocol Layer shall set the corresponding bit of TX_ProtDORDY to “1” for each TX_SymbolClk cycle.</p> <p>When a TX_Symbol bus range does not contain new data, the Protocol Layer shall set the corresponding bit of TX_ProtDORDY to “0” for each TX_SymbolClk cycle.</p> <p>When the M-TX 8b10b encoding function is bypassed, the Protocol Layer shall set the applicable bits of TX_ProtDORDY to “1” for each TX_SymbolClk cycle during a BURST.</p> | TX_ProtDORDY Bits | TX_Symbol bits (8b10b enabled) | 0 | bits[9:0] (bits[7:0]) | 1 | bits[19:10] (bits[15:8]) | 2 | bits[29:20] (bits[23:16]) | 3 | bits[39:30] (bits[31:24]) | 4 | bits[49:40] (bits[39:32]) | 5 | bits[59:50] (bits[47:40]) | 6 | bits[69:60] (bits[55:48]) | 7 | bits[79:70] (bits[63:56]) |
| TX_ProtDORDY Bits | TX_Symbol bits (8b10b enabled) |                |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |                                |   |                       |   |                          |   |                           |   |                           |   |                           |   |                           |   |                           |   |                           |
| 0                 | bits[9:0] (bits[7:0])          |                |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |                                |   |                       |   |                          |   |                           |   |                           |   |                           |   |                           |   |                           |   |                           |
| 1                 | bits[19:10] (bits[15:8])       |                |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |                                |   |                       |   |                          |   |                           |   |                           |   |                           |   |                           |   |                           |   |                           |
| 2                 | bits[29:20] (bits[23:16])      |                |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |                                |   |                       |   |                          |   |                           |   |                           |   |                           |   |                           |   |                           |   |                           |
| 3                 | bits[39:30] (bits[31:24])      |                |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |                                |   |                       |   |                          |   |                           |   |                           |   |                           |   |                           |   |                           |   |                           |
| 4                 | bits[49:40] (bits[39:32])      |                |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |                                |   |                       |   |                          |   |                           |   |                           |   |                           |   |                           |   |                           |   |                           |
| 5                 | bits[59:50] (bits[47:40])      |                |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |                                |   |                       |   |                          |   |                           |   |                           |   |                           |   |                           |   |                           |   |                           |
| 6                 | bits[69:60] (bits[55:48])      |                |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |                                |   |                       |   |                          |   |                           |   |                           |   |                           |   |                           |   |                           |   |                           |
| 7                 | bits[79:70] (bits[63:56])      |                |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |                                |   |                       |   |                          |   |                           |   |                           |   |                           |   |                           |   |                           |   |                           |

| Signal Name     | Direction | Detection Type | Width        | Signal Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------|-----------|----------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TX_DataNCtrl    | I         | Level          | 1, 2, 4 or 8 | <p>TX_DataNCtrl indicates the type of symbol on the indicated range of TX_Symbol.</p> <p>The width of the TX_DataNCtrl is one, two, four or eight bits depending on the TX_Symbol bus width of 10, 20, 40 or 80 bits, respectively.</p> <p>The bits of TX_DataNCtrl are mapped the same as the bits of TX_ProtDORDY.</p> <p>The Protocol Layer shall set the corresponding bit of TX_DataNCtrl to “0” when the related TX_Symbol bus range carries a data symbol.</p> <p>The Protocol Layer shall set the corresponding bit of TX_DataNCtrl to “1” when the related TX_Symbol bus range carries a control symbol.</p> <p>The Protocol Layer should set all bits of TX_DataNCtrl to “0” when 8b10b encoding is bypassed.</p> <p>The M-TX shall ignore all bits of TX_DataNCtrl when 10b8b decoding is bypassed.</p> |
| TX_Burst        | I         | Transition     | 1            | <p>TX_Burst initiates a BURST.</p> <p>The Protocol Layer shall set TX_Burst to “1” to initiate a BURST, and hold the value for the duration of the BURST.</p> <p>Once TX_Burst is set to “1”, the M-TX shall send the PREPARE sequence (and SYNC sequence in the case of a HS-BURST), followed by data or FILLER symbols.</p> <p>If any bit of TX_ProtDORDY is set to “1”, the M-TX shall send the data present on the corresponding TX_Symbol bus range.</p> <p>If any bit of TX_ProtDORDY is set to “0”, the M-TX shall send one FILLER for each TX_ProtDORDY bit set to 0.</p> <p>Once TX_Burst is set to “0”, the M-TX shall send the TAIL-OF-BURST sequence (see <b>Section 4.7.2.4</b>).</p> <p>TX_Burst shall not be asserted while TX_ADAPT_ACTIVE or TX_ADAPT_REQ are asserted.</p>                       |
| TX_ADAPT_ACTIVE | O         | Level          | 1            | <p>TX_ADAPT_ACTIVE = 1 indicates to the Protocol that M-TX is in ADAPT sub-state. The signal is asserted after PREPARE. The signal is de-asserted with the end of the ADAPT sequence.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| TX_ADAPT_REQ    | I         | Level          | 1            | <p>The Protocol Layer shall set TX_ADAPT_REQ to “1” to initiate an ADAPT sequence.</p> <p>Once set to “1”, the M-TX shall send the PREPARE sequence followed by the ADAPT sequence.</p> <p>The Protocol shall de-assert TX_ADAPT_REQ with TX_ADAPT_ACTIVE = 1 earliest, but latest with the end of the ADAPT sequence when the M-TX returns to STALL. The Protocol Layer shall ensure RX_ADAPT_Control set to ADAPT prior to assertion of this signal.</p>                                                                                                                                                                                                                                                                                                                                                         |

**Table 54 M-TX Test Extensions**

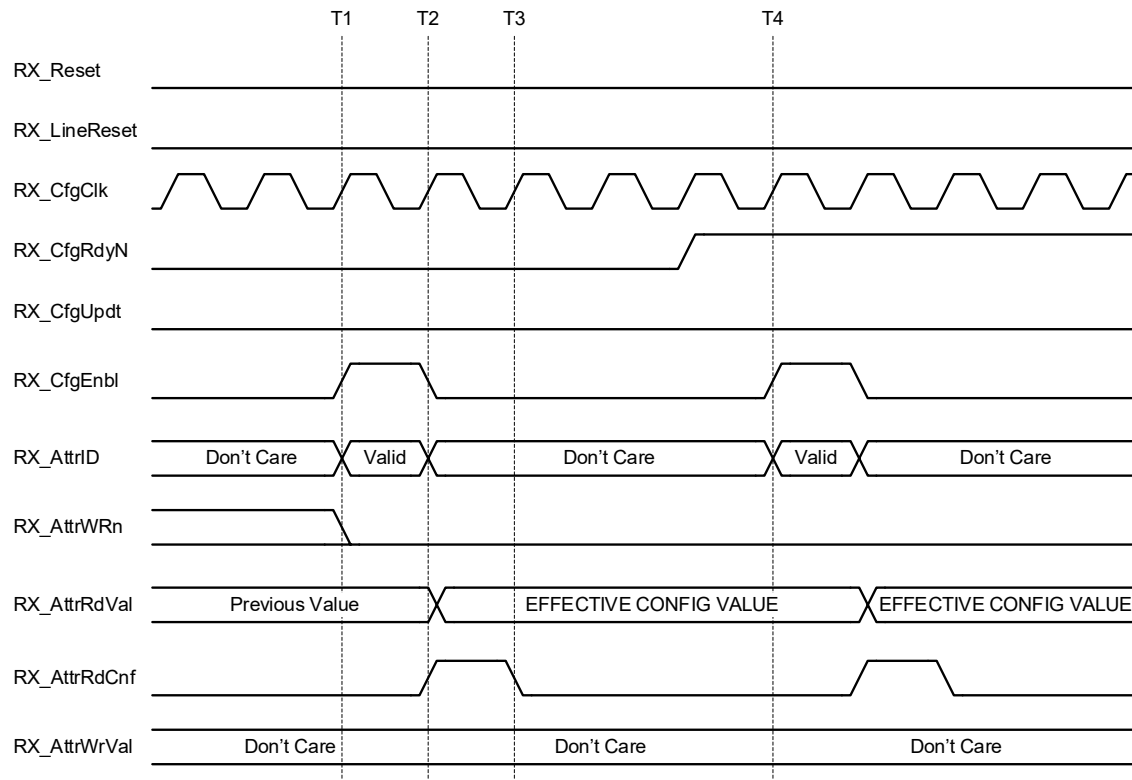
| <b>Signal Name</b> | <b>Direction</b> | <b>Detection Type</b> | <b>Width</b> | <b>Signal Description</b>               |
|--------------------|------------------|-----------------------|--------------|-----------------------------------------|
| TST_RTObserve      | O                | Asynch                | 8            | Identical behavior as in M-RX interface |
| TST_RTControl      | I                | Asynch                | 8            | Identical behavior as in M-RX interface |

## A.4 Interface Usage Examples

To aid in the design of a conformant implementation, the following use-cases are provided depicting the required interface behavior.

### A.4.1 Attribute Read from Effective Configuration

**Figure 57** illustrates an example of an attribute read from the M-RX. The example shows the M-RX effective configuration bank being read regardless of RX\_CfgRdyN value.



**Figure 57 Interface Behavior for Attribute Read Operations**

At T1, on the rising edge of RX\_CfgClk, the Protocol Layer sets RX\_CfgEnbl to “1”, sets RX\_AttrWRn to “0”, and sets the value of RX\_AttrID to the attribute identifier.

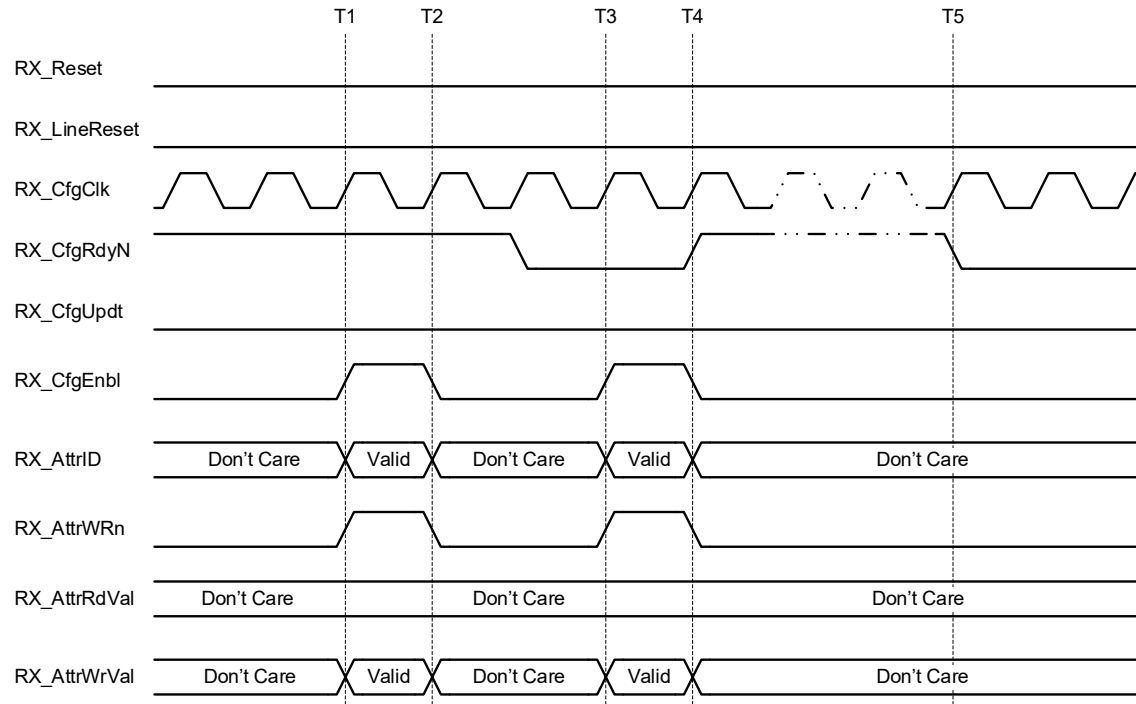
At T2, on the rising edge of RX\_CfgClk, the M-RX captures the command. In response, the M-RX updates RX\_AttrRdVal with the effective configuration bank attribute value and, if implemented, asserts Rx\_AttrRdCnf for one RX\_CfgClk cycle. Also at T2, the Protocol Layer sets RX\_CfgEnbl to “0” on the rising edge of RX\_CfgClk.

At T3, the Protocol Layer can capture RX\_AttrRdVal. The M-RX holds the value on RX\_AttrRdVal until a subsequent read operation, or local RESET.

At T4, on the rising edge of RX\_CfgClk, the Protocol Layer initiates a second read operation. In this instance, the M-RX has set RX\_CfgRdyN set to “1” indicating it cannot process a write operation. Note that the read operation is unaffected by the RX\_CfgRdyN signal.

#### A.4.2 Attribute Write to Shadow Memory and Effective Configuration

**Figure 58** shows two attribute writes to the M-RX. In this use-case, an attribute in the shadow memory bank is updated independently of RX\_CfgRdyN, then an effective configuration bank attribute is updated only when RX\_CfgRdyN is “0”.



**Figure 58 Interface Behavior for Attribute Write Operations**

At T1, on the rising edge of RX\_CfgClk, the Protocol Layer sets RX\_CfgEnbl and RX\_AttrWRn to “1”, and sets the value of RX\_AttrID and RX\_AttrWrVal.

At T2, the M-RX samples these signals on the rising edge of RX\_CfgClk and performs the requested operation, in this case updating its shadow memory bank. Since the effective configuration bank is not changed, the M-RX performs the requested operation even though RX\_CfgRdyN is “1” at this time. The Protocol Layer, on the rising edge of RX\_CfgClk at T2, sets RX\_CfgEnbl and RX\_AttrWRn to “0”, and optionally sets to “0” RX\_AttrID and RX\_AttrWrVal.

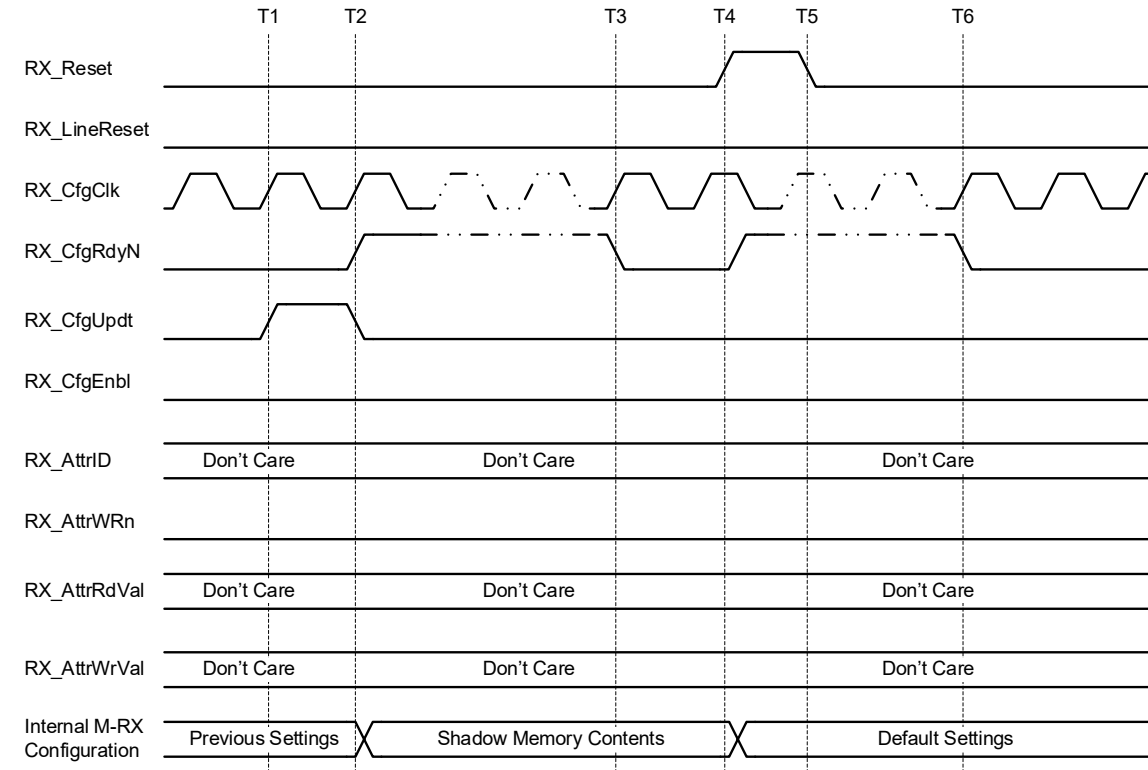
At T3, another write operation is performed in the same manner as the first, which causes the M-RX to write either to the effective configuration bank or to the shadow memory bank, depending on the implementation as this operation is performed by the M-RX when RX\_CfgRdyN is “0” as illustrated in this use-case.

As a result of the operation, the M-RX optionally sets RX\_CfgRdyN to “1” at T4, when the write operation is processed. The M-RX optionally holds RX\_CfgRdyN at “1” until the change in the configuration is complete. The M-RX then sets RX\_CfgRdyN to “0” synchronously with RX\_CfgClk at T5. The M-RX is then ready to perform any subsequent write operation.



### A.4.3 Effective Configuration Single-step Update and Local RESET

**Figure 59** shows a single-step (atomic) update of the effective configuration bank followed by a local RESET.



**Figure 59 Interface Behavior for RX\_CfgUpdt and RX\_Reset**

At T1, the Protocol Layer sets RX\_CfgUpdt to “1” for one cycle of RX\_CfgClk to upload the entire shadow memory bank into the effective configuration bank in one step. The Protocol Layer holds RX\_CfgEnbl at “0” for this operation. RX\_AttrID, RX\_AttrWRn, and RX\_AttrWrVal are ignored by the M-RX.

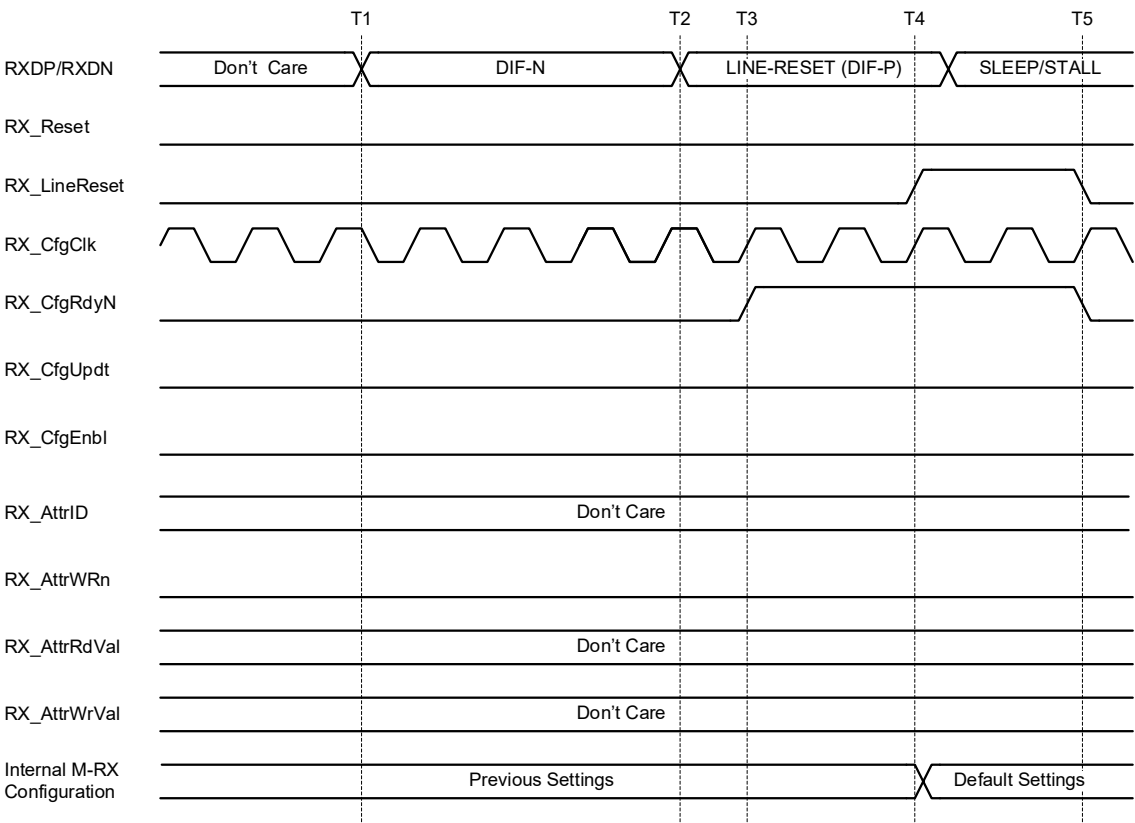
The M-RX processes the command on the rising edge of RX\_CfgClk at T2, when the entire shadow memory is uploaded into the effective configuration bank. The M-RX then sets RX\_CfgRdyN to “1” and holds the value until the change in the M-RX configuration is complete and the M-RX is ready to perform subsequent write operations.

At T3, the M-RX sets RX\_CfgRdyN to “0” on the rising edge of RX\_CfgClk.

At T4, the Protocol Layer sets RX\_Reset to “1”, asynchronous to RX\_CfgClk, causing a local RESET. The M-RX asynchronously sets RX\_CfgRdyN to “1” in response, and holds the value until the Protocol Layer sets RX\_Reset to “0”, which occurs at T5, and it finishes processing the local RESET. Once the M-RX is ready to perform subsequent write operations, it sets RX\_CfgRdyN to “0”, which occurs synchronously at T6.

A.4.4 Received LINE-RESET

2361 **Figure 60** shows a Type-I M-RX receiving a LINE-RESET.



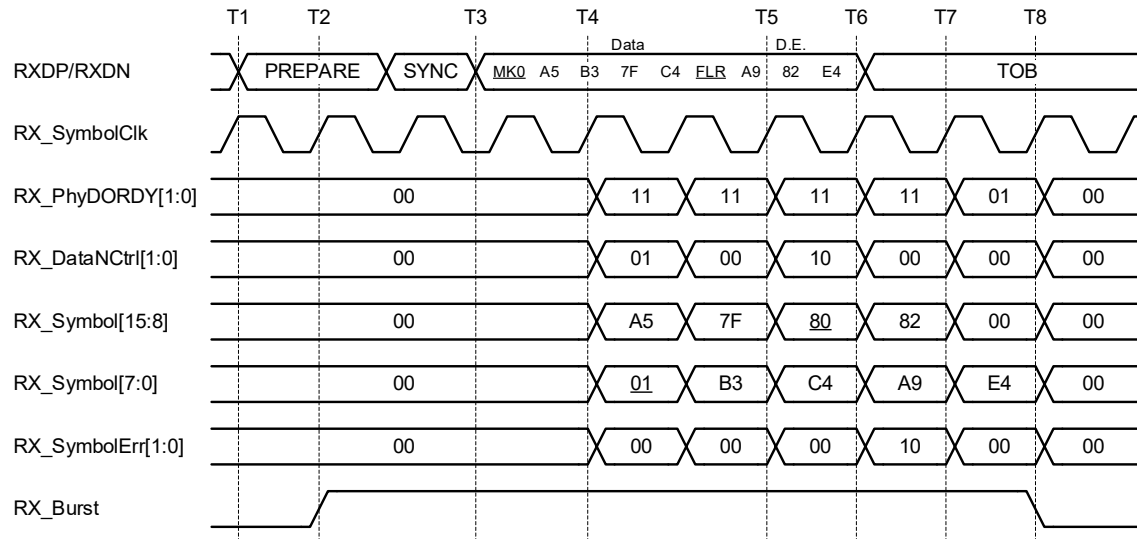
2362

**Figure 60 Interface Behavior for LINE-RESET**

#### A.4.5 HS Data Reception with 20-bit RX\_Symbol Bus

**Figure 61** shows the interface behavior for an M-RX with a 20-bit interface during HS data reception. 10b8b decoding is enabled in this use-case.

In this use-case, the M-RX receives a data transmission from the attached M-TX. An RD error occurs near the end of the transmission.



D.E. = Disparity Error

**Figure 61 Example 20 bit Interface Behavior for HS Data Reception**

At T1, the M-RX detects the PREPARE sequence and sets RX\_Burst to “1” on the rising edge of RX\_SymbolClk at T2.

At T3, the SYNC sequence ends. The M-RX receives the first two symbols, a MARKER0 (MK0) and A5 (data).

At T4, on the rising edge of RX\_SymbolClk, the M-RX sets RX\_Symbol[7:0] to “01” (MARKER0) and RX\_Symbol[15:8] to “A5”. The M-RX also sets RX\_DataNCtrl[0] to “1” indicating a control symbol is on RX\_Symbol[7:0], and sets RX\_DataNCtrl[1] to “0” indicating data is on RX\_Symbol[15:8]. RX\_SymbolErr[1:0] is held at “00” indicating no errors on RX\_Symbol. Finally, the M-RX sets RX\_PhyDORDY[1:] to “11” indicating data is available on RX\_Symbol. On the next rising edge of RX\_SymbolClk, the M-RX sets RX\_Symbol[7:0] and RX\_Symbol[15:8] to the next two symbols received, “B3” and “7F”, respectively. The M-RX sets RX\_DataNCtrl[1:0] to “00” indicating both symbols are data. The M-RX sets the remaining signals the same as at T4.

At T5, the M-RX sets RX\_DataNCtrl[1:0] to “10” indicating it received another control symbol. The M-RX also sets RX\_Symbol[7:0] to “C4” (data) and RX\_Symbol[15:8] to “80” (FILLER). The M-RX sets the remaining signals the same as at T4.

#### Note:

By itself, the FILLER symbol does not cause the M-RX to set RX\_PhyDORDY[1] to “0”. However, a midstream deassertion of RX\_PhyDORDY is possible in plesiochronous Type-I systems due to, e.g. internal FIFO refills in an M-RX implementation.

2387 The M-RX receives the next two symbols, “A9” and “82”, in the same manner as the first six symbols.  
2388 However, as shown in **Figure 61**, the “82” symbol has an RD error.

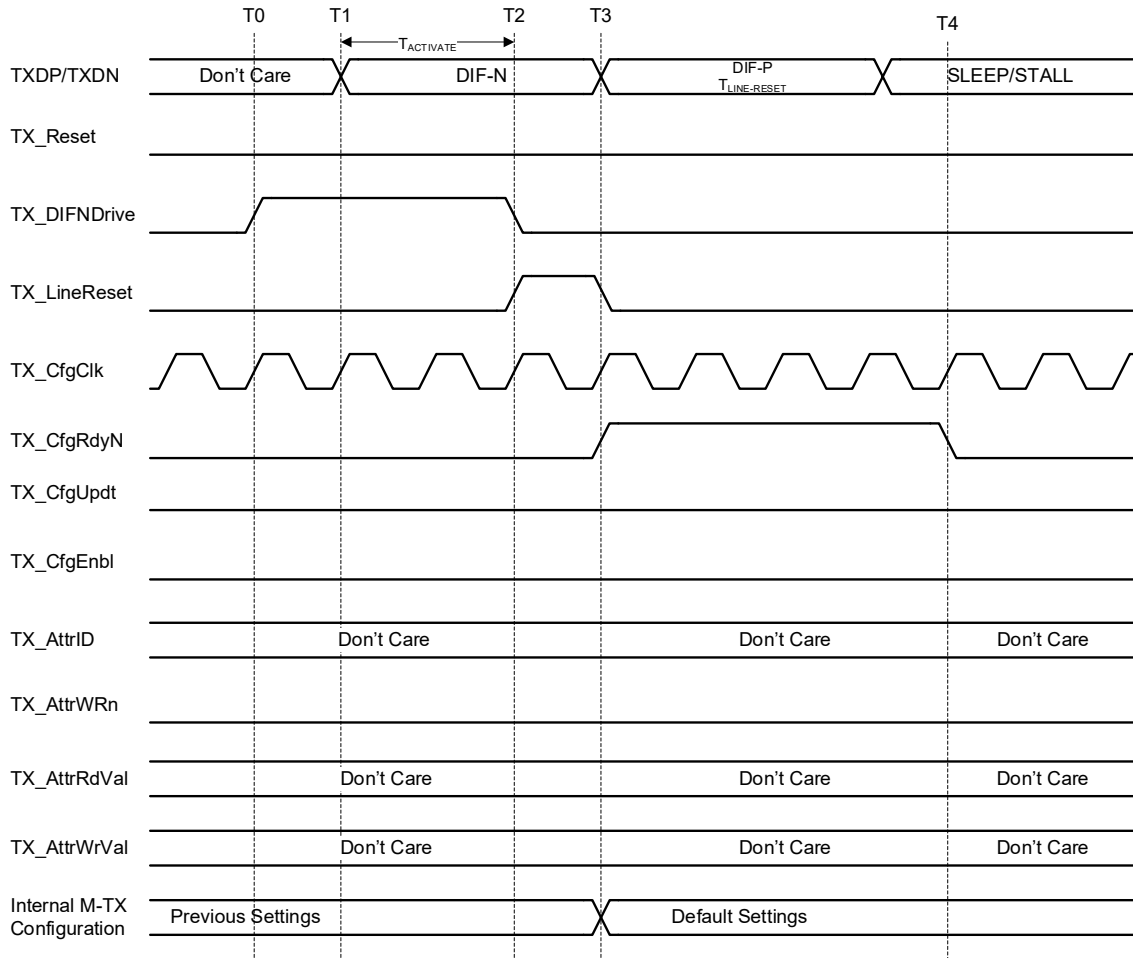
2389 At T6, on the rising edge of RX\_SymbolClk, the M-RX sets RX\_Symbol[7:0] to “A9”, RX\_Symbol[15:8]  
2390 to “82”, and RX\_SymbolErr[1:0] to “10” indicating an error in the data on RX\_Symbol[15:8]. Finally, the  
2391 M-RX sets RX\_PhyDORDY[1:0] to “11” indicating data is available on RX\_Symbol.

2392 At T7, the M-RX detects the end of the BURST and determines it has received an odd number of symbols.  
2393 It sets RX\_Symbol[7:0] to “E4”, RX\_Symbol[15:8] to “00”, and RX\_PhyDORDY[1:0] to “01” indicating  
2394 RX\_Symbol[15:8] does not contain data. The M-RX also sets RX\_DataNCtrl[1:0] to “00” indicating  
2395 RX\_Symbol does not contain any control symbols. Finally, the M-RX sets RX\_SymbolErr[1:0] to “00”  
2396 indicating there are no errors.

2397 At T8, on the rising edge of RX\_SymbolClk, the M-RX sets RX\_Burst to “0” indicating the end of the Burst.  
2398 “00” on RX\_Symbol[15:0] is a don't care condition.

#### A.4.6 TX\_LineReset Behavior

**Figure 62** shows a LINE-RESET use-case. In this use-case, the Protocol Layer sends a LINE-RESET to initialize the M-TX and M-RX attached to the LINE.



**Figure 62 Interface Behavior for a TX\_LineReset Command**

At T0, the Protocol Layer sets TX\_DIFNDrive to “1” and holds it to “1” for  $T_{ACTIVATE}$  before issuing LINE-RESET.

At T1, the Protocol Layer ensures the M-TX is in SLEEP or STALL state, and waits for  $T_{ACTIVATE}$  before issuing LINE-RESET. The M-TX detects TX\_DIFNDrive at “1” and drives DIF-N on the LINE.

At T2, the Protocol Layer sets TX\_LineReset to “1” on the rising edge of TX\_CfgClk, and optionally sets it to “0” one TX\_CfgClk cycle later at T3.

At T3, the M-TX sets TX\_CfgRdyN to “1”, updates its internal configuration registers to their default values, and starts driving the LINE-RESET condition.

The M-TX holds TX\_CfgRdyN at “1” while it is processing the LINE-RESET.

At T4, on the rising edge of TX\_CfgClk, the M-TX sets TX\_CfgRdyN to “0” to signal its internal FSM exit to SLEEP or STALL state. At this time, the M-TX is ready for any subsequent write command or TX\_LineReset pulse.

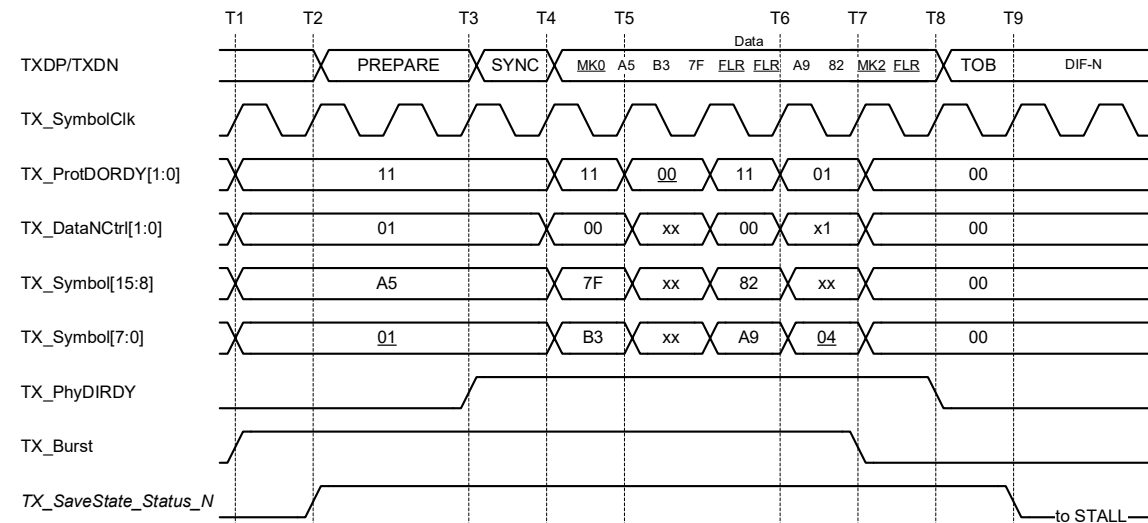
**Note:**

The M-TX only monitors the 0-to-1 transition on TX\_LineReset to interpret the command. Consequently, the M-TX does not detect whether the Protocol Layer leaves TX\_LineReset at “1” or sets it to “0” at T3.

**A.4.7 HS Transmission on 20-bit TX\_Symbol Bus with Data Throttled by Protocol Layer**

Figure 63 shows an HS transmission with the Protocol Layer controlling the data throughput. 8b10b encoding is enabled in this use-case.

In this use-case, the Protocol Layer cannot supply transmission requests as fast as the M-TX transmissions on the LINE. The Protocol Layer throttles the data throughput by changing the value on TX\_ProtDORDY. The M-TX continues to transmit, but inserts FILLER symbols whenever the Protocol Layer does not have new data to send.



**Figure 63 Interface Behavior for HS Transmission with Protocol Layer Throttling Data**

At T1, on the rising edge of TX\_SymbolClk, the Protocol Layer sets TX\_ProtDORDY[1:0] to “11”, indicating both TX\_Symbol[7:0] and TX\_Symbol[15:8] contain data; TX\_DataNCtrl[1:0] to “01”, indicating the value on TX\_Symbol[7:0] (01) is a control symbol (MARKER0), and the value on TX\_Symbol[15:8] (A5) is a data symbol. Finally, the Protocol Layer initiates the HS transmission by setting TX\_Burst to “1”.

At T2, on the rising edge of TX\_SymbolClk, the M-TX reads the Protocol Layer request and issues PREPARE and SYNC sequences. The M-TX also sets TX\_SaveState\_Status\_N to “1”.

At T3, on the rising edge of TX\_SymbolClk, the M-TX sets TX\_PhyDIRDY to “1”, far enough in advance of the start of data transmission for the Protocol Layer to read TX\_PhyDIRDY at T4.

At T4, on the rising edge of TX\_SymbolClk, the Protocol Layer holds TX\_ProtDORDY[1:0] at “11”, indicating new data is available, and sets TX\_DataNCtrl[1:0] to “00”, indicating the values on TX\_Symbol[7:0] (B3) and TX\_Symbol[15:8] (7F) are data symbols.

At T5, on the rising edge of TX\_SymbolClk, the Protocol Layer sets TX\_ProtDORDY to “00” indicating it does not have new data to send. The M-TX ignores the values on TX\_DataNCtrl[1:0] and TX\_Symbol[15:0], and inserts two FILLER symbols on the LINE.

At T6, on the rising edge of TX\_SymbolClk, the Protocol Layer sets TX\_ProtDORDY[1:0] to “01”, indicating only TX\_Symbol[7:0] has available data, and sets TX\_DataNCtrl[1:0] to “01”, indicating the value on TX\_Symbol[7:0] (04) is a control symbol (MARKER2).

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2442 At T7, on the rising edge of TX\_SymbolClk, the Protocol Layer sets TX\_Burst to “0” indicating the end of  
2443 the HS-BURST. In this use-case, the M-TX inserts a FILLER symbol after the MARKER2 symbol. “00” on  
2444 TX\_Symbol[15:0] is a don't care condition.

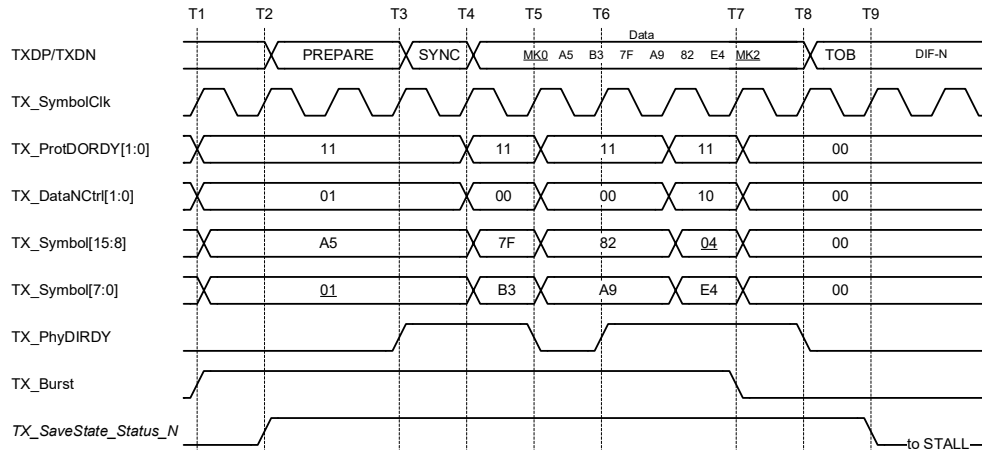
2445 At T8, on the rising edge of TX\_SymbolClk, the M-TX reads the TX\_Burst signal as “0” and begins  
2446 transmitting the TAIL-OF-BURST sequence on the LINE. The M-TX sets TX\_PhyDIRDY to “0”, indicating  
2447 it is no longer prepared to accept new data to transmit.

2448 At T9, on completion of TOB, the M\_TX sets *TX\_SaveState\_Status\_N* to “0” and enters STALL state, leaving  
2449 *TX\_SaveState\_Status\_N* at “1”.

#### A.4.8 HS Transmission on 20-bit TX\_Symbol Bus with Data Throttled by M-TX

**Figure 64** shows an HS transmission with the M-TX controlling the data throughput. 8b10b encoding is enabled in this use-case.

In this use-case, the M-TX transmissions on the LINE lag the Protocol Layer requests so the M-TX needs to slow down the transfer from the Protocol Layer. The M-TX throttles the data throughput by changing the value on TX\_PhyDIRDY.



**Figure 64 Interface Behavior for HS Transmission with M-TX Throttling Data**

At T1, on the rising edge of TX\_SymbolClk, the Protocol Layer sets TX\_ProtDORDY[1:0] to “11”, indicating both TX\_Symbol[7:0] and TX\_Symbol[15:8] contain data; TX\_DataNCtrl[1:0] to “01”, indicating the value on TX\_Symbol[7:0] (01) is a control symbol (MARKER0), and the value on TX\_Symbol[15:8] (A5) is a data symbol. Finally, the Protocol Layer initiates the HS transmission by setting TX\_Burst to “1”.

At T2, on the rising edge of TX\_SymbolClk, the M-TX reads the Protocol Layer request and issues PREPARE and SYNC sequences. The M-TX also sets TX\_SaveState\_Status\_N to “1”.

At T3, on the rising edge of TX\_SymbolClk, the M-TX sets TX\_PhyDIRDY to “1”, far enough in advance of the start of data transmission for the Protocol Layer to read TX\_PhyDIRDY.

At T4, on the rising edge of TX\_SymbolClk, the Protocol Layer sets TX\_ProtDORDY[1:0] to “11”, indicating new data is available, and sets TX\_DataNCtrl[1:0] to “00”, indicating the values on TX\_Symbol[7:0] (B3) and TX\_Symbol[15:8] (7F) are data symbols.

At T5, on the rising edge of TX\_SymbolClk, the M-TX sets TX\_PhyDIRDY to “0”, indicating the M-TX is busy. The Protocol Layer holds TX\_ProtDORDY at “11” indicating it has new data to send.

At T6, on the rising edge of TX\_SymbolClk, the M-TX sets TX\_PhyDIRDY to “1” indicating it is again available to accept new data. However, the Protocol Layer reads TX\_PhyDIRDY as “0”, and consequently holds the values on TX\_ProtDORDY[1:0], TX\_DataNCtrl[1:0], and TX\_Symbol[15:0].

On the next rising edge of TX\_SymbolClk, the Protocol Layer sets TX\_ProtDORDY[1:0] to “11”, and sets TX\_DataNCtrl[1:0] to “10”, indicating the value on TX\_Symbol[7:0] (E4) is a data symbol and the value on TX\_Symbol[15:8] (04) is a control symbol (MARKER2).

At T7, on the rising edge of TX\_SymbolClk, the Protocol Layer sets TX\_Burst to “0” indicating the end of the HS-BURST.

At T8, the M-TX reads the TX\_Burst signal as “0” on the rising edge of TX\_SymbolClk, and begins transmitting the TAIL-OF-BURST sequence on the LINE. The M-TX sets TX\_PhyDIRDY to “0”, indicating it is no longer prepared to accept new data to transmit.

At T9, on completion of TOB, the M-TX sets TX\_SaveState\_Status\_N to “0” and enters STALL state, leaving TX\_SaveState\_Status\_N at “1”.



## Annex B Recommended Test Functionality (informative)

2482 The purpose of this annex is to provide guidelines for testability features for M-PHY applications. Because  
2483 explicit test modes are not defined within the Physical Layer, most test functionality is left to higher layers  
2484 to implement. However, this must be done in a manner that produces the necessary behavior at the Physical  
2485 Layer interface that is needed for performing physical layer measurements with standard laboratory  
2486 equipment.

2487 This annex describes the functional behavior that should be provided at the Physical Layer interface in order  
2488 for various classes of measurements to be performed. The behavior is described in an abstract manner,  
2489 without reference to specific protocols or applications. Because multiple applications of M-PHY technology  
2490 exist, options for different architectures are discussed. Applications that use M-PHY technology should  
2491 ensure that sufficient functionality is designed into the higher layer specifications to allow the necessary test  
2492 functionality to be supported at the Physical Layer interface. Note that this functionality may be supported  
2493 within the normal operating capabilities of the protocol, or may be implemented via specialized test modes  
2494 if necessary.

2495 This annex is divided into two main sections, test pattern generation and test pattern verification. Test pattern  
2496 generation is primarily applicable to transmitter measurements, and test pattern verification is applicable to  
2497 receiver tolerance measurements. A brief section on interoperability testing is also discussed.

## B.1 Test Pattern Generation

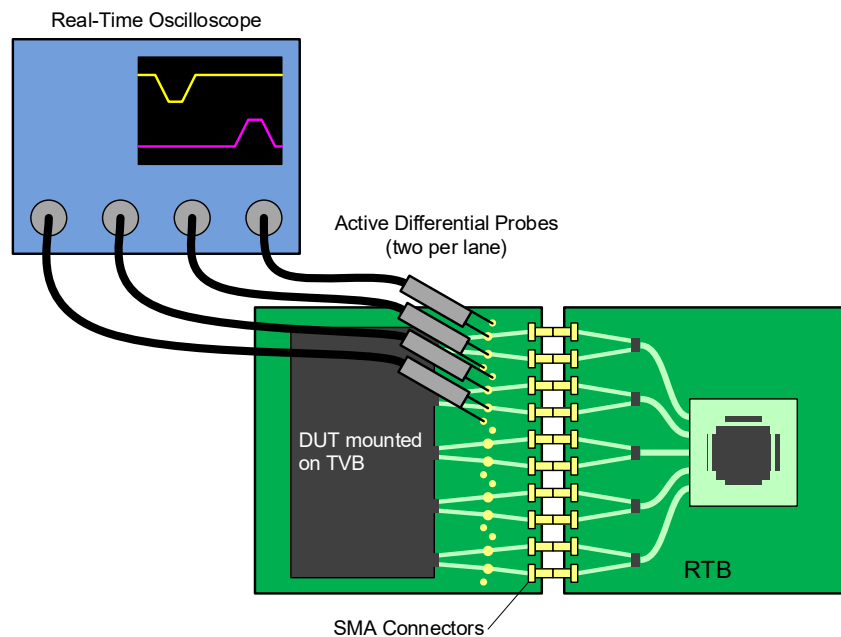
### B.1.1 General Transmitter Test Approach

In order to perform transmitter signaling measurements such as amplitude (swing), rise and fall times, skew, jitter, etc, it is necessary for the M-PHY Device Under Test (DUT) to transmit known test patterns into a reference termination load. The signals observed at this reference load are captured using an oscilloscope, and measured for conformance.

The reference termination may consist of an external fixture that contains a precision reference termination structure, which is then probed using high-bandwidth active probes. Or in some cases the oscilloscope itself may be used as the reference termination (in cases where a  $100\ \Omega$  differential termination is required), in which case the signal is sent directly into the instrument, using coaxial cables.

In the case of M-PHY technology, where signals must also be measured into an open (unterminated) termination, active probing must be used, as it is the only way to observe signals under these conditions. Active probing is also preferable for terminated measurements, as it allows the signal to be observed as close to the TX PINs as possible, and with minimal capacitive loading.

An example transmitter test setup is shown in **Figure 65**, where the DUT is mounted on an SMA-based Test Vehicle Board (TVB), and is connected to a Reference Termination Board (RTB). Each signaling Lane is probed using two active differential probes.



**Figure 65 Transmitter Test Setup**

### B.1.2 Test Patterns

2514 Some transmitter measurements, e.g., rise and fall times, are typically performed on short repeating patterns  
2515 consisting of a single repeated 10-bit code word, e.g., D30.7, D10.2, etc. Other measurements, such as  
2516 transmitter jitter, are required to be performed on longer repeating patterns such as CJTPAT and CRPAT (see  
2517 *[INC01]*).

2518 As a result, it is desirable for M-PHY devices to support a mode that allows a test pattern (up to 2320 bits in  
2519 length for CJTPAT, and 1960 bits for CRPAT). For maximum flexibility, this mode should allow arbitrary  
2520 sequences of validly encoded 8b10b 10-bit codewords up to several thousand bits in length.

### B.1.3 Signaling Type and Speed

2521 Two types of signaling are used for an M-PHY implementation, NRZ and PWM. A Type-I MODULE uses  
2522 NRZ signaling for HS transmission and uses PWM signaling for LS transmission. A Type-II MODULE uses  
2523 NRZ signaling for LS and HS transmission. In addition, different speed ranges (GEARs) are defined for both  
2524 HS and LS transmission.

2525 DUTs should provide a mechanism that allows both the signaling type and GEAR to be controlled for test  
2526 purposes.

### B.1.4 Continuous vs. Burst Modes

2527 Under normal operation, data transmission occurs in bursts, with power-saving states occurring between  
2528 bursts.

2529 Most transmitter measurements can be performed on burst-mode signaling using a real-time oscilloscope.  
2530 These instruments can capture individual burst waveforms, which can then be post-processed to extract the  
2531 required measurements.

2532 Note that a second class of oscilloscope exists, known as a sampling oscilloscope, which requires a  
2533 continuous, repeating pattern in order to observe and measure a signal. These instruments sample multiple  
2534 instances of the same repeating waveform at different time offsets in order to build a picture of the transmitted  
2535 signal. These types of instruments are typically capable of higher bandwidths and greater vertical precision  
2536 than real-time oscilloscopes, however they require a continuous, repeating pattern, and cannot measure  
2537 burst-mode signaling.

2538 In order to support the widest range of test instruments and greatest measurement flexibility, M-PHY devices  
2539 should support both burst-based and continuous transmission modes for test pattern generation.

### B.1.5 Disconnect

2540 Mechanisms may exist within the protocol to allow configuration of desired test modes and capabilities  
2541 through the Physical Layer interface. However in these instances, capability must be provided that allows the  
2542 DUT to remain in the configured test mode once the test mode has been entered, such that it may be  
2543 disconnected from a protocol-aware LINK partner (that may have been used to perform all or part of the  
2544 configuration), and reconnected to the test setup. This implies that the DUT maintains the configured  
2545 transmitter test mode even when no signaling is present at the DUT's receiver. This functionality is often  
2546 informally referred to as 'disconnect' in the test community, in that if a DUT supports "disconnect", it will  
2547 maintain its test modes after being disconnected from a LINK partner.

2548 M-PHY devices should support disconnect for all test modes.

**B.1.6 Configuration**

2549 One method for implementing such a feature would be to define a special protocol mechanism, which would  
2550 allow a special frame or command containing the desired pattern to be sent to the DUT via the Physical Layer  
2551 interface. Upon reception of this packet, the DUT would transmit the provided pattern continuously, using  
2552 the desired signaling type, gear, and any other desired settings (which could also be specified along with the  
2553 pattern.) The test pattern could be transmitted continuously until a separate reset packet is received, or the  
2554 DUT is power cycled.

## B.2 Test Pattern Verification

### B.2.1 General Receiver Test Approach

The general approach used for verifying receiver conformance involves using a laboratory-grade signal generator to generate signaling that contains controlled amounts of degradation, of various types, per the specification requirements. The signal generator is calibrated by measuring the specified characteristics into a reference termination (which is the same reference termination used for the transmitter conformance measurements). Once the required amount of degradation is calibrated, the signal is removed from the reference termination and applied to the DUT's receiver.

At this point, some observable mechanism must be used to determine whether or not the DUT can successfully decode the received signaling without error. There are several ways that this can be achieved.

### B.2.2 Loopback Mode

Loopback mode is one of the most common mechanisms used for receiver testing. In this mode, data that is received at the RX is retransmitted out the TX. The TX signal can then be observed to verify whether or not any bits were received in error (as the error would be propagated to the TX). Note however that different types of loopback modes exist, and the subtleties of these differences can impact their ability to be used with different types of test instruments. The important differences are discussed below:

#### B.2.2.1 Synchronous vs. Plesiochronous

One of the most important characteristics of a loopback mode pertains to how the clocking architecture is defined with respect to the receiver and transmitter. For a synchronous loopback, the recovered clock from the RX is used to retransmit the signal on the TX. This means there is a bit-for-bit relationship between receiver and transmitter, and the exact bit sequence that was sent into the receiver will appear at the transmitter.

Typically, this type of loopback mode is implemented outside the scope of normal operation, where the standard protocol operation is no longer applicable, and the DUT will simply forward any data received to the transmitter. The received data is typically not 8b10b decoded and re-encoded in the loopback path, which ensures that a single error at the receiver translates to a single error at the transmitter. This behavior allows traditional Bit Error Rate Tester (BERT) instruments to be used to test the receiver (as these instruments typically require a bit-for-bit correlation between the transmitted and received data patterns.)

This document actually specifies this exact type of loopback. The LOOPBACK feature defined in **Section 4.10.1** is intended for symmetric architectures that support the same MODE and GEAR settings for the M-RX and M-TX. If this feature is supported, it can actually be used for both receiver and transmitter verification, as most transmitter measurements can be performed on the TX output while the desired test pattern is transmitted into the RX. Note however that this case is not ideal for all transmitter tests, particularly jitter, as measured jitter and frequency while in LOOPBACK are not necessarily the same as during normal operation, as the clock reference is not the same.

Other types of loopback include a plesiochronous loopback (sometimes referred to as a “far-end retimed loopback”), which is similar to the synchronous loopback, except the transmitter and receiver run on separate clock domains, i.e., have separate clock references. This means that the RX and TX are operating at almost the same rate, but are not exactly matched. This is still considered a test mode that operates outside the scope of normal protocol operation, where data must be inserted or deleted from the data being looped back in order to account for the rate difference between RX and TX. This is typically accomplished by inserting or deleting specifically defined control codewords that are not considered part of the CRC-checked frame data stream.

In this scenario, a BERT or other signal source may be used to generate the test signal that is sent into the receiver, however the signal that is retransmitted by the DUT must be checked using a Frame Error Counter, which is a device that can receive the framed data patterns, and compute or check the CRC (which is included as part of the defined pattern.)

### B.2.3 Receiver Pattern Checking

Note that the loopback described above can only be used for symmetric architectures, and requires the same MODEs and GEARs to be supported by both the M-RX and M-TX. For M-PHY applications and architectures that are not bidirectional and symmetric, a different approach must be used to verify received data for the purposes of conformance testing.

One option consists of a dedicated RX test mode, whereby a predefined test pattern can be transmitted into the M-RX, and the checking operation is actually performed by the receiver itself. This can be done on a bit-for-bit level (if the expected pattern is known by the receiver). However, an easier approach is to use the CRC functionality that already exists in most devices.

Such a dedicated RX test mode must be simple enough that a majority of the protocol is bypassed. The DUT must be placed into a mode where simple, framed patterns containing valid CRC's can be sent into the receiver, using a non-protocol-aware signal generator. Note that most current lab signal sources contain some degree of sequencing capability that can be used to send startup or configuration information prior to a repeating test sequence. The only limitation to these instruments however is that they cannot be “interactive” in that they cannot detect and react to transmissions coming from the DUT, if timing-sensitive handshaking is required as part of the protocol. In some cases where the timings are known and repeatable, it may be possible to create sequences that can mimic an interactive protocol exchange, however these typically must be created on a per-DUT basis, and require knowledge of the exact timings required.

If a mode exists where a receiver is able to verify CRC-checked frame data, a mechanism must be provided that allows for observation of the results of the checking operation. While this may be achieved through internal vendor-specific registers and counters, it is also possible (and preferable) to allow this to be performed through the Physical Layer interface.

Several options exist to enable this, which are all based on acknowledgement mechanisms, provided the DUT contains a low-speed TX, which may be used to communicate information about the received data.

If sufficient bandwidth exists, the DUT could transmit some form of defined positive acknowledgement for each successfully received frame, and a negative acknowledgement for each frame received in error. If sufficient bandwidth does not exist, the positive acknowledgements can be omitted, and only the negative acknowledgements sent in the error cases (which are assumed to be few). The acknowledgements may be as simple as a single codeword or short pattern, or any other sequence that can be detected and counted using non-protocol-specific laboratory instruments (or possibly a simple FPGA).

In the extreme case, the DUT technically only needs to indicate if any errors were observed over a given period in order for a test to be designed that can verify conformance. If a known amount of data is transmitted to the DUT over a given interval, and the DUT indicates provides a single acknowledgement that no errors were observed, this is a sufficient observable to determine conformance. While knowing an exact error count may certainly be useful for debugging and troubleshooting purposes, such level of detail is not necessary for determining conformance.

Applications that do not or cannot implement LOOPBACK should implement some form of dedicated pattern-checking mode, which is capable of verifying a CRC-checked, framed pattern, and which can provide some form of acknowledgement-based observation mechanism.

#### B.2.4 Receiver Configuration – Termination

2635 Note that for the dedicated RX pattern checking test mode (and also potentially loopback modes as well),  
2636 some level of configuration of the receiver must occur. This includes the MODE and GEAR operation of the  
2637 receiver, as well as the termination mode (terminated or unterminated).

2638 Configuration of the termination mode is another important mechanism. The receiver HS termination is either  
2639 disabled during normal operation, or enabled such that it is only active during the reception of an HS burst.  
2640 However, another mode is needed for test purposes, in which the termination can be manually forced into an  
2641 enabled state.

2642 This mode is necessary in order to perform S-parameter measurements of the receiver termination. Because  
2643 the measurement cannot be made during reception of an HS burst, the receiver must be placed into a mode  
2644 where the termination is permanently enabled for the duration of the measurement.

2645 Applications should provide a mechanism that allows manual enabling and disabling of the receiver HS  
2646 termination.

#### B.3 Interoperability Testing

2647 Note that the mentioned transmitter and receiver test mechanisms all have been discussed in the context of  
2648 conformance testing. However, it is important to note that the same mechanisms, e.g., dedicated pattern  
2649 generation and checking modes, loopback, etc., can also be used to perform physical layer interoperability  
2650 verification as well.

2651 This is performed in the same manner as conformance testing, however instead of using a lab signal generator  
2652 to generate the test signals, another M-PHY device is used, which is placed into pattern generation mode.  
2653 This allows vendor-to-vendor physical layer interoperability testing to be performed using the same  
2654 methodologies that are used for conformance testing. (Note that this only verifies interoperability of the  
2655 physical layer, however isolation and verification of just the physical layer functionality is an important  
2656 component of any interoperability test strategy.)

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## Annex C SI Dithering (informative)

When constructing systems using the high speed interface to connect a baseband IC (BBIC) with a radio frequency IC (RFIC) noise coupling between the high speed interface and sensitive LNA inputs of the RFIC is a concern. Interface bit rates are at frequencies that may cause EMI near some of the air interface frequencies. The least destructive EMI would occur if the interface data appeared as a random UI rate bit stream with no repeating sub-UI rate patterns. However, the encoding of the interface data into 8b10b symbols causes repetitive 10 UI patterns in an HS-BURST. Analysis has shown that these repeating SI rate patterns can cause spectral peaking in the EMI that exacerbates the noise coupling problem.

SI rate symbol timing can not be changed during a BURST. Symbol boundaries are established at the start of each BURST and must remain on the same 10 UI boundary for the remainder of the BURST. However, 10 UI symbol boundaries may be changed from HS-BURST to HS-BURST. Analysis shows that dithering of the SI starting locations, BURST to BURST, by some fraction of an SI, spreads SI rate EMI enough to offer some EMI benefit.

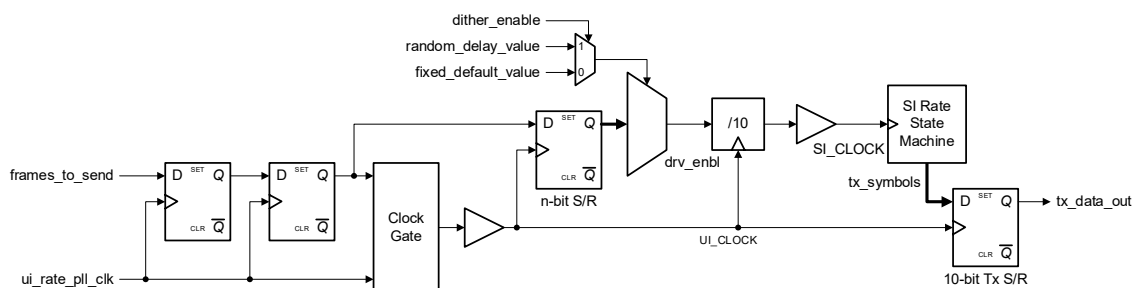
### C.1 Dither Method

Delaying the start of each HS-BURST with reference to the last BURST, some random number of UI, accomplishes the desired dithering. This happens naturally in many implementations, but forced dithering ensures a good distribution of starting locations in any system.

Within the physical interface there is a UI rate divide by ten counter to produce the SI rate symbol boundaries. If this counter is left running during STALL states, then all HS-BURSTs have the same SI boundaries. That is, the SI clock will be coherent from BURST to BURST, producing maximum EMI. In order to accomplish dithering, this counter should be stopped and restarted from BURST to BURST. Stopping the counter during STALL may be a good practice for power efficiency as well. However, even when the counter is restarted for each HS-BURST, it is possible that the “frames to send”, or “start” signal to the physical interface is generated in a way that produces a poor distribution of symbol boundaries from BURST to BURST, the worst case being the same symbol boundary every BURST. To guarantee a good distribution of BURST to BURST SI starting locations, the “start” signal may be delayed a random number of UI intervals before starting the divide by ten counter to establish the new symbol boundary.

In order to adequately randomize the dither delay value, some type of pseudorandom value is needed from BURST to BURST. For example, an 8-bit PRBS might be used to provide the random dither locations. This can be done by ensuring that the PRBS is clocked at least once per HS-BURST. The recommended method is to clock it once at the EOT symbol of each BURST.

**Figure 66** is an example of a circuit that accomplishes this BURST to BURST starting location dither.



**Figure 66 Dithering Circuit Example**

### C.1.1 Dither Magnitude

Since the SI rate patterns repeat every 10 UI, the maximum useful dithering spreads starting locations over a 10 UI range. The minimum dither possible is two locations. Spreading the starting locations over just two locations showed significant benefit in simulations. **Table 55** shows all of the possible useful dithering ranges. Because of the reduced complexity required to produce a flat dithering distribution when using a power-of-2 ( $2^x$ ) number of starting locations, dithering control is limited to four settings; one location (no dithering), two, four and eight locations. In this case, one, two or three bits of the eight bit PRBS generator can be used directly, with no division of the random number by the dither amount necessary.

**Table 55 Dithering Ranges**

| Number of Random Start Positions | Starting UI Delay Range | Range from Default Delay                         | Divide Required? |
|----------------------------------|-------------------------|--------------------------------------------------|------------------|
| 1 (no dither)                    | 4 (default)             | [0]                                              | No               |
| 2                                | 4-5                     | [0] [+1]                                         | No               |
| 3                                | 3-4-5                   | [-1] [0] [+1]                                    | Yes              |
| 4                                | 3-4-5-6                 | [-1] [0] [+1] [+2]                               | No               |
| 5                                | 2-3-4-5-6               | [-2] [-1] [0] [+1] [+2]                          | Yes              |
| 6                                | 2-3-4-5-6-7             | [-2] [-1] [0] [+1] [+2] [+3]                     | Yes              |
| 7                                | 1-2-3-4-5-6-7           | [-3] [-2] [-1] [0] [+1] [+2] [+3]                | Yes              |
| 8                                | 1-2-3-4-5-6-7-8         | [-3] [-2] [-1] [0] [+1] [+2] [+3] [+4]           | No               |
| 9                                | 0-1-2-3-4-5-6-7-8       | [-4] [-3] [-2] [-1] [0] [+1] [+2] [+3] [+4]      | Yes              |
| 10                               | 0-1-2-3-4-5-6-7-8-9     | [-4] [-3] [-2] [-1] [0] [+1] [+2] [+3] [+4] [+5] | Yes              |

In case a HS-BURST is started to issue a real time critical message over the interface, then the random delay inserted between the “start” signal to the physical interface and the actual start of the BURST adds uncertainty to the delivery time of the message. In order to produce the least uncertainty for this message, a default start delay of half of the maximum dither range should be used when dither is disabled. The range of dither delays is then spread equally around this default delay to produce an uncertainty of approximately plus or minus one half of the maximum dither range.

## Annex D Setting of Attributes Values (informative)

2701 The purpose of this informative annex is to provide guidelines for setting attribute values of a MODULE at  
2702 one end of a LANE to their corresponding attribute values of a MODULE at the other end of the LANE in  
2703 M-PHY applications. Value assignment to the attributes need to be performed in a careful manner in order to  
2704 have successful operation of a LANE and thus a LINK. Though every effort has been made to keep attribute  
2705 names identical for M-TX and M-RX thus providing guidance to which attribute pairs need to be matched,  
2706 for some attributes it might be difficult to deduce values to be set or control. To ensure successful operational  
2707 behavior of a LANE after reconfiguration, the protocol above M-PHY needs to analyze the capabilities of  
2708 M-PHY MODULEs (through capability attributes) before setting required values to configurable attributes.

2709 This annex is divided into two main sections, a set of attributes that needs to be matched between M-RXs  
2710 and M-TXs of a LANE, and a set of attributes that should to be changed when M-PHY speed needs to be  
2711 changed with implications for changing certain attributes.

## D.1 Attribute Pair Matching for MODULEs of a LANE

**Table 56** provides a list of attribute pairs that need to be set to the same value for successful LANE operation. If an attribute value of a pair is changed to a new value, then the value of matching attribute in the pair also needs to be changed to the same value. The TX\_HIBERN8\_Control and RX\_Enter\_HIBERN8 attributes need to be matched only when the attribute value is set to one of the corresponding values specified after the “==” sign, otherwise the attribute values do not need to be matched.

**Table 56 Attribute Pairs of a LANE to be Matched**

| M-TX Configuration Attribute         | M-RX Configuration Attribute |
|--------------------------------------|------------------------------|
| TX_MODE                              | RX_MODE                      |
| TX_HSRATE_Series                     | RX_HSRATE_Series             |
| TX_HSGEAR                            | RX_HSGEAR                    |
| TX_PWMGEAR                           | RX_PWMGEAR                   |
| TX_BYPASS_8B10B_Enable               | RX_BYPASS_8B10B_Enable       |
| TX_HS_Unterminated_LINE_Drive_Enable | RX_HS_Unterminated_Enable    |
| TX_LS_Terminated_LINE_Drive_Enable   | RX_LS_Terminated_Enable      |
| TX_HIBERN8_Control == ENTER          | RX_Enter_HIBERN8 == YES      |

**Table 57** lists the M-TX configuration attribute values and their recommended logical relationship with the corresponding M-RX capability attribute values.

**Table 57 Relationship between M-TX Configuration and M-RX Capability Attributes**

| M-TX Configuration Attribute   | Logical Relationship | M-RX Capability Attribute                                                       |
|--------------------------------|----------------------|---------------------------------------------------------------------------------|
| TX_HS_SYNC_LENGTH              | ≥                    | RX_HS_Gn_SYNC_LENGTH_Capability <sup>1</sup>                                    |
| TX_HS_PREPARE_LENGTH           | ≥                    | RX_HS_Gn_PREPARE_LENGTH_Capability <sup>1</sup>                                 |
| TX_LS_PREPARE_LENGTH           | ≥                    | RX_LS_PREPARE_LENGTH_Capability                                                 |
| TX_PWM_BURST_Closure_Extension | ≥                    | RX_PWM_BURST_Closure_Length_Capability                                          |
| TX_Min_ActivateTime            | ≥                    | Min[RX_Min_ActivateTime_Capability,<br>RX_Advanced_Min_ActivateTime_Capability] |
| TX_HS_ADAPT_Length             | ≥                    | RX_HS_Gx_ADAPT_INITIAL_Capability <sup>2</sup>                                  |
| TX_HS_ADAPT_Length             | ≥                    | RX_HS_Gx_ADAPT_REFRESH_Capability <sup>2</sup>                                  |

**Note:**

1.  $n = 1, 2, \text{ or } 3$ , and depends on the value of RX\_HSGEAR.
2.  $x=4 \text{ or } 5$ ; Applicable only to G4 and G5

## D.2 Attribute Values Changed with LANE Speed Setting

### D.2.1 Intra-MODE GEAR Change

- 2719 The following attribute pairs should be changed when a LANE needs to be switched from one HS\_GEAR to  
2720 another HS\_GEAR (Intra-MODE change), while accommodating the restrictions in **Table 56**:
- 2721 • TX\_HSGEAR and RX\_HSGEAR – for a different HS-GEAR in the same RATE series
  - 2722 • TX\_HSRATE\_Series and RX\_HSRATE\_Series – for the same HS-GEAR in a different RATE series
  - 2723 • TX\_HSGEAR and RX\_HSGEAR, TX\_HSRATE\_Series and RX\_HSRATE\_Series – for a different  
2724 HS-GEAR in a different RATE series

### D.2.2 Inter-MODE Gear Change

- 2725 When LANE settings need to be changed from one MODE to another, the TX\_MODE and RX\_MODE  
2726 attribute values should be changed, while accommodating the restrictions in **Table 56**, along with the attribute  
2727 pairs listed in **Section D.2.1** based on the MODE and GEAR being requested.

## D.3 Interpretation of Certain Attributes

### D.3.1 TX\_PWM\_BURST\_Closure\_Extension

2728 The protocol above the M-TX determines the value of the PWM BURST Closure length for the M-TX from  
 2729 the RX\_PWM\_BURST\_Closure\_Length\_Capability value of the M-RX and requirements of the protocols  
 2730 above the M-RX and M-TX. The RX\_PWM\_BURST\_Closure\_Length\_Capability value of the M-RX  
 2731 communicates to the local protocol any extra cycles needed by the PHY to flush the pipeline. In case the  
 2732 remote protocol requires additional clock cycles for symbol or PAYLOAD processing, the protocol may adopt  
 2733 any of the following methods.

2734 In the first method, the local protocol at M-TX lengthens PWM-BURST by setting  
 2735 TX\_PWM\_BURST\_Closure\_Extension. The value of TX\_PWM\_BURST\_Closure\_Extension should be set  
 2736 to greater than, or equal to, the sum of the value of RX\_PWM\_BURST\_Closure\_Length\_Capability and the  
 2737 number of additional clock cycles required by the remote protocol. The sum of the additional clock cycles  
 2738 needed by the remote protocol at M-RX and the maximum limit of  
 2739 RX\_PWM\_BURST\_Closure\_Length\_Capability set by the protocol cannot exceed 255 SI.

2740 In addition, the local protocol at M-TX should get the value of  
 2741 RX\_PWM\_BURST\_Closure\_Length\_Capability of the remote M-RX from the remote protocol. Also, the  
 2742 local protocol at M-TX should get any additional clock cycles required by the remote protocol through  
 2743 protocol level communication.

2744 In the second method, the local protocol at M-TX inserts the needed number of FILLERS before requesting  
 2745 a TAIL-OF-BURST sequence, i.e. before issuing M-LANE-BurstEnd.request, at the end of the last  
 2746 PAYLOAD of a PWM-BURST. The local protocol at M-TX should set the value of  
 2747 TX\_PWM\_BURST\_Closure\_Extension to greater than, or equal to, the value of  
 2748 RX\_PWM\_BURST\_Closure\_Length\_Capability.

### D.3.2 M-TX and M-RX Polarity Control

2749 With the TX\_DRIVER\_POLARITY and RX\_RECEIVER\_POLARITY attributes, MPHY provides the  
 2750 ability to adapt to any potential MPHY LINE-interface pin-swap at the device level and reduces constraints  
 2751 for routing the high-speed LINE traces on a PCB. Pin swaps and LINE crossing on PCB traces can be  
 2752 compensated by programming TX\_DRIVER\_POLARITY and/or RX\_RECEIVER\_POLARITY according  
 2753 to the PCB trace polarity.

2754 TX\_DRIVER\_POLARITY and RX\_RECEIVER\_POLARITY are system-dependent attributes that are  
 2755 independent of LINE-RESET. If TX\_DRIVER\_POLARITY or RX\_RECEIVER\_POLARITY are changed  
 2756 to new values by the Protocol after the initial local RESET, the Protocol should change these attribute values  
 2757 to the new values after a subsequent local RESET is applied and de-asserted, and before M-TX or M-RX is  
 2758 requested to exit HIBERN8 state.

## Annex E Guidance for Protocols on Managing LANE-to-LANE Skew (informative)

This annex provides recommendations for skew parameters in a SUB-LINK with multiple LANEs. As shown in **Figure 67**, a SUB-LINK subsystem consists of a LANE management controller and M-TX on the transmitting side of the SUB-LINK, interconnect, and LANE management controller and M-RX on the receiving side of the SUB-LINK. The interconnect can be only completely galvanic.

LANE-to-LANE skew (L2L skew) must be considered in the case where there are multiple LANEs transmitting data, and where it is desired to optimally reuse hardware in the LANE management controller. Architecture decisions for the LANE management controller are based on the use of an optimized clocking mechanism, maximum skew possible between Symbol clocks of data LANEs of multiple RMMIs, shift register depth requirements, and possibly a de-skewing mechanism to be adopted along with throughput desired.

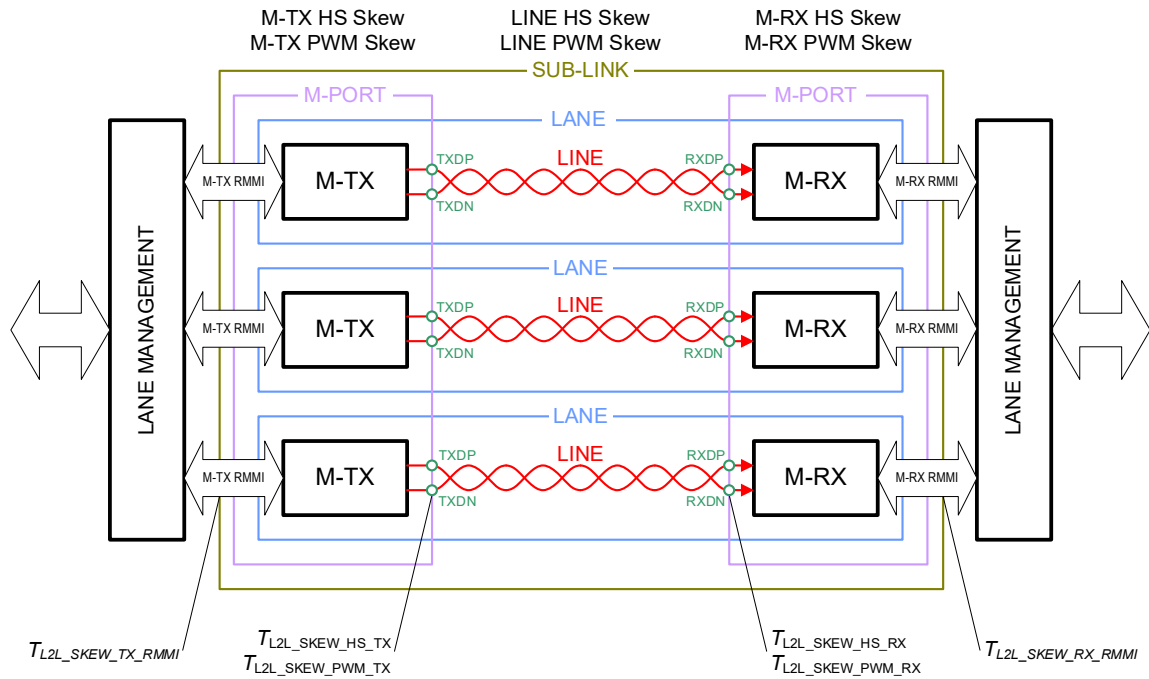
M-TX and M-RX skew parameters originate due to clock generation, clock routing skews and analog block skews due to device mismatches. The total skew can be a combination of  $UI_{HS}$  lengths and fixed delays in terms of propagation time. Interconnect induced skew parameters are generally independent of GEAR, and are in terms of propagation time. Although this annex does not discuss use-cases involving implementations with a large number of LANEs within a SUB-LINK, a higher number of LANEs tends to correlate with increased skew.

Parameters defined in this document pertaining to skew parameters, and the sections where the definitions can be found, are shown in **Table 58**.

**Table 58 LANE to LANE Skew Parameters**

| Parameter | Section        | Description      |
|-----------|----------------|------------------|
| HS-TX     | <b>5.1.2.4</b> | M-TX in HS-MODE  |
| PWM-TX    | <b>5.1.3.3</b> | M-TX in PWM-MODE |
| SYS-TX    | <b>5.1.4.2</b> | M-TX in SYS-MODE |
| HS-RX     | <b>5.2.2.1</b> | M-RX in HS-MODE  |
| PWM-RX    | <b>5.2.3.4</b> | M-RX in PWM-MODE |
| SYS-RX    | <b>5.2.4.1</b> | M-RX in SYS-MODE |

A graphical representation of the point of measurement for each parameter is shown in **Figure 67**. A skew parameter is the aggregate skew possible at the indicated interface in the figure.



**Figure 67 Measurement Points**

### HS-MODE Skew Equations

$$\text{M-TX HSSkew} = T_{L2L\_SKEW\_HS\_TX} - T_{L2L\_SKEW\_TX\_RMMI} \quad (\text{Equation 45})$$

$$\text{Interconnect HS Skew} = T_{L2L\_SKEW\_HS\_RX} - T_{L2L\_SKEW\_HS\_TX} \quad (\text{Equation 46})$$

$$\text{M-RX HSSkew} = T_{L2L\_SKEW\_RX\_RMMI} - T_{L2L\_SKEW\_HS\_RX} \quad (\text{Equation 47})$$

### PWM-MODE Skew Equations

$$\text{M-TX PWM Skew} = T_{L2L\_SKEW\_PWM\_TX} - T_{L2L\_SKEW\_TX\_RMMI} \quad (\text{Equation 48})$$

$$\text{Interconnect PWM Skew} = T_{L2L\_SKEW\_PWM\_RX} - T_{L2L\_SKEW\_PWM\_TX} \quad (\text{Equation 49})$$

$$\text{M-RX PWM Skew} = T_{L2L\_SKEW\_RX\_RMMI} - T_{L2L\_SKEW\_PWM\_RX} \quad (\text{Equation 50})$$



23-Jun-2021

Two new parameters are defined in this annex for discussing the LANE-to-LANE skew. The first parameter,  $T_{L2L\_SKEW\_TX\_RMMI}$ , is the LANE-to-LANE skew on the transmitting side of the SUB-LINK, and is defined as the time difference between the TX\_SymbolClk at the M-TX RMMIs in a SUB-LINK. For the purposes of this annex, an M-TX RMMI is assumed to be a synchronous interface to the M-TX, and as such is considered the T0 reference plane. Any symbol skew accumulated before  $T_{L2L\_SKEW\_TX\_RMMI}$  is not considered.

$T_{L2L\_SKEW\_RX\_RMMI}$  is defined as the time difference between two RX\_SymbolClk that qualify reference data points, e.g., MARKER0, on M-RX-DATA SAP of the M-RX RMMIs in a SUB-LINK.  $T_{L2L\_SKEW\_RX\_RMMI}$  is a design parameter. The LANE management controller needs to respect  $T_{L2L\_SKEW\_RX\_RMMI}$  for implementation of adequate LANE recomposition.

LANE management controllers at M-RX RMMI and M-TX RMMI have the option to exercise the de-skew mechanism of multiple LANEs in a given SUB-LINK by using training sequences and offsetting the phase of the reference clock either at M-RX or at M-TX.

**Table 59** shows the L2L skew parameter values for a galvanic-only interconnect, in a tightly coupled use-case where latency requirements are stringent.

**Table 59 L2L Skew Parameters for Tightly Coupled Use-case**

| Parameter                 | Value                             |                   |
|---------------------------|-----------------------------------|-------------------|
| $T_{L2L\_SKEW\_HS\_TX}$   | $2 U_{I_{HS}}$                    |                   |
| $T_{L2L\_SKEW\_PWM\_TX}$  | $2 * T_{PWM\_TX}$                 |                   |
| $T_{L2L\_SKEW\_HS\_RX}$   | $2 U_{I_{HS}} + 33 \text{ ps}$    |                   |
| $T_{L2L\_SKEW\_PWM\_RX}$  | $2 * T_{PWM\_TX} + 33 \text{ ps}$ |                   |
| $T_{L2L\_SKEW\_RX\_RMMI}$ | HS-MODE                           | $5 U_{I_{HS}}$    |
|                           | PWM-MODE                          | $5 * T_{PWM\_RX}$ |

In order to ensure interoperability between an M-PORT and a LANE management controller, the LANE management controller should be able to de-skew by at least  $\pm 1$  SI at the M-RX RMMI.

In this first example, a short interconnect ( $\leq 10$  cm) using galvanic LINES on a FR4-class PCB, travel time is about 6 ps/mm. From **Table 59**, the skew interconnect margin is 33 ps, which provides about 5 mm of physical length mismatch ( $33 \text{ ps} \div 6 \text{ ps/mm} \approx 5 \text{ mm}$ ) over 10 cm, or about 5%, for a worst-case interconnect. Interconnect skew parameters are GEAR independent. Note, the  $U_{I_{HS}}$  values shown in **Table 59** apply for all modes of communication.

2809 **Table 60** shows the L2L skew parameter values for another galvanic-only interconnect in a nominally coupled  
 2810 use-case where latency requirements are less stringent than in the previous example.

**Table 60 L2L Skew Parameters for Nominally Coupled Use-case**

| Parameter                   | Value            |                                    |
|-----------------------------|------------------|------------------------------------|
| $T_{L2L\_SKEW\_HS\_TX}$     | 10 $U_{IHS}$     |                                    |
| $T_{L2L\_SKEW\_HS\_G4\_TX}$ | 20 $U_{IHS}$     |                                    |
| $T_{L2L\_SKEW\_PWM\_TX}$    | 10 $T_{PWM\_TX}$ |                                    |
| $T_{L2L\_SKEW\_HS\_RX}$     | 30 $U_{IHS}$     |                                    |
| $T_{L2L\_SKEW\_HS\_G4\_RX}$ | 60 $U_{IHS}$     |                                    |
| $T_{L2L\_SKEW\_PWM\_RX}$    | 30 $T_{PWM\_TX}$ |                                    |
| $T_{L2L\_SKEW\_HS\_G5\_TX}$ | 40 $U_{IHS}$     |                                    |
| $T_{L2L\_SKEW\_HS\_G5\_RX}$ | 120 $U_{IHS}$    |                                    |
| $T_{L2L\_SKEW\_RX\_RMMI}$   | HS-MODE          | 40 $U_{IHS}$ @ HS-G1, HS-G2, HS-G3 |
|                             |                  | 80 $U_{IHS}$ @ HS-G4               |
|                             |                  | 160 $U_{IHS}$ @ HS-G5              |
|                             | PWM-MODE         | 40 $T_{PWM\_RX}$                   |

## Annex F Guidance for Protocols on Managing ADAPT Sub-State and RX Equalization (informative)

2811 The purpose of the ADAPT sequence is to train an M-RX equalizer to adapt its filter characteristics to the  
2812 channel. The properties of a channel depend predominantly on the material characteristics of the printed  
2813 circuit board, the package and bonding material, the line length, and potential connectors involved. In  
2814 addition to these mostly time-invariant properties, the channel characteristics may be influenced by  
2815 temperature and voltage variations.

### F.1 When to Use ADAPT

2816 The specification does not mandate the use of ADAPT sequences. A protocol running on top of M-PHY  
2817 should implement provisions to initiate the ADAPT sub-state, but should not enforce the use of ADAPT.

2818 In general, a long length Initial ADAPT sequence is intended only for the first M-RX calibration after power  
2819 up, to train the M-RX equalizer onto the channel and M-TX characteristics. The Initial ADAPT sequence can  
2820 also be used by the protocol when handling fatal LINK error recovery.

2821 A Refresh ADAPT sequence is much shorter than an Initial ADAPT sequence. Refresh ADAPT is intended  
2822 to compensate for temperature and voltage drifts in the channel by incrementally adjusting the equalizer  
2823 settings. However, a protocol may not always need to do this, particularly under the following conditions:

- 2824 • The M-RX implementation can guarantee signal reception under worst-case channel conditions,  
2825 specifically when an application uses a low-loss channel.
- 2826 • Temperature and voltage drifts are compensated sufficiently by the implementation.

2827 An embedded application may not require an Initial ADAPT training sequence. For example, if the optimal  
2828 equalizer settings are determined during laboratory bring-up and restored using vendor-specific sideband  
2829 programming any time the LINK enters operation in HS-G4 and HS-G5.

2830 At the expense of additional power consumption, an M-RX may not need Refresh ADAPT training. It may  
2831 continuously or periodically re-adapt its equalizer using information from toggling activities during an HS-  
2832 BURST.

2833 There is no requirement to initiate ADAPT after HIBERN8 exit, as the equalizer settings are retained during  
2834 HIBERN8. However, a Refresh ADAPT can be required after a lengthy HIBERN8 due to temperature and  
2835 voltage drift.

2836 A protocol should initiate ADAPT after a change, which affects the signal waveform, e.g., the M-TX  
2837 equalizer settings, the M-TX amplitude setting, or a HS RATE series change.

### F.2 Detecting the Need for ADAPT

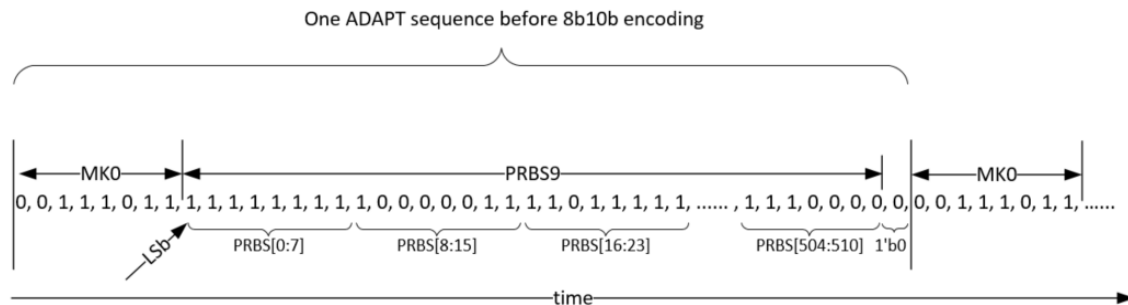
2838 The M-PHY ADAPT sequence is an open-loop training. This specification does not define a means for an  
2839 M-TX to detect if the M-RX equalizer training is successful. It is assumed that an M-RX receives data with  
2840 the defined *BER* at the end of an ADAPT sequence. However, the protocol needs to ensure that channel  
2841 degradations are monitored at the M-RX and that this information may be used by the peer M-TX to trigger  
2842 an ADAPT sequence.

### F.3 ADAPT Sequences in Lower HS-GEARS

2843 This specification defines ADAPT for use in the HS-G4 and HS-G5 RATE series A and B only.

## F.4 ADAPT Sequence

This section provides additional guidance for the ADAPT sequence specified in **Section 4.7.2.3**. The ADAPT sequence is defined as an 8b10b encoded pattern of MK0, followed by a PRBS9 sequence 511 bits in length, beginning with 9 b1 bits and ending with a single b0 bit. This is shown in **Figure 68**.



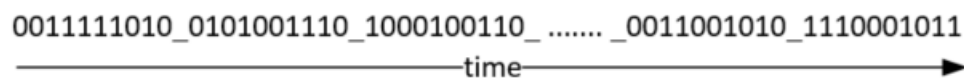
### Figure 68 One ADAPT Sequence Before 8b10b Encoding

The line coding is defined in **Section 4.5**, which also specifies that the LSB of an 8b10b encoded symbol is transmitted first. The ADAPT sequence before and after 8b10b encoding is shown in **Figure 69**, based on the definitions in **Section 4.5**.

|                       | Data Byte | Encoded symbol   | [a,b,c,d,e, i, f,g, h, j], note „a“ is transmitted first |
|-----------------------|-----------|------------------|----------------------------------------------------------|
| time<br>↓<br>MK0[7:0] | = 8'hBC   | MK0' RD- [9:0]   | = [0,0,1,1,1,1,0,1,0]                                    |
| PRBS9[7:0]            | = 8'hFF   | D31.7' RD+ [9:0] | = [0,1,0,1,0,0,1,1,1,0]                                  |
| PRBS9[15:8]           | = 8'hC1   | D1.6' RD- [9:0]  | = [1,0,0,0,1,0,0,1,1,0]                                  |
| ....                  |           |                  |                                                          |
| PRBS9[503:496]        | = 8'hB8   | D24.5' RD+ [9:0] | = [0,0,1,1,0,0,1,0,1,0]                                  |
| {1'b0,PRBS9[510:504]} | = 8'h07   | D7.0' RD- [9:0]  | = [1,1,1,0,0,0,1,0,1,1]                                  |

**Figure 69 Symbol Encoding Within ADAPT Sequence (Before Enc. -> Data Byte)**

The start and end fragments of the serialized 8b10b encoded ADAPT sequence are shown in **Figure 70**.



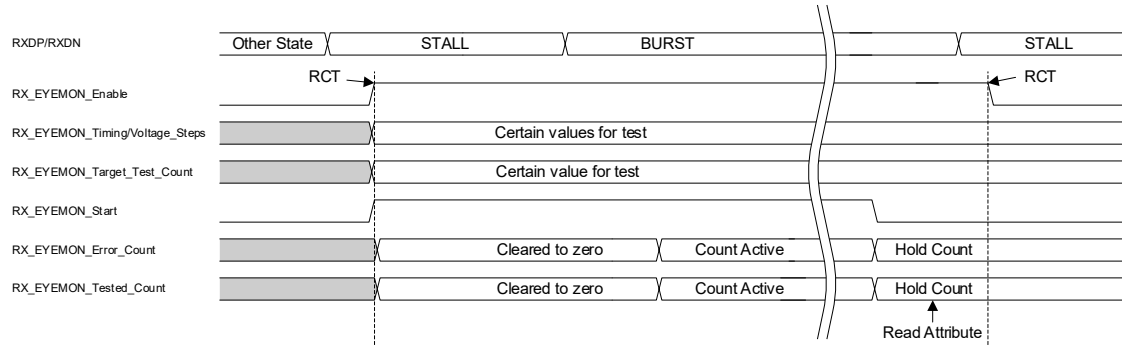
### Figure 70 Start and End of Serialized 8b10b Encoded ADAPT Sequence

## Annex G Eye Monitor Sequence Examples

**Figure 71** illustrates the expected operating sequence for the Eye Monitor function.

**Note:**

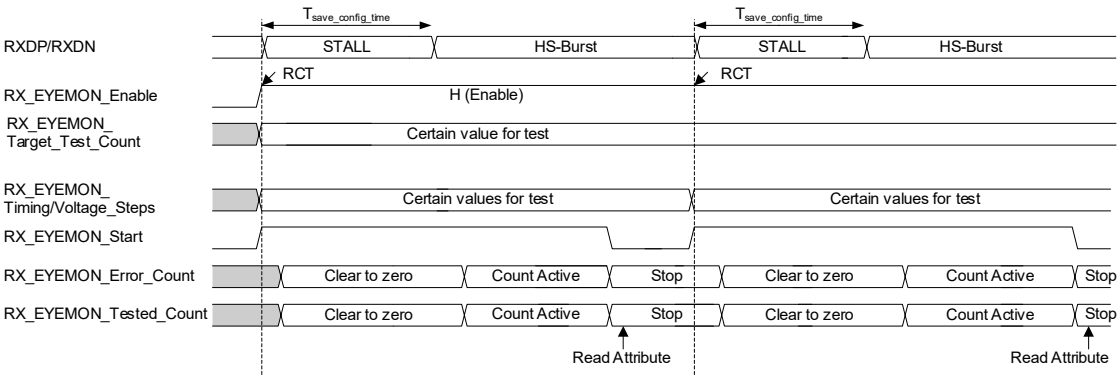
The Eye Monitor function is specified in **Section 5.2.2.5**, and the EYEMON Attributes referenced in this section are defined in **Table 45**, **Table 46**, and **Table 47**.



**Figure 71 Eye Monitor Operation**

- First, attribute **RX\_EYEMON\_Enable** is set by RCT, and the Eye Monitor data path is activated. The M-RX shall not use any EYEMON configuration attribute while **RX\_EYEMON\_Enable** is disabled.
  - RX\_EYEMON\_Start** is activated at the same time as RCT, i.e., when **RX\_EYEMON\_Enable** is activated.
  - Before **RX\_EYEMON\_Enable** is activated, the attributes **RX\_EYEMON\_Timing\_Steps**, **RX\_EYEMON\_Voltage\_Steps**, and **RX\_EYEMON\_Target\_Test\_Counts** shall be sent. They will be updated at the RCT.
  - The Eye Monitor measurement is executed in the HS-Burst after evaluation is ready during **RX\_EYEMON\_Start=1**.
  - When the Eye Monitor test is finished, **RX\_EYEMON\_Start** is cleared. The system should read the results when **RX\_EYEMON\_Start=0**.
  - Lastly, **RX\_EYEMON\_Enable** is disabled by the RCT after finishing all operations.
- Note that HS-GEAR and RATE shall not be changed while **RX\_EYEMON\_Enable=1**. **RX\_EYEMON\_Enable** and other configurations are defined as INLINE-CR, which is asserted and de-asserted by RCT.
- While the Eye Monitor function is activated, the BURST state should be continued until the end of a single Eye Monitor measurement.
- Note that Hibern8 or STALL entry might interrupt the Eye Monitor measurement. The behavior in such a case is implementation-specific, but the measurement result data should be held until the next new measurement starts.

2880 **Figure 72** illustrates continuous Eye Monitor operations, i.e., repeatedly issuing RCT while dynamically  
2881 changing the measurement offset settings. In this case, before each RCT is issued, the configurations for the  
2882 next measurement are set; then, the configurations are updated at the RCT.



**Figure 72 Eye Monitor Continuous Operation**

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